

Consistent forward-curve families with scheduled jumps

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Abstract

We study which finite-dimensional forward-curve families remain consistent with arbitrage-free dynamics when the curve jumps at scheduled policy dates, as the SOFR curve does at central-bank meetings. Under the drift and jump martingale conditions for term structures with predetermined jump dates, the meeting-date step family admits only the trivial dynamics, and its extension by ramps is consistent under deterministic step volatility exactly when each ramp slope is the squared step volatility and each level jump is centered Gaussian under the exponential tilt accumulated over earlier intervals; joint Gaussianity is not forced. Consistency alone imposes no restriction on meeting variances across meetings. In a Gaussian model with ideal payoffs, contracts settled after a date see the earlier variances only through two aggregates, and on the listed calendar of three-month SOFR futures options every meeting variance of 2026 and 2027 is identified in principle, though not to tick precision for distant meetings. For realizations in time to maturity that serve every schedule, one factor with a quasi-exponential shape and linearly recurrent meeting loadings is realized for every bounded Lipschitz scale, and with a constant scale the recurrence is also necessary. With deterministic or bounded predictable volatilities, a smooth exponential-polynomial block can be attached to a scheduled-step front end only by an independence splice or by indicator-times-exponential terms, up to a stated counting convention, and in the uncorrelated splice the front end frees only the level direction the two families share. We also correct a published sufficient condition for the number of correlated factors of a biexponential-polynomial family. The results are machine-checked in Lean, with the coverage in Appendix A.

1. Introduction

This paper proves four things about forward curves that jump at scheduled policy dates. The meeting-date step family admits only the trivial dynamics, and its extension by ramps is consistent exactly under the differentiated drift and tilted centered Gaussian level jumps (Theorems 1 and 11). Consistency alone gives no cross-meeting restriction on event variances, and in a scalar Gaussian model ideal option panels identify them exactly when a rank condition holds, which the listed SOFR option calendar satisfies in principle (Remark 12, Propositions 19, 20 and 24). Realizations in time to maturity that serve every schedule exist for one factor when the shape is quasi-exponential and the meeting loadings are linearly recurrent, and with a constant scale the recurrence is also necessary (Propositions 30, 31 and 33). And attaching a smooth block to a scheduled-step front end is possible only through the two known constructions, up to a stated counting convention (Propositions 34, 35, 41, 43, 44 and 47).

Which finite-dimensional forward-curve families are compatible with arbitrage-free dynamics when the curve jumps at scheduled policy dates, and what does compatibility force about the sizes of the jumps across meetings? The classical program ([9], [24], [25], [26]) asks the first half of this question for parametric families such as Nelson–Siegel and the exponential-polynomial families, under Wiener-driven or stochastically continuous jump-driven Heath–Jarrow–Morton dynamics. Every jump in that literature is stochastically continuous. We found no paper that poses or solves the consistency or finite-dimensional-realization question for curve families with breakpoints at scheduled dates, or for term-structure models with predictable jump times; Section 4 records what the literature does contain.

The motivation is the front end of the SOFR curve. The policy rate moves at the scheduled meetings of the central bank, so the forward curve has steps at known calendar dates; desks build it as a meeting-date step function spliced onto a smooth long-end family, and [37] is the static version of that construction fitted to futures. The constructive precedents are the SOFR models of [34] and [12]. The results are meant to be usable in three ways: as a consistent front-end curve for carry and roll-down (Sections 3.1 to 3.9), as a test of whether consistency restricts option-implied event variances across meetings (Sections 3.10 to 3.23), and as a classification of how the scheduled front end can carry stochastic volatility and be attached to a smooth long-end block (Sections 3.24 to 3.41).

The front end. Theorem 1 says that the pure maturity-step family supports only the trivial dynamics: under the two no-arbitrage conditions of Section 2 its drift, volatility and curve jumps vanish, so $f(t, T) = f(0, T)$ (Corollary 2), the scheduled-jump analogue of the negative result for Nelson–Siegel ([24]). Its diffusive leg is not new in mechanism: [34] §3.1.5 and §5 state informally that piecewise-constant short-rate paths constrain the initial curve, and the ramp drift behind the argument is their eq. 28 and [12] eq. 18; Theorem 1 gives the statement for the curve family with a proof and adds the jump leg, which neither paper treats. Theorem 11 classifies the extension by ramps, the step-plus-ramp family (2.5), under deterministic step volatility: consistency is equivalent to the differentiated HJM drift (Proposition 8) and to tilted centered Gaussian level jumps (Proposition 9), and Proposition 10 is an explicit multi-date instance. Sufficiency is the content of the models of [34], [12], [30], [40] and [41]; the necessity directions, including the equivalence of the integrated and differentiated drift conditions without a differentiability hypothesis, and the non-Gaussian example of Proposition 9(d), are new. Proposition 48 shows that on the interval after a scheduled date the compensating shape of the curve jump is the derivative of the conditional cumulant generating function of the level jump, so that a straight ramp forces a Gaussian level jump, and proves the pre-meeting forward-rate formula (5.1). Section 3.9 shows that the forced ramps are two orders of magnitude below the diffusive convexity and below quote precision, by a factor of about fifteen to thirty per meeting and less in aggregate over a year of meetings.

Event variances. Remark 12 separates state dimension from parameter dimension: scalar-state models fill the nonnegative orthant of meeting variances as their parameters vary, so consistency alone gives no cross-meeting restriction. Propositions 13, 14, 16 and 26 give the restrictions of the fixed Gaussian and stochastic-volatility specifications of [12], and Proposition 25 those that survive every recalibration of the scales. In the scalar Gaussian model, Propositions 15, 17, 19, 20 and 23, Corollaries 21 and 22 and Lemma 18 determine which meeting variances ideal bond calls, compounded-rate futures and European, American and futures-style calls identify; Proposition 20 isolates the mechanism, two aggregates of the earlier variances, and Proposition 24 evaluates the rank criterion on the listed SOFR option calendar. Proposition 49 adds a diffusion between meetings and an arbitrary initial curve, which the surfaces do not see: the diffusion variance is read off the expiry gaps without a meeting, so the listed calendar identifies a constant diffusion together with every meeting variance and at most eight diffusion cells, and the diffusion’s error, entering each meeting in proportion to its gap, degrades practical identification. Proposition 50 lets the diffusion depend on maturity through a known shape: the rank criterion keeps its form with window weights, and on the listed calendar the answers survive for a loading that starts at the next meeting and, for exponential decay, at every rate except one critical value.

Stochastic volatility and realizations. Proposition 27 gives a bounded three-exponent family with one scheduled breakpoint whose variance state enters the curve. Proposition 28 gives the finite representation of the curve under separable volatilities with meeting-dependent loadings, and Proposition 29 a time-inhomogeneous Markov realization with a Lipschitz scale state; both extend known calendar-maturity results ([10] §3.3 and Proposition 6.2, [30] Example 3.10) to meeting restarts. For realizations in time to

maturity that serve every schedule, Propositions 30 and 33 show that the ordinal meeting loadings must satisfy a linear recurrence, and Proposition 31 gives the converse for one factor and every bounded Lipschitz scale, adding meeting restarts to the quasi-exponential answer of [10] and [11]. On a finite horizon, Proposition 51 bounds the forward-curve and bond-price errors of a recurrent approximation, each model with its own drift, by the loading error, with constants free of the number and spacing of meetings, and Proposition 31 realizes the approximating model.

The splice. Propositions 34, 35, 41, 43 and 44 show that a step factor with maturity steps can be attached to a fixed exponential-polynomial block only through the independence splice of [34] §3 or the indicator-times-exponential terms of [12] eq. 36: with deterministic volatilities, with bounded predictable scales, with a block volatility driven by the block's own state, and with several factors, where the condition is orthogonality of the block's noise to the aggregate maturity-step noise. Propositions 37, 38, 40, 45, 46 and 47 treat the block's own restrictions: in the uncorrelated splice the front end frees only the level direction that the two families share, and in the correlated splice with one step factor only that level may correlate with the step noise. Proposition 52 combines several step factors with a full polynomial part and states the answer without a counting convention: only the level may covary with the aggregate maturity-step noise, the current front-end noise acts as extra level noise, and consistency is the block-alone condition for that effective level. Proposition 53 extends it, without a stopping argument, to locally square-integrable noises, which admit Heston-type scales. Two of these statements rest on counting conventions, stated where they are used: a block uncorrelated with the aggregate step noise counts as the independence splice, and a correlated level counts with the front end. Lemma 36 and Proposition 39 settle, at a parameter point, the number of correlated factors of a biexponential-polynomial block and show that a published sufficient condition for it ([53] §4.2) is false as stated.

What is conditional. Assumptions 2.1 and 2.2 are hypotheses throughout. Every result that uses stochastic integration, Itô's formula, solutions of stochastic differential equations or their Markov property takes these from cited sources rather than proving them, and several take solutions of stated equations as given; each such result is marked conditional, and its hypotheses are stated with it. The corresponding hypothesis structures of the formalization have only degenerate or vacuous instances, which show that their fields are consistent but give no nontrivial model of them. Section 5.1 lists what the results rest on, and Section 5 as a whole what is not done.

Every result in Section 3 is proved in the text, conditional on the cited inputs stated with it. Appendix A lists the exact Lean coverage: Theorem 1 to Remark 12 are formalized, except Corollary 2; Propositions 13 and 14 have conditional formalizations, Proposition 15 is fully formalized in its stated ideal model, Proposition 16 is formalized using the cited stochastic-calculus fields, and Proposition 26 also uses the cited predictability field; both assume solutions of their variance equations. Proposition 27 is formalized conditional on the two supplied calculus structures, Proposition 28 conditional on the supplied calculus, except the word "semimartingale", Proposition 29 conditional on the supplied calculus, strong-existence, predictability and Markov inputs, except its descriptive paragraph, Proposition 30 conditional on the supplied calculus, Proposition 31 conditional on the supplied calculus and predictability input, Lemma 32 and Proposition 33 conditional on the supplied calculus, Propositions 34 and 35 conditional on the supplied calculus, Lemma 36 without conditions, Propositions 37 and 38 conditional on the supplied calculus, Propositions 39 and 40 without conditions, Proposition 41, Lemma 42 and Propositions 43 and 44 conditional on the supplied calculus, Proposition 45 as a statement at a parameter point, and Proposition 46 conditional on the supplied calculus and, for (b), on the cited results of [25], Proposition 47 conditional on the supplied calculus, Proposition 48 without conditions, Proposition 49 with (a) and (b) conditional on a stated Gaussian law of the Wiener integrals, satisfied in the formalization only without diffusion, and

(c) to (e) without conditions, Proposition 50 likewise, with (a) and (b) conditional and (c) and (d) without conditions, Proposition 51 with its deterministic estimates without conditions, (a) under the supplied calculus, and (b) and (c) also under a Brownian driver, Proposition 52 conditional on Assumption 2.1 for the representation, stated as an explicit hypothesis, and Proposition 53 likewise, with its integrability class entering only the converse. Lemma 18, Propositions 17, 19, 20 and 23 to 25 and Corollaries 21 and 22 are fully formalized for their stated ideal prices, exercise problems, calendar and cones. Propositions 27, 29 and 46(b) rest on hypothesis structures whose instances are vacuous, with separate checks that their premises and conclusions are inhabited; the other calculus structures have degenerate instances, and the structure for the standing assumptions a nonzero one; no calculus structure is given a nontrivial model of its fields.

Section 2 fixes the setting and the standing assumptions, Section 3 states and proves the results, Section 4 surveys related work, Section 5 states what is not done, and Appendix A records the machine verification.

2. Setting and standing assumptions

Throughout, $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, Q)$ is a filtered probability space, Q the pricing measure, and the scheduled dates are

$$0 = T_0 < T_1 < \dots < T_N, \quad I_k := [T_k, T_{k+1}) \ (k < N), \quad I_N := [T_N, \infty). \quad (2.1)$$

The intervals $I_0, \dots, I_N$ partition $[0, \infty)$; for $t \geq 0$, $j(t)$ denotes the index with $t \in I_{j(t)}$. The forward rate $f(t, T)$, $T \geq t$, has the short rate $r_t = f(t, t)$ and the bank account $B_t = \exp(\int_0^t r_s ds)$, and is assumed to satisfy, for $T \geq t$,

$$f(t, T) = f(0, T) + \int_0^t \alpha(s, T) ds + \int_0^t \langle \sigma(s, T), dW_s \rangle + \sum_{n: T_n \leq t} \xi_n(T), \quad (2.2)$$

with W a d -dimensional Brownian motion, α real, σ $\mathbb{R}^d$ -valued, and $\xi_n(\cdot)$ an $\mathcal{F}_{T_n}$ -measurable random function, the jump of the whole curve at the scheduled date T_n . Between scheduled dates the curve is continuous in t ; at T_n it jumps by ξ_n . Because the discontinuities sit at fixed calendar dates, families are parametrized in (t, T) rather than in time to maturity alone, and are time-inhomogeneous; the same parametrization appears in [41] Proposition 1 and [12] eqs. 2 to 5.

Regularity. The coefficients vanish before the current date, $\alpha(t, T) = 0$ and $\sigma(t, T) = 0$ for $T < t$ and $\xi_n(u) = 0$ for $u < T_n$; α and σ are progressively measurable in (t, T) , ξ_n is $\mathcal{F}_{T_n} \otimes \mathcal{B}(\mathbb{R}_+)$ -measurable, and all three are locally integrable in the maturity variable almost surely. These are the standing assumptions of [30] (their Assumption 3.3) in the special case used here.

The two no-arbitrage conditions are taken from [30] Theorem 3.5, which characterizes the risk-neutral measures of a term structure with an overnight numéraire, roll-over dates at which the numéraire may jump, and expected jump dates at which the curve may jump. We use the case with no roll-over dates, so the term-structure measure has no atoms, the bank account is absolutely continuous, the policy rate jumps but nothing pays a lump sum at T_n ; with no random measure, so no totally inaccessible jumps; with finitely many fixed jump dates; and with the overnight rate equal to the short end, $r_t = f(t, t)$. This is the case the source itself exercises in its Example 3.10. The characterization is an equivalence; only its two conditions are used, as hypotheses, and bond prices, the numéraire and local martingales do not appear below.

Assumption 2.1 (drift condition between scheduled dates, integrated form). For every $T > 0$, for $Q \otimes dt$ -almost every $(\omega, t) \in \Omega \times [0, T]$,

$$\int_t^T \alpha(t, u) du = \frac{1}{2} \left| \int_t^T \sigma(t, u) du \right|^2, \quad (2.3)$$

with $|\cdot|$ the Euclidean norm on $\mathbb{R}^d$ and both integrals Lebesgue integrals in u . Source: [30] Theorem 3.5(ii), whose integrated coefficients reduce to the two integrals in (2.3) once the roll-over set is empty and the random measure is absent; the original drift condition is [36], and an equivalent statement in affine-semimartingale form, allowing a deterministic clock with atoms, is [40] Proposition 6.11 eq. 74. The pointwise form $\alpha(t, T) = \langle \sigma(t, T), \int_t^T \sigma(t, u) du \rangle$ is the T -derivative of (2.3); the two agree for almost every T at each fixed (ω, t) , and exactly when $\alpha(t, \cdot)$ and $\sigma(t, \cdot)$ are piecewise constant on the same finite partition. Only the integrated form is assumed.

Assumption 2.2 (jump martingale condition at scheduled dates). For every $n = 1, \dots, N$ and every real $T \geq T_n$, Q -almost surely,

$$E \left[\exp \left(- \int_{T_n}^T \xi_n(u) du \right) \mid \mathcal{F}_{T_n-} \right] = 1, \quad (2.4)$$

where $\mathcal{F}_{T_n-}$ is the sigma-algebra generated by the $\mathcal{F}_t$, $t < T_n$, and the conditional expectation of the nonnegative random variable is taken in $[0, \infty]$; (2.4) forces that variable to be integrable. $T_0 = 0$ is an interval endpoint, not a jump date. Source: [30] Theorem 3.5(iv) together with the sigma-integrability of their (3.6), simplified by their Assumption 3.6; with no atoms and the overnight rate equal to $f(t, t)$, the display of their Remark 3.9 reads exactly (2.4) with $\xi_n = \Delta V(T_n, \cdot)$ in their notation, and their Example 3.10 deduces its compensating drift from this condition. Its content is that the size and direction of the jump at a scheduled date cannot be predicted, as the source says after its Theorem 3.5.

Assumptions 2.1 and 2.2 are hypotheses, not results of this paper. Each result below says in words which of them it uses; the lemmas of Section 3 use neither, and Theorem 1 uses both.

The step-plus-ramp family. The curves S^+ consist of a step and a ramp attached to each scheduled date: at time $t \in I_j$, for every maturity $T \in I_k$ with $k \geq j$,

$$f(t, T) = \sum_{m \leq k} [M_m(t) + C_m(t)(T - T_m)], \quad (2.5)$$

with M_m the level and C_m the ramp coefficient attached to T_m , so that on the current interval $f(t, T) = r_t + \sum_{m \leq j} C_m(t)(T - t)$. The case $C_m \equiv 0$ is the maturity-step family S of Section 3.1. In the bookkeeping of [34] eq. 28 the ramps are indexed by pairs of scheduled dates, with pairwise coefficients C_{nm} entered with a minus sign; only their sums attached to a date are identifiable (Section 3.8). We call S^+ the extension of S by ramps; no minimality is claimed.

Scope. Every result below is a statement about the model just described, with prices generated exactly by it. Sections 3.10 to 3.23 use two fixed specifications: a scalar Gaussian model with zero initial curve whose only parameters are the meeting variances, and the Gaussian and stochastic-volatility specifications of [12] with loadings, decay and scales fixed. The payoffs priced there are ideal: European bond calls, backward-looking futures with continuously compounded or exact discrete bond-implied accrual, and cash-settled European or American calls on those futures with stated exercise windows. Section 3.21 takes the listed contracts' expiry calendar and tick size from the exchange rules as inputs to that ideal

model, not their payoffs. Sections 3.43 and 3.44 add to the scalar Gaussian model a Gaussian diffusion with piecewise-constant variance, parallel or with a known maturity shape, and an arbitrary bounded initial curve. Not modelled: listed SOFR contracts with their settlement, margin and exercise-into-futures rules; accrual conventions other than those stated; quoted or noisy prices; an unknown or fitted initial curve outside Sections 3.43 and 3.44; additional diffusion or stochastic volatility where none is stated; and calibration to market data. An identification result therefore says which meeting variances the stated ideal observations determine within the stated specification, and a non-identification pair says that they do not determine them there; neither is a statement about listed instruments, and an infeasible panel is a rejection of the specification, not an arbitrage certificate. Section 5.3 states what a transfer to listed instruments would require.

3. Main results

3.1 The maturity-step family

Definition 3.1. Coefficients $(\alpha, \sigma, (\xi_n))$ as in Section 2 belong to the *maturity-step family* S if there are functions $\mu_k(t, \omega) \in \mathbb{R}$ and $s_k(t, \omega) \in \mathbb{R}^d$ ($k = 0, \dots, N$) and random variables $X_{n,k}$ ($1 \leq n \leq k \leq N$) such that, for every ω ,

$$\alpha(t, T, \omega) = \mu_k(t, \omega) \quad \text{and} \quad \sigma(t, T, \omega) = s_k(t, \omega) \quad \text{whenever } t \leq T \text{ and } T \in I_k, \quad (3.1)$$

$$\xi_n(u, \omega) = X_{n,k}(\omega) \quad \text{whenever } u \in I_k \text{ and } n \leq k. \quad (3.2)$$

That is, at each time the drift and the volatility, and at each scheduled date the curve jump, are step functions of maturity on the scheduled intervals. No measurability or integrability of $\mu_k, s_k, X_{n,k}$ is assumed beyond what (3.1), (3.2) and the regularity of Section 2 imply. At the level of the curve, S is the family $f(t, T) = b_k(t)$ for $T \in I_k, T \geq t$, with each level an Itô process $db_k = \mu_k dt + \langle s_k, dW \rangle$ between scheduled dates and a jump $X_{n,k}$ of b_k at T_n ; (3.1) and (3.2) are the coefficient form of that description, and we take them as the definition. Desks build the front-end SOFR curve as such a step function.

Theorem 1. Let the coefficients belong to S and satisfy the regularity conditions of Section 2 together with Assumptions 2.1 and 2.2. Then:

- (i) (jump leg) for every $1 \leq n \leq k \leq N$, $X_{n,k} = 0$ Q -almost surely; hence for every n there is an almost-sure event on which $\xi_n(u) = 0$ for all $u \in \mathbb{R}$;
- (ii) (drift and volatility leg) for $Q \otimes dt$ -almost every $(\omega, t) \in \Omega \times [0, \infty)$, $\mu_k(t, \omega) = 0$ and $s_k(t, \omega) = 0$ for every $k \geq j(t)$; hence $\alpha(t, T, \omega) = 0$ and $\sigma(t, T, \omega) = 0$ for all $T \in \mathbb{R}$.

The theorem uses both standing assumptions: (i) uses Assumption 2.2 only and (ii) uses Assumption 2.1 only. The values $\mu_k(t), s_k(t)$ for $k < j(t)$ concern maturities before t and are not constrained.

Corollary 2. If in addition W is a d -dimensional Brownian motion for $(\mathcal{F}_t)$ and f is given by (2.2), then for every $T \geq 0$ and $t \in [0, T]$, $f(t, T) = f(0, T)$ Q -almost surely: the maturity-step family supports no randomness at all.

Proof of Corollary 2 from Theorem 1. Fix $T \geq 0$ and $t \leq T$. By (ii) and Fubini's theorem, for Q -almost every ω , $\alpha(s, T, \omega) = 0$ for ds -almost every $s \in [0, t]$, so the drift integral in (2.2) vanishes almost surely; the integrand $s \mapsto \sigma(s, T)$ vanishes $Q \otimes ds$ -almost everywhere on $[0, t]$, so its stochastic integral vanishes almost surely; and by (i) the jump sum vanishes almost surely. $\square$

The corollary uses two standard facts, Fubini's theorem and the vanishing of the Itô integral of an integrand that vanishes almost everywhere; it is the one statement of this section not among the machine-checked ones (Appendix A).

3.2 Four lemmas

The proof of Theorem 1 rests on four lemmas about a fixed step curve: a closed form for the integral of a step function on the partition (Lemma 3), a two-point moment lemma (Lemma 4), the jump leg for one scheduled date (Lemma 5) and the drift leg for one time (Lemma 6). None of them uses Assumption 2.1 or 2.2.

Lemma 3 (integral of a step function). Let the dates be as in (2.1), let $s_0, \dots, s_N \in \mathbb{R}^d$ be fixed vectors, and let $\sigma : [0, \infty) \rightarrow \mathbb{R}^d$ be the step function $\sigma(u) := s_k$ for $u \in I_k$. For $0 \leq t \leq T < \infty$ with $t \in I_j$ and $T \in I_k$ one has $j \leq k$ and

$$\int_t^T \sigma(u) du = c_{jk}(t) + s_k(T - \max(T_k, t)), \quad (3.3)$$

where

$$c_{jj}(t) := 0, \quad c_{jk}(t) := s_j(T_{j+1} - t) + \sum_{m=j+1}^{k-1} s_m(T_{m+1} - T_m) \quad (k > j); \quad (3.4)$$

in particular $c_{jk}(t)$ does not depend on T . Consequently $T \mapsto \langle s_k, \int_t^T \sigma(u) du \rangle$ is affine on $I_k \cap [t, \infty) = [\max(T_k, t), T_{k+1})$ with slope $|s_k|^2$:

$$\left\langle s_k, \int_t^T \sigma(u) du \right\rangle = a_{jk}(t) + |s_k|^2 T, \quad a_{jk}(t) := \langle s_k, c_{jk}(t) \rangle - |s_k|^2 \max(T_k, t). \quad (3.5)$$

Proof. Split $[t, T]$ at the scheduled dates it contains, $t < T_{j+1} \leq \dots \leq T_k \leq T$. On each piece σ is constant except at one endpoint, a null set, so each piece integrates to the constant times the length; adding the pieces gives (3.3), and the T -dependence sits only in the last piece $s_k(T - T_k)$. When $k = j$ there is a single piece and (3.3) reads $s_j(T - t)$. Linearity of the inner product turns (3.3) into (3.5). $\square$

The identity is displayed but not stated in [34], whose eq. 28 contains the ramp terms and whose appendix eqs. 65 to 67 evaluate the integral of a step indicator case by case; [29] Theorem 3.7 and Remark 3.8 give the general drift condition without specializing to step volatilities.

Lemma 4 (two exponential moments). Let X be a real random variable on a probability space $(\Omega, \mathcal{F}, P)$ and let $\lambda \in \mathbb{R}$, $\lambda \neq 0$. If

$$E[\exp(-\lambda X)] \text{ is finite and equals } 1, \quad (3.6)$$

$$E[\exp(-\lambda X/2)] \text{ is finite and equals } 1, \quad (3.7)$$

then $X = 0$ P -almost surely.

Proof. Put $Y := \exp(-\lambda X/2)$, so $Y > 0$ and $Y^2 = \exp(-\lambda X)$. By (3.6), $E[Y^2] = 1$, so Y is square-integrable; by (3.7), $E[Y] = 1$. Then $E[(Y - 1)^2] = E[Y^2] - (E[Y])^2 = 0$, so $Y = 1$ almost surely, and injectivity of $\exp$ with $\lambda \neq 0$ gives $X = 0$ almost surely. $\square$

This is the equality case of the Cauchy–Schwarz inequality $(E[Y])^2 \leq E[Y^2]$, or equivalently the strict convexity of the cumulant generating function of X , which vanishes at the three collinear points $0, -\lambda/2, -\lambda$. Both conditions are needed: a Gaussian X with mean $\lambda\sigma^2/2$ and variance σ^2 satisfies (3.6) alone with $X \neq 0$. The finiteness half of (3.7) follows from (3.6).

Lemma 5 (a maturity-step jump at a scheduled date vanishes). Let $(\Omega, \mathcal{F}, P)$ be a probability space, let the dates be as in (2.1) with $N \geq 1$, fix $n \in \{1, \dots, N\}$ and real random variables $X_n, \dots, X_N$, and let ξ be the jump that is a step in maturity with these values,

$$\xi(u, \omega) := X_k(\omega) \text{ for } u \in I_k, \ k \geq n, \quad \xi(u, \omega) := 0 \text{ for } u < T_n. \quad (3.8)$$

For real $T \geq T_n$ put $Y_T := \exp(-\int_{T_n}^T \xi(u, \cdot) du)$, a nonnegative random variable. If

$$Y_T \text{ is integrable and } E[Y_T] = 1 \quad \text{for every real } T \geq T_n, \quad (3.9)$$

then $X_k = 0$ P -almost surely for every $k = n, \dots, N$. The conclusion already follows when (3.9) is assumed only at the $2(N - n + 1)$ maturities

$$T_k + h_k \text{ and } T_k + h_k/2, \quad k = n, \dots, N, \quad h_k := (T_{k+1} - T_k)/2 \ (k < N), \quad h_N := 1. \quad (3.10)$$

Conditional form: if $\mathcal{G} \subseteq \mathcal{F}$ is a sub-sigma-algebra and

$$E[Y_T | \mathcal{G}] = 1 \quad P\text{-almost surely, for every real } T \geq T_n, \quad (3.11)$$

with the conditional expectation of the nonnegative Y_T taken in $[0, \infty]$, then (3.9) holds and the same conclusion follows.

Proof. Fix ω . Then $\xi(\cdot, \omega)$ is the step function of Lemma 3 with $d = 1$, values $X_k(\omega)$ for $k \geq n$ and 0 before T_n , so Lemma 3 with $t = T_n$ gives, for $T \in I_k$ with $k \geq n$, $\int_{T_n}^T \xi(u, \omega) du = c_k(\omega) + X_k(\omega)(T - T_k)$, where $c_n = 0$ and c_k is a fixed linear combination of $X_n, \dots, X_{k-1}$ with the interval lengths as coefficients. Induct on k from n to N . If $X_m = 0$ almost surely for $n \leq m < k$, then $c_k = 0$ almost surely, and (3.9) at the two maturities (3.10), which lie in I_k , reads $E[\exp(-h_k X_k)] = 1$ and $E[\exp(-h_k X_k/2)] = 1$, both finite. Lemma 4 with $\lambda = h_k > 0$ gives $X_k = 0$ almost surely. For the conditional form, $Y_T \geq 0$ and the tower property in $[0, \infty]$ give $E[Y_T] = E[E[Y_T | \mathcal{G}]] = 1 < \infty$, which is (3.9). $\square$

Both maturities in each interval are needed, and no interval can be skipped: a Gaussian X_k with mean $h_k s^2/2$ and variance s^2 satisfies (3.9) at $T_k + h_k$ and fails it at $T_k + h_k/2$, and if X_{k_0} is Gaussian with the mean tuned to the length of I_{k_0} while all other X_k vanish, (3.9) holds at every maturity outside I_{k_0} with $X_{k_0} \neq 0$; any two distinct maturities in I_k would serve in place of (3.10). With $\mathcal{G} = \mathcal{F}_{T_n-}$ and $\xi = \xi_n$, (3.11) is Assumption 2.2 for the jump at T_n , specialized to a jump that is a step in maturity; that specialization is the definition of S , not a consequence of the assumption. The finite form (3.10) is what lets Theorem 1 pass from "for every T , almost surely" in (2.4) to a single almost-sure event.

[30] Example 3.10 is the boundary case: a Gaussian jump of the curve at a scheduled date is consistent only when accompanied by a ramp in maturity whose slope is the jump variance, so a jump that is a pure step in maturity is forced to vanish; the direct computation in the Gaussian case gives the same. [41] Proposition 1 and [40] Examples 6.15 and 6.16 are Gaussian scheduled jumps of an affine state with compensating ramps, and [3] computes its curve by pricing from a short-rate model with discretely distributed expected jumps rather than assuming a step family; none contains the statement.

Lemma 6 (a maturity-step drift and volatility under the drift condition vanish). Let the dates be as in (2.1), fix $j \in \{0, \dots, N\}$ and $t \in I_j$, real numbers $\mu_0, \dots, \mu_N$ and vectors $s_0, \dots, s_N \in \mathbb{R}^d$, and let $\alpha : [0, \infty) \rightarrow \mathbb{R}$ and $\sigma : [0, \infty) \rightarrow \mathbb{R}^d$ be the step functions $\alpha(u) := \mu_k$, $\sigma(u) := s_k$ for $u \in I_k$. Put $m_k := \max(T_k, t)$, so $m_j = t$ and $m_k = T_k$ for $k > j$. If

$$\int_t^T \alpha(u) du = \frac{1}{2} \left| \int_t^T \sigma(u) du \right|^2 \quad \text{for every real } T \geq t, \quad (3.12)$$

then $\mu_k = 0$ and $s_k = 0$ for every $k = j, \dots, N$; the values for $k < j$ never enter (3.12) and are unconstrained. The conclusion already follows when (3.12) is assumed only for T in a set $M \subset [t, \infty)$ such that

$$\text{for every } k = j, \dots, N, \ M \text{ contains two distinct points of the open interval } (m_k, T_{k+1}); \quad (3.13)$$

for instance M the rationals in (t, ∞) , or the $2(N-j+1)$ points $m_k + h_k, m_k + h_k/2$ with $h_k := (T_{k+1} - m_k)/2$ for $k < N$ and $h_N := 1$.

Proof. Fix $k \geq j$ and write $T = m_k + \tau$ with $0 \leq \tau < T_{k+1} - m_k$, so that $T \in I_k$ and $T \geq t$. Lemma 3, applied to σ and, with $d = 1$, to α , gives $\int_t^T \sigma = C_k + s_k \tau$ and $\int_t^T \alpha = A_k + \mu_k \tau$, where C_k and A_k are the constants (3.4) for σ and for α : zero for $k = j$, and for $k > j$ fixed linear combinations of $s_j, \dots, s_{k-1}$ and of $\mu_j, \dots, \mu_{k-1}$. Substituting into (3.12) and expanding the square,

$$A_k + \mu_k \tau = \frac{1}{2} |C_k|^2 + \langle C_k, s_k \rangle \tau + \frac{1}{2} |s_k|^2 \tau^2 \quad (3.14)$$

at every τ for which (3.12) is assumed. Induct on k from j to N . If $s_m = 0$ and $\mu_m = 0$ for $j \leq m < k$, then $C_k = 0$ and $A_k = 0$ exactly, and (3.14) at two distinct $\tau_1, \tau_2 > 0$, supplied by (3.13) or by the two points $h_k, h_k/2$, reads $\mu_k \tau_i = \frac{1}{2} |s_k|^2 \tau_i^2$. Dividing by τ_i and subtracting, $\frac{1}{2} |s_k|^2 (\tau_1 - \tau_2) = 0$, so $s_k = 0$, and then $\mu_k = 0$. $\square$

One maturity per interval does not suffice, since $\mu_k := \frac{1}{2} |s_k|^2 \tau_1$ with $s_k \neq 0$ satisfies (3.14) at τ_1 alone; skipping an interval does not suffice either, since with the volatility nonzero on one interval only and the drift tuned to its length both sides of (3.12) agree at every maturity outside that interval; and the point m_k itself is uninformative, which is why (3.13) asks for points of the open interval. For fixed (ω, t) , condition (3.12) is Assumption 2.1 with $\alpha(u) = \alpha(t, u, \omega)$ and $\sigma(u) = \sigma(t, u, \omega)$; that assumption holds for every $T > 0$ outside a null set of (ω, t) that may depend on T , and the version of the lemma with M the rationals is what lets Theorem 1 use it at countably many fixed maturities, so that a single null set suffices.

[34] §3.1.5, eqs. 29 and 30 and the sentences after them, is the short-rate shadow of this lemma: with step volatilities the short rate acquires a deterministic term depending on t which must cancel against the initial curve, so piecewise-constant short-rate paths preclude free interpolation of the initial term structure, and their footnote 18 identifies the term as the steepening of a Gaussian model without mean reversion; no statement about the curve family and no proof are given. [12] eqs. 18 and 19 carry the same drift with the integral of the step function left unevaluated. The classical analogues are [24] Proposition 3.2 and Theorem 4.1, where a consistent Itô process for the Nelson–Siegel family is quasi-deterministic, and [25] Theorem 3.2, which forces only the exponent coordinates of an exponential-polynomial family to be locally constant; as Remark 4.3 of [24] says, wider exponential-polynomial families do admit restricted nontrivial consistent processes, so total vanishing is specific to Nelson–Siegel there and to the maturity-step family here.

3.3 Proof of Theorem 1

Proof of (i). Fix n with $1 \leq n \leq N$ and ω . By (3.2) and the vanishing of ξ_n before T_n , the function $u \mapsto \xi_n(u, \omega)$ is the step function (3.8) with $X_k := X_{n,k}$ for $k = n, \dots, N$. Assumption 2.2 says that for every real $T \geq T_n$, (3.11) holds with $\mathcal{G} := \mathcal{F}_{T_n-}$. The conditional form of Lemma 5 gives $X_{n,k} = 0$ almost surely for every $k = n, \dots, N$. There are finitely many pairs (n, k) , so the event on which all $X_{n,k}$ vanish has probability one, and on it $\xi_n(u) = 0$ for every u : for $u < T_n$ by regularity, for $u \geq T_n$ by (3.2). $\square$

Proof of (ii). Step 1, one null set. Let $M := \mathbb{Q} \cap (0, \infty)$. For each $T \in M$, Assumption 2.1 provides a $Q \otimes dt$ -null set $N_T \subset \Omega \times [0, T]$ outside which (2.3) holds at T . The countable union $N := \bigcup_{T \in M} N_T$ is null. Fix $(\omega, t) \notin N$ with $t \geq 0$ and put $j := j(t)$; for every $T \in M$ with $T \geq t$, $(\omega, t) \in \Omega \times [0, T]$ and $(\omega, t) \notin N_T$, so (2.3) holds at T .

Step 2, Lemma 6 applies. Define step functions on $[0, \infty)$ by $\tilde{\alpha}(u) := \mu_k(t, \omega)$ and $\tilde{\sigma}(u) := s_k(t, \omega)$ for $u \in I_k$. By (3.1), $\tilde{\alpha}(u) = \alpha(t, u, \omega)$ and $\tilde{\sigma}(u) = \sigma(t, u, \omega)$ for every $u \geq t$, so the integrals over $[t, T]$ agree and, by Step 1, (3.12) holds for $\tilde{\alpha}, \tilde{\sigma}$ at every $T \in M$ with $T \geq t$. For each $k = j, \dots, N$ the open interval (m_k, T_{k+1}) is nonempty and contained in (t, ∞) , so it contains two distinct rationals; thus $M \cap [t, \infty)$ satisfies (3.13), and Lemma 6 gives $\mu_k(t, \omega) = 0$ and $s_k(t, \omega) = 0$ for every $k = j, \dots, N$.

Step 3. Step 2 holds for every (ω, t) outside the null set N . On the same set, $\alpha(t, T, \omega) = 0$ for all T : for $T < t$ by regularity, for $T \geq t$ by (3.1) and Step 2; likewise σ . $\square$

The proof uses Assumption 2.1 only at rational maturities and Assumption 2.2 only at two maturities per scheduled interval; the finite and countable forms of Lemmas 5 and 6 exist for this bookkeeping. The hypotheses of Theorem 1 are consistent: the zero coefficients satisfy everything, and a nonzero example lying outside S is given in Appendix A.

3.4 Piecewise-affine curves

The step-plus-ramp family (2.5) replaces steps by steps plus ramps. The lemma below is the analogue of Lemma 3 one degree up, with the identification statement that reads ramp slopes and levels off an identity between integrals; it is deterministic and uses no standing assumption.

Lemma 7 (integral of a piecewise-affine function, and identification). Let the dates be as in (2.1), fix vectors $a_0, \dots, a_N$ and $b_0, \dots, b_N$ in $\mathbb{R}^d$, and define the piecewise-affine function

$$g(u) := a_k + b_k(u - T_k) \quad \text{for } u \in I_k. \quad (3.15)$$

(a) For $0 \leq t \leq T < \infty$ with $t \in I_j$, $T \in I_k$, and $m_k := \max(T_k, t)$, one has $j \leq k$ and

$$\int_t^T g(u) du = e_{jk}(t) + a_k(T - m_k) + \frac{1}{2}b_k[(T - T_k)^2 - (m_k - T_k)^2], \quad (3.16)$$

where $e_{jj}(t) := 0$ and, for $k > j$,

$$e_{jk}(t) := a_j(T_{j+1} - t) + \frac{1}{2}b_j[(T_{j+1} - T_j)^2 - (t - T_j)^2] + \sum_{m=j+1}^{k-1} [a_m(T_{m+1} - T_m) + \frac{1}{2}b_m(T_{m+1} - T_m)^2]; \quad (3.17)$$

$e_{jk}(t)$ does not depend on T . Equivalently, with $\tau := T - m_k \in [0, T_{k+1} - m_k)$,

$$\int_t^T g(u) du = e_{jk}(t) + g(m_k) \tau + \frac{1}{2} b_k \tau^2, \quad (3.18)$$

so on $[m_k, T_{k+1})$ the integral is a polynomial of degree at most two in τ , with τ^2 -coefficient $\frac{1}{2}b_k$, τ -coefficient $g(m_k)$ and constant term $e_{jk}(t)$.

(b) If $P(\tau) := c_0 + c_1 \tau + c_2 \tau^2$ with $c_0, c_1, c_2 \in \mathbb{R}^d$ vanishes at three distinct real numbers, then $c_0 = c_1 = c_2 = 0$; if $c_2 = 0$ and P vanishes at two distinct real numbers, then $c_0 = c_1 = 0$. Consequently two functions of the form $p + q\tau + r\tau^2$ that agree at three distinct points have the same coefficients, so the three coefficients in (3.18) are determined by the values of the integral at any three distinct maturities of $[m_k, T_{k+1})$; and two functions of the form (3.15) that agree at two distinct points of every I_k have the same a_k and b_k , so the representation (3.15) on the fixed partition is unique.

Proof. (a) On a piece $[p, q) \subset I_k$ the integrand is $a_k + b_k(u - T_k)$ except at the null point q , so $\int_p^q g = a_k(q - p) + \frac{1}{2}b_k[(q - T_k)^2 - (p - T_k)^2]$. For $k = j$ the single piece $[t, T)$ gives (3.16) with $m_j = t$ and $e_{jj} = 0$. For $k > j$, additivity along $t < T_{j+1} \leq \dots \leq T_k \leq T$ gives the sum of the piece integrals; the first $k - j$ pieces are $e_{jk}(t)$ by (3.17) and the last is $a_k(T - T_k) + \frac{1}{2}b_k(T - T_k)^2$, which is the rest of (3.16) since $m_k = T_k$. For (3.18), write $T - T_k = \tau + (m_k - T_k)$, so the bracket in (3.16) is $\tau^2 + 2\tau(m_k - T_k)$, and collect terms using $g(m_k) = a_k + b_k(m_k - T_k)$, valid because $m_k \in I_k$. (b) With τ_1, τ_2, τ_3 distinct, $0 = P(\tau_i) - P(\tau_1) = (\tau_i - \tau_1)[c_1 + c_2(\tau_1 + \tau_i)]$ for $i = 2, 3$; dividing by $\tau_i - \tau_1 \neq 0$ and subtracting gives $c_2(\tau_2 - \tau_3) = 0$, so $c_2 = 0$, then $c_1 = 0$, then $c_0 = P(\tau_1) = 0$. The affine case is the first half of the same computation. The two consequences follow by applying this to differences. $\square$

With all $b_k = 0$, Lemma 7 reduces to Lemma 3. Three points are needed in (b), since $(\tau - \tau_1)(\tau - \tau_2)$ is a nonzero quadratic vanishing at two. The step-plus-ramp curves (2.5) are of the form (3.15) on each I_k with $a_k = \sum_{m \leq k} [M_m + C_m(T_k - T_m)]$ and $b_k = \sum_{m \leq k} C_m$. The partition bookkeeping is stated nowhere in the literature; [34] eqs. 65 to 67 and [12] eqs. 18 and 19 integrate or leave unevaluated the step case only.

3.5 The drift of a step-plus-ramp curve

In the step-plus-ramp family the volatility at a fixed time is a step function of maturity and the drift is piecewise affine. The next result identifies the drift from the volatility: the drift condition holds if and only if the drift is the differentiated HJM drift, so that on each scheduled interval the ramp slope equals the squared step volatility. Like Lemmas 6 and 7 it is deterministic and pointwise in (ω, t) , and it uses no standing assumption; the passage from Assumption 2.1 to its hypothesis for almost every (ω, t) is the bookkeeping of Section 3.3 with three rational maturities per interval in place of two.

Let the dates be as in (2.1), fix $j \in \{0, \dots, N\}$ and $t \in I_j$, and put $m_k := \max(T_k, t)$ for $k \geq j$. Let σ be the step function with values $s_0, \dots, s_N \in \mathbb{R}^d$ as in Lemma 3, let $C_k := c_{jk}(t)$ be its constants (3.4), so that $\int_t^u \sigma = C_k + s_k(u - m_k)$ for $u \in [m_k, T_{k+1})$, and let α be a piecewise-affine function with data $a_0, \dots, a_N, b_0, \dots, b_N \in \mathbb{R}$ as in (3.15). The *differentiated HJM drift* of σ from time t is

$$\alpha^*(u) := \left\langle \sigma(u), \int_t^u \sigma(v) dv \right\rangle, \quad u \geq t, \quad (3.19)$$

which is piecewise affine on $[t, \infty)$ with level and slope

$$b_k^* := |s_k|^2, \quad a_k^* := \langle s_k, C_k \rangle - |s_k|^2(m_k - T_k), \quad k \geq j, \quad (3.20)$$

on the pieces $[m_k, T_{k+1})$: $\alpha^*(u) = \langle s_k, C_k \rangle + |s_k|^2(u - m_k) = a_k^* + b_k^*(u - T_k)$ there.

Proposition 8. The following are equivalent.

(A) The drift condition holds for every real $T \geq t$:

$$\int_t^T \alpha(u) du = \frac{1}{2} \left| \int_t^T \sigma(u) du \right|^2. \quad (3.21)$$

(A') Condition (3.21) holds for every T in some set $M \subset [t, \infty)$ such that

$$\text{for every } k = j, \dots, N, \text{ } M \text{ contains three distinct points of the open interval } (m_k, T_{k+1}). \quad (3.22)$$

(B) $a_k = a_k^*$ and $b_k = b_k^*$ for every $k = j, \dots, N$.

(B') $\alpha(u) = \alpha^*(u)$ for every $u \geq t$.

The values a_k, b_k for $k < j$ enter none of these and are unconstrained.

Proof. Fix $k \geq j$, let $L_k := T_{k+1} - m_k \in (0, \infty]$ be the length of the piece $[m_k, T_{k+1})$, and for T in the piece write $\tau := T - m_k$. By Lemma 7(a) in the form (3.18) with $d = 1$, and by (3.4) with bilinearity of the inner product,

$$\int_t^T \alpha(u) du = e_{jk}(t) + \alpha(m_k)\tau + \frac{1}{2}b_k\tau^2, \quad \frac{1}{2} \left| \int_t^T \sigma(u) du \right|^2 = \frac{1}{2}|C_k|^2 + \langle C_k, s_k \rangle \tau + \frac{1}{2}|s_k|^2\tau^2, \quad (3.23)$$

with $\alpha(m_k) = a_k + b_k(m_k - T_k)$ and $e_{jk}(t)$ the constant (3.17). The constants obey, for $k < N$, the recursions $e_{j,k+1}(t) = e_{jk}(t) + \alpha(m_k)L_k + \frac{1}{2}b_kL_k^2$, the increment being the integral of α over the piece, and $C_{k+1} = C_k + s_kL_k$, hence $\frac{1}{2}|C_{k+1}|^2 = \frac{1}{2}|C_k|^2 + \langle C_k, s_k \rangle L_k + \frac{1}{2}|s_k|^2L_k^2$.

(B) implies (A): under (B), $\alpha(m_k) = \langle s_k, C_k \rangle$ and $\frac{1}{2}b_k = \frac{1}{2}|s_k|^2$, so the τ - and τ^2 -coefficients in (3.23) agree on every piece; the constants agree by induction on k , since $e_{jj}(t) = 0 = \frac{1}{2}|C_j|^2$ and the two recursions then carry $e_{jk}(t) = \frac{1}{2}|C_k|^2$ to $k+1$. So the two sides of (3.21) coincide as polynomials in τ on every piece, which is (A). (A) implies (A') with $M = [t, \infty)$. (A') implies (B): on each piece the two quadratics in (3.23) agree at three distinct points of $(0, L_k)$, so by Lemma 7(b) their coefficients agree; the τ^2 -coefficients give $b_k = |s_k|^2$, and the τ -coefficients, with $\alpha(m_k) = a_k + b_k(m_k - T_k)$, give $a_k = a_k^*$. This step is interval-local and uses no induction. (B) and (B') are equivalent because the pieces $[m_k, T_{k+1})$, $k \geq j$, partition $[t, \infty)$ and on each piece α and α^* are affine in $u - T_k$: equal data give equal functions, and equal functions agree at two distinct points of a piece of positive length, so by the affine case of Lemma 7(b) the data agree. $\square$

Three maturities per interval cannot be reduced to two: a quadratic with the wrong τ^2 -coefficient, hence the wrong slope b_k , can match the right side of (3.23) at two points of an interval, and Lemma 6 needed only two because there the slope was known to vanish. With all $b_k = 0$ imposed, Proposition 8 recovers Lemma 6. The induction in the sufficiency direction is not vacuous: a deviation from (B) on one interval changes the constants $e_{jk}(t)$ of every later interval. The proposition determines the drift from the volatility with no differentiability hypothesis on α in the maturity beyond membership in the family, which is what makes (A) and (B') equivalent: [36] state the drift condition in differentiated form under differentiability in maturity, and [30] Theorem 3.5(ii) in integrated form; within this family the two forms are the same statement. Sufficiency, (B) implies (A), is the content of the explicit models of [34] eqs. 28 to 30, [12] eqs. 18 and 19, [30] Example 3.10 eq. 3.14 with constant jump shape, [40] Examples 6.15 and 6.16

and [41] Proposition 1, all of which construct members of the family and verify the drift condition; none states the converse. The classical analogue is the consistency calculus of [9] §2 and §3 and [25] §2 and §3, which match the drift generated by the volatility against the tangent space of the family, and which do not treat a piecewise-affine family with breakpoints at scheduled dates. In the parametrization (2.5) with drifts $\dot{M}_m, \dot{C}_m$, the slope of the drift on I_k is $\sum_{m \leq k} \dot{C}_m(t)$, so (B) reads $\sum_{m \leq k} \dot{C}_m(t) = |s_k(t)|^2$ for every $k \geq j(t)$ (a computation, not part of the proposition).

3.6 The jump of a step-plus-ramp curve at a scheduled date

At a scheduled date the curve of the step-plus-ramp family jumps by a function that is affine on each later scheduled interval. The next result characterizes the level jumps that the jump martingale condition allows, given the slope jumps. It is stated for one scheduled date and an arbitrary sub-sigma-algebra in place of $\mathcal{F}_{T_n-}$, like the conditional form of Lemma 5; the assembly over all dates and with the drift half is left to the classification of Section 5.2.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{G} \subseteq \mathcal{F}$ a sub-sigma-algebra. Let the dates be as in (2.1) with $N \geq 1$, fix $n \in \{1, \dots, N\}$, real random variables $X_n, \dots, X_N$ (the level jumps) and $\mathcal{G}$ -measurable real random variables $y_n, \dots, y_N$ (the slope jumps), put $L_k := T_{k+1} - T_k$ for $k < N$, and let ξ be the piecewise-affine jump

$$\xi(u, \omega) := X_k(\omega) + y_k(\omega)(u - T_k) \text{ for } u \in I_k, k \geq n, \quad \xi(u, \omega) := 0 \text{ for } u < T_n. \quad (3.24)$$

The hypothesis is the jump martingale condition for this jump: for every real $T \geq T_n$,

$$E\left[\exp\left(-\int_{T_n}^T \xi(u, \cdot) du\right) \middle| \mathcal{G}\right] = 1 \quad P\text{-almost surely}, \quad (3.25)$$

the conditional expectation of the nonnegative variable taken in $[0, \infty]$. By Lemma 7(a) with $d = 1$ and $t = T_n$, for $T \in I_k, k \geq n$, and $\tau := T - T_k$,

$$\int_{T_n}^T \xi(u, \cdot) du = S_k + X_k \tau + \frac{1}{2} y_k \tau^2, \quad S_n := 0, \quad S_k := \sum_{m=n}^{k-1} (X_m L_m + \frac{1}{2} y_m L_m^2) \quad (k > n). \quad (3.26)$$

Put $R_k := \exp(-\sum_{m=n}^{k-1} L_m X_m)$ and $Z_k := \exp(\sum_{m=n}^{k-1} \frac{1}{2} y_m L_m^2)$, so that $\exp(-S_k) = R_k/Z_k$, with $R_k > 0$ and Z_k positive and $\mathcal{G}$ -measurable; $R_n = Z_n = 1$.

Proposition 9. (a) If (3.25) holds, then for every $k \in \{n, \dots, N\}$:

$$E[R_k | \mathcal{G}] = Z_k \quad P\text{-almost surely}, \quad (3.27)$$

so that $P_k(A) := E[(R_k/Z_k)\mathbf{1}_A]$ is a probability measure on $(\Omega, \mathcal{F})$ agreeing with P on $\mathcal{G}$; $y_k \geq 0$ almost surely; and the regular conditional distribution κ_k of X_k given $\mathcal{G}$ under P_k satisfies

$$\kappa_k(\omega, \cdot) = N(0, y_k(\omega)) \quad \text{for } P\text{-almost every } \omega, \quad (3.28)$$

with $N(0, 0) := \delta_0$. In words: given $\mathcal{G}$, the level jump on the date's own interval is centered Gaussian with variance the slope jump, and on each later interval the level jump is centered Gaussian with variance the slope jump under the tilt by the exponent accumulated over the earlier intervals.

(b) Conversely, if $y_k \geq 0$ almost surely and (3.27) and (3.28) hold for every k , then (3.25) holds.

(c) If conditionally on $\mathcal{G}$ the vector $(X_n, \dots, X_N)$ is Gaussian with $\mathcal{G}$ -measurable conditional mean μ and covariance Σ , then (3.25) holds if and only if, almost surely, for every k ,

$$\mu_k = \sum_{m=n}^{k-1} L_m \Sigma_{km} \quad \text{and} \quad \gamma_k = \Sigma_{kk}. \quad (3.29)$$

(d) There are a probability space, $N = 2$, $n = 1$, trivial $\mathcal{G}$, deterministic $y_1, y_2 > 0$ and (X_1, X_2) satisfying (3.25) with $X_1 \sim N(0, y_1)$ and X_2 not Gaussian: under a measure Q carrying independent $X_1 \sim N(-L_1 y_1, y_1)$ and $Z \sim N(0, y_2)$, put $s := \text{sign}(X_1 + L_1 y_1)$, $X_2 := s|Z|$ and $dP/dQ := \exp(L_1 X_1 + \frac{1}{2} L_1^2 y_1)$. Under P , $X_1 \sim N(0, y_1)$, (3.25) holds for every $T \geq T_1$, and X_2 has the density $x \mapsto 2\varphi_{y_2}(x) [p\mathbf{1}\{x > 0\} + (1-p)\mathbf{1}\{x < 0\}]$ with $p := \Phi(L_1 \sqrt{y_1}) > \frac{1}{2}$, which is discontinuous at 0; under the tilt $P_2 = Q$ the law of X_2 is $N(0, y_2)$, as (3.28) requires.

The proposition uses no standing assumption. With $\mathcal{G} = \mathcal{F}_{T_n-}$ and $\xi = \xi_n$, (3.25) is Assumption 2.2 for the jump at T_n , specialized to a jump of the form (3.24) with slopes measurable before the date; that specialization is a hypothesis on the family. Condition (3.29) is the "Gaussian jump laws with matching covariances" of the literature: the level jump on a later interval has conditional mean equal to its covariances with the earlier level jumps weighted by the earlier interval lengths, and each slope jump equals the corresponding variance. Part (d) shows that (3.25) does not force joint Gaussianity; what it forces is exactly the per-interval structure (3.28), of which (3.29) is the jointly Gaussian case. With zero drift and volatility and no further jumps the example of (d) is a consistent member of the step-plus-ramp family (a statement, not proved here); Theorem 11 draws the consequence for the classification, and Proposition 48 treats shapes that are not affine.

Proof. (a) Step 1, a change of measure. If $D \geq 0$ is $\mathcal{F}$ -measurable with $E[D \mid \mathcal{G}] = 1$, then $P_D(A) := E[D\mathbf{1}_A]$ is a probability measure agreeing with P on $\mathcal{G}$, and for every real random variable Y and measurable $h \geq 0$, $E_{P_D}[h(Y) \mid \mathcal{G}] = E_P[Dh(Y) \mid \mathcal{G}]$ almost surely, both sides taken in $[0, \infty]$; hence a regular conditional distribution κ of Y given $\mathcal{G}$ under P_D satisfies $\int h d\kappa(\omega, \cdot) = E_P[Dh(Y) \mid \mathcal{G}](\omega)$ for almost every ω . Step 2. Condition (3.25) at $T = T_k$, where $\tau = 0$, reads $E[R_k/Z_k \mid \mathcal{G}] = 1$; pulling out the $\mathcal{G}$ -measurable Z_k gives (3.27), so $D_k := R_k/Z_k$ qualifies for Step 1 and defines P_k . Step 3. For $\tau \in [0, L_k)$, (3.25) at $T = T_k + \tau$ with (3.26) reads, after pulling out the $\mathcal{G}$ -measurable factors, $E[D_k e^{-\tau X_k} \mid \mathcal{G}] = \exp(\frac{1}{2} \gamma_k \tau^2)$; by Step 1 the left side is the Laplace transform $M_k(\omega, \tau) := \int e^{-\tau x} \kappa_k(\omega, dx)$ of the regular conditional distribution of X_k under P_k , so $M_k(\omega, \tau) = \exp(\frac{1}{2} \gamma_k(\omega) \tau^2)$ for almost every ω , for each fixed τ . Step 4. Taking the union of the exceptional sets over rational τ gives one null set; off it, $M_k(\omega, \cdot)$ is the Laplace transform of a probability measure, hence convex with values in $(0, \infty]$, finite at the rationals of $[0, L_k)$, therefore finite and continuous on $(0, L_k)$, and equal to $\exp(\frac{1}{2} \gamma_k \tau^2)$ there by density. Step 5. Its logarithm is convex (Hölder), while $\frac{1}{2} \gamma_k \tau^2$ is strictly concave if $\gamma_k < 0$; so $\gamma_k \geq 0$ off the null set. Step 6. Fix ω off the null set, $y := \gamma_k(\omega)$, and $\tau_0 \in (0, L_k)$, and tilt: $\nu(dx) := e^{-\tau_0 x} \kappa_k(\omega, dx)/M_k(\omega, \tau_0)$ has Laplace transform $\exp(y\tau_0 s + \frac{1}{2} y s^2)$ for s in the open interval $(-\tau_0, L_k - \tau_0)$ around 0, which is the Laplace transform of $N(-y\tau_0, y)$. Two probability measures on $\mathbb{R}$ whose Laplace transforms are finite and agree on an open interval containing 0 coincide ([19]; the statement is standard), so $\nu = N(-y\tau_0, y)$, and tilting back by $e^{\tau_0 x}$ gives $\kappa_k(\omega, \cdot) = N(0, y)$, with δ_0 when $y = 0$. This is (3.28).

(b) Given (3.27), Step 1 applies to D_k , and (3.28) gives $E[D_k e^{-\tau X_k} \mid \mathcal{G}] = \exp(\frac{1}{2} \gamma_k \tau^2)$; multiplying by $\exp(-\frac{1}{2} \gamma_k \tau^2)$ and using (3.26) returns (3.25) at $T = T_k + \tau$, and every $T \geq T_n$ is of this form for one k .

(c) Conditionally on $\mathcal{G}$, tilting the Gaussian law $N(\mu, \Sigma)$ by $\exp(-\langle \ell, x \rangle)$ with $\ell_m = L_m$ for $m < k$ and 0 otherwise gives $N(\mu - \Sigma \ell, \Sigma)$, whose k -th marginal, $N(\mu_k - \sum_{m < k} L_m \Sigma_{km}, \Sigma_{kk})$, is κ_k . By (a) and (b), (3.25) holds if and only if this marginal is $N(0, \gamma_k)$ for every k and (3.27) holds; the first is (3.29), and (3.29) implies

(3.27) because $-\langle \ell, \mu \rangle + \frac{1}{2} \langle \ell, \Sigma \ell \rangle$ reduces, after the cross terms cancel against (3.29) for the earlier indices, to $\frac{1}{2} \sum_{m < k} L_m^2 y_m = \log Z_k$.

(d) Under Q , $X_1 + L_1 y_1$ is symmetric and independent of Z , so s is a fair sign independent of $|Z|$ and $X_2 \sim N(0, y_2)$; dP/dQ integrates to one. Under P , completing the square gives $X_1 \sim N(0, y_1)$; (3.25) on I_1 is the Gaussian Laplace transform, and on I_2 the density dP/dQ cancels $\exp(-S_2)$ exactly, leaving $E_Q[e^{-\tau X_2}] e^{-\frac{1}{2} y_2 \tau^2} = 1$. Since dP/dQ depends on X_1 alone, $(s, |Z|)$ stay independent under P with $P(s = 1) = \Phi(L_1 \sqrt{y_1}) = p$, which gives the stated density of X_2 ; its jump at 0 rules out a Gaussian law. Finally $D_2 = (dP/dQ)^{-1}$, so $P_2 = Q$. $\square$

The one analytic input is the uniqueness of a law from its Laplace transform on an interval; everything else is the change of measure and the quadratic (3.26). The mechanism of (a) is the one the literature anticipates for a single Gaussian jump, "a linear ramp forces a quadratic cumulant generating function", made precise for a whole sequence of intervals, and (d) shows that it yields Gaussianity on later intervals only under the accumulated tilt. [30] Example 3.10 is the sufficiency instance for one date with a Gaussian jump, whose compensator slope for a constant jump shape is the variance, which is (3.29) for $k = n$; their Proposition 4.5 carries the general-law tilted-mean mechanism for the overnight rate. [40] Examples 6.15 and 6.16 and [41] Proposition 1 are the jointly Gaussian multi-date instances, sufficiency in case (c). [20] and [13] have non-Gaussian scheduled jump laws in short-rate models whose curves are not piecewise affine, so they do not bear on (d). The characterization and the example are stated nowhere.

3.7 An explicit step-plus-ramp model

The one-date version of the following construction is [30] Example 3.10 with constant jump shape and centered Gaussian jump; [40] Example 6.16 and [41] Proposition 1 give multi-date Gaussian affine constructions. Here the coefficients are given directly on the scheduled partition and both standing assumptions are verified for every date and maturity.

Fix deterministic vectors $s_0, \dots, s_N \in \mathbb{R}^d$ and numbers $y_{n,k} \geq 0$, $1 \leq n \leq k \leq N$. Let

$$\mathcal{P} := \{(n, k) : 1 \leq n \leq k \leq N\}, \quad \Omega := \mathbb{R}^{\mathcal{P}}, \quad Q := \bigotimes_{(n,k) \in \mathcal{P}} N(0, y_{n,k}), \quad (3.30)$$

where $N(0, 0) := \delta_0$, and write $X_{n,k}$ for the coordinate maps. Thus the $X_{n,k}$ are independent and $X_{n,k} \sim N(0, y_{n,k})$. Take the filtration

$$\mathcal{F}_t := \sigma(X_{m,k} : (m, k) \in \mathcal{P}, T_m \leq t). \quad (3.31)$$

For $T \geq t$, $T \in I_k$, and $1 \leq n \leq k$, define

$$\sigma(t, T) := s_k, \quad \alpha(t, T) := \left\langle s_k, \int_t^T \sigma(t, u) du \right\rangle, \quad (3.32)$$

$$\xi_n(T) := X_{n,k} + y_{n,k}(T - T_k), \quad (3.33)$$

and set each coefficient to zero before its current date as in Section 2. Let $f(0, \cdot)$ be Borel measurable and bounded on bounded intervals.

Proposition 10. The coefficient system (3.30)–(3.33) satisfies the regularity conditions of Section 2, Assumption 2.1 pointwise for every $0 \leq t \leq T$, and Assumption 2.2 for every scheduled date and every $T \geq T_n$.

Proof. The initial curve is locally integrable. The volatility is deterministic, Borel measurable and bounded by $S := \max_k |s_k|$. Lemma 3 makes $(t, T) \mapsto \int_t^T \sigma(t, u) du$, hence α , Borel measurable on each of the finitely many regions determined by the scheduled intervals. The bounds $|\sigma| \leq S$ and $|\alpha(t, T)| \leq S^2(T - t)$ give the required local integrability. Each ξ_n is a finite sum of indicator-times-affine functions with $\mathcal{F}_{T_n}$ -measurable coefficients and is locally integrable pathwise. This proves the regularity conditions.

Equation (3.32) is the differentiated HJM drift. Proposition 8, (B') implies (A), therefore gives (2.3) for every $0 \leq t \leq T$.

Fix n and let $T \in I_k$, $k \geq n$, with $\tau := T - T_k$. Lemma 7 gives

$$\int_{T_n}^T \xi_n(u) du = \sum_{m=n}^{k-1} (L_m X_{n,m} + \frac{1}{2} y_{n,m} L_m^2) + \tau X_{n,k} + \frac{1}{2} y_{n,k} \tau^2. \quad (3.34)$$

The left limit of (3.31) at T_n is generated by the coordinates $X_{m,k}$ with $m < n$, while (3.34) uses only coordinates whose first index is n . The two sets of coordinates are independent. The conditional expectation in (2.4) is therefore the unconditional expectation, and independence together with the Gaussian Laplace transform gives

$$\prod_{m=n}^{k-1} \exp(\frac{1}{2} y_{n,m} L_m^2) \exp(-\frac{1}{2} y_{n,m} L_m^2) \exp(\frac{1}{2} y_{n,k} \tau^2) \exp(-\frac{1}{2} y_{n,k} \tau^2) = 1. \quad (3.35)$$

This is (2.4). □

Choosing some s_k or $y_{n,k}$ nonzero makes the coefficient system nonzero. On an extension carrying a Brownian motion independent of the coordinates, insert (3.32) and (3.33) into (2.2). If the initial curve is affine on each I_k , then the curve remains affine there. Its slope on $I_k \cap [t, \infty)$ is the initial slope plus $|s_k|^2 t + \sum_{n: T_n \leq t} y_{n,k}$: the drift contributes $|s_k|^2 t$, the diffusion is constant in T on the interval, and each realized jump contributes $y_{n,k}$. Thus the construction is a member of the step-plus-ramp family, not merely a coefficient system satisfying the two standing conditions.

3.8 Classification of the step-plus-ramp family

We now assemble Propositions 8 and 9 under the quantifiers of the standing assumptions. Coefficients belong to the *step-plus-ramp family* S^+ with deterministic step volatility if there are deterministic functions $s_k : [0, \infty) \rightarrow \mathbb{R}^d$, coefficient processes a_k, b_k , level jumps $X_{n,k}$, and $\mathcal{F}_{T_n-}$ -measurable slope jumps $y_{n,k}$ such that, for $T \geq t$, $T \in I_k$, and $u \in I_k$, $k \geq n$,

$$\sigma(t, T) = s_k(t), \quad \alpha(t, T) = a_k(t) + b_k(t)(T - T_k), \quad (3.36)$$

$$\xi_n(u) = X_{n,k} + y_{n,k}(u - T_k). \quad (3.37)$$

The coefficients also satisfy the regularity conditions of Section 2. Put $j := j(t)$, $m_k := \max(T_k, t)$, and $C_k(t) := c_{jk}(t)$, with $c_{jk}(t)$ defined by (3.4) for the step values $s_0(t), \dots, s_N(t)$. Thus, for $u \in [m_k, T_{k+1})$,

$$\int_t^u \sigma(t, v) dv = C_k(t) + s_k(t)(u - m_k). \quad (3.38)$$

For $1 \leq n \leq k \leq N$, set

$$R_{n,k} := \exp\left(-\sum_{m=n}^{k-1} L_m X_{n,m}\right), \quad Z_{n,k} := \exp\left(\sum_{m=n}^{k-1} \frac{1}{2} y_{n,m} L_m^2\right). \quad (3.39)$$

Sufficiency and explicit Gaussian instances occur in [34] eqs. 28 to 30, [12] eqs. 18 and 19, [30] Example 3.10, [40] Examples 6.15 and 6.16, and [41] Proposition 1. The theorem adds necessity and states the exact jump law outside the jointly Gaussian class.

Theorem 11. For coefficients in S^+ with deterministic step volatility, Assumptions 2.1 and 2.2 hold if and only if both of the following conditions hold.

(i) Outside one $Q \otimes dt$ -null set, for every $t \geq 0$ and every $k \geq j(t)$,

$$b_k(t) = |s_k(t)|^2, \quad a_k(t) = \langle s_k(t), C_k(t) \rangle - |s_k(t)|^2(m_k - T_k). \quad (3.40)$$

Equivalently, outside that null set,

$$\alpha(t, T) = \left\langle \sigma(t, T), \int_t^T \sigma(t, u) du \right\rangle \quad \text{for every } T \geq t. \quad (3.41)$$

In particular, the drift is determined by the deterministic volatility and is deterministic outside the null set; its slope on I_k is $|s_k(t)|^2$.

(ii) For every $1 \leq n \leq k \leq N$, $y_{n,k} \geq 0$ almost surely,

$$E[R_{n,k} \mid \mathcal{F}_{T_n-}] = Z_{n,k} \quad Q\text{-almost surely}, \quad (3.42)$$

and, under the probability measure $Q_{n,k}(A) := E[(R_{n,k}/Z_{n,k})\mathbf{1}_A]$, the conditional distribution of $X_{n,k}$ given $\mathcal{F}_{T_n-}$ is $N(0, y_{n,k})$. Thus the level jump is centered Gaussian with variance the slope jump on the date's own interval and under the accumulated tilt on every later interval.

If, conditionally on $\mathcal{F}_{T_n-}$, $(X_{n,n}, \dots, X_{n,N})$ is jointly Gaussian with mean $\mu^{(n)}$ and covariance $\Sigma^{(n)}$, condition (ii) is equivalent to

$$\mu_k^{(n)} = \sum_{m=n}^{k-1} L_m \Sigma_{km}^{(n)}, \quad y_{n,k} = \Sigma_{kk}^{(n)}, \quad k = n, \dots, N. \quad (3.43)$$

Proof. For necessity in (i), let M be the positive rationals. Assumption 2.1 at each $T \in M$ fails only on a $Q \otimes dt$ -null set in $\Omega \times [0, T]$. Their countable union is one null set in $\Omega \times [0, \infty)$. Fix (ω, t) outside it. Every open piece (m_k, T_{k+1}) , including the unbounded last piece, contains three distinct points of M . Equations (3.36) and (3.38) therefore put the fixed functions $T \mapsto \alpha(t, T, \omega)$ and $T \mapsto \sigma(t, T)$ under condition (A') of Proposition 8. Its implication (A') to (B) gives (3.40), and its equivalence of (B) and (B') gives (3.41).

For necessity in (ii), fix n . Equations (3.37) and (2.4) are (3.24) and (3.25) with $\mathcal{G} = \mathcal{F}_{T_n-}$. Proposition 9(a) gives nonnegativity, (3.42), and the conditional Gaussian law for every $k \geq n$.

Conversely, (3.40) is condition (B) of Proposition 8 for almost every (ω, t) . The implication (B) to (A) gives (2.3) outside that null set for every maturity. For each n , condition (ii) is the hypothesis of Proposition

9(b), which gives (2.4) for every $T \geq T_n$. Hence both standing assumptions hold. The jointly Gaussian statement is Proposition 9(c) at each date. $\square$

In the curve parametrization (2.5), the ramp coefficient C_m attached to T_m is, in the pairwise bookkeeping of [34] eq. 28 with pairwise coefficients C_{nm} entered with a minus sign, the identifiable combination $C_m = -\sum_n C_{nm}$ over the pairs whose later date is T_m ; suppose these coefficients have drifts $\dot{C}_m$. The slope of the curve drift on I_k is $\sum_{m \leq k} \dot{C}_m$, so (3.40) gives $\sum_{m \leq k} \dot{C}_m = |s_k|^2$, or $\dot{C}_k = |s_k|^2 - |s_{k-1}|^2$ for $k \geq 1$. With volatility increments $s_k = \sum_{m \leq k} v_m$, this is the corresponding sum of the pairwise relations $dC_{nm}/dt = -\langle v_n, v_m \rangle$. This last identification assumes uniqueness of the coefficient semimartingale decomposition and is not part of Theorem 11.

3.9 Magnitudes

The classification of Theorem 11 is structural, not a pricing correction. Take a meeting at which the policy rate moves by 25 basis points or not at all, with equal probability. The level jump has variance

$$v = \frac{1}{4} (0.0025)^2 \approx 1.6 \times 10^{-6} \quad (3.44)$$

in units of rate squared. A consistent model gives the jump a Gaussian law of that variance and puts on the interval after the meeting a ramp of slope v (Theorem 11(ii)), so the forward rate at time to maturity τ past the meeting is corrected by $v\tau$. Six weeks past the meeting, $\tau \approx 0.115$, the correction is 1.8×10^{-7} , about 0.002 basis points; one year past it, 1.6×10^{-6} , about 0.016 basis points, and eight such meetings spread over a year contribute in total less than 0.1 basis point at the one-year point. A year of diffusion with volatility 100 basis points per year, $\sigma = 0.01$, gives by the same theorem the drift correction $\sigma^2\tau = 10^{-4}$ at $\tau = 1$, about one basis point. The ramps that consistency forces at meetings are two orders of magnitude below the diffusive convexity at the one-year point and below quoting precision at the front end. What Theorem 11 fixes is the form of the family and the law of its jumps; the bound (3.185) of Proposition 48 shows that a two-point jump law and its Gaussian ramp are indistinguishable at pricing precision within one meeting interval.

3.10 The scalar Gaussian model

The following construction is the constant-shape, zero-diffusion specialization of [30] Example 3.10, eq. 3.14. Fix $N \geq 3$, the dates (2.1), and deterministic $v_1, \dots, v_N \geq 0$. On the product space carrying independent coordinates $Z_n \sim N(0, v_n)$, with $N(0, 0)$ the point mass at zero, let $\mathcal{F}_t$ reveal Z_n at T_n , and put

$$X_t := \sum_{T_n \leq t} Z_n, \quad f(t, T) := X_t + \sum_{T_n \leq t} v_n(T - T_n), \quad r_t := f(t, t). \quad (3.45)$$

The initial curve is zero, the drift and volatility vanish, and the curve jump at T_n is

$$\xi_n(u) := \mathbf{1}\{u \geq T_n\} [Z_n + v_n(u - T_n)]. \quad (3.46)$$

Write $B_t := \exp(\int_0^t r_s ds)$ and $P(t, T) := \exp(-\int_t^T f(t, u) du)$. For $h = T - T_n \geq 0$, $\int_{T_n}^T \xi_n = hZ_n + \frac{1}{2}v_nh^2$, whose exponential has conditional expectation one given the earlier coordinates, so Assumption 2.2 holds, including at $v_n = 0$; the zero drift satisfies Assumption 2.1; the regularity conditions are immediate; and

$$\frac{P(t, T)}{B_t} = \prod_{T_n \leq t} \exp(-(T - T_n)Z_n - \frac{1}{2}v_n(T - T_n)^2) \quad (3.47)$$

is a finite product of independent unit-mean factors, so the discounted bond is a true martingale. Every current curve lies in S^+ ; on a later interval the level and slope coefficients are $Z_n + v_n(T_k - T_n)$ and v_n , whose Gaussian mean and covariance satisfy (3.43), so the model is in the jointly Gaussian class of Theorem 11. This is the model of Sections 3.13 and 3.15 to 3.21.

Remark 12 (state dimension and parameter dimension). The scalar X_t is a time-inhomogeneous Markov state for the current curve, the affine line $\{T \mapsto x + \sum_{T_n \leq t} v_n(T - T_n)\}$, and for the conditional law of the future short-rate path, which is X_t plus independent future coordinates plus a deterministic function of time; the diffusion-factor count is zero. The short-rate jump at T_n is $\Delta r_{T_n} = Z_n$, so for every $t < T_n$ its risk-neutral event variance is v_n , and the ideal call paying $(\Delta r_{T_n} - K)^+$ at T_n , with price $C_n(t, K) := E_Q[(B_t/B_{T_n})(\Delta r_{T_n} - K)^+ | \mathcal{F}_t]$, factors because the discount factor does not contain Z_n :

$$\frac{C_n(t, K)}{P(t, T_n)} = E[(Z_n - K)^+], \quad v_n = 2\pi \left[\frac{C_n(t, 0)}{P(t, T_n)} \right]^2. \quad (3.48)$$

As the parameters $v_1, \dots, v_N$ vary, the vector of event variances ranges over all of $[0, \infty)^N$, and the same holds for the remaining meetings after any number of them have passed; in each fixed model it is constant on the state line, so the state image is the single point $(v_1, \dots, v_N)$. State dimension one and diffusion-factor count zero therefore give no restriction across meetings by themselves: a cross-meeting restriction has to fix the model's parameters and say which payoffs are observed, which is what the rest of this section does. The construction is known; the distinction between its fixed-model state image and the union over its parameters is not stated in the source.

3.11 Gaussian meeting-variance cones

Fix the Gaussian specialization of [12] eqs. 35 to 39. Let $W_1, \dots, W_d$ be independent Brownian motions, $\gamma_{i,j}$ the ordinal meeting loadings, $G_{i,j} := \sum_{h=1}^i \gamma_{h,j}$, $\lambda_j \geq 0$ the decay parameters, and $a_j(s)$ deterministic scales; put

$$h_j(s, T) := a_j(s)e^{-\lambda_j(T-s)}G_{j(T)-j(s),j}, \quad g_{n,j}(s) := a_j(s)e^{-\lambda_j(T_n-s)}\gamma_{n-j(s),j}. \quad (3.49)$$

The volatility is $\sigma_j = h_j$, the drift is the differentiated HJM drift, and there is no curve jump at fixed maturity; the short-rate jump at T_n is the maturity discontinuity of a curve continuous in time. Its Brownian part is $\sum_j \int_0^{T_n} g_{n,j} dW_j$, so its conditional variance at $t < T_n$ is

$$V_n(t) := \text{Var}_Q(\Delta r_{T_n} | \mathcal{F}_t) = \sum_{j=1}^d \int_t^{T_n} a_j(s)^2 e^{-2\lambda_j(T_n-s)} \gamma_{n-j(s),j}^2 ds. \quad (3.50)$$

Proposition 13. Fix t , the schedule, the loadings and the decay parameters, and retain the M meetings after t . If $a_j(s) = a_j$ and $q_j = a_j^2$, then

$$V(t) = A(t)q, \quad q \in [0, \infty)^d, \quad A_{n,j}(t) = \sum_{k=j(t)}^{n-1} \gamma_{n-k,j}^2 H_j(T_n; \max(t, T_k), T_{k+1}), \quad (3.51)$$

where $H_j(T; a, b) = b - a$ if $\lambda_j = 0$ and $H_j(T; a, b) = [e^{-2\lambda_j(T-b)} - e^{-2\lambda_j(T-a)}]/(2\lambda_j)$ otherwise. The attainable variance vectors form the cone $A(t)[0, \infty)^d$, the linear equalities common to it are exactly $w^\top V(t) = 0$ for $w \in \ker A(t)^\top$, and its containing column space has dimension $\text{rank } A(t) \leq d$; with three factors and

$M > 3$ meetings there are at least $M - 3$ independent equalities. If instead $a_j(s)^2 = q_{j,l}$ on a deterministic partition $[b_l, b_{l+1})$, the matrix has dK columns,

$$A_{n,(j,l)}(t) = \int_{[t, T_n] \cap [b_l, b_{l+1}]} e^{-2\lambda_j(T_n-s)} \gamma_{n-j(s),j}^2 ds, \quad (3.52)$$

and for one Brownian factor, $t = 0$, $b_k = T_k$ and $\gamma_{1,1} \neq 0$ this N -by- N matrix is lower triangular with $\det A = \prod_{n=1}^N \gamma_{1,1}^2 H_1(T_n; T_{n-1}, T_n) > 0$, so the cone has nonempty interior and no nonzero linear equality survives all meeting-dependent scale choices.

Proof. Splitting the integral in (3.50) at the meeting dates and evaluating the exponential integral gives (3.51); the cone and equality statements are $(\text{range } A)^\perp = \ker A^\top$. Splitting also at the scale partition gives (3.52). In the one-factor case an interval beginning at or after T_n cannot enter row n , each diagonal entry is the displayed positive factor, and invertibility gives the interior and the trivial common left nullspace. $\square$

Loadings and decay are fixed in (3.51); recalibrating them changes A , and Section 3.22 determines what survives when the scales, and then the decays, vary. The derivation of (3.50) from the Brownian construction takes the maturity limits and the conditional-variance identity as given.

3.12 The stochastic-volatility variance map

Retain the stochastic-volatility specification of [12] eqs. 35 to 41: Brownian motions W_j, U_j with

$$d[W_i, W_j]_s = d[U_i, U_j]_s = \mathbf{1}_{\{i=j\}} ds, \quad d[W_i, U_j]_s = \mathbf{1}_{\{i=j\}} \rho_j(s) ds, \quad (3.53)$$

and continuous nonnegative solutions, starting from one, of

$$dv_j(s) = \theta_j(s)(1 - v_j(s)) ds + \alpha_j(s) \sqrt{v_j(s)} dU_j(s), \quad (3.54)$$

with deterministic, locally bounded, piecewise constant coefficients, $\theta_j, \alpha_j \geq 0$, $|\rho_j| \leq 1$, and $E[\sup_{s \leq H} v_j(s)^2] < \infty$ on finite horizons. Existence, nonnegativity and this moment bound are hypotheses; the stochastic calculus used below is cited in Section 4 and enters the formal development as the fields of a hypothesis structure audited against those sources. Put

$$\begin{aligned} h_j(s, T) &:= a_j(s) e^{-\int_s^T \lambda_j(u) du} G_{j(T)-j(s),j}, & \sigma_j(s, T) &:= h_j(s, T) \sqrt{v_j(s)}, \\ g_{n,j}(s) &:= a_j(s) e^{-\int_s^{T_n} \lambda_j(u) du} \gamma_{n-j(s),j}, & b_{n,j}(s) &:= g_{n,j}(s) \int_s^{T_n} h_j(s, u) du, \\ R_{n,j}(u) &:= \int_u^{T_n} b_{n,j}(s) e^{-\int_u^s \theta_j(q) dq} ds, & F_{n,j}(u) &:= g_{n,j}(u)^2 + 2\rho_j g_{n,j} \alpha_j R_{n,j} + \alpha_j^2 R_{n,j}^2, \end{aligned} \quad (3.55)$$

all unlabelled functions in the last expression being evaluated at u .

Proposition 14 (conditional). Under the preceding stochastic hypothesis, the short-rate jump and its conditional variance are

$$\begin{aligned}\Delta r_{T_n} &= \sum_{j=1}^d \left[\int_0^{T_n} b_{n,j}(s) v_j(s) ds + \int_0^{T_n} g_{n,j}(s) \sqrt{v_j(s)} dW_j(s) \right], \\ V_n(t) &= \sum_{j=1}^d \int_t^{T_n} F_{n,j}(u) \left[1 + (v_j(t) - 1) e^{-\int_t^u \theta_j(q) dq} \right] du,\end{aligned}\tag{3.56}$$

with the kernel nonnegative because $F_{n,j} = (g_{n,j} + \rho_j \alpha_j R_{n,j})^2 + (1 - \rho_j^2)(\alpha_j R_{n,j})^2$. For fixed parameters, dates and observation time, with $E_j(t, u) := e^{-\int_t^u \theta_j}$,

$$A_{n,j}(t) := \int_t^{T_n} F_{n,j}(u) E_j(t, u) du, \quad C_n(t) := \sum_j \int_t^{T_n} F_{n,j}(u) (1 - E_j(t, u)) du, \quad V(t) = C(t) + A(t)v(t),\tag{3.57}$$

with $C_n(t) \geq 0$ and $A_{n,j}(t) \geq 0$. Hence $V(t) \in C(t) + A(t)[0, \infty)^d$, every $w \in \ker A(t)^\top$ gives $w^\top(V(t) - C(t)) = 0$, and the containing affine space has dimension at most d .

Proof. The maturity difference gives the jump. Since $R' = \theta R - b$ and $R(T_n) = 0$, the product rule gives

$$\int_t^{T_n} b_{n,j}(s) v_j(s) ds = R_{n,j}(t) v_j(t) + \int_t^{T_n} \theta_j R_{n,j} ds + \int_t^{T_n} \alpha_j R_{n,j} \sqrt{v_j} dU_j,\tag{3.58}$$

so after subtracting the conditional mean the future random part is $\sum_j \int_t^{T_n} \sqrt{v_j} (g_{n,j} dW_j + \alpha_j R_{n,j} dU_j)$; the conditional isometry and (3.53) give the kernel F , and the integrating-factor solution of (3.54) gives $E[v_j(u) | \mathcal{F}_t] = 1 + (v_j(t) - 1) E_j(t, u)$, proving (3.56). The rest is algebra. $\square$

The random HJM drift and its covariance with the Brownian term are in F ; replacing F by g^2 is valid only when every volatility-of-volatility coefficient vanishes. The short-rate jump is measurable immediately before its meeting in this continuous filtration and is not an additional curve jump.

3.13 Identification from ideal bond calls

In the model of Section 3.10 at observation time zero, for expiry S , bond maturity $U > S$ and strike $K > 0$, define the ideal European bond call and the cumulative variance

$$C(S, U, K) := E_Q[B_S^{-1}(P(S, U) - K)^+], \quad q(S, U) := (U - S)^2 \sum_{T_n \leq S} v_n.\tag{3.59}$$

The forward-measure calculation is that of [30] Lemma 4.11 and Proposition 4.12.

Proposition 15. For $q > 0$,

$$C(S, U, K) = \Phi(d_1) - K\Phi(d_2), \quad d_1 = \frac{-\log K + q/2}{\sqrt{q}}, \quad d_2 = d_1 - \sqrt{q},\tag{3.60}$$

and $C(S, U, K) = (1 - K)^+$ for $q = 0$; in particular $C(S, U, 1) = 2\Phi(\sqrt{q}/2) - 1$. At one expiry the complete call surface identifies exactly the cumulative variance $\sum_{T_n \leq S} v_n$: two parameter vectors give the same prices for all $U > S$ and $K > 0$ if and only if those sums agree, so for $N = 3$ and $\varepsilon > 0$ the vectors $(\varepsilon, 2\varepsilon, 3\varepsilon)$ and $(2\varepsilon, \varepsilon, 3\varepsilon)$ give the same full surface at $S = T_3$ with different first and second event variances. Conversely,

expiries $T_n \leq S_n < T_{n+1}$ with maturities $U_n > S_n$ and the strike-one prices at each S_n identify every v_n , including zero:

$$z_0 := 0, \quad z_n := \frac{4}{(U_n - S_n)^2} \left[\Phi^{-1} \left(\frac{1 + C(S_n, U_n, 1)}{2} \right) \right]^2, \quad v_n = z_n - z_{n-1}. \quad (3.61)$$

Proof. By (3.47), $E[B_S^{-1}] = E[B_S^{-1}P(S, U)] = 1$, so $dQ^S = B_S^{-1}dQ$ is a probability measure on $\mathcal{F}_S$ and the discounted payoff is integrable. Completing the square in each revealed coordinate preserves independence and shifts Z_n to mean $-v_n(S - T_n)$; substitution in the bond formula cancels the meeting-date terms and gives $\log P(S, U) \sim N(-q/2, q)$ under Q^S . Integrating the lognormal positive part proves (3.60), and strict monotonicity of Φ makes the strike-one price injective in q . Equal sums give equal surfaces. At S_n exactly the first n meetings enter q , so (3.61) recovers the cumulative sums and their differences. $\square$

3.14 Exact support with zero variance mean reversion

Specialize the setting of Section 3.12 to $\theta_j = 0$ and constant $\alpha_j \geq 0$, so that the variance states solve

$$dv_j(s) = \alpha_j \sqrt{v_j(s)} dU_j(s), \quad v_j(0) = 1. \quad (3.62)$$

Fix $0 < t < T_N$, retain the future meetings, take $A(t)$ from (3.57) with the full kernel, and put $J := \{j : \alpha_j > 0\}$, $J' := \{j \in J : A_j(t) \neq 0\}$.

Proposition 16. Assume adapted, nonnegative solutions of (3.62) with almost surely continuous paths on a filtered probability space satisfying the usual conditions, driven by Brownian motions with the co-variations (3.53); the second-moment bound of Section 3.12 is not needed. For nonnegative λ_j ,

$$E_Q \left[e^{-\sum_j \lambda_j v_j(t)} \right] = \exp \left(- \sum_j \frac{\lambda_j}{1 + \alpha_j^2 t \lambda_j / 2} \right). \quad (3.63)$$

If $j \notin J$ then $v_j(t) = 1$; if $j \in J$ then, with $c_j = \alpha_j^2 t / 2$, $v_j(t)$ has the law of $\sum_{k=1}^{L_j} E_{j,k}$ with $L_j \sim \text{Poisson}(1/c_j)$ and $E_{j,k} \sim \text{Exponential}(\text{mean } c_j)$, independently across j , an empty sum being zero. The support and affine hull of the meeting-variance vector are

$$\text{supp } V(t) = \sum_{j \notin J} A_j(t) + A_J(t)[0, \infty)^J, \quad \text{aff supp } V(t) = \sum_{j \notin J} A_j(t) + \text{range } A_J(t), \quad (3.64)$$

so the affine-hull dimension is $\text{rank } A_J(t) \leq |J|$ and the affine equalities about the deterministic contribution are exactly $w \in \ker A_J(t)^\top$; moreover

$$\text{Cov}_Q(V(t)) = \sum_{j \in J} \alpha_j^2 t A_j(t) A_j(t)^\top, \quad Q(V(t) = \sum_{j \notin J} A_j(t)) = \exp \left(- \sum_{j \in J'} \frac{2}{\alpha_j^2 t} \right), \quad (3.65)$$

with $\text{rank Cov}_Q(V(t)) = \text{rank } A_J(t)$. For deterministic $0 \leq s < t$ and $J_s := \{j \in J : v_j(s) > 0\}$, almost surely the conditional law of $V(t)$ given $\mathcal{F}_s$ has support $\sum_{j \notin J} A_j(t) + A_{J_s}(t)[0, \infty)^{J_s}$, affine hull $\sum_{j \notin J} A_j(t) + \text{range } A_{J_s}(t)$, mean $A(t)v(s)$, and

$$\text{Cov}_Q(V(t) \mid \mathcal{F}_s) = \sum_{j \in J} \alpha_j^2(t-s) v_j(s) A_j(t) A_j(t)^\top, \quad Q\left(V(t) = \sum_{j \in J} A_j(t) \mid \mathcal{F}_s\right) = \exp\left(- \sum_{j \in J} \frac{2v_j(s)}{\alpha_j^2(t-s)}\right); \quad (3.66)$$

a coordinate that is zero at s stays zero and supplies no later support direction. For the printed constant parameters of [12] Table 2, $\theta = (0, 0, 0)$ and $\alpha = (1.57, 0.82, 0)$, so $\text{supp } V(t) = A_3(t) + \text{cone}\{A_1(t), A_2(t)\}$, whose affine hull has dimension at most two; rank two is not asserted.

Proof. For terminal time t , set $q_j(s) = \lambda_j/[1 + \alpha_j^2 \lambda_j(t-s)/2]$; Itô's formula gives $\exp(-\sum_j q_j(s) v_j(s))$ as its initial value plus stochastic integrals whose drift cancels, and stopping at the hitting times of compact intervals makes the stopped process a bounded martingale, so the exponential is a martingale on $[0, t]$, proving (3.63). The compound Poisson sum has the same Laplace transform, and multivariate Laplace uniqueness, [43] Theorem 1.1, identifies the law; its support is $[0, \infty)$ with an atom at zero and its variance is $\alpha_j^2 t$. The continuous linear map in (3.57) gives (3.64), with [32] Lemma A.3 the general continuous-image statement; nonnegativity of the columns makes the image cone closed. Independence gives (3.65), and repeating the bounded-martingale argument from $v(s)$ gives the regular conditional distribution and (3.66). $\square$

The existence of the solutions of (3.62) is a hypothesis of the statement and is not constructed; the stochastic calculus in the proof is the cited standard one (Section 4), and the short-rate jump whose variance vector $V(t)$ collects is the jump of the forward-rate construction of Section 3.12 with the differentiated HJM drift. At $t = 0$ and at $s = t$ the laws are points; the support can have faces and an atom at its deterministic contribution. Proposition 26 extends the exact-support calculation to nonnegative mean reversion and piecewise-constant volatility of volatility.

3.15 Deterministic price intervals

Use the model and the meeting-separated strike-one calls of Proposition 15, with $h_n = U_n - S_n$, $z_n = \sum_{i \leq n} v_i$ and $C_n(v) = 2\Phi(h_n \sqrt{z_n}/2) - 1$. For deterministic intervals $0 \leq a_n \leq b_n \leq 1$ with $a_n < 1$, define

$$H_n(c) := \frac{4}{h_n^2} \left[\Phi^{-1} \left(\frac{1+c}{2} \right) \right]^2, \quad l_n := H_n(a_n), \quad u_n := H_n(b_n) \text{ if } b_n < 1, \quad u_n := +\infty \text{ if } b_n = 1, \quad (3.67)$$

where $+\infty$ means no finite upper restriction, and set $L_n = \max_{i \leq n} l_i$, $R_n = \min_{j \geq n} u_j$, $L_0 = R_0 = 0$.

Proposition 17. A finite nonnegative vector v has $C_n(v) \in [a_n, b_n]$ for every n if and only if $l_n \leq z_n \leq u_n$ for every n , and such a vector exists if and only if $L_n \leq R_n$ for $n = 1, \dots, N$. When feasible, the attainable values of v_n form the interval with endpoints

$$\underline{v}_n = \max(0, L_n - R_{n-1}) \text{ (or 0 if } R_{n-1} = +\infty), \quad \bar{v}_n = R_n - L_{n-1} \text{ (or } +\infty \text{ if } R_n = +\infty), \quad (3.68)$$

every finite endpoint and intermediate value being attained by a full feasible vector; the coordinate intervals are projections, and choosing their values independently need not preserve the joint constraints. On a price range $[0, \beta_n]$ with $\beta_n < 1$, H_n is Lipschitz with derivative $H'_n(c) = 4y/[h_n^2 \varphi(y)]$, $y = \Phi^{-1}((1+c)/2)$, φ the standard normal density, so two price vectors in these ranges differing by at most ε_n coordinatewise give

$$|v_n - v'_n| \leq k_n \varepsilon_n + k_{n-1} \varepsilon_{n-1}, \quad k_n = \frac{4y_n}{h_n^2 \varphi(y_n)}, \quad k_0 = \varepsilon_0 = 0, \quad (3.69)$$

and there is no finite Lipschitz constant up to price one.

Proof. Strict monotonicity of C_n makes each price interval equivalent to $l_n \leq z_n \leq u_n$. Since the cumulative sums are nondecreasing, every feasible sequence satisfies $L_n \leq z_n \leq R_n$; conversely $z_n = L_n$ is finite, nondecreasing and feasible exactly when $L_n \leq R_n$. The differences $z_n - z_{n-1}$ give (3.68): $z_{n-1} = \min(L_n, R_{n-1})$ attains the lower endpoint, $z_n = R_n$, or any large finite value when $R_n = +\infty$, the upper one, and convexity of the constraints fills the intermediate values. Differentiating H_n and using that $y/\varphi(y)$ increases for $y \geq 0$, the mean value theorem gives (3.69); H_n and its derivative diverge as $c \uparrow 1$. $\square$

3.16 Identification from compounded-rate futures and calls

In the model of Section 3.10 at time zero, for $0 \leq a < b$, $\delta = b - a$, define the ideal futures rate for the continuously compounded backward-looking rate, with the undiscounted conditional-expectation convention of [54] (13), (19) and (20), and the European cash-call price with expiry $0 \leq S \leq b$ and strike $K > 0$:

$$G_t(a, b) := E_Q \left[\exp \left(\int_a^b r_u du \right) \middle| \mathcal{F}_t \right], \quad F(t; a, b) := \frac{G_t(a, b) - 1}{\delta}, \quad (3.70)$$

$$C(S, a, b, K) := E_Q[B_S^{-1}(G_S(a, b) - K)^+].$$

Dividing C by δ gives the price of a call on $F(S; a, b)$ with strike $(K - 1)/\delta$. For each meeting set

$$\begin{aligned} w_i &:= (b - T_i)^+ - (a - T_i)^+, & d_i &:= \frac{1}{2}((b - T_i)^{+2} - (a - T_i)^{+2}), & h_i &:= d_i + \frac{1}{2}w_i^2, \\ j_i &:= \mathbf{1}\{T_i \leq S\}w_i(S - T_i), & k_i &:= \mathbf{1}\{T_i \leq S\}w_i^2, \end{aligned} \quad (3.71)$$

and $p := \sum_i h_i v_i$, $z := \sum_i j_i v_i$, $q := \sum_i k_i v_i$, $m := e^{p-z}$. Write, for $m > 0$, $s \geq 0$ and $K > 0$,

$$c_K(m, s) := m\Phi(D_1) - K\Phi(D_2), \quad D_1 = \frac{\log(m/K) + s^2/2}{s}, \quad D_2 = D_1 - s \quad (s > 0); \quad (3.72)$$

$$c_K(m, 0) := (m - K)^+.$$

The two-strike argument is the following lemma.

Lemma 18. For any fixed $K_* > 1$, the map $(m, s) \mapsto (c_1(m, s), c_{K_*}(m, s))$ is injective on $\{m \geq 1, s \geq 0\}$.

Proof. For $s > 0$,

$$\frac{\partial c_K}{\partial m} = \Phi(D_1) > 0, \quad \frac{\partial c_K}{\partial s} = m\varphi(D_1) > 0, \quad \left(\frac{\varphi}{\Phi}\right)'(x) < 0. \quad (3.73)$$

Along a level set $c_1(m, s) = \text{const} > 0$, implicit differentiation with the last inequality shows that c_{K_*} is strictly increasing in s , so the pair determines (m, s) on $\{c_1 > 0\}$. A zero strike-one price forces $m = 1$ and $s = 0$, since $c_1(m, s) \geq (m - 1)^+$ with equality only at $s = 0$; continuity at $s = 0$ handles the remaining boundary. $\square$

Proposition 19. (a) $G(a, b)$ is an integrable Q -martingale,

$$G_t(a, b) = \exp\left(\sum_{T_i \leq t} w_i Z_i + \sum_i d_i v_i + \frac{1}{2} \sum_{T_i > t} w_i^2 v_i\right), \quad 1 + \delta F(0; a, b) = e^p, \quad (3.74)$$

$\log G_S \sim N(\log m - q/2, q)$ under $dQ^S = B_S^{-1} dQ$, and $C(S, a, b, K) = c_K(m, \sqrt{q})$. The complete surface $K > 0$ determines (m, q) , since $m = \lim_{K \downarrow 0} C(S, a, b, K)$ and $q = 4[\Phi^{-1}((1 + C(S, a, b, m)/m)/2)]^2$; with the futures quote it determines (p, z, q) , and by Lemma 18 the two calls at $K = 1$ and any fixed $K_* > 1$ do the same, including at $q = 0$.

(b) Fix L triples (S_ℓ, a_ℓ, b_ℓ) and let M be the $3L$ -by- N matrix whose rows for contract ℓ are $h^{(\ell)}, j^{(\ell)}, k^{(\ell)}$. For $v, v' \in [0, \infty)^N$, equality of the L futures quotes and either the L complete surfaces or the $2L$ fixed-strike calls is equivalent to $Mv = Mv'$. The panel identifies every v if and only if $\text{rank } M = N$; if $\text{rank } M < N$, every strictly positive v has a distinct strictly positive v' with the same observations, and the set compatible with a model-generated panel is $\{x \in [0, \infty)^N : Mx = Mv\}$.

(c) If one accrual interval lies after all meetings and $T_n \leq S_n < T_{n+1}$, with $S_N = T_N$, then $q_n = \delta^2 \sum_{i \leq n} v_i$, so $v_n = (q_n - q_{n-1})/\delta^2$ with $q_0 = 0$. Futures alone also identify: with $a_n = T_n$, $T_n < b_n < T_{n+1}$ for $n < N$, $b_N > T_N$, and $\delta_n = b_n - T_n$,

$$v_n = \frac{p_n - \delta_n \sum_{i < n} (b_n - T_i) v_i}{\delta_n^2}, \quad p_n = \log(1 + \delta_n F(0; T_n, b_n)). \quad (3.75)$$

(d) Full later-window panels need not identify: for $N = 3$, $T_i = i$ and $\varepsilon > 0$, the vectors $v = \varepsilon(2, 3, 2)$ and $v' = \varepsilon(3, 1, 3)$ give the same futures quote for every $3 \leq a < b$ and the same call price for every $3 \leq S \leq a < b$ and $K > 0$, although their event variances differ.

Proof. Integrating the short rate in (3.45) gives $\int_a^b r_u du = \sum_i (w_i Z_i + d_i v_i)$; independence and the Gaussian exponential moment give (3.74), also during accrual, and the tower property the martingale assertion. The discount factor tilts every revealed coordinate to mean $-v_i(S - T_i)$ without changing variances or independence, so $\log G_S$ has the displayed Q^S -law with first moment m , which gives integrability and the price; monotone convergence as $K \downarrow 0$ recovers m , and at $K = m$ normal symmetry gives $C/m = 2\Phi(\sqrt{q}/2) - 1$. Since $h_i - j_i \geq 0$, $m \geq 1$, so Lemma 18 applies. Each contract thus identifies (p, z, q) , three linear forms in v , which is (b); full column rank gives injectivity, and under rank deficiency a kernel vector gives a small two-sided perturbation of every strictly positive v . For (c), a post-meeting accrual interval has $w_i = \delta$, so the rows of k at successive expiries are lower triangular; with $a_n = T_n$ the futures coefficient vanishes for $i > n$ and equals $\delta_n(b_n - T_i)$ for $i \leq n$, giving (3.75). For (d), every triple's coefficients depend only on $\sum_i v_i$ and $\sum_i T_i v_i$, which are 7ε and 14ε for both vectors. $\square$

3.17 Contracts settled after a date

Replace the continuously compounded convention of Section 3.16 by exact discrete bond-implied compounding and the European option by an American cash call. Fix $A \leq S \leq a < b$, a partition $a = u_0 < \dots < u_J = b$, $d_j = u_{j+1} - u_j$, and define the bond-implied simple overnight rates, the exact compounded rate, its futures process and the cash call

$$L_j = \frac{P(u_j, u_{j+1})^{-1} - 1}{d_j}, \quad R = \frac{\prod_{j=0}^{J-1} (1 + d_j L_j) - 1}{\delta}, \quad F(t; a, b) = E_Q[R \mid \mathcal{F}_t], \quad (3.76)$$

$$U(A, S, a, b, K) = \sup_{\tau \in \mathcal{T}[A, S]} E_Q[B_\tau^{-1}(F(\tau; a, b) - K)^+],$$

with $\delta = b - a$ and $\mathcal{T}[A, S]$ the stopping times valued in $[A, S]$; this is the exact product convention of [54], and exercise pays cash rather than entering a futures position.

The following result shows, for every contract settled after a fixed date and with any number of unrevealed meetings, that the revealed variances enter only through their sum and their date-weighted sum, and that the compatible set is a polytope with explicit sharp coordinate bounds. Corollaries 21 and 22 specialize it to American cash exercise after, and across, the last meeting.

In the model of Section 3.10 fix a deterministic date $A \geq 0$, let $k = j(A)$ be the number of meetings at or before A , and write

$$V_k = \sum_{i \leq k} v_i, \quad H_k = \sum_{i \leq k} T_i v_i, \quad h_i = A - T_i \ (i \leq k), \quad y = r_A, \quad Z_{>k} = (Z_{k+1}, \dots, Z_N). \quad (3.77)$$

Proposition 20. (a) For $t \geq A$ and $U \geq t$,

$$f(t, U) = y + V_k(U - A) + \sum_{k < i \leq j(t)} [Z_i + v_i(U - T_i)], \quad \frac{B_t}{B_A} = \exp\left(\int_A^t r_s ds\right), \quad (3.78)$$

and $Z_i = r_{T_i} - r_{T_{i-}}$ for $i > k$. Every forward rate, short rate, bond price, bank-account ratio and bond-implied fixing at or after A is therefore a fixed Borel function of the state $(y, Z_{>k})$, of $(V_k, v_{k+1}, \dots, v_N)$, and of the dates $T_{k+1}, \dots, T_N$ and A ; the individual $v_1, \dots, v_k$ and $T_1, \dots, T_k$ enter only through V_k . Conversely, for $t \geq A$, $\sigma(f(s, \cdot) : A \leq s \leq t) = \sigma(y, Z_i : k < i \leq j(t))$.

(b) $B_A^{-1} = \exp(-\sum_{i \leq k} [h_i Z_i + v_i h_i^2 / 2])$ has $E_Q[B_A^{-1}] = 1$. Under $\tilde{Q} := B_A^{-1} \cdot Q$ the coordinates stay independent, $Z_i \sim N(-v_i h_i, v_i)$ for $i \leq k$ and $Z_i \sim N(0, v_i)$ for $i > k$, so $y \sim N(0, V_k)$ is independent of $Z_{>k}$, and for every Borel $\Phi \geq 0$

$$E_Q[B_A^{-1} \Phi(y, Z_{>k})] = E[\Phi(Y, Z')], \quad Y \sim N(0, V_k), \quad Z'_i \sim N(0, v_i) \ (i > k), \quad \text{all independent}, \quad (3.79)$$

and likewise for real Φ with $E_Q[B_A^{-1} |\Phi|]$ finite. If $V_k > 0$, the residuals $\xi_i := Z_i + v_i h_i - (v_i / V_k) y$, $i \leq k$, are under $\tilde{Q}$ jointly Gaussian and independent of $(y, Z_{>k})$, and $\sigma(Z_1, \dots, Z_k) = \sigma(y, \xi_1, \dots, \xi_k)$; if $V_k = 0$ then $Z_1 = \dots = Z_k = 0$ and $y = 0$ almost surely.

(c) Let $S \geq A$ be deterministic and $X = \Phi(y, Z_{k+1}, \dots, Z_{j(S)})$ a cash flow paid at S , with Φ Borel and allowed to depend on $(V_k, v_{k+1}, \dots, v_N)$ and the contract data. If $E_Q[B_S^{-1} |X|] < \infty$, then

$$E_Q[B_S^{-1} X] = E\left[\exp\left(-\int_A^S \tilde{r}_s ds\right) \Phi(Y, Z')\right], \quad (3.80)$$

with $\tilde{r}$ the short rate of (3.78) evaluated at (Y, Z') ; the right side depends on the revealed variances only through V_k . This covers every European payoff on bonds, bond-implied fixings and compounded rates over accrual windows $[a, b]$ with $a \geq A$, windows crossing unrevealed meetings included, and every partition.

(d) Let $A \leq S$ and let $\Phi : [A, S] \times \mathbb{R} \times \mathbb{R}^{N-k} \rightarrow \mathbb{R}$ be Borel with $\Phi(t, y, z)$ depending on z only through the coordinates $i \leq j(t)$, so that $X_t := \Phi(t, y, Z_{>k})$ is a function of the curve on $[A, t]$, and assume a Borel $D \geq 0$ with $|\exp(-\int_A^t r_s ds) \Phi(t, y, Z_{>k})| \leq D(y, Z_{>k})$ for $t \in [A, S]$ and $E[D(Y, Z')] < \infty$. Then the time-zero American value

$$U := \sup_{\tau \in \mathcal{T}[A,S]} E_Q[B_\tau^{-1} X_\tau] = \sup_{\rho \in \tilde{\mathcal{T}}[A,S]} E\left[\exp\left(-\int_A^\rho \tilde{r}_s ds\right) \Phi(\rho, Y, Z')\right] \quad (3.81)$$

is finite, where $\tilde{\mathcal{T}}[A, S]$ are the stopping times valued in $[A, S]$ of the filtration generated by Y and by the Z'_i revealed at T_i . Hence U depends on $(v_1, \dots, v_k)$ only through V_k . Every cash call (3.76), in particular those of Corollaries 21 and 22, with any exercise window $[A, S]$ crossing any number of unrevealed meetings and any accrual window $[a, b]$ with $a \geq A$, satisfies the domination and is included; no formula for the value is asserted.

(e) For an accrual window $a < b$ with $a \geq T_k$, a partition $a = u_0 < \dots < u_J = b$ with no meeting strictly inside a fixing interval, and the exact compounding of (3.76), $\prod_j (1 + d_j L_j) = \exp(\int_a^b r_s ds)$ and

$$F(0; a, b) = \frac{1}{\delta} \left[\exp\left(\delta(bV_k - H_k) + \delta \sum_{k < i, T_i \leq a} v_i(b - T_i) + \sum_{a < T_i \leq b} v_i(b - T_i)^2\right) - 1 \right], \quad (3.82)$$

so the time-zero quote depends on $(v_1, \dots, v_k)$ only through (V_k, H_k) , also for windows that cross unrevealed meetings at partition points.

(f) Fix $v_{k+1}, \dots, v_N$. Two vectors $(v_1, \dots, v_k)$ and $(v'_1, \dots, v'_k)$ with the same V_k and H_k give the same quote (3.82) for every window and partition in (e), the same price (3.80) for every contract in (c) and the same value (3.81) for every contract in (d). If the observation family contains two quotes (3.82) with $a \geq T_k$ and distinct b , then conversely equal observations force equal (V_k, H_k) , so the set of $(v_1, \dots, v_k)$ consistent with exact observation of the family is the polytope

$$\Pi(V_k, H_k) = \left\{ x \in [0, \infty)^k : \sum_i x_i = V_k, \sum_i T_i x_i = H_k \right\}, \quad (3.83)$$

nonempty if and only if $T_1 V_k \leq H_k \leq T_k V_k$, over which each coordinate attains its minimum and maximum at a point with at most two nonzero coordinates. For $k = 3$, with $V = V_3$ and $H = H_3$, the sharp attained bounds are

$$\begin{aligned} s_* &= \min\left\{ \frac{H - T_1 V}{T_2 - T_1}, \frac{T_3 V - H}{T_3 - T_2} \right\} \geq 0, & x_2 &\in [0, s_*], \\ x_1 &\in \left[\frac{T_3 V - H - (T_3 - T_2)s_*}{T_3 - T_1}, \frac{T_3 V - H}{T_3 - T_1} \right], & x_3 &\in \left[\frac{H - T_1 V - (T_2 - T_1)s_*}{T_3 - T_1}, \frac{H - T_1 V}{T_3 - T_1} \right]. \end{aligned} \quad (3.84)$$

For dates $(1, 2, 3)$ and every $\varepsilon > 0$ the vectors $\varepsilon(2, 3, 2)$ and $\varepsilon(3, 1, 3)$, completed by any common $v_4, \dots, v_N$, are two such points, with bounds $x_1, x_3 \in [0, 7\varepsilon/2]$ and $x_2 \in [0, 7\varepsilon]$.

Proof. (a) For $t \geq A$, $j(t) \geq k$, $y = \sum_{i \leq k} [Z_i + v_i h_i]$ and $\sum_{i \leq k} v_i (U - T_i) = \sum_{i \leq k} v_i h_i + V_k (U - A)$, which gives (3.78); the path of r on $[A, t]$ is right-continuous, piecewise affine and jumps by Z_i at T_i , which is the jump identity. For the sigma-algebras, $y = f(A, A)$, each Z_i with $A < T_i \leq t$ equals $f(T_i, T_i) - f(\max(A, T_{i-1}), T_i)$, and each $f(s, U)$ with $A \leq s \leq t$ is a function of $(y, Z_i : k < i \leq j(s))$ by (3.78).

(b) Only $i \leq k$ enter $\int_0^A r_s ds = \sum_{i \leq k} [h_i Z_i + v_i h_i^2/2]$. The density is a product of unit-mean factors, and completing the square tilts the earlier coordinates and leaves the later ones, which gives the laws, the independence and (3.79). The vector (y, ξ) is a linear image of a Gaussian vector with $\text{Cov}_{\tilde{Q}}(\xi_i, y) = v_i - (v_i/V_k)V_k = 0$, so ξ is independent of y and, being a function of $Z_{\leq k}$, of $Z_{>k}$; $Z_i = \xi_i + (v_i/V_k)y - v_i h_i$ gives the sigma-algebra identity, and $V_k = 0$ forces every earlier variance to vanish.

(c) $B_S^{-1} = B_A^{-1}(B_S/B_A)^{-1}$, and $(B_S/B_A)^{-1}X$ is a Borel function of $(y, Z_{>k})$ by (3.78); apply (3.79) to its absolute value and to itself. Fixings are ratios of bond prices at times at or after A , and the compounded rate is a finite product of them.

(d) For $\tau \in \mathcal{T}[A, S]$, $B_\tau^{-1}X_\tau = B_A^{-1}Y_\tau$ with $Y_t := (B_t/B_A)^{-1}\Phi(t, y, Z_{>k})$ and $|Y_\tau| \leq D(y, Z_{>k})$, so $U = \sup_\tau E_{\tilde{Q}}[Y_\tau]$ is finite by (3.79). For $t \geq A$ the completed filtration is $\mathcal{F}_t = \sigma(y, \xi, Z_i : k < i \leq j(t)) \vee \mathcal{N}$, with y, ξ and $Z_{>k}$ independent under $\tilde{Q}$, and Y is adapted to $\mathcal{G}_t := \sigma(y, Z_i : k < i \leq j(t)) \vee \mathcal{N}$; both filtrations are right-continuous because j is right-continuous with finitely many jumps. Realize $\tilde{Q}$ on the product of the laws of y, ξ and $Z_{>k}$. Given τ , for each rational $q \in [A, S]$ and for $q = S$ there is $C_q \in \sigma(y, \xi, Z_i : k < i \leq j(q))$ with $\tilde{Q}(\{\tau \leq q\} \triangle C_q) = 0$; by Fubini, for ξ outside a null set every section τ_ξ has $\{\tau_\xi \leq q\} \in \mathcal{G}_q$, and right-continuity extends this to every t , so τ_ξ is a $\mathcal{G}$ -stopping time valued in $[A, S]$, a rule on the auxiliary space. By Fubini,

$$E_{\tilde{Q}}[Y_\tau] = \int E[Y_{\tau_\xi}] (\text{law of } \xi)(d\xi) \leq \sup_{\rho \in \tilde{\mathcal{T}}[A, S]} E\left[\exp\left(-\int_A^\rho \tilde{r}_s ds\right) \Phi(\rho, Y, Z')\right], \quad (3.85)$$

and conversely every $\rho \in \tilde{\mathcal{T}}[A, S]$ is an $\mathcal{F}$ -stopping time on the model with $E_{\tilde{Q}}[Y_\rho]$ equal to the auxiliary expectation by (3.79); when $V_k = 0$ the residuals are dropped and τ is almost surely its own section. The auxiliary problem is specified by $(V_k, v_{k+1}, \dots, v_N)$, the later dates, A, S and Φ . For the cash calls (3.76), R is a Borel function of $(y, Z_{>k})$ by (a) and the telescoping in (e) on each fixing interval, $F(t; a, b)$ has the version $E_Q[R | \mathcal{G}_t]$ because ξ is independent of $(y, Z_{>k})$ under Q as well, and the bound $\max_{0 \leq u \leq d} e^{-zu - qu^2/2} \leq e^{d|z|}$ for the discount factors gives $D = C \exp(c(|y| + \sum_{i>k} |Z_i|))$.

(e) On a fixing interval with no meeting inside, $r_u = f(u_j, u)$, so $P(u_j, u_{j+1})^{-1} = B_{u_{j+1}}/B_{u_j}$ and the product telescopes. With $m = (a + b)/2$, $\int_a^b r_s ds = \sum_{T_i \leq a} [\delta Z_i + v_i \delta(m - T_i)] + \sum_{a < T_i \leq b} [(b - T_i)Z_i + v_i(b - T_i)^2/2]$, and the Gaussian exponential moment with $m + \delta/2 = b$ gives (3.82); since $a \geq T_k$, the $i \leq k$ part of the first sum is $\delta(bV_k - H_k)$.

(f) The first sentence collects (c), (d) and (e). Two quotes with distinct b, b' give $bV_k - H_k$ and $b'V_k - H_k$, hence (V_k, H_k) . The polytope is the nonnegative orthant cut by two affine equations; it is nonempty if and only if (V_k, H_k) lies in the cone generated by the $(1, T_i)$, and a linear function attains its extrema at vertices, which have at most two nonzero coordinates because the constraint matrix has two rows. For $k = 3$, eliminating x_1 and x_3 gives $x_1 = (T_3V - H - (T_3 - T_2)x_2)/(T_3 - T_1)$ and $x_3 = (H - T_1V - (T_2 - T_1)x_2)/(T_3 - T_1)$, both decreasing in x_2 , and nonnegativity of all three coordinates is exactly $0 \leq x_2 \leq s_*$; both endpoints are attained. For dates $(1, 2, 3)$, $V = 7\varepsilon$ and $H = 14\varepsilon$ give $s_* = 7\varepsilon$. $\square$

Specialized to $k = N$ and $k = N - 1$, (d) gives the dependence through V in Corollary 21 and through (V_-, w) in Corollary 22, whose value formulas follow. The randomization step in (d) is the classical fact that a stopping time of a filtration enlarged by an independent variable is a randomized stopping time in the sense of [5]; (3.85) proves the value inequality directly. The proposition says nothing about the unrevealed variances $v_{k+1}, \dots, v_N$, jointly or individually; when they also vary, the equal-observation set is larger and is not described; and a fixing interval with a meeting strictly inside is excluded from (e) while its fixing remains covered by (c).

3.18 Late American cash exercise after the last meeting

Take $T_N \leq A \leq S \leq a < b$ in (3.76). Write $V = \sum_{i=1}^N v_i$ and $H = \sum_{i=1}^N T_i v_i$. After the last meeting, with $X = \sum_i Z_i$, (3.45) gives $r_t = X + tV - H$ and $f(t, U) = X + UV - H$, so $1 + d_j L_j = P(u_j, u_{j+1})^{-1} = B_{u_{j+1}}/B_{u_j}$, the product telescopes, and by the Gaussian exponential moment

$$\prod_{j=0}^{J-1} (1 + d_j L_j) = \exp\left(\int_a^b r_s ds\right), \quad F(0; a, b) = \frac{\exp\{\delta(bV - H)\} - 1}{\delta}. \quad (3.86)$$

For $L = S - A$ put $D_{V,L}(x) := \max_{0 \leq u \leq L} e^{-xu - Vu^2/2}$, which equals 1 for $x \geq 0$, $e^{x^2/(2V)}$ for $-VL < x < 0$ and $e^{-xL - VL^2/2}$ for $x \leq -VL$ when $V > 0$.

Corollary 21. Since every Z_i is revealed by A , R is $\mathcal{F}_A$ -measurable and $F(t; a, b) = R = [\exp\{\delta r_A + \delta V((a+b)/2 - A)\} - 1]/\delta$ for $t \in [A, b]$. If $V > 0$, the time $\tau_* = A + \min\{L, \max\{0, -r_A/V\}\}$ is optimal and

$$U(A, S, a, b, K) = E_{Y \sim N(0, V)} \left[\left(\frac{e^{\delta Y + \delta V((a+b)/2 - A)} - 1}{\delta} - K \right)^+ D_{V,L}(Y) \right], \quad (3.87)$$

so every late American cash value depends on the meeting variances only through V ; if $A < S$ it strictly exceeds the value under mandatory exercise at S ; if $V = 0$ then $F = 0$, $B = 1$, $U = (-K)^+$ and A is optimal. For $N = 3$, $T_i = i$ and $\varepsilon > 0$, the vectors $v = \varepsilon(2, 3, 2)$ and $v' = \varepsilon(3, 1, 3)$ give every futures quote (3.86) with $a \geq 3$ and every value (3.87) with $3 \leq A \leq S \leq a < b$, for every partition and strike, although their event variances differ.

Proof sketch. The payoff is constant along each path over the exercise interval, while $B_{A+u}/B_A = \exp\{r_A u + Vu^2/2\}$ for $0 \leq u \leq L$; minimizing the convex quadratic over $[0, L]$ gives τ_* and $D_{V,L}$, τ_* is $\mathcal{F}_A$ -measurable and hence a stopping time, and pathwise domination shows it attains the supremum. Under $dQ^A = B_A^{-1}dQ$, $r_A \sim N(0, V)$, which gives (3.87), and $D_{V,L}(x) \leq e^{L|x|}$ gives integrability. A Gaussian set of positive probability has an interior optimizer when $A < S$, which is the strict premium; $V = 0$ is direct. Both vectors have $V = 7\varepsilon$ and $H = 14\varepsilon$, and the dependence through V alone is Proposition 20(d) with $k = N$. Appendix A records the formal proof. $\square$

3.19 American cash exercise across one unrevealed meeting

Keep the contract (3.76) and move the exercise window across the last meeting: fix

$$T_{N-1} \leq A < T_N < S \leq a < b, \quad (3.88)$$

so that exactly one meeting is unrevealed when exercise opens and falls strictly inside the window, while accrual still follows the last meeting; the partition, strike and dates are fixed before models are compared. Write $T = T_N$, $\ell = T - A$, $L = S - T$, $m = (a+b)/2$, $V_- = \sum_{i < N} v_i$, $w = v_N$, so $V = V_- + w$, and

$$f_-(x) = \frac{\exp\{\delta[x + V_-(m - A) + w(b - T)]\} - 1}{\delta}, \quad f_+(y) = \frac{\exp\{\delta[y + V(m - T)]\} - 1}{\delta}. \quad (3.89)$$

For $q \geq 0$ and $d \geq 0$ let $u_{q,d}(z) = \min\{d, \max\{0, -z/q\}\}$ if $q > 0$, and $u_{0,d}(z) = 0$ for $z \geq 0$, $u_{0,d}(z) = d$ for $z < 0$; then

$$D_{q,d}(z) := \exp\{-zu_{q,d}(z) - qu_{q,d}(z)^2/2\} = \max_{0 \leq u \leq d} e^{-zu - qu^2/2}, \quad (3.90)$$

which extends the discount maximizer of Section 3.18 to $q = 0$. Put

$$p(x) = (f_-(x) - K)^+ D_{V_-, \ell}(x), \quad c(x) = e^{-x\ell - V_- \ell^2/2} E_{Z \sim N(0, w)}[(f_+(x + V_- \ell + Z) - K)^+ D_{V_-, L}(x + V_- \ell + Z)]. \quad (3.91)$$

Corollary 22. With the dates (3.88), exact compounding and the initial quote are as in (3.86), and there is a simultaneous version of the futures process on the exercise window with

$$F(t; a, b) = f_-(r_A) \text{ for } A \leq t < T, \quad F(t; a, b) = f_+(r_T) \text{ for } T \leq t \leq S, \quad (3.92)$$

so the underlying jumps at T . The American cash value is finite and equals

$$U(A, S, a, b, K) = E_{Y \sim N(0, V_-)}[\max\{p(Y), c(Y)\}]; \quad (3.93)$$

for fixed contract data it depends on the variance vector only through (V_-, w) , while the futures quote also depends on H . The rule that exercises at $A + u_{V_-, \ell}(r_A)$ on $E = \{p(r_A) > c(r_A)\}$ and otherwise waits for the revelation at T and exercises at $T + u_{V_-, L}(r_T)$ is a stopping time valued in $[A, S]$ attaining the supremum; on E its time is strictly less than T , ties wait, and no decision before T uses Z_N . For $N = 4$, $T_i = i$ and $\varepsilon > 0$, the strictly positive vectors

$$v = \varepsilon(2, 3, 2, 1), \quad v' = \varepsilon(3, 1, 3, 1) \quad (3.94)$$

give every futures quote with $4 \leq a < b$ and every value (3.93) with $3 \leq A < 4 < S \leq a < b$, for every partition and strike, although their first three event variances differ; the crossed variance is the same in the pair.

Proof sketch. Set $x = r_A$. For $0 \leq u < \ell$, $r_{A+u} = x + V_- u$; at T , $y := r_T = x + V_- \ell + Z_N$; for $0 \leq u \leq L$, $r_{T+u} = y + V u$; and $R = f_+(y)$. Since Z_N is independent of $\mathcal{F}_A$, the Gaussian exponential moment gives

$$E_Q[f_+(x + V_- \ell + Z_N) \mid \mathcal{F}_A] = f_-(x), \quad (3.95)$$

and since the filtration is constant on $[A, T)$ and after T , (3.92) is a simultaneous adapted version of the futures process. Minimizing $zu + qu^2/2$ over $[0, d]$ gives (3.90). The bank-account ratios are

$$\frac{B_{A+u}}{B_A} = e^{xu + V_- u^2/2}, \quad 0 \leq u \leq \ell, \quad \frac{B_{T+u}}{B_T} = e^{yu + V u^2/2}, \quad 0 \leq u \leq L, \quad (3.96)$$

so, measured at A , every payoff before T is at most $p(x)$, and the largest discounted payoff after T has conditional expectation $c(x)$. Since $D_{V_-, L} \geq 1$, (3.95) and convexity of the positive part give

$$c(x) \geq e^{-x\ell - V_- \ell^2/2} (f_-(x) - K)^+, \quad (3.97)$$

whose right side is $p(x)$ when $u_{V_-, \ell}(x) = \ell$; so $p(x) > c(x)$ forces $u_{V_-, \ell}(x) < \ell$, and early exercise, when strictly preferable, is attained at an allowed time before T . For any $\tau \in \mathcal{T}[A, S]$, $\{\tau < T\} = \bigcup_{q \in \mathbb{Q} \cap [A, T)} \{\tau \leq q\}$ belongs to $\mathcal{F}_A$, because the filtration is constant on $[A, T)$; so conditioning on $\mathcal{F}_A$ bounds every admissible payoff by $\max\{p(x), c(x)\}$. The stated rule attains the bound, and it is a stopping time: E and its early exercise time are $\mathcal{F}_A$ -measurable and its late exercise time is $\mathcal{F}_T$ -measurable, so each event $\{\tau \leq t\}$ lies in $\mathcal{F}_t$. The value is therefore $E_Q[B_A^{-1} \max\{p(r_A), c(r_A)\}]$. With $h_i = A - T_i \geq 0$,

$$B_A^{-1} = \exp\left\{-\sum_{i < N} h_i Z_i - \frac{1}{2} \sum_{i < N} v_i h_i^2\right\}, \quad E_Q[B_A^{-1}] = 1, \quad (3.98)$$

and under the measure with this density the earlier Z_i are independent $N(-v_i h_i, v_i)$, so $r_A \sim N(0, V_-)$, which is (3.93); the bounds $1 \leq D_{q,d}(z) \leq e^{d|z|}$ give finiteness. For (3.94) both vectors have $V_- = 7\varepsilon$, $w = \varepsilon$, $V = 8\varepsilon$ and $H = 18\varepsilon$; the dependence through (V_-, w) alone is Proposition 20(d) with $k = N - 1$. Appendix A records the formal proof, including the measurability and integrability steps. $\square$

Before T the holder knows every earlier shock and faces one revelation; the futures rate is constant until then while discounting evolves deterministically, so the choice at A is between exercising before T and waiting, the latter valued by a Gaussian expectation. Discounting to time zero centres the law of r_A and removes the separate dependence of the cash value on the dates of the revealed shocks, which is what (3.94) exploits while keeping V_- , w and H fixed. The proposition says nothing about the crossed variance w on its own, gives no necessary and sufficient equality condition for arbitrary parameter pairs, and does not cover windows opening before T_{N-1} , exercise across two unrevealed meetings or accrual spanning a meeting.

3.20 Futures-style exercise

The cash calls of Sections 3.17 to 3.19 pay cash at the exercise date and are discounted. Under futures-style margining the premium is paid at expiry and variation margin is exchanged daily without remuneration, so the option price is the undiscounted conditional expectation of the payoff on the futures price, [48] §2.2 eq. 3, the same convention as the futures rate itself in [54] (20); [38] footnote 2 leaves the discounting of cash-settled SOFR futures options open. That a futures-style option is never exercised early is known: [44] Properties B and C, pp. 330 to 331, prove it in the Black lognormal futures model with fixed rates, and [15] inequalities (1) and (2), pp. 15 to 16, by Jensen's inequality under the undiscounted risk-adjusted expectation with stochastic rates. Part (b) below is the scheduled-jump instance of that result, proved directly on the piecewise-constant filtration, which is in neither source; the identification content of parts (c) to (e) is new. Nothing is asserted about which convention a listed contract follows.

In the model of Section 3.10 fix a window $a < b$, $\delta = b - a$, $m = (a + b)/2$, a partition $a = u_0 < \dots < u_J = b$ with no meeting strictly inside a fixing interval, the exact compounding (3.76), and the coefficients w_i , d_i of (3.71). Fix deterministic $0 \leq A \leq S \leq b$ and a real strike K , and define the futures-style American value

$$\tilde{U}(A, S, a, b, K) := \sup_{\tau \in \mathcal{T}[A, S]} E_Q[(F(\tau; a, b) - K)^+], \quad (3.99)$$

with no discount factor.

Proposition 23. (a) The product telescopes, $\int_a^b r_s ds = \sum_i (w_i Z_i + d_i v_i)$, and for $0 \leq t \leq b$

$$F(t; a, b) = \frac{1}{\delta} \left[\exp\left(\sum_{T_i \leq t} w_i Z_i + \sum_i d_i v_i + \frac{1}{2} \sum_{T_i > t} w_i^2 v_i\right) - 1 \right] \quad (3.100)$$

is a version of $E_Q[R \mid \mathcal{F}_t]$ that is constant on each $[T_n, T_{n+1}) \cap [0, b]$, measurable at $T_{j(t)}$, square integrable and a martingale, with $1 + \delta F(0; a, b) = E_Q[1 + \delta F(t; a, b)]$.

(b) For every $\tau \in \mathcal{T}[A, S]$, $E_Q[(F(\tau; a, b) - K)^+] \leq E_Q[(F(S; a, b) - K)^+]$; exercise at S is optimal and

$$\tilde{U}(A, S, a, b, K) = E_Q[(F(S; a, b) - K)^+], \quad (3.101)$$

for every $A \leq S$, including windows crossing any number of unrevealed meetings. This is the scheduled-jump instance of the known result of [44] Properties B and C and [15] inequalities (1) and (2), whose mechanism, a martingale futures price and a convex payoff, is the same; it is proved here by the discrete optional sampling theorem over meeting indices on the scheduled filtration. By contrast, the cash value of Corollary 21 strictly exceeds its mandatory-exercise value whenever $V > 0$ and $A < S$.

(c) With $n = j(S)$, $\sigma_S^2 = \sum_{i \leq n} w_i^2 v_i$, $\mu_S = \sum_i d_i v_i + \frac{1}{2} \sum_{i > n} w_i^2 v_i$, $G = e^{\mu_S + \sigma_S^2/2} = 1 + \delta F(0; a, b)$ and $K' = 1 + \delta K$, the value is $c_{K'}(G, \sigma_S)/\delta$ in the notation (3.72):

$$\begin{aligned} \tilde{U} &= \frac{1}{\delta} [G\Phi(d_1) - K'\Phi(d_2)], \quad d_{1,2} = \frac{\log(G/K') \pm \sigma_S^2/2}{\sigma_S} \quad (\sigma_S > 0, K' > 0); \\ \tilde{U} &= \frac{G - K'}{\delta} \quad (K' \leq 0); \quad \tilde{U} = \frac{(e^{\mu_S} - K')^+}{\delta} \quad (\sigma_S = 0). \end{aligned} \quad (3.102)$$

It is finite, continuous and nonincreasing in K .

(d) Let $k = j(A)$, suppose $a \geq T_k$, and take V_k, H_k from (3.77). Then $w_i = \delta$ and $d_i = \delta(m - T_i)$ for $i \leq k$, so

$$\sigma_S^2 = \delta^2 V_k + \sum_{k < i \leq n} w_i^2 v_i, \quad \mu_S = \delta(mV_k - H_k) + \sum_{i > k} d_i v_i + \frac{1}{2} \sum_{i > n} w_i^2 v_i, \quad (3.103)$$

and every value (3.101), every futures-style European value and the quote $F(0; a, b)$ depend on $(v_1, \dots, v_k)$ only through (V_k, H_k) , while variances with $T_i > b$ do not enter. Conversely, fix $v_{k+1}, \dots, v_N$. Two quotes with $a \geq T_k$ and distinct b identify (V_k, H_k) . For a window with $a \geq T_k$, let n_b be the largest n with $T_n < b$ and take one expiry S in each interval $[T_n, T_{n+1}) \cap [A, b]$, $n = k, \dots, n_b$, with $S = b$ allowed; the surface $\{\tilde{U}(A, S, a, b, K) : K \in \mathbb{R}\}$ at S identifies σ_S^2 , and successive differences give $w_n^2 v_n$ for $k < n \leq n_b$, so every unrevealed variance with $T_n < b$ is identified, while a variance at $T_n \geq b$ enters neither the quote nor the surfaces of this window. Among vectors sharing the later block, the set consistent with exact observation of the quotes and surfaces is the polytope (3.83), as for the cash-settled contracts.

(e) For dates $(1, 2, 3, 4, 5)$, any $3 \leq A < 4$, any window with $a \geq A$ whose partition contains the meetings inside it, and every allowed S and K , the vectors $(2\varepsilon, 3\varepsilon, 2\varepsilon, v_4, v_5)$ and $(3\varepsilon, \varepsilon, 3\varepsilon, v_4, v_5)$, $\varepsilon > 0$, $v_4, v_5 \geq 0$, give identical quotes and values. The vectors $\varepsilon(2, 3, 2, 3, 2)$ and $\varepsilon(2, 3, 2, 1, 2)$, which differ only at the fourth meeting, are separated by every surface with expiry $S \in [4, 5)$ and window $a < 4 < b$: σ_S^2 differs by $2(b-4)^2\varepsilon$ and μ_S by $(b-4)^2\varepsilon$, both larger for the first vector, the quotes differ, and the value differs at every strike.

Proof. (a) Telescoping and the accrual integral are as in Proposition 20(e) with the coefficients of (3.71). For $t \leq b$ and $n = j(t)$, $\mathcal{F}_t = \sigma(Z_1, \dots, Z_n) \vee \mathcal{N}$, the Z_i with $i > n$ are independent of it, and $E_Q[e^{w_i Z_i}] = e^{w_i^2 v_i/2}$, which gives (3.100); it is a function of n and $(Z_1, \dots, Z_n)$, hence constant between meetings and measurable at T_n , square integrable by Gaussian exponential moments, and a martingale as a family of conditional expectations of the integrable R ; the mean identity is the tower property.

(b) The futures price is a martingale for the filtration generated at the meetings, so $(F - K)^+$ is a submartingale by Jensen's inequality and optional sampling at the bounded meeting index of any exercise time makes exercise at S optimal, as in [44] and [15]; the comparison with the cash call is Corollary 21.

(c) By (3.100) at $t = S$, $F(S; a, b) = [e^{Y + \mu_S} - 1]/\delta$ with $Y = \sum_{i \leq n} w_i Z_i \sim N(0, \sigma_S^2)$, so $\tilde{U} = E[(e^{Y + \mu_S} - K')^+]/\delta$, the Gaussian call integral (3.72) with forward G ; the three cases are its evaluation, and $K \mapsto E[(W - K)^+]$ is continuous and nonincreasing for integrable W .

(d) For $i \leq k$, $T_i \leq T_k \leq a$, so (3.71) gives $w_i = \delta$ and $d_i = \delta(m - T_i)$; summing gives (3.103), and $w_i = 0$ for $T_i > b$. For the converse, the quote exponent is $\delta(bV_k - H_k)$ plus terms fixed by the later block, so two

distinct b give (V_k, H_k) . Next, $\partial\tilde{U}/\partial K = -Q(Y + \mu_S > \log K')$, so the surface determines the law of $Y + \mu_S$, a Gaussian, hence σ_S^2 , also when it is a point mass. Expiries S_n with $j(S_n) = n$ give $\sigma_{S_n}^2 - \sigma_{S_{n-1}}^2 = w_n^2 v_n$ with $w_n = b - T_n > 0$ for $T_n < b$, while $T_n = b$ gives $w_n = 0$. Equal observations therefore force equal (V_k, H_k) and equal identified later coordinates; with the later block fixed, equal observations are equivalent to equal (V_k, H_k) , and the equal-observation set is the polytope by Proposition 20(f).

(e) The first pair has $V_3 = 7\varepsilon$ and $H_3 = 14\varepsilon$ with a common later block; apply (d) with $k = 3$. The second pair shares V_3, H_3 and v_5 and has $v_4 = 3\varepsilon$ against ε ; for $S \in [4, 5)$ and $a < 4 < b$, $w_4 = b - 4$, $d_4 = (b - 4)^2/2$ and $n = 4$, so by (3.103) σ_S^2 differs by $2(b - 4)^2\varepsilon$ and μ_S by $(b - 4)^2\varepsilon$, both larger for the first vector, and G by the factor $e^{2(b-4)^2\varepsilon}$. By (3.73) the value $c_{K'}(G, \sigma_S)/\delta$ is strictly increasing in G and nondecreasing in σ_S for $K' > 0$, and for $K' \leq 0$ it is $(G - K')/\delta$; so it is strictly larger for the first vector at every strike. $\square$

Under this convention the exercise payoff is a convex function of a martingale and no discounting rewards waiting, so the payoff process is a submartingale and the last allowed date is optimal, which is the argument of [15] and the content of [44] Properties B and C; the scheduled model makes it discrete, since every stopping rule reduces to a choice of meeting index. The European value is a Black-type formula whose only variance input is σ_S^2 , which separates each unrevealed meeting inside the window as expiries pass it, while the revealed shocks enter σ_S^2 and μ_S only through their sum and date-weighted sum, exactly as the quote does; the polytope ambiguity survives the change of convention, and the cash convention's timing value, which depends on V alone by Corollary 21, is absent. [54] footnote 8 notes that the listed options are American and carry an early-exercise premium, which is the cash convention's timing value. Nothing is asserted about which settlement convention describes a listed SOFR futures option, and the hybrid with the premium paid at trade is not modelled; the identification in (d) uses expiries in every meeting interval of the window and partitions containing the meetings inside it; a variance at a meeting at or after b is invisible to that window's quote and surfaces, and which later windows identify it is not addressed here.

3.21 The listed calendar

The rank criterion of Proposition 19 is now evaluated on the actual expiry calendar of the listed three-month SOFR futures options, with the ideal European cash payoffs of that proposition, exact prices, the zero initial curve and a stated quote precision; no market data enter. The contract rules are those of 'cme2026sr3futures' (Chapter 460) and 'cme2026sr3options' (Chapter 460A), the exercise-price rule also in 'cme2020sr3filing' and 'cme2020sr3optionsfiling', and the meeting days are the decision days of [49]. The Reference Quarter of a future is the interval from the third Wednesday of the third month before its delivery month to the third Wednesday of the delivery month, settled on compounded daily SOFR over it; a quarterly option's underlying is the future whose Reference Quarter begins on the third Wednesday of the expiry month, a serial option's the future of the next quarterly month, a mid-curve option's the future twelve or more months later; every option other than a weekly mid-curve stops trading on the Friday before the third Wednesday of its expiry month; prices are quoted in index points, one hundredth of a point being one basis point, and the minimum fluctuation is one quarter or one half of a basis point depending on the expiry and the premium. Which expiries are listed is set by the exchange; the 2020 filings show four serial and the nearest quarterly expiries listed.

Observation is on 2 January 2026, time is measured in years on an actual-over-360 basis, the scheduled dates are the sixteen decision days of 2026 and 2027, and an option expiring on a Friday S sees a meeting T_i if and only if $T_i \leq S$. The decision days are 28 January, 18 March, 29 April, 17 June, 29 July, 16 September, 28 October and 9 December 2026, and 27 January, 17 March, 28 April, 9 June, 28 July, 15 September, 27

October and 8 December 2027. The futures and options in the window are as follows; every Reference Quarter spans exactly 91 days, $\delta = 91/360$.

Future	Reference Quarter	Option expiries (two serials and the quarterly)
Jun 2026	18 Mar to 17 Jun 2026	16 Jan, 13 Feb, 13 Mar 2026
Sep 2026	17 Jun to 16 Sep 2026	10 Apr, 15 May, 12 Jun 2026
Dec 2026	16 Sep to 16 Dec 2026	10 Jul, 14 Aug, 11 Sep 2026
Mar 2027	16 Dec 2026 to 17 Mar 2027	16 Oct, 13 Nov, 11 Dec 2026
Jun 2027	17 Mar to 16 Jun 2027	15 Jan, 12 Feb, 12 Mar 2027
Sep 2027	16 Jun to 15 Sep 2027	16 Apr, 14 May, 11 Jun 2027
Dec 2027	15 Sep to 15 Dec 2027	16 Jul, 13 Aug, 10 Sep 2027
Mar 2028	15 Dec 2027 to 15 Mar 2028	15 Oct, 12 Nov, 10 Dec 2027

The assumptions are: (A1) every serial and quarterly expiry in the window is listed; (A2) no exchange holiday moves these Fridays; (A3) the at-the-money premium of each option is known to within $\varepsilon \in \{0.25, 0.5\}$ basis points, one half to one minimum fluctuation; (A4) a reference meeting standard deviation $s_0 \in \{10, 15, 25\}$ basis points for every meeting, a size against which the uncertainty is compared and not a market estimate. A meeting is called not practically identified when the propagated uncertainty of its variance is at least s_0^2 .

Proposition 24. In the model of Section 3.10 on this calendar, under (A1) to (A4), the following hold.

(a) Every listed option expires before its underlying accrues, $S < a$, so for every meeting $T_i \leq S$ the coefficients of (3.71) are $w_i = \delta$ and $k_i = \delta^2$, and $k_i = 0$ for $T_i > S$; each complete surface at (S, a, b) determines, by Proposition 19(a),

$$q(S) = \delta^2 \sum_{T_i \leq S} v_i, \quad (3.104)$$

the accumulated variance of the meetings revealed by S . Meetings inside the accrual window enter no option surface.

(b) Merge the expiries in date order, $S_1 < \dots < S_L$, with S_0 the observation date. The surfaces identify every meeting variance with $T_i \leq S_L$, globally on the orthant, if and only if each gap $(S_{\ell-1}, S_\ell]$ contains at most one meeting; a gap with two or more meetings identifies only their sum, and meetings after S_L are identified by no option. On the calendar above every one of the sixteen meetings lies in its own gap, eleven within one future's strip and five only by a difference across futures (18 March, 17 June and 16 September 2026, 17 March and 15 September 2027), so all sixteen are identified, by

$$v_i = \frac{q_\ell}{\delta_\ell^2} - \frac{q_{\ell-1}}{\delta_{\ell-1}^2}, \quad T_i \text{ the meeting in } (S_{\ell-1}, S_\ell], \quad q_0 = 0, \quad (3.105)$$

with δ_ℓ the accrual length of the future underlying S_ℓ . Mid-curve options share the Fridays and revealed sets of the standard options and add proportional rows; weekly mid-curves subdivide gaps; both add redundancy only. With one row per meeting the coefficient matrix is $K = \delta^2 L$, L the lower triangular matrix of ones, whose singular values are $\delta^2 / (2 \sin((2k-1)\pi/66))$, $k = 1, \dots, 16$:

$$\sigma_{\min}(K) = \frac{\delta^2}{2 \cos(\pi/33)} = 0.0321, \quad \sigma_{\max}(K) = \frac{\delta^2}{2 \sin(\pi/66)} = 0.671, \quad \kappa(K) = 20.9. \quad (3.106)$$

(c) At the at-the-money strike the cash price is $c = m[2\Phi(\sqrt{q}/2) - 1]$, so $q = 4[\Phi^{-1}((1 + c/m)/2)]^2$ and

$$\frac{dq}{dc} = \frac{2\sqrt{q}}{m\varphi(\sqrt{q}/2)}, \quad (3.107)$$

nondecreasing in q . A perturbation of ε in rate units of the at-the-money premium, whose cash price is δ times the premium, perturbs q by at most $(dq/dc)\delta\varepsilon$ at the larger price, and by (3.105) the variance of the meeting in gap ℓ by at most

$$\Delta v_i \leq 2\varepsilon \left[\frac{\sigma_\Sigma(S_\ell)}{m_\ell \varphi(\delta_\ell \sigma_\Sigma(S_\ell)/2)} + \frac{\sigma_\Sigma(S_{\ell-1})}{m_{\ell-1} \varphi(\delta_{\ell-1} \sigma_\Sigma(S_{\ell-1})/2)} \right], \quad \sigma_\Sigma(S)^2 = \sum_{T_i \leq S} v_i. \quad (3.108)$$

For accumulated standard deviations below one percentage point both $\varphi(\delta \sigma_\Sigma/2)/\varphi(0)$ and m are within 2×10^{-5} of one, so

$$\Delta v_i \approx \frac{2}{\varphi(0)} \varepsilon [\sigma_\Sigma(S_\ell) + \sigma_\Sigma(S_{\ell-1})] = 5.01 \varepsilon [\sigma_\Sigma(S_\ell) + \sigma_\Sigma(S_{\ell-1})] : \quad (3.109)$$

the amplification grows with the accumulated standard deviation, not with the meeting's own variance. Under (A4), $\sigma_\Sigma(S) = s_0 \sqrt{n(S)}$ with $n(S)$ the number of meetings revealed by S , and the n -th meeting has

$$\frac{\Delta v_n}{s_0^2} \approx 5.01 \frac{\varepsilon}{s_0} (\sqrt{n} + \sqrt{n-1}), \quad (3.110)$$

nondecreasing in n . Evaluating (3.108) exactly on the calendar: at $\varepsilon = 0.5$ and $s_0 = 10$ basis points, meetings 1 to 4 (28 January to 17 June 2026) are identified to within their size and meetings 5 to 16 (29 July 2026 onward) are not practically identified; at $\varepsilon = 0.5$ and $s_0 = 15$, meetings 1 to 9 are identified and 10 to 16 (17 March 2027 onward) are not; at $\varepsilon = 0.5$ and $s_0 = 25$, and at $\varepsilon = 0.25$ for all three sizes, all sixteen are identified, the closest case being the sixteenth meeting at $\varepsilon = 0.25$, $s_0 = 10$ with ratio 0.98. In ℓ^2 , $\|\Delta v\|_2 \leq \|\Delta q\|_2/\sigma_{\min}(K)$.

(d) The futures quote of each future gives $p = \sum_i h_i v_i$ with $h_i = \delta(b - T_i)$ for $T_i \leq a$ and $h_i = (b - T_i)^2$ for $a < T_i < b$. The quotes are redundant for the meetings identified in (b), and for a meeting inside an accrual window after the last option expiry the quote would be the only observation, with a small coefficient: for the 8 December 2027 meeting inside the December 2027 Reference Quarter, which the 10 December 2027 expiry does identify, $h_i = (7/360)^2 = 3.78 \times 10^{-4}$ against $h = \delta(b - T) = 0.482$ for the 28 January 2026 meeting in the same row, a ratio of about 1270; a half-basis-point perturbation of that futures rate moves p by 1.26×10^{-5} and hence v_i by 0.033, a standard deviation of about 18 percentage points.

(e) If instead only the four nearest serials are listed on the observation date, the serial expiries available are those of January, February, April and May 2026 and consecutive listed expiries are quarterly thereafter; the gap from 12 June to 11 September 2026 then contains the meetings of 17 June and 29 July, and the gap from 11 September to 11 December 2026 those of 16 September, 28 October and 9 December, so by (b) only their sums are identified from mid-2026 on, unless mid-curve or weekly expiries fill the gaps.

Proof. (a) For a quarterly option expiring in month M the Reference Quarter begins on the third Wednesday of M and the expiry is the Friday before it; for a serial option the underlying's Reference Quarter begins on the third Wednesday of the next quarterly month, later still; for mid-curves twelve or more months later. Hence $S < a$, and (3.71) gives $w_i = (b - T_i)^+ - (a - T_i)^+ = \delta$ for $T_i \leq a$, the indicator selecting the revealed meetings; Proposition 19(a) recovers (m, q) from the surface.

(b) By (3.104) the map $v \mapsto (q_1, \dots, q_L)$ is linear with rows $\delta_\ell^2 \mathbf{1}\{T_i \leq S_\ell\}$, prefix indicators, so it has full column rank if and only if the prefix sets grow by at most one element per step and reach every meeting; when a gap holds two meetings the vector with +1 and -1 on them is in the kernel and Proposition 19(b) gives strictly positive indistinguishable pairs. The gap assignment is read off the calendar; the cross-future cases are those whose bounding expiries belong to different futures, for which (3.105) uses the two accrual lengths, here equal. With one row per meeting, L^{-1} is the bidiagonal difference matrix, whose Gram matrix is tridiagonal with 2 on the diagonal except 1 in the last entry and -1 off the diagonal, with eigenvalues $4 \sin^2((2k-1)\pi/(2(2n+1)))$, $k = 1, \dots, n$, the eigenvectors having entries $\sin(j\theta_k)$ with $\theta_k = (2k-1)\pi/(2n+1)$; the singular values of L are their reciprocal square roots, and $n = 16$ gives (3.106).

(c) Proposition 19(a) at $K = m$ gives $c = m[2\Phi(\sqrt{q}/2) - 1]$ by normal symmetry, and differentiating the inverse as in Proposition 17 gives (3.107), nondecreasing in q because $y/\varphi(y)$ increases; the mean value theorem turns a premium perturbation into (3.108) with $dc = \delta\varepsilon$ and the triangle inequality on (3.105). The approximation (3.109) uses $\varphi(y)/\varphi(0) = e^{-y^2/2}$ with $y = \delta\sigma_\Sigma/2 \leq 0.0013$ for $\sigma_\Sigma \leq 0.01$, and $m = \exp(\delta(b-S) \sum_{T_i \leq S} v_i) \leq 1 + 1.1 \times 10^{-5}$ for sixteen meetings of 25 basis points, $b-S$ being at most about 0.44 years for a serial expiry. Substituting $\sigma_\Sigma(S) = s_0\sqrt{n}$ gives (3.110); the exact expression (3.108) dominates it by a factor at most $1 + 2 \times 10^{-6}$ at these sizes, and its evaluation at every meeting, tick and size gives the thresholds. The ℓ^2 bound is the definition of the smallest singular value.

(d) The coefficients are (3.71) with $d_i = (b - T_i)^2/2$ and $w_i = b - T_i$ inside the window, so $h_i = (b - T_i)^2$; the ratio is exactly 1274, and $\Delta p = \Delta \log(1 + \delta F) \approx \delta \Delta F$.

(e) The reduced list of expiries is a sub-list of the calendar in date order whose gaps up to 12 June 2026 hold at most one meeting each, and the two later gaps hold two and three meetings; (b) applies. $\square$

One calendar fact carries the structure: every listed option stops trading on the Friday before the third Wednesday on which its underlying's Reference Quarter starts, so an option never sees a meeting inside its own accrual window, and its surface reports the accumulated variance of the meetings it has seen. Differences between consecutive expiries peel off one meeting at a time as long as no two meetings fall between consecutive Fridays, which on this calendar never happens, the expiries falling in mid-month and the meetings late in the month. Rank is the easy part: a tick-size error in the at-the-money premium becomes an error in the accumulated variance proportional to the accumulated standard deviation, and a single meeting's variance is the difference of two such numbers, so the sixteenth meeting is recovered as the difference of two variances of order sixteen times its own, each known to a fixed fraction of a tick. On the full calendar the listed surfaces therefore identify every meeting variance before the last expiry exactly in principle; beyond the listed serials identification collapses to sums; and beyond roughly the first year it is empirical in practice. The result assumes (A1) to (A4); real premiums carry bid-ask spreads and stale quotes beyond one tick; the model has no diffusion between meetings, and a diffusive variance would enter (3.108) in the same way and worsen the amplification; the listed options are American and exercise into a futures position, and Corollaries 21 and 22 and Proposition 23 say that American exercise and either settlement convention add no identification among revealed variances, which is cited here and not used; nothing is asserted about meetings after the last listed expiry, unknown initial curves, calibration or noisy prices beyond the stated perturbation; and the precision statement is a worst-case single-quote propagation, which using all strikes or overlapping futures can reduce in its random part but not in a common tick-level bias.

3.22 Restrictions surviving scale calibration

Proposition 13 fixes the scales and asks which linear equalities the cone leaves; here the scales vary freely and the question is which restrictions survive. Use the model of Section 3.11 with constant decays $\lambda_j \geq 0$, fix an observation time $t < T_N$ with $k_0 = j(t)$, retain the rows $n = k_0 + 1, \dots, N$, and write $\Delta_n = T_n - T_{n-1}$. Admissible scales are nonnegative Borel functions bounded on bounded intervals. Let $A(t)$ be the matrix (3.52) with the partition at the meeting dates, its columns indexed by (j, k) , $k = k_0, \dots, N - 1$,

$$A_{n,(j,k)}(t) = \gamma_{n-k,j}^2 H_j(T_n; \max(t, T_k), T_{k+1}) \text{ for } n > k, \quad A_{n,(j,k)}(t) = 0 \text{ for } n \leq k, \quad (3.111)$$

and put $K(t) := \{A(t)q : q \geq 0\}$.

Proposition 25. (a) The set of vectors $V(t)$ of (3.50) attained as the scales range over all admissible functions is exactly the closed convex cone $K(t)$. Refining the partition or allowing measurable scales adds nothing: on each meeting interval the kernel $s \mapsto (e^{-2\lambda_j(T_n-s)} \gamma_{n-j(s),j}^2)_n$ has a fixed direction, so every sub-interval contributes a multiple of the same column.

(b) A linear inequality $w^\top V(t) \geq 0$ holds for every admissible scale choice if and only if $w^\top A(t) \geq 0$ componentwise, and $V(t)$ is attainable if and only if it satisfies all of them; the surviving equalities are those with $A(t)^\top w = 0$, as in Proposition 13. For one factor with $\gamma_{1,1} \neq 0$, $A(t)$ is lower triangular with positive diagonal and the surviving inequalities are exactly the $N - k_0$ conditions $q = A(t)^{-1}V(t) \geq 0$, the recursion

$$q_{k_0+1} = \frac{V_{k_0+1}}{A_{k_0+1,k_0}} \geq 0, \quad q_{k+1} = \frac{V_{k+1} - \sum_{l < k} A_{k+1,l} q_{l+1}}{A_{k+1,k}} \geq 0, \quad k = k_0 + 1, \dots, N - 1, \quad (3.112)$$

whose rows are the rows of $A(t)^{-1}$; they are irredundant, and every surviving inequality is a nonnegative combination of them.

(c) If $\gamma_{i,j} = \gamma_j$ for all i , each factor loading every future meeting equally, with $\gamma_j \neq 0$, then for each factor $A_{n,(j,k)} = e^{-2\lambda_j \Delta_n} A_{n-1,(j,k)}$ for $k < n - 1$, and with $\lambda_{\max} = \max_j \lambda_j$

$$V_n(t) \geq e^{-2\lambda_{\max} \Delta_n} V_{n-1}(t), \quad n = k_0 + 2, \dots, N, \quad V_{k_0+1}(t) \geq 0, \quad (3.113)$$

hold for every admissible scale choice and characterize $K(t)$: every vector satisfying (3.113) is attained, already by the scales of one factor with $\lambda_j = \lambda_{\max}$. As the scales are recalibrated freely, the event variance of a meeting can fall below that of the preceding meeting at most by the factor $e^{-2\lambda_{\max} \Delta_n}$; with zero decay, $V(t)$ is nondecreasing in the meeting index.

(d) If the loadings may vary over all real arrays, nothing survives beyond $V(t) \geq 0$: the loading $\gamma_{1,j} \neq 0$, $\gamma_{i,j} = 0$ for $i \geq 2$ makes $A(t)$ diagonal with positive diagonal, so every nonnegative vector is attained. If the decays may vary without bound, the closure of the union of the cones is the nonnegative orthant; if they are bounded by Λ , then (3.113) with Λ in place of $\lambda_{\max}$ survives every joint recalibration of scales and decays under constant ordinal loadings and characterizes the union.

Proof. (a) Fix j and a meeting interval $[\max(t, T_k), T_{k+1})$. On it $j(s) = k$, so the row- n entry of the kernel is $e^{2\lambda_j s}$ times the fixed vector $(e^{-2\lambda_j T_n} \gamma_{n-k,j}^2 \mathbf{1}_{\{n > k\}})_n$; for any admissible a_j , (3.50) restricted to the interval is $\int a_j(s)^2 e^{2\lambda_j s} ds$ times that vector, a nonnegative multiple of the column (3.111), with the multiplier ranging over $[0, \infty)$, the column itself being the case $a_j \equiv 1$ and any multiple attained by a constant scale on the

interval. Summing over intervals and factors gives $V(t) = A(t)q$ with $q \geq 0$, every such q attained by piecewise-constant scales; the cone is closed because it is finitely generated.

(b) For a finitely generated cone $\{Aq : q \geq 0\}$, $w^\top V \geq 0$ on the cone if and only if $w^\top A \geq 0$, and V lies in the cone if and only if $w^\top V \geq 0$ for all such w , by the separating hyperplane theorem applied to the closed convex cone. Equalities are the inequalities holding with both signs. For $d = 1$ the matrix is square and lower triangular with positive diagonal by Proposition 13, so $q = A^{-1}V$ is unique and V is in the cone if and only if $q \geq 0$, which is (3.112); the rows of A^{-1} are linearly independent, so none is implied by the others, and a surviving w satisfies $w^\top = (w^\top A)A^{-1}$ with $w^\top A \geq 0$.

(c) With constant loadings, $A_{n,(j,k)} = \gamma_j^2 H_j(T_n; \max(t, T_k), T_{k+1})$, and $H_j(T_n; a, b) = e^{-2\lambda_j(T_n - T_{n-1})} H_j(T_{n-1}; a, b)$ whenever $b \leq T_{n-1}$, directly from the definition of H_j ; this gives the row relation for $k \leq n - 2$ and, for each factor's contribution, $V_n^{(j)} = e^{-2\lambda_j \Delta_n} V_{n-1}^{(j)} + A_{n,(j,n-1)} q_{j,n-1}$ with a nonnegative diagonal term. Summing over j , $V_n \geq \sum_j e^{-2\lambda_j \Delta_n} V_{n-1}^{(j)} \geq e^{-2\lambda_{\max} \Delta_n} V_{n-1}$, which is (3.113). Conversely, given V satisfying (3.113), take a factor j^* with $\lambda_{j^*} = \lambda_{\max}$ and set the other scales to zero; the one-factor matrix is lower triangular and (3.112) reads $q_{j^*,n-1} = [V_n - e^{-2\lambda_{\max} \Delta_n} V_{n-1}] / A_{n,(j^*,n-1)} \geq 0$, with $q_{j^*,k_0} = V_{k_0+1} / A_{k_0+1,(j^*,k_0)} \geq 0$. Irredundancy: the inequalities (3.113) are the rows of the one-factor A^{-1} up to positive scaling, by (b).

(d) With $\gamma_{1,j} \neq 0$ and $\gamma_{i,j} = 0$ for $i \geq 2$, (3.111) vanishes unless $n = k + 1$, so $A(t)$ is diagonal with positive entries and $q = A^{-1}V \geq 0$ for every $V \geq 0$. For unbounded decays the constants $e^{-2\lambda \Delta_n}$ tend to zero, so a nonnegative V is the limit of $V + \eta \mathbf{1}$, which satisfies (3.113) for a large enough decay and so lies in that cone by (c); with $\lambda_j \leq \Lambda$ each cone satisfies (3.113) with Λ , and the cone with decay Λ attains every such vector by (c). $\square$

Remark 12 removed the restrictions one might infer from state dimension alone, and Proposition 13 showed that with time-dependent scales no linear equality survives. What remains are inequalities, and they are exact: within an interval the decay and loading kernels move every meeting's coefficient in the same proportion, so the variance vector is a nonnegative combination of finitely many fixed directions, one per meeting interval and factor, and the attainable set is a polyhedral cone whatever the scales do; its facets are the restrictions that no scale calibration can violate. When each factor loads all future meetings equally, the cone is generated by the shifted copies of one direction and its facets are the consecutive-meeting ratio bounds: a vector of event variances that falls faster than the largest decay allows between two meetings cannot be produced by this family at any scale calibration. The ordinal loadings of [12] are not the constant pattern of (c); for them the surviving restrictions are the triangular ones of (3.112), which depend on the loadings, and no closed form is given beyond the recursion. Under the stochastic volatility of Section 3.12 the fixed-model vector is random and nothing is claimed here about which restrictions on its support survive scale variation. A violation of (3.113) by estimated variances would show only that no scale calibration of this family reproduces them. Farkas' lemma and the closedness of finitely generated cones are used as standard facts.

3.23 Exact support with positive variance mean reversion

Return to the stochastic-volatility construction of Section 3.12. Suppose each θ_j is constant and nonnegative, and each $\alpha_j(s)$ is nonnegative and piecewise constant with finitely many pieces on bounded intervals. In addition to the solution and supremum-square hypotheses of Section 3.12, assume that the noise integrands $\alpha_j \sqrt{v_j}$ and the forward-rate loading integrands of (3.55) are predictable and have almost surely finite integrated square on every finite horizon. The SDE and forward curve use their Itô integrals. These

domain conditions are imposed on the given almost surely continuous state process; existence of such a process is an assumption.

The one-piece square-root transition law with positive mean reversion is stated in [47] Proposition 1, and the time-dependent parameters used below are those of [12] Table 2. Neither source gives the support of the meeting-variance vector after composition over pieces.

Proposition 26 (conditional). Under the preceding hypotheses, let $0 \leq s < t < T_N$, retain M future meetings, and take $C(t)$ and $A(t)$ from (3.57). On a single piece of length h , with starting value $x \geq 0$, constant $\alpha > 0$ and $\theta > 0$, the conditional Laplace transform at the piece's end is

$$E_Q[e^{-\lambda v_j(s+h)} \mid v_j(s) = x] = \left[1 + \frac{\alpha^2 \lambda}{2\theta}(1 - e^{-\theta h})\right]^{-2\theta/\alpha^2} \exp\left(-\frac{\lambda e^{-\theta h} x}{1 + \frac{\alpha^2 \lambda}{2\theta}(1 - e^{-\theta h})}\right), \quad \lambda \geq 0. \quad (3.114)$$

For $\theta = 0$ and $\alpha > 0$ this is $\exp[-\lambda x/(1 + \alpha^2 h \lambda/2)]$. For $\alpha = 0$, the coordinate follows $v_j(s+h) = 1 + (x-1)e^{-\theta h}$. Composing these one-piece laws over the breakpoints gives the joint conditional law of $v(t)$ given $\mathcal{F}_s$: its coordinates are independent, and their joint Laplace transform is the product of the composed coordinate transforms.

Let $J = \{j : \alpha_j > 0 \text{ on some piece of } (0, t]\}$, and let ℓ_j be the lower endpoint of the support of $v_j(t)$. If the last piece before t has $\alpha_j > 0$, then $\ell_j = 0$. If an earlier piece is stochastic but the last piece is deterministic, ℓ_j is the flow of zero under $u \mapsto 1 + (u-1)e^{-\theta_j h}$ across the final deterministic pieces. If every piece is deterministic, $v_j(t) = \ell_j = 1$. The support, affine hull and covariance of the actual vector of conditional short-rate jump variances are

$$\begin{aligned} V(t) &= C(t) + A(t)v(t), \\ \text{supp } V(t) &= C(t) + A(t)\ell + A_J(t)[0, \infty)^J, \\ \text{aff supp } V(t) &= C(t) + A(t)\ell + \text{range } A_J(t), \\ \text{Cov}_Q(V(t)) &= \sum_{j \in J} \text{Var}_Q(v_j(t)) A_{j,j}(t) A_{j,j}(t)^\top. \end{aligned} \quad (3.115)$$

This support is a closed translated cone. Every variance in the sum is positive; hence the affine-hull dimension and covariance rank both equal $\text{rank } A_J(t)$, and the almost-sure affine equalities are exactly those given by $w \in \ker A_J(t)^\top$. The mass at the vertex $C(t) + A(t)\ell$ is the product of the lower-endpoint atom masses for active coordinates with nonzero columns. A factor with $\theta_j > 0$ has no such atom. A factor with $\theta_j = 0$ is absorbed when its starting value is zero; on a single stochastic piece of length h starting at $x > 0$, its atom at zero is $\exp[-2x/(\alpha_j^2 h)]$.

For the conditional law given $\mathcal{F}_s$, replace ℓ by the lower endpoints obtained by starting at $v(s)$ and using the pieces on $(s, t]$, and replace J by

$$J_s = \{j : \alpha_j > 0 \text{ on some piece of } (s, t], \quad \theta_j > 0 \text{ or } v_j(s) > 0\}. \quad (3.116)$$

Then (3.115) holds for the regular conditional support, affine hull and covariance with A_{J_s} and conditional coordinate variances. In particular,

$$E_Q[V(t) \mid \mathcal{F}_s] = C(t) + A(t)(1 + (v_j(s) - 1)e^{-\theta_j(t-s)}), \quad \text{Cov}_Q(V(t) \mid \mathcal{F}_s) = \sum_{j \in J_s} \text{Var}_Q(v_j(t) \mid \mathcal{F}_s) A_{j,t}(t) A_{j,t}(t)^\top. \quad (3.117)$$

A positive-speed coordinate that is zero at s becomes active on any later stochastic piece; a zero-speed coordinate at zero stays absorbed.

For the time-dependent column of [12] Table 2, the printed speeds are (0.1, 0, 11.0) and every printed volatility-of-volatility value on its four pieces is strictly positive. At any observation time whose last piece carries one of those positive values, for any completion of the missing calibration inputs satisfying the hypotheses,

$$\text{supp } V(t) = C(t) + A(t)[0, \infty)^3, \quad \text{aff supp } V(t) = C(t) + \text{range } A(t), \quad \text{rank Cov}_Q(V(t)) = \text{rank } A(t) \leq 3. \quad (3.118)$$

Thus the two-active-factor bound of Proposition 16 for the printed constant column does not extend to the time-dependent column. For $M > 3$, at least $M - 3$ affine equalities remain. The vertex has no atom if either positive-speed factor has a nonzero column. Equation (3.118) does not assert $\text{rank } A(t) = 3$; the loadings, dates and time conventions needed to determine that rank were not published.

Proof. On one coefficient piece, use the exponential whose terminal value is $e^{-\lambda v_j(s+h)}$ and whose two backward coefficients are the coefficient of x and the constant term in the exponent of (3.114). Differentiation in actual time gives for the first coefficient its mean-reversion multiple plus half its squared value times α_j^2 , and for the second minus θ_j times the first. Itô's formula then cancels the drift against these derivatives and the drift and quadratic variation of (3.54). Localize where the nonnegative state and coefficients are bounded; the stopped stochastic integral is a true martingale. The exponential lies in $(0, 1]$, so bounded convergence removes the localization. Its conditional expectation gives (3.114), and the same argument across each finite common coefficient partition gives the product transform. The integrand assumptions make the integral identities valid for the original state; an everywhere-continuous adapted version gives predictable square-root integrands, and indistinguishability transfers the SDE, the forward curve and its short-rate jumps between the versions.

For positive speed and volatility of volatility, (3.114) is the Laplace transform of a scaled noncentral chi-square law, equivalently a Poisson mixture of laws with strictly positive density on the positive half-line. It has a positive density on $(0, \infty)$, support $[0, \infty)$ and no atom at zero even when $x = 0$. At zero speed it reduces to the absorbed law of Proposition 16; a deterministic piece applies the increasing affine flow to its input support. Composing these facts proves the coordinate supports, lower endpoints and atoms, and Laplace uniqueness gives the conditional product law.

Proposition 14 gives the affine identity for the actual conditional short-rate jump variances. The image of the product of coordinate half-lines is (3.115); nonnegative columns make this finitely generated image closed, and the continuous-image support identity then makes it exact. Conditional independence makes the covariance a sum of positive weighted column outer products, so its kernel is $\ker A_J(t)^\top$ and its rank equals $\text{rank } A_J(t)$. The same argument applied to the regular conditional product law proves (3.116) and (3.117); differentiating the one-piece transform at zero gives the displayed conditional mean, and iteration gives it across pieces. Finally all three printed factors have a positive last piece, so all three support half-lines contribute to (3.118), while positive mean reversion in factors one and three removes their lower-endpoint atoms. $\square$

The result is conditional on the stipulated nonnegative solutions, their supremum-square moment bound and the admissible Itô integrands. It does not construct the solutions, prove a calibrated rank, identify variances from listed prices, or establish restrictions surviving variation of the stochastic-volatility parameters.

3.24 A bounded curve with a variance state and a scheduled jump

Fix $\lambda > 0$ and one scheduled date $\tau > 0$. Let W_1, W_2 be independent standard Brownian motions under Q , with the completed right-continuous joint filtration, and let the smaller filtration be the completed natural filtration of W_2 with the same null sets. Assume the cited Itô calculus is available on this basis in both filtrations: a supplied one-driver structure R for W_2 in the smaller filtration and a supplied two-driver structure S for (W_1, W_2) in the joint filtration, with the same measure and actual W_2 path, unit diagonal and zero off-diagonal covariation densities, and the integral, Itô formula, Lipschitz SDE, continuous-adapted predictability and left-endpoint integral-approximation properties recorded in AX-03 to AX-06, AX-09 and AX-11. Assume also the bounded-quadratic-variation exponential-martingale property AX-10 in the joint calculus. These are cited stochastic-calculus inputs from [4]; constructing both calculus structures on this Brownian basis is not part of the result.

Proposition 27 (conditional). Under these inputs there is, up to indistinguishability, a unique continuous solution with zero initial coefficients, stopped at τ , of

$$\begin{aligned} v_t &= 2 + \tanh Z_2(t), & Z_1(t) &= \frac{t \wedge \tau}{\lambda} + W_1(t \wedge \tau), \\ Z_2(t) &= \int_0^{t \wedge \tau} \frac{v_s - 2}{2\lambda} ds + \int_0^{t \wedge \tau} \sqrt{v_s} dW_2(s), & Z_4(t) &= - \int_0^{t \wedge \tau} \frac{v_s}{2\lambda} ds. \end{aligned} \quad (3.119)$$

Define, with the right value at $T = \tau$,

$$f(t, T) = \begin{cases} 0, & T < \tau, \\ Z_1(t)e^{-\lambda(T-\tau)} + Z_2(t)e^{-2\lambda(T-\tau)} + Z_4(t)e^{-4\lambda(T-\tau)}, & T \geq \tau. \end{cases} \quad (3.120)$$

Each curve is bounded in maturity by $|Z_1(t)| + |Z_2(t)| + |Z_4(t)|$, is exponential-polynomial of degree zero to the right of the scheduled breakpoint, and has fixed exponents $\lambda, 2\lambda, 4\lambda$. Boundedness is per realized curve in the sense of [25] Definition 3.1, not uniform over all coefficient vectors or sample points. The fixed exponents and the two doubling pairs satisfy, rather than contradict, that source's exponent restrictions. Its fixed-maturity HJM drift equals the sum of volatility times integrated volatility. With $r_t = f(t, t)$, $B_t = \exp(\int_0^t r_s ds)$ and $P(t, T) = \exp(-\int_t^T f(t, u) du)$, every $P(\cdot, T)/B$ is a positive true Q -martingale on $[0, T]$. The curve is continuous in time at fixed maturity, while the short rate has an almost surely nonzero jump at τ :

$$\Delta r_\tau = Z_1(\tau) + Z_2(\tau) + Z_4(\tau) = W_1(\tau) + \int_0^\tau \sqrt{v_s} dW_2(s) \neq 0 \quad Q\text{-a.s.} \quad (3.121)$$

Before τ , $1 < v_t < 3$, $d[Z_2]_t = v_t dt$, and $d[v]_t = (1 - \tanh^2 Z_2(t))^2 v_t dt > 0$. The three forward rates at maturities $\tau, \tau + \log(2)/\lambda, \tau + 2 \log(2)/\lambda$ recover $Z_1(t), Z_2(t), Z_4(t)$ for $t \leq \tau$, hence also v_t . Thus the stochastic variance state enters the curve itself, including the coefficient of a shape generated by the first driver's drift. This bounded family gives a counterexample to the exclusion conjecture for curve-entering variance with scheduled breakpoints.

Proof. The scalar drift $\tanh z/(2\lambda)$ and diffusion $\sqrt{2 + \tanh z}$ are bounded and globally Lipschitz: their derivatives have absolute values at most $1/(2\lambda)$ and $1/2$. The cited Lipschitz theorem in the smaller filtration supplies Z_2 , and the other coordinates follow from (3.119). Continuous-adapted predictability and boundedness put the integrands in the stated Itô domains. Left-endpoint approximation identifies the W_2 integrals in the two filtrations; uniqueness of their limits at each rational time and path continuity give one almost-sure identity at all times. Stopping at τ and uniqueness in the joint filtration give the asserted stopped system.

For $t < \tau \leq T$, set $x = T - \tau$. The two volatility shapes are $\sigma_1 = e^{-\lambda x}$ and $\sigma_2 = \sqrt{v_t} e^{-2\lambda x}$; their maturity integrals are $(1 - e^{-\lambda x})/\lambda$ and $\sqrt{v_t}(1 - e^{-2\lambda x})/(2\lambda)$. Equation (3.119) gives

$$\alpha(t, T) = \frac{e^{-\lambda x} - e^{-2\lambda x}}{\lambda} + \frac{v_t(e^{-2\lambda x} - e^{-4\lambda x})}{2\lambda} = \sum_{j=1}^2 \sigma_j(t, T) \int_t^T \sigma_j(t, u) du. \quad (3.122)$$

For $T < \tau$ or $t \geq \tau$, the fixed-maturity drift and volatility vanish. There is no fixed-maturity curve jump. For $T \geq \tau$, put $A_k(T) = (1 - e^{-k\lambda(T-\tau)})/(k\lambda)$, $k = 1, 2, 4$. Before τ , $P(t, T) = \exp(-\sum_k Z_k(t)A_k(T))$ and $B_t = 1$. Itô's formula and

$$\frac{A_1}{\lambda} + \frac{(v_t - 2)A_2}{2\lambda} - \frac{v_t A_4}{2\lambda} = \frac{A_1^2 + v_t A_2^2}{2} \quad (3.123)$$

give $d(P/B)/(P/B) = -A_1 dW_1 - \sqrt{v_t} A_2 dW_2$. After τ the coefficients are frozen, $d \log P(t, T)/dt = r_t$, and the discounted bond is constant; the values match at τ . Its stochastic logarithm has quadratic variation at most $7\tau/(4\lambda^2)$, since $v < 3$. AX-10 makes the resulting exponential a true martingale. If $T < \tau$, the discounted bond is one.

Itô's formula gives the two bracket densities in the statement and the inverse $Z_2 = \frac{1}{2} \log[(v-1)/(3-v)]$. The evaluation matrix at the three maturities has rows $(1, 1, 1)$, $(1/2, 1/4, 1/16)$, $(1/4, 1/16, 1/256)$, with determinant $-21/1024$, so the curve recovers v . The $e^{-2\lambda x}$ coefficient Z_2 has martingale part $\sqrt{v} dW_2$, whereas Z_4 has finite variation and closes the HJM drift without further exponents. Finally the drift of $Z_1 + Z_2 + Z_4$ cancels. Its remaining W_2 integral is measurable for the completed W_2 -path sigma-algebra, by the smaller-filtration construction and integral comparison. The independent $W_1(\tau)$ has a non-atomic Gaussian law, proving (3.121). $\square$

The proposition is conditional on the supplied two-filtration calculus. Their existence on the Brownian basis is not constructed here or established by the degenerate Upstream instances. It covers one scheduled date with a maturity breakpoint and short-rate jump, but no fixed-maturity curve jump, no post-meeting diffusion and no classification of other exponent sets or variance dynamics.

3.25 Separable volatility with meeting loadings

Fix a horizon $H > T_N$; all times and maturities in this section lie in $[0, H]$, and I_N is used only up to H . Let $W^1, \dots, W^d$ be independent standard Brownian motions on a filtered probability space satisfying the usual conditions. Assume the cited Itô calculus of this basis: for predictable integrands h with $\int_0^H h(s)^2 ds < \infty$ almost surely, the integral $\int_0^t h dW^j$ has a continuous adapted version, is linear in h , and commutes with stopping at deterministic times, as recorded in [4] §2.2 and §3.1 to §3.3; the driver covariations are the identity. Constructing this calculus is not part of the result. Let each scale χ_j be predictable with $\int_0^H \chi_j(s)^2 ds < \infty$ almost surely; no equation for χ_j is assumed. Let $\varphi_j : [0, H] \rightarrow \mathbb{R}$ be continuous deterministic maturity shapes, which may vanish or change sign, and $a_{j,k,m}$ real constants, $1 \leq j \leq d$, $0 \leq k, m \leq N$. Put

$$g_{j,k}(T) := a_{j,k,m} \varphi_j(T) \quad (T \in I_m), \quad G_{j,k}(T) := \int_0^T g_{j,k}(u) du, \quad \Phi_{j,m}(T) := \int_{T_m}^T \varphi_j(u) du. \quad (3.124)$$

Here $\Phi_{j,m}$ is a deterministic primitive, not the accumulated variance Φ of [12] eq. 28, and the functions $G_{j,k}(\cdot)$ are not the constants $G_{i,j}$ of Section 3.11. The constants let the maturity loading change when time crosses a meeting. The ordinal loadings of Section 3.11 are the case $a_{j,k,m} = G_{m-k,j}$ for $m \geq k$ and $a_{j,k,m} = 0$ otherwise, which includes zero volatility on the current maturity interval. For $t \in I_k$ and $t \leq T$ set

$$\sigma_j(t, T) := \chi_j(t) g_{j,k}(T), \quad \alpha(t, T) := \sum_{j=1}^d \chi_j(t)^2 g_{j,k}(T) [G_{j,k}(T) - G_{j,k}(t)], \quad (3.125)$$

take every curve jump ξ_n to be zero, take a bounded deterministic initial curve $f(0, \cdot)$, continuous within each I_m with finite one-sided limits, and define f by (2.2). Values at the finitely many meeting times do not affect any integral. Put

$$A_{j,k}(t) := \int_0^t \mathbf{1}_{I_k}(s) \chi_j(s)^2 ds, \quad M_{j,k}(t) := \int_0^t \mathbf{1}_{I_k}(s) \chi_j(s) dW_s^j - \int_0^t \mathbf{1}_{I_k}(s) \chi_j(s)^2 G_{j,k}(s) ds, \quad (3.126)$$

and, for each maturity interval I_m ,

$$C_{j,m}(t) := \sum_k a_{j,k,m} [M_{j,k}(t) + G_{j,k}(T_m) A_{j,k}(t)], \quad V_{j,m}(t) := \sum_k a_{j,k,m}^2 A_{j,k}(t). \quad (3.127)$$

Proposition 28 (conditional). Under the stated calculus inputs:

(a) The coefficients (3.125) satisfy Assumption 2.1 for every $t \leq T$, not only almost everywhere, and Assumption 2.2 holds trivially because there are no curve jumps. There is a version of the curve, continuous in t , such that simultaneously for all $t \leq T \leq H$

$$f(t, T) = f(0, T) + \sum_{j,k} g_{j,k}(T) [M_{j,k}(t) + G_{j,k}(T) A_{j,k}(t)], \quad (3.128)$$

and for each fixed T it agrees with the process (2.2). The current curve is therefore determined by the initial curve and at most $2d(N+1)$ real coefficients, which do not depend on T . Each $A_{j,k}$ is adapted, continuous, nonnegative, nondecreasing and of finite variation; each $M_{j,k}$ is an adapted continuous semi-martingale. Both vanish up to T_k and are constant after T_{k+1} when $k < N$: crossing a meeting retains the earlier coefficients and starts the next integrals at zero.

(b) For $T \in I_m$ and $t \leq T$,

$$f(t, T) = f(0, T) + \sum_{j=1}^d \varphi_j(T) [C_{j,m}(t) + \Phi_{j,m}(T) V_{j,m}(t)]. \quad (3.129)$$

The two shapes per factor are φ_j and $\varphi_j \Phi_{j,m}$, restarted at each maturity meeting. The coefficient $V_{j,m}$ is realized accumulated variance, adapted, continuous, nonnegative and nondecreasing, and has no martingale part; it is not the conditional variance of a future short-rate jump. For $t \in I_k$, in integral form,

$$dC_{j,m} = a_{j,k,m}\chi_j dW^j + a_{j,k,m}\chi_j^2[G_{j,k}(T_m) - G_{j,k}(t)]dt, \quad dV_{j,m} = a_{j,k,m}^2\chi_j^2 dt. \quad (3.130)$$

(c) If $\varphi_j = 1$ for every j and $f(0, \cdot)$ is piecewise affine, (3.129) is a step-plus-ramp curve. On I_m its level at T_m is $f(0, T_m) + \sum_j C_{j,m}(t)$, and its slope is the initial slope plus $\sum_j V_{j,m}(t)$. The additional slope is adapted and of finite variation with derivative $\sum_j a_{j,k,m}^2\chi_j^2$ on I_k ; it need not be deterministic. This is the local slope on I_m , not the coefficient of one ramp in a sum over overlapping maturity intervals.

The separable device $\sigma(t, T) = \chi(t)\varphi(T)$ with a stochastic scale, and its exponential meeting instance with ordinal indicators, are [12] eqs. 20 to 22 and 36 to 43. Finite-dimensional realizations of calendar-fixed shapes, with running time as a state, are known: [10] §3.3 with Theorem 3.4 and Proposition 6.2, and, for the Gaussian case with scheduled jumps, [30] Example 3.10. Proposition 28 extends that known computation to loadings that restart at the meetings, stochastic scales and continuous shapes, and is not new in substance. No exponential restriction on the shapes is needed for (a) to (c).

Proof. Continuity of φ_j on $[0, H]$ and finitely many loadings make every $g_{j,k}$ bounded, and $G_{j,k}$ bounded and absolutely continuous with derivative $g_{j,k}$ almost everywhere. Masking χ_j by a deterministic time interval and multiplying by bounded deterministic functions preserves predictability and almost sure finiteness of $\int_0^H \chi_j^2 ds$, so every stochastic integrand below lies in the domain of the cited integral, and the drift integrals exist pathwise. Each masked stochastic integral is the difference of the integral stopped at T_{k+1} and at T_k , hence zero up to T_k and constant after T_{k+1} ; its finite-variation companion in (3.126) has the same property. There are finitely many such integrals, so their continuous versions can be chosen on one event of probability one.

For $t \in I_k$ put $D_j(T) := G_{j,k}(T) - G_{j,k}(t)$. Absolute continuity gives

$$\int_t^T g_{j,k}(u) D_j(u) du = \frac{1}{2} D_j(T)^2, \quad (3.131)$$

even where $g_{j,k}$ jumps at a maturity meeting or changes sign, since $G_{j,k}$ is continuous and its square satisfies the ordinary chain rule. Multiplying by $\chi_j(t)^2$ and summing over j gives $\int_t^T \alpha(t, u) du = \frac{1}{2} \sum_j (\int_t^T \sigma_j(t, u) du)^2$, which is (2.3) for every $t \leq T$; independent drivers give the sum of squares without cross terms.

Split each time integral in (2.2) into the finitely many intervals I_k . On the k -th piece $g_{j,k}(T)$ and $G_{j,k}(T)$ do not depend on the integration variable, so

$$\int_0^t \mathbf{1}_{I_k}(s) \chi_j(s)^2 g_{j,k}(T) [G_{j,k}(T) - G_{j,k}(s)] ds = g_{j,k}(T) G_{j,k}(T) A_{j,k}(t) - g_{j,k}(T) \int_0^t \mathbf{1}_{I_k}(s) \chi_j(s)^2 G_{j,k}(s) ds, \quad (3.132)$$

and the diffusion term is $g_{j,k}(T)$ times the stochastic integral in (3.126), by linearity of the integral in the integrand. Summing gives (3.128) with finite sums only; no stochastic Fubini theorem is used. The right side of (3.128) is one version for all maturities; for each fixed T it agrees with the integral process (2.2), so no uncountable family of almost sure events is intersected.

For $T \in I_m$ the primitive splits as

$$G_{j,k}(T) = G_{j,k}(T_m) + a_{j,k,m}\Phi_{j,m}(T). \quad (3.133)$$

Substituting (3.133) and $g_{j,k}(T) = a_{j,k,m}\varphi_j(T)$ into (3.128) and collecting the two shapes gives (3.127) and (3.129); the constants $G_{j,k}(T_m)$ carry the contributions of all earlier maturity intervals and are retained in

$C_{j,m}$. Differentiating (3.126) in integral form gives (3.130). The coefficients of $V_{j,m}$ are squares and $A_{j,k}$ is an integral of a square, which gives the monotonicity and finite-variation assertions. Finally $\varphi_j = 1$ gives $\Phi_{j,m}(T) = T - T_m$, and (3.129) with (3.130) gives (c). $\square$

The proposition is conditional on the cited calculus and on the given scales; it constructs neither, and no stochastic-volatility equation for χ_j is solved. It is a finite representation of the current curve, not a finite-dimensional Markov realization: finitely many current coefficients do not determine the conditional law of the future scales, and a model for χ_j together with a proof that an augmented state closes is required; Proposition 29 supplies both for a Lipschitz scale state, with a time-inhomogeneous realization. The statement that every deterministic shape gives a finite-dimensional realization is therefore established only in this weaker sense. The deterministic time dependence in (3.130) is not removed, so the proposition says nothing about time homogeneity and neither proves nor refutes Conjecture 2. The bound $2d(N + 1)$ is not a minimal dimension, and the coefficients are not claimed to be recoverable from the curve. Not covered: nonzero fixed-maturity curve jumps and restart laws; the law of the short-rate jump that crossing a maturity breakpoint produces; true bond martingales and bond prices; correlated drivers, and several independently scaled shapes within one driver on one time interval, which would add cross terms.

3.26 A Markov realization with a Lipschitz scale state

Keep the setting and notation of Section 3.25. Assume in addition that the drivers are $m \geq d$ independent standard Brownian motions $B^1, \dots, B^m$ whose increments after each time s are independent of $\mathcal{F}_s$, with $W^j = B^j$ for $j \leq d$. Assume that the cited calculus supplies, besides the integral of Section 3.25, strong solutions of equations with Lipschitz coefficients ([4] §5.2), predictability of real adapted processes all of whose paths are continuous ([4] p. 14), and the Markov property of solutions of equations whose coefficients are Borel in time and Lipschitz in the state ([57] Theorem 5.2.2). Constructing this calculus is not part of the result. Fix $p \geq 1$, $z_0 \in \mathbb{R}^p$, and coefficients $\beta : \mathbb{R}_{\geq 0} \times \mathbb{R}^p \rightarrow \mathbb{R}^p$ and $\Sigma : \mathbb{R}_{\geq 0} \times \mathbb{R}^p \rightarrow \mathbb{R}^{p \times m}$ that are Borel in (t, z) , Lipschitz in z with one constant for all t , and bounded on every set $[0, T'] \times \{|z| \leq R\}$. The scale state solves

$$Z_t = z_0 + \int_0^t \beta(u, Z_u) du + \sum_{l=1}^m \int_0^t \Sigma_{\cdot l}(u, Z_u) dB_u^l, \quad (3.134)$$

and may use the curve drivers, so leverage is allowed. Let $\psi_j : \mathbb{R}_{\geq 0} \times \mathbb{R}^p \rightarrow \mathbb{R}$ be jointly continuous, bounded by K_ψ and Lipschitz in z uniformly in t , and take the scales of Section 3.25 to be $\chi_j(t) = \psi_j(t, Z_t)$. With the coefficients (3.126), put $n = p + 2d(N + 1)$ and

$$X_t = (Z_t, (M_{j,k}(t))_{j,k}, (A_{j,k}(t))_{j,k}) \in \mathbb{R}^n. \quad (3.135)$$

Proposition 29 (conditional). Under these inputs:

(a) Equation (3.134) has a solution every path of which is continuous, and every solution is indistinguishable from one; with such a Z , each χ_j is predictable and bounded, so Proposition 28 applies.

(b) Write $e_H = \mathbf{1}_{[0,H]}$. The frozen state $X_{t \wedge H}$ solves, from $x_0 = (z_0, 0, 0)$, the equation whose Z -block has the coefficients $e_H \beta$ and $e_H \Sigma$, and whose other blocks have, at $x = (z, \mu, a)$, the drifts and the diffusion against B^j

$$\begin{aligned} \beta_{M_{j,k}}(t, x) &= -\mathbf{1}_k(t) e_H(t) \psi_j(t, z)^2 G_{j,k}(t \wedge H), & \beta_{A_{j,k}}(t, x) &= \mathbf{1}_k(t) e_H(t) \psi_j(t, z)^2, \\ \Sigma_{M_{j,k}, j}(t, x) &= \mathbf{1}_k(t) e_H(t) \psi_j(t, z), \end{aligned} \quad (3.136)$$

with every other diffusion entry zero. These coefficients are Borel in (t, x) , Lipschitz in x uniformly in t , bounded on bounded sets, and do not depend on (μ, a) ; they switch at the meetings.

(c) The frozen state is Markov for the filtration: for deterministic $s \leq t_1, \dots, t_r$ and bounded Borel F , $E[F(X_{t_1 \wedge H}, \dots, X_{t_r \wedge H}) \mid \mathcal{F}_s]$ is almost surely a Borel function of $X_{s \wedge H}$.

(d) The affine map, which does not depend on t ,

$$\Lambda(x)(T) = f(0, T) + \sum_{j,k} g_{j,k}(T) [\mu_{j,k} + G_{j,k}(T) a_{j,k}] \quad (3.137)$$

satisfies $f(t, T) = \Lambda(X_t)(T)$ simultaneously for all $0 \leq t \leq T \leq H$. Hence, for finitely many pairs $s \leq t_i \leq T_i \leq H$ and a bounded Borel function F of the values $f(t_i, T_i)$, $E[F \mid \mathcal{F}_s]$ is almost surely a Borel function of X_s . The state dimension, running time excluded, is at most $p + 2d(N + 1)$, for every choice of continuous shapes.

(e) If $a_{j,k,k} = 0$ for all j, k , as for the ordinal loadings of Section 3.11, and β, Σ and ψ_j do not depend on t , then on each $I_k \cap [0, H]$ the coefficients of (b) are functions of x alone: their only time dependence is the switch at each meeting and the freeze at H , for every continuous φ_j . If instead ψ_j is not identically zero and $a_{j,k,k}\varphi_j$ is not identically zero on $I_k \cap [0, H]$, the drift of $M_{j,k}$ depends on t within I_k .

In calendar maturity the map Λ is fixed and the time dependence sits in the coefficients (3.136), through the meeting indicators and $G_{j,k}(t)$; in time to maturity $x = T - t$ the curve is $\Lambda(X_t)(t + x)$, which depends on t explicitly. Realizations of time-varying systems with running time as a state are [10] §3.3 (Theorem 3.4) with Proposition 6.2, [30] Example 3.10 is the Gaussian case with scheduled jumps, and [12] §3.5 and §3.6 is the exponential instance with a Heston scale. Proposition 29 extends these known calendar-maturity realizations to meeting restarts, arbitrary continuous shapes and a Lipschitz scale state, and is not new in substance.

Proof. (a) A solution with almost surely continuous paths exists by the cited strong-existence result. Its paths are continuous and start at z_0 off a null set, which lies in $\mathcal{F}_0$ by the usual conditions; setting the process equal to z_0 on that set gives an adapted process with every path continuous, indistinguishable from the original. Each coordinate of it, and running time, are then predictable by the cited predictability result, so $\Sigma(s, Z_s)$, $\beta(s, Z_s)$ and $\psi_j(s, Z_s)$ are predictable, and the box bounds make them square integrable in time on every path. The replaced and original diffusion integrands differ by an admissible integrand that vanishes off a null set, so its integral vanishes by the isometry and the two stochastic integrals agree almost surely at each time; both sides of (3.134) are continuous in t , so agreement at rational times gives agreement at all times. Finally $|\chi_j| \leq K_\psi$.

(b) Each block of (3.136) is a product of Borel functions of t with a continuous function of (t, z) . From $|\psi_j(t, z)^2 - \psi_j(t, z')^2| \leq 2K_\psi L_\psi |z - z'|$, with L_ψ the Lipschitz constant of ψ_j , and the bound $\bar{G}$ on every $|G_{j,k}|$ over $[0, H]$, the blocks are Lipschitz in z with constants $2K_\psi L_\psi \bar{G}$, $2K_\psi L_\psi$ and L_ψ , uniformly in t , and bounded by $K_\psi^2(1 + \bar{G}) + K_\psi$. The integrands are admissible by (a). On $[0, H]$ the definitions (3.126) with $\chi_j = \psi_j(\cdot, Z)$ are, coordinate by coordinate, the equations of the blocks (3.136); the drivers other than B^j enter with the zero integrand. The Z -block is (3.134) at $t \wedge H$, by the stopped-integral identity at the deterministic time H . After H the coefficients vanish, which is the freeze; the identities at each time extend to all times by continuity, as in (a).

(c) The cited Markov property applies to the equation of (b), since its coefficients are Borel in time and Lipschitz in the state and its initial value is deterministic; it gives the case $r = 1$, and a Borel function of $X_{s \wedge H}$ by the Doob–Dynkin lemma. For products $F_1(x_1) \cdots F_r(x_r)$ with the times ordered, the tower property at the next-to-last time reduces r to $r - 1$, with F_{r-1} replaced by F_{r-1} times the bounded Borel function

from the case $r = 1$. Products of indicators of Borel sets form a π -system generating the Borel sets of $(\mathbb{R}^n)^r$, and the functional monotone class theorem extends the property to every bounded Borel F .

(d) Formula (3.128) is $f(t, T) = \Lambda(X_t)(T)$ with the coordinates of X_t substituted, and $X = X_{\wedge H}$ on $[0, H]$; each $\Lambda(\cdot)(T_i)$ is affine, so (c) applies. The dimension is that of (3.135).

(e) For $t \in I_k \cap [0, H]$, (3.133) gives $G_{j,k}(t) = G_{j,k}(T_k) + a_{j,k,k}\Phi_{j,k}(t)$, which is constant when $a_{j,k,k} = 0$; the indicators are then constant on that set, and every block of (3.136) is a function of x alone. Conversely, choose z with $\psi_j(z) \neq 0$; $\Phi_{j,k}$ is constant on the nondegenerate interval $I_k \cap [0, H]$ only if φ_j vanishes there, by continuity. $\square$

The proposition is conditional on the cited calculus, the Brownian drivers and the three cited results above; it constructs none of them. It is a finite-dimensional, time-inhomogeneous Markov realization of the current curve, with running time as a state. It is not the realization in time to maturity with a schedule entering only through the distances to the meetings that Conjecture 2 concerns, and it proves no nonexistence of such a realization. Part (e) concerns the coordinates (3.135) in calendar maturity: it answers only the weaker reading in which the state equations between meetings have no explicit time dependence, and says nothing about other realizations. Not covered: transition functions indexed by the initial state, their composition law, the Feller or strong Markov property; a scale built from a solution whose paths are continuous only almost surely, unless its masked scales are separately known to be admissible integrands; square-root or unbounded scales, such as the Heston scale of [12] eqs. 29 and 30, whose coefficients are not globally Lipschitz; minimality of the dimension and recovery of the state from the curve, since frozen coordinates remain in the state; true bond martingales; and correlated curve drivers.

3.27 Meeting loadings obstruct realizations in time to maturity

Realizations in time to maturity. Anchor the curve at the current time, with time to maturity $x = T - t$. On the maturities between consecutive remaining meetings, the volatility of factor j is an ordinal loading, indexed by the number of meetings in $(t, t + x]$, times a fixed continuous function $\lambda_j(x)$ times a function of a scale state that solves an autonomous Lipschitz equation. A realization is a finite-dimensional state with the same equation coefficients on every scheduled interval, a fixed restart map applied at each meeting, and a fixed map from the state and the distances to the remaining meetings to the curve; these are the same for every schedule, which enters only through those distances, and the state dimension counts every coordinate, running time included. The curve map is required to be locally Lipschitz in the state for each fixed distance vector and maturity, as the smoothness in [10] Definition 3.1 implies. [12] eqs. 36 to 38 with constant mean reversion is the case $\lambda_j(x) = e^{-\kappa_j x}$.

This section treats one factor, a constant scale and no fixed-maturity curve jumps. Fix real ordinal loadings $a_0, a_1, a_2, \dots$, independent of the schedule, and a continuous $\lambda : [0, \infty) \rightarrow \mathbb{R}$. For a schedule $0 = T_0 < T_1 < \dots < T_N$ and $t \geq 0$, let $i(t, T)$ be the number of meetings in $(t, T]$ and $D(t)$ the vector of distances from t to the remaining meetings. The volatility is

$$\sigma(t, T) = a_{i(t, T)} \lambda(T - t), \quad 0 \leq t \leq T. \quad (3.138)$$

A realization with state dimension q consists of a state Y_t in a set $Z \subseteq \mathbb{R}^q$ and a map G , the same for every schedule, with

$$f(t, t + x) = G(Y_t, D(t))(x) \quad Q\text{-a.s., for every } t \geq 0 \text{ and } x \geq 0, \quad (3.139)$$

with the null set allowed to depend on (t, x) , and such that $y \mapsto G(y, D)(x)$ is locally Lipschitz on Z for each fixed D and x , as the smoothness of the curve map in [10] Definition 3.1 implies. The identity is required in this version form because the curve at each maturity involves its own stochastic integral, which the cited calculus fixes only up to a null set of its own; one null set for all x does not follow from it. Nothing about the state dynamics or the restart map is used. For $R, n \geq 0$ the Hankel section $H_{R,n}$ is the $(R+1) \times (n+1)$ matrix with entries $(H_{R,n})_{i,l} = a_{i+l}$.

Proposition 30 (conditional). Let W be a standard Brownian motion on a filtered probability space with the usual conditions, with the cited Itô calculus of Section 3.25, and let $\lambda \equiv 1$.

(a) Fix $R, n \geq 0$, the schedule $T_m = m$ for $m = 1, \dots, n+R$, and $t = n + \frac{1}{2}$. The curve (2.2) with volatility (3.138), the drift that satisfies Assumption 2.1, and any initial curve satisfies, at the maturities $T^{(i)} = n+i+\frac{3}{4}$, $i = 0, \dots, R$,

$$(f(t, T^{(i)}))_{i=0}^R = c + H_{R,n} \Delta W \quad Q\text{-a.s.}, \quad (3.140)$$

with c deterministic, $\Delta W_0 = W_t - W_{T_n}$ and $\Delta W_l = W_{T_{n-l+1}} - W_{T_{n-l}}$ for $1 \leq l \leq n$. If a realization with state dimension q reproduces this curve at t , then $\text{rank } H_{R,n} \leq q$.

(b) If a realization with state dimension q serves every schedule, then $\text{rank } H_{R,n} \leq q$ for all R, n , and there are $L \leq q$ and real $c_0, \dots, c_{L-1}$ with

$$a_{i+L} = \sum_{l=0}^{L-1} c_l a_{i+l} \quad \text{for every } i \geq 0. \quad (3.141)$$

(c) For the cumulative ordinal loadings of [12] eq. 36 with meeting weights $\gamma_h = 1/h$, that is $a_0 = 0$ and $a_i = \sum_{h=1}^i 1/h$, no recurrence (3.141) holds. Hence for $\lambda \equiv 1$, which is quasi-exponential and even exponential ($\kappa = 0$), no realization serves every schedule, for any q .

The obstruction comes from the loadings, not from the shape: without meetings only a_0 enters and the Hankel condition is vacuous, and the answer is the quasi-exponential condition on λ of [10] Proposition 5.1 and Corollary 5.1. None of the realization results cited in Section 4 has scheduled maturity breakpoints. Finite Hankel rank and linear recurrences are equivalent by classical linear algebra; the direction needed is proved below. The condition (3.141) is the discrete analogue, for the meeting loadings, of the quasi-exponential condition on the shape.

Proof. (a) At each fixed maturity the volatility $\sigma(\cdot, T)$ is deterministic and piecewise constant. For $s \in I_k$ and $T \in I_m$, $m \geq k$, the number of meetings in $(s, T]$ is $m - k$. The maturity $T^{(i)}$ lies strictly inside I_{n+i} , and $t \in I_n$, so the driver integral up to t is $\sum_{k=0}^n a_{n+i-k}$ times the increment of W over $I_k \cap [0, t]$, by the elementary values and linearity of the integral; changing the integrand at the finitely many meeting times changes the integral by one of zero variance, hence not at all. Putting $l = n - k$ gives (3.140); the initial curve and the drift integral enter only through c . The pieces are disjoint intervals of positive length, so ΔW is a vector of independent centered Gaussian variables with positive variances, and its law is absolutely continuous on $\mathbb{R}^{n+1}$. Let $r = \text{rank } H_{R,n}$ and choose r independent rows, with coordinate projection P . Then $PH_{R,n}$ maps onto $\mathbb{R}^r$, so the law of $V = P(c + H_{R,n}\Delta W)$ is absolutely continuous on $\mathbb{R}^r$. By (3.139), $V = \Phi(Y_t)$ almost surely, with $\Phi(y) = (G(y, D(t))(T^{(i)} - t))_i$ over the chosen rows and $D(t)$ deterministic; Φ is locally Lipschitz on Z . If $r > q$, cover Z by countably many compact cubes on each of which Φ is Lipschitz; splitting a cube into K^q subcubes of side $1/K$ covers its image by K^q balls of total volume of order $K^{q-r} \rightarrow 0$. So $\Phi(Z)$ lies in a Lebesgue-null set, which contradicts the absolute continuity of the law of V . Hence $r \leq q$.

(b) Apply (a) to every (R, n) ; one q serves all, because the realization is the same for every schedule. Fix $n = q$ and let K_R be the set of unit vectors $c \in \mathbb{R}^{q+1}$ with $\sum_{l=0}^q c_l a_{i+l} = 0$ for all $i \leq R$. It is nonempty because the $q+1$ columns of $H_{R,q}$ are dependent, compact, and decreasing in R , so the K_R share a unit vector c . Let L be the largest index with $c_L \neq 0$. If $L = 0$, then $a \equiv 0$ and (3.141) holds with $L = 0$; otherwise dividing by $-c_L$ gives (3.141).

(c) A recurrence (3.141) for a_i passes to $y_{i+1} = a_{i+1} - a_i$: $1/(i+L+1) = \sum_{l < L} c_l/(i+l+1)$ for every $i \geq 0$. Both sides are rational functions of i that agree at infinitely many points, hence identical; but the left side has a pole at $-L-1$ and the right side has poles only in $\{-1, \dots, -L\}$. The case $L = 0$ would force $a \equiv 0$. By (b), no q works. $\square$

The same argument excludes every ordinal loading whose meeting weights satisfy no linear recurrence. The proposition is conditional on the cited calculus. It treats only $\lambda \equiv 1$, one factor and a constant scale; Proposition 33 extends necessity of (3.141) to every nonzero quasi-exponential shape with a constant scale; for other shapes, several factors and stochastic scales it is not shown. Sufficiency, whether a quasi-exponential shape with loadings satisfying (3.141) always admits a realization, is not shown here. The obstruction needs the regularity of G : with a merely measurable curve map and restart map dimension arguments fail. It uses (3.139) at one time and so does not cover local realizations up to a stopping time that may precede t . It needs one realization to serve schedules with arbitrarily many meetings: on a single schedule with N meetings the Hankel sections that arise have rank at most $N+1$. Whether that requirement suits the SOFR use, where only a few years of meetings are listed, is a question of scope. Nonzero fixed-maturity curve jumps and restart laws are not treated.

3.28 Quasi-exponential shapes with recurrent loadings are realized

Keep the realization notion and the notation $i(t, T)$, $D(t)$ of Section 3.27, and write $i(x, D)$ for the number of distances in D that are at most x , so that $i(t, t+x) = i(x, D(t))$. Assume

- (H1) $\lambda(x) = c e^{Ax} b$ for an $r \times r$ matrix A , a row vector c and a column vector b , that is, λ is quasi-exponential ([10] Remark 5.1);
- (H2) $a_i = u M^i v$ for all $i \geq 0$, for a $p \times p$ matrix M , a row vector u and a column vector v .

Condition (H2) is equivalent to a recurrence (3.141), through the companion matrix of its coefficients. The matrices A and M of this section are unrelated to the coefficients $A_{j,k}$, $M_{j,k}$ of Section 3.25. Write $m = pr$, $\otimes$ for the Kronecker product, $\hat{A} = I_p \otimes A$, $\hat{M} = M \otimes I_r$, $\hat{b} = v \otimes b$ and $\hat{c} = u^\top \otimes c^\top$ in $\mathbb{R}^m$, and for $x \geq 0$ and a distance vector D the row vectors

$$k(x, D) = (u M^{i(x, D)}) \otimes (c e^{Ax}), \quad K(x, D) = \int_0^x k(y, D) dy. \quad (3.142)$$

The basis is that of Section 3.26, with the curve driven by $W = B^1$ on a finite horizon H and the cited calculus of Sections 3.25 and 3.26 together with Itô's product rule ([4] §3.5). The scale state $Z \in \mathbb{R}^{p'}$ solves an autonomous equation $dZ = \mu_Z(Z) dt + \Sigma_Z(Z) dB$ from a deterministic z_0 , with μ_Z , Σ_Z globally Lipschitz and bounded on bounded sets, and is taken with every path continuous, as Proposition 29(a) provides. It may use B^1 , so any correlation between the scale and the curve driver is allowed. Let $\psi : \mathbb{R}^{p'} \rightarrow \mathbb{R}$ be bounded by K_ψ and Lipschitz, the initial curve zero, and

$$\sigma(t, T) = \psi(Z_t) a_{i(t, T)} \lambda(T - t), \quad 0 \leq t \leq T \leq H. \quad (3.143)$$

Proposition 31 (conditional). Under (H1), (H2) and these inputs:

(a) The drift $\alpha(t, T) = \sigma(t, T) \int_t^T \sigma(t, u) du$ satisfies Assumption 2.1 on every path, and the curve (2.2) with zero initial curve and no curve jumps is well defined for $0 \leq t \leq T \leq H$.

(b) Let $Y = (Z, \Xi, \Pi, \Theta) \in \mathbb{R}^{p'} \times \mathbb{R}^m \times \mathbb{R}^{m \times m} \times \mathbb{R}^m$, with $(\Xi, \Pi, \Theta)_0 = 0$, solve between meetings

$$d\Xi = \hat{A}\Xi dt + \psi(Z)\hat{b} dW, \quad d\Pi = (\hat{A}\Pi + \Pi\hat{A}^\top + \psi(Z)^2\hat{b}\hat{b}^\top)dt, \quad d\Theta = (\hat{A}\Theta + \Pi\hat{c})dt, \quad (3.144)$$

together with the equation of Z , and at each meeting apply the restart map

$$\mathcal{R}(Z, \Xi, \Pi, \Theta) = (Z, \hat{M}\Xi, \hat{M}\Pi\hat{M}^\top, \hat{M}\Theta). \quad (3.145)$$

Then, for every t and x with $t + x \leq H$, almost surely, with a null set that may depend on x and one random state for all x ,

$$f(t, t+x) = G(Y_t, D(t))(x), \quad G((Z, \Xi, \Pi, \Theta), D)(x) = k(x, D)[\Xi + \Pi K(x, D)^\top + \Theta], \quad (3.146)$$

with Ξ on its version continuous between meetings. The coefficients of (3.144) are globally Lipschitz in Y , the same on every interval and free of t , D and the schedule; $\mathcal{R}$ is a fixed linear map; G is linear in the state, does not depend on Z , and depends on the schedule only through D . Since the horizon H is arbitrary, this gives (3.139) for every t and x , so this is a realization in the sense of Section 3.27, serving every schedule, with state dimension $p' + m + m(m+1)/2 + m$, counting Π as symmetric. The constant scale $\psi \equiv 1$ is the case in which Z may be dropped.

(c) Consequently, for one factor and every scale of this class, a quasi-exponential shape with linearly recurrent ordinal loadings has a realization serving every schedule, with no condition on the correlation between the scale and the curve driver.

Without meetings, [11] Proposition 5.2 gives sufficiency of quasi-exponential shapes for $\sigma = \sum_j \varphi_{ij}(r, y) \lambda_j(x)$, and its Proposition 4.16 inserts a scale independent of the driving noise when the curve map does not depend on it, under smoothness and genericity assumptions; the constant-scale case without meetings is [10] Proposition 5.1. Proposition 31 adds the meeting breakpoints and restarts, with the recurrent loadings as the new ingredient; its argument is pathwise in the drift and uses only predictability of the scale in the diffusion, so no independence is needed.

Proof. For $s \leq t$ let $n(s, t)$ be the number of meetings in $(s, t]$, so that $i(s, T) = n(s, t) + i(t, T)$, and put $w(s, t) = (M^{n(s, t)}v) \otimes (e^{A(t-s)}b) \in \mathbb{R}^m$. By (H1), (H2) and the mixed-product rule, for $s \leq t \leq T = t + x$,

$$\sigma(s, T) = \psi(Z_s) k(x, D(t)) w(s, t). \quad (3.147)$$

(a) The scale $\psi(Z_s)$ is adapted with continuous paths, hence predictable, and bounded; the coordinates of $w(\cdot, t)$ are bounded deterministic Borel functions, so every integrand is admissible. For fixed s , $\sigma(s, \cdot) = \psi(Z_s) \tilde{\sigma}(s, \cdot)$ with $\tilde{\sigma}$ bounded and piecewise continuous, so $S(s, T) = \int_s^T \sigma(s, u) du$ is absolutely continuous and $\int_s^T \alpha(s, u) du = \frac{1}{2} S(s, T)^2$, which is (2.3) for one driver.

(b), diffusion. By (3.147) and linearity, $\int_0^t \sigma(s, t+x) dW_s = k(x, D(t)) \Xi_t$ with $\Xi_t = \int_0^t \psi(Z_s) w(s, t) dW_s$, one random vector for all x . With $\tau \leq t$ the last meeting, $w(s, t) = e^{\hat{A}(t-\tau)} w(s, \tau)$ for $s \leq \tau$ and $w(s, t) = e^{\hat{A}(t-s)} \hat{b}$

for $\tau < s \leq t$, so $\Xi_t = e^{\hat{A}(t-\tau)}\Xi_\tau + e^{\hat{A}t} \int_\tau^t e^{-\hat{A}s} \psi(Z_s) \hat{b} dW_s$. The right side is continuous in t , and the product rule gives the first equation of (3.144) between meetings. At a meeting T_j , $w(s, T_j) = \hat{M}w(s, T_j^-)$ for $s < T_j$; the two integrands differ only at $s = T_j$, so their integrals agree almost surely, which is the restart (3.145) for Ξ .

(b), drift. Fix a path. Split $S(s, T) = \psi(Z_s)[\tilde{J}(s, t) + K(x, D(t))w(s, t)]$ with $\tilde{J}(s, t) = \int_s^t \tilde{\sigma}(s, u) du$. Then

$$\int_0^t \alpha(s, T) ds = k(x, D(t))[\Pi_t K(x, D(t))^\top + \Theta_t], \quad \Pi_t = \int_0^t \psi(Z_s)^2 w w^\top ds, \quad \Theta_t = \int_0^t \psi(Z_s)^2 w \tilde{J} ds, \quad (3.148)$$

with $w = w(s, t)$ and $\tilde{J} = \tilde{J}(s, t)$. Between meetings $\partial_t w = \hat{A}w$, $w(t, t) = \hat{b}$, $\tilde{J}(t, t) = 0$ and $\partial_t \tilde{J}(s, t) = \hat{c}^\top w(s, t)$, because $k(0, D) = \hat{c}^\top$; since $s \mapsto \psi(Z_s)^2$ is continuous, Leibniz's rule gives the equations of Π and Θ in (3.144). At a meeting $\tilde{J}(s, \cdot)$ is continuous and w jumps by $\hat{M}$, which gives (3.145). With the zero initial state this proves (3.146) at each t ; (3.146) defines the version of the curve, which for each fixed T agrees almost surely with (2.2).

(b), conditions. The drift of Ξ is linear and its diffusion $\psi(Z)\hat{b}$ is bounded and Lipschitz in Z ; the source of Π is Lipschitz with constant $2K_\psi L_\psi |\hat{b}|^2$, with L_ψ the Lipschitz constant of ψ ; the drift of Θ is linear. None involves t , D or the schedule. Π is symmetric, which gives the dimension. (c) restates (b). $\square$

The proposition is conditional on the cited calculus and on the scale state of Section 3.26; it constructs neither. It gives sufficiency only: necessity of the recurrence and of quasi-exponential shapes under a stochastic scale is not shown, and Propositions 30 and 33 cover only a constant scale. Not covered: several factors, several scales per factor, and scales that move the shape, which admit no realization even without meetings ([11] Propositions 4.11 and 4.12); scales outside the stated class, such as unbounded or non-Lipschitz ψ and square-root or CIR scale states, so the Heston-type scale of [12] eqs. 29 and 30 is excluded; nonzero initial curves, nonzero fixed-maturity curve jumps, and minimality of the dimension; true bond martingales.

3.29 Recurrent loadings are necessary for every quasi-exponential shape

Keep the realization notion and the notation of Section 3.27, with one factor, the constant scale $\psi \equiv 1$ and no fixed-maturity curve jumps, and let

$$\sigma(t, T) = a_{i(t, T)} \lambda(T - t), \quad \lambda(x) = c e^{Ax} b, \quad (3.149)$$

with λ quasi-exponential and not identically zero. The basis is that of Proposition 30, and the initial curve is any deterministic curve.

Lemma 32. Let $F : [0, \tau] \rightarrow \mathbb{R}^N$ be bounded, with finitely many pieces on each of which it is continuous. Then $\int_0^\tau F(s) dW_s$, taken coordinatewise, is centered Gaussian with covariance $\int_0^\tau F(s)F(s)^\top ds$.

Proof. Left-endpoint step functions F_k on refining partitions converge to F in $L^2([0, \tau])$. The integral of a step function is a linear combination of Brownian increments by the elementary values and linearity of the integral, hence centered Gaussian with covariance $\int F_k F_k^\top ds$. The isometry gives $E|\int (F - F_k) dW|^2 = \int |F - F_k|^2 ds \rightarrow 0$; along an almost surely convergent subsequence the characteristic functions converge by bounded convergence, and their limit $\exp(-u^\top C u/2)$, $C = \int F F^\top ds$, identifies the law. $\square$

Fix $n \geq 1$, $R \geq 0$, the schedule $T_m = m$ for $m = 1, \dots, n + R$, $t = n + \frac{1}{2}$ and $T^{(i)} = t + \frac{1}{4} + i$ for $i = 0, \dots, R$. After the reduction to a controllable pair (A, b) in the proof, put $h_k = a_k c e^{kA} \in \mathbb{R}^{1 \times r}$ and let $\mathcal{H}_{R,n}$ be the $(R + 1) \times nr$ block Hankel matrix with $1 \times r$ blocks

$$(\mathcal{H}_{R,n})_{i,l} = h_{i+l}, \quad 0 \leq i \leq R, \quad 1 \leq l \leq n. \quad (3.150)$$

Proposition 33 (conditional). Under these assumptions:

- (a) If a realization with state dimension q reproduces the curve at t on this schedule, then $\text{rank } \mathcal{H}_{R,n} \leq q$.
- (b) If a realization with state dimension q serves every schedule, then the ordinal loadings satisfy a linear recurrence (3.141), of order at most $q + 1$.

(c) Consequently, for one factor, the constant scale and a nonzero quasi-exponential λ , a realization serving every schedule exists, from the zero initial curve, if and only if the ordinal loadings satisfy a linear recurrence. Necessity holds for every deterministic initial curve; sufficiency is Proposition 31 with $\psi \equiv 1$. In particular, for $\lambda \equiv 1$, Propositions 30 and 31 together characterize the ordinal loadings that admit a realization as the linearly recurrent ones. This settles the one-factor constant-scale case of Conjecture 2.

Proposition 30 is the case $\lambda \equiv 1$, where the pieces carry Brownian increments. Without meetings the answer is the quasi-exponential condition alone ([10] Proposition 5.1). The reduction to a controllable pair is the standard restriction to the reachable subspace.

Proof. Reduction. The span of $b, Ab, A^2b, \dots$ is A -invariant and contains b , and λ is unchanged when A and c are restricted to it; so (A, b) may be taken controllable, and then $c \neq 0$. If $y^\top e^{Au}b = 0$ on an interval, differentiation gives $y^\top A^k b = 0$ for all k , so $y = 0$; hence

$$\Gamma_\delta = \int_0^\delta e^{Au} b b^\top e^{A^\top u} du \quad (3.151)$$

is positive definite for every $\delta > 0$.

(a) As in Proposition 30, split $[0, t]$ into piece 0, $[n, t]$, and pieces $l \geq 1$, $[n-l, n-l+1]$, with right ends $\tau_0 = t$ and $\tau_l = n-l+1$; on piece l , $i(s, T^{(i)}) = i+l$, and changing the integrand at the finitely many piece ends does not change the integral. Put $\zeta_l = \int_{\text{piece } l} e^{A(\tau_l-s)} b dW_s \in \mathbb{R}^r$. The constant matrix $e^{A(T^{(i)}-\tau_l)}$ comes out of the integral, and $T^{(i)} - \tau_l = i+l - \frac{1}{4}$ for $l \geq 1$, so

$$f(t, T^{(i)}) = \bar{f}_i + a_i c e^{A/4} e^{iA} \zeta_0 + \sum_{l=1}^n h_{i+l} e^{-A/4} \zeta_l \quad (3.152)$$

with $\bar{f}_i$ deterministic. By Lemma 32, $(\zeta_0, \dots, \zeta_n)$ is centered Gaussian with covariance $\text{diag}(\Gamma_{1/2}, \Gamma_1, \dots, \Gamma_1)$, positive definite by (3.151), so its law is absolutely continuous. The coefficient matrix of $(\zeta_1, \dots, \zeta_n)$ in (3.152) is $\mathcal{H}_{R,n}(I_n \otimes e^{-A/4})$, of rank $\text{rank } \mathcal{H}_{R,n}$, and adding the columns of ζ_0 cannot lower the rank ρ of the whole linear map. So the vector of curve values has an absolutely continuous law on a ρ -dimensional affine subspace, and the covering argument of Proposition 30(a) gives $\text{rank } \mathcal{H}_{R,n} \leq \rho \leq q$.

(b) By (a), $\text{rank } \mathcal{H}_{R,n} \leq q$ for all R and n . For each n the first n blocks of the $q+1$ rows $(h_{i+1}, h_{i+2}, \dots)$, $i = 0, \dots, q$, are rows of $\mathcal{H}_{q,n}$ and hence dependent; the annihilating unit vectors form nonempty compact sets decreasing in n , so a unit $d \in \mathbb{R}^{q+1}$ satisfies $\sum_i d_i h_{i+l} = 0$ for every $l \geq 1$. Right multiplication by e^{-lA} gives $\sum_i d_i a_{i+l} c e^{iA} = 0$. For i_0 with $d_{i_0} \neq 0$, $c e^{i_0 A} \neq 0$, so some coordinate gives scalars $e_i = d_i (c e^{iA})_s$, not all zero, with $\sum_i e_i a_{i+l} = 0$ for all $l \geq 1$. This is a recurrence of order at most q for $(a_{k+1})_{k \geq 0}$, or $a_l = 0$ for all $l \geq 1$; prepending a_0 with a zero coefficient gives (3.141) of order at most $q+1$.

(c) Sufficiency is Proposition 31 with $\psi \equiv 1$, necessity is (b). $\square$

The proposition is conditional on the cited calculus. It treats one factor and the constant scale; necessity under a stochastic scale and with several factors remains open: with a stochastic scale independent of W the law is a Gaussian mixture, and with leverage it is not Gaussian, and neither case is treated. Several factors, necessity of quasi-exponential shapes when meetings are present, realizations without the Lipschitz regularity of the curve map, and local realizations are not covered. The order bound $q + 1$ is not claimed to be sharp.

3.30 The splice with a correlated step factor

Fix a finite horizon $H > T_N$, with I_N used up to H . Let B^1, B^2 be independent standard Brownian motions with the cited Itô calculus of Section 3.25, and for a fixed $\rho \in [-1, 1]$ put $W^S = B^1$ and $W^B = \rho B^1 + \sqrt{1 - \rho^2} B^2$. The step factor has the deterministic volatility $\sigma^S(t, T) = s_m(t)$ for $T \in I_m$, with bounded Borel s_m , and the block factor $\sigma^B(t, T) = b e^{-a(T-t)}$ with $a \neq 0$ and $b \neq 0$. The curve is (2.2) with these two factors, no curve jumps, and the drift of Assumption 2.1 written with the independent drivers:

$$\alpha = \sigma^S S^S + \sigma^B S^B + \rho(\sigma^S S^B + \sigma^B S^S), \quad S^X(u, T) = \int_u^T \sigma^X(u, v) dv. \quad (3.153)$$

Write $\Delta_m = s_m - s_{m-1}$, $1 \leq m \leq N$, for the jump of the step volatility at the maturity T_m , and call the step factor genuine if some Δ_m is nonzero on a subset of $[0, T_m)$ of positive measure. A block is a finite-dimensional space $\mathcal{E}$ of real-analytic functions on $\mathbb{R}$, invariant under the shifts $g \mapsto g(\cdot + h)$, $h \geq 0$, and containing e^{-ax} and e^{-2ax} ; a forward shift is injective on $\mathcal{E}$ by analyticity, so $\mathcal{E}$ is invariant under every shift. The family at time t is $\mathcal{M}_t(\mathcal{E}) = \{p + g(\cdot - t) : p \in S^+, g \in \mathcal{E}\}$ restricted to $[t, H]$, with S^+ the functions affine on each $I_m \cap [0, H]$; it is the restriction of one space $\mathcal{M}(\mathcal{E})$ of functions of calendar maturity. The curve is consistent with $\mathcal{M}(\mathcal{E})$ if $f(0, \cdot) \in \mathcal{M}(\mathcal{E})$ and for every $t \in [0, H]$ some random element g_t of $\mathcal{M}_t(\mathcal{E})$ satisfies $g_t(T) = f(t, T)$ almost surely for every $T \in [t, H]$. This is the invariance of [9] Definition 2.2 from the initial time, read in the version form of (3.139).

Proposition 34 (conditional). Under the cited calculus:

(a) For $t \leq T$ with $T \in I_m$, writing $S^S(u, T) = s_m(u)T + \kappa_m(u)$ on I_m ,

$$\int_0^t \alpha(u, T) du = p_t(T) + g_t^B(T) + e^{-aT}(\eta_m(t) + \theta_m(t)T), \quad (3.154)$$

with $p_t \in S^+$, $g_t^B \in \text{span}\{e^{-aT}, e^{-2aT}\}$, $\theta_m(t) = \rho b \int_0^t e^{au} s_m(u) du$ and $\eta_m(t) = \rho b \int_0^t e^{au} (\kappa_m(u) - s_m(u)/a) du$. For $t < T_m$,

$$\theta_m(t) - \theta_{m-1}(t) = \rho b \int_0^t e^{au} \Delta_m(u) du, \quad \eta_m(t) - \eta_{m-1}(t) = -\rho b \int_0^t e^{au} (T_m + \frac{1}{a}) \Delta_m(u) du. \quad (3.155)$$

For every t and T , almost surely, the random part of $f(t, T)$ equals $\sum_m \mathbf{1}_{I_m}(T) \int_0^t s_m dW^S + b e^{-aT} \int_0^t e^{au} dW^B$, one version for all T , which lies in $S^+ + \text{span}\{e^{-aT}\}$.

(b) If the curve is consistent with $\mathcal{M}(\mathcal{E})$ for some block $\mathcal{E}$, then $\rho \Delta_m = 0$ almost everywhere on $[0, T_m)$ for every m . A genuine step factor therefore splices onto a fixed block only if $\rho = 0$.

(c) Let $\mathcal{E}_0 = \text{span}\{e^{-ax}, x e^{-ax}, e^{-2ax}\}$, a block, and $f(0, \cdot) \in \mathcal{M}(\mathcal{E}_0)$. (i) If $\rho \Delta_m = 0$ almost everywhere on $[0, T_m)$ for every m , the curve is consistent with $\mathcal{M}(\mathcal{E}_0)$. (ii) For every ρ , the curve is consistent with the enlarged family $\mathcal{M}(\mathcal{E}_0) + \text{span}\{\mathbf{1}_{I_m}(T) e^{-aT}, \mathbf{1}_{I_m}(T) T e^{-aT} : 0 \leq m \leq N\}$.

(d) If the step factor is genuine, $\rho \neq 0$ and $\rho\Delta_m$ is not almost everywhere zero on $[0, T_m)$, then for every block $\mathcal{E}$ there is a nonempty open set of times $t < T_m$ at which $f(t, \cdot)$ has no version in $\mathcal{M}_t(\mathcal{E})$, and almost surely no element of $\mathcal{M}_t(\mathcal{E})$ agrees with $f(t, \cdot)$ at every rational T . At such t the coefficient θ of Te^{-aT} in (3.154) jumps across T_m .

Case (c)(i) with $\rho = 0$ is the independence splice of [34] §3, and the indicator-times-exponential terms of (c)(ii) are those of [12] eq. 36. Proposition 34 shows that, with deterministic volatilities, one step factor and one exponential block factor, these are the only two ways: a step factor with maturity steps and nonzero correlation forces the indicator terms into any fixed block. A correlation that vanishes wherever the step factor has maturity steps is the independence splice locally in time.

Proof. (a) With $S^B(u, T) = b(1 - e^{-a(T-u)})/a$, the cross part of (3.153) is $\rho b[\sigma^S(u, T)/a + e^{-a(T-u)}(S^S(u, T) - \sigma^S(u, T)/a)]$. On I_m , $\sigma^S(u, T) = s_m(u)$ and $S^S(u, \cdot)$ is continuous and piecewise affine, so $S^S(u, T) = s_m(u)T + \kappa_m(u)$ with κ_m bounded. Integrating over $u \in [0, t]$, the first term is constant in T on I_m , hence in S^+ , and the second gives $e^{-aT}(\eta_m(t) + \theta_m(t)T)$. The term $\sigma^S S^S$ is affine in T on I_m , and $\sigma^B S^B = (b^2/a)(e^{-a(T-u)} - e^{-2a(T-u)})$. This is (3.154). Continuity of $S^S(u, \cdot)$ at T_m gives $\kappa_m - \kappa_{m-1} = -\Delta_m T_m$, hence (3.155). The integrands of the random part are bounded deterministic Borel functions, and the integral splits by linearity; the result involves finitely many random variables, so it is one version for all T .

(b) Fix t and a version $g_t \in \mathcal{M}_t(\mathcal{E})$. On one event of probability one, g_t agrees at every rational T with the version from (a) plus the deterministic parts; both are affine plus real-analytic on each $I_j \cap [t, H]$, hence they agree there. The version of the random part, p_t, g_t^B and $f(0, \cdot)$ lie in $\mathcal{M}_t(\mathcal{E})$, so $X_t(T) = e^{-aT}(\eta_m(t) + \theta_m(t)T)$ on $I_m \cap [t, H]$ lies in $\mathcal{M}_t(\mathcal{E})$. For $t < T_m$, write $X_t = p + g(\cdot - t)$; the real-analytic $e(T) = g(T - t)$ differs from $e^{-aT}(\eta_m + \theta_m T)$ by an affine function on I_m , hence on $\mathbb{R}$, and similarly on I_{m-1} . Subtracting, $e^{-aT}((\eta_m - \eta_{m-1}) + (\theta_m - \theta_{m-1})T)$ is affine on I_{m-1} , and since $1, T, e^{-aT}, Te^{-aT}$ are linearly independent on every interval, both differences vanish. By (3.155) and Lebesgue's differentiation theorem, $\rho b e^{au} \Delta_m(u) = 0$ for almost every $u < T_m$.

(c) Under (i) the differences (3.155) vanish for $t < T_m$, and for $t \geq T_m$ the interval I_{m-1} does not meet $(t, H]$; so on $[t, H]$, X_t is $e^{-aT}(\eta + \theta T)$ with common coefficients, which lies in $\text{span}\{e^{-a(T-t)}, (T-t)e^{-a(T-t)}\} \subseteq \mathcal{E}_0(\cdot - t)$. With the version from (a) this is a version in $\mathcal{M}_t(\mathcal{E}_0)$. Under (ii) the terms $\mathbf{1}_{I_m}(T)e^{-aT}(\eta_m(t) + \theta_m(t)T)$ lie in the added span.

(d) The function $\varphi(t) = \int_0^t e^{au} \Delta_m(u) du$ is continuous and, by Lebesgue's differentiation theorem, not identically zero on $[0, T_m)$, so $\{t < T_m : \varphi(t) \neq 0\}$ is open and nonempty. There $\theta_m \neq \theta_{m-1}$, and the argument of (b), applied on any event of positive outer probability on which an element of $\mathcal{M}_t(\mathcal{E})$ agrees with $f(t, \cdot)$ at every rational T , gives a contradiction. $\square$

The proposition is conditional on the cited calculus. It is a classification only in its setting: deterministic volatilities, one step factor, one block factor with a single exponent, a correlation constant in time, and families of the form S^+ plus a block, possibly enlarged by indicator terms. Not covered: stochastic volatilities, under which the ramp slope and the cross-term coefficients are adapted (Proposition 28(c); Proposition 41 treats predictable bounded scales); blocks with several exponents or polynomial factors, several block or step factors, and time-varying correlation; blocks with constant or linear terms, where the front end could absorb part of the block's equation; fixed-maturity curve jumps; consistency from every initial time; and the exponent restrictions of [53] inside the block.

3.31 The splice with a quasi-exponential block factor

Keep the setting and notation of Section 3.30, and replace the block factor by

$$\sigma^B(t, T) = \lambda(T - t), \quad \lambda(x) = c e^{Ax} b, \quad (3.156)$$

with an invertible $r \times r$ matrix A , so that no exponent is zero, (A, b) controllable and $c \neq 0$. Put $\Lambda(y) = \int_0^y \lambda = c A^{-1}(e^{Ay} - I)b$. Section 3.30 is the case $r = 1$, $A = -a$. A block is as there, now required to contain λ and $\lambda\Lambda$, and $\mathcal{E}_1$ is the smallest block containing λ , $x\lambda(x)$ and $\lambda\Lambda$; it is finite-dimensional because each of these functions has finitely many independent translates. Consistency is that of Section 3.30. For $1 \leq m \leq N$ put

$$v_m(t) = \int_0^t \Delta_m(u) e^{-Au} b \, du \in \mathbb{R}^r. \quad (3.157)$$

Proposition 35 (conditional). Under the cited calculus:

(a) The time integral of the drift is as in (3.154), with $e^{-aT}(\eta_m + \theta_m T)$ replaced by a function $X_{t,m}$ on I_m that extends to an entire function of T . For $t < T_m$ the jump of the extensions across T_m is

$$X_{t,m}(T) - X_{t,m-1}(T) = \rho c e^{AT} (A^{-1} + (T - T_m)I) v_m(t) - \rho c A^{-1} b \int_0^t \Delta_m(u) \, du. \quad (3.158)$$

The random part has, for every t and T almost surely, the version $\sum_m \mathbf{1}_{I_m}(T) \int_0^t s_m \, dW^S + c e^{AT} \int_0^t e^{-Au} b \, dW_u^B$, one for all T .

(b) If the curve is consistent with $\mathcal{M}(\mathcal{E})$ for some block $\mathcal{E}$, then $\rho\Delta_m = 0$ almost everywhere on $[0, T_m)$ for every m .

(c) Let $f(0, \cdot) \in \mathcal{M}(\mathcal{E}_1)$. (i) If $\rho\Delta_m = 0$ almost everywhere on $[0, T_m)$ for every m , the curve is consistent with $\mathcal{M}(\mathcal{E}_1)$. (ii) For every ρ , the curve is consistent with $\mathcal{M}(\mathcal{E}_1) + \text{span}\{\mathbf{1}_{I_m}(T)g(T) : g \in \text{span}\{c e^{AT}y, Tc e^{AT}y : y \in \mathbb{R}^r\}, 0 \leq m \leq N\}$.

(d) If the step factor is genuine and $\rho \neq 0$, the curve is consistent with $\mathcal{M}(\mathcal{E})$ for no block $\mathcal{E}$.

Controllability and $c \neq 0$ are no loss of generality: restricting A and c to the span of $b, Ab, A^2b, \dots$ leaves λ unchanged and A invertible, and then $\lambda \neq 0$ means $c \neq 0$. Proposition 35 extends Proposition 34 from one exponent to every quasi-exponential block factor without an exponent zero: a correlated step factor with maturity steps still forces indicator-times-quasi-exponential terms into the family, and the independence splice of [34] §3 remains the only alternative. Quasi-exponential functions are those of [10] Remark 5.1.

Proof. (a) For $T \in I_m$, $S^S(u, T) = s_m(u)T + \kappa_m(u)$ and $S^B(u, T) = \Lambda(T - u)$, so the cross part of the drift is $\rho[s_m(u)\Lambda(T - u) + \lambda(T - u)(s_m(u)T + \kappa_m(u))]$, entire in T and bounded on compacts uniformly in u ; its integral $X_{t,m}$ is entire. Subtracting the expressions for m and $m - 1$, with $\kappa_m - \kappa_{m-1} = -\Delta_m T_m$, gives the integrand $\rho\Delta_m(u)[\Lambda(T - u) + \lambda(T - u)(T - T_m)]$, and $e^{A(T-u)} = e^{AT}e^{-Au}$ gives (3.158). The term $\sigma^S S^S$ lies in S^+ . The term $\sigma^B S^B = \lambda\Lambda(T - u)$ integrates into $\mathcal{E}(\cdot - t) = \mathcal{E}$: a forward shift is bijective on a finite-dimensional space of analytic functions, and the coordinates of $g(\cdot + h)$ in a basis are continuous in h . The random part is as in Proposition 34(a).

(b) As in Proposition 34(b), consistency puts the deterministic cross term in $\mathcal{M}_t(\mathcal{E})$ for every t , and analytic continuation from I_{m-1} and I_m makes (3.158) affine in T for $t < T_m$. Its constant part is affine, and $c e^{AT}(A^{-1} + (T - T_m)I)v_m(t)$ is an exponential polynomial whose exponents are eigenvalues of A , all nonzero; since $x^k e^{\mu x}$ for distinct nonzero μ , together with 1 and x , are linearly independent, it vanishes. So $v_m(t) \in \mathcal{K} = \{v : c e^{AT}(TI + N_m)v = 0 \text{ for all } T\}$ with $N_m = A^{-1} - T_m I$, for every $t < T_m$. The function v_m is absolutely continuous with derivative $\Delta_m(t)e^{-At}b$ almost everywhere, and a $\mathcal{K}$ -valued absolutely continuous function has its derivative in $\mathcal{K}$ almost everywhere; so $e^{-At}b \in \mathcal{K}$ for almost every t with

$\Delta_m(t) \neq 0$. If that set had positive measure, the projection of $e^{-At}b$ onto $\mathcal{K}^\perp$, real-analytic in t , would vanish on a set with an accumulation point, hence identically; differentiating at 0 puts every $A^k b$ in $\mathcal{K}$, so $\mathcal{K} = \mathbb{R}^r$ by controllability. Then $c e^{AT}(TI + N_m) = 0$ for all T ; at $T = 0$, $cA^{-1} = T_m c$, and the derivative at 0 gives $c(AN_m + I) = 0$, that is $2c = T_m cA$. Multiplying the first identity by A gives $c = T_m cA = 2c$, so $c = 0$, a contradiction. Hence $\rho\Delta_m = 0$ almost everywhere on $[0, T_m)$.

(c) Under (i) the jump (3.158) vanishes for $t < T_m$, and for $t \geq T_m$ the interval I_{m-1} does not meet $(t, H]$; so on $[t, H]$ the cross term is one function, the integral of $\rho[s(u)\Lambda(T-u) + \lambda(T-u)(s(u)T + \kappa(u))]$ with coefficients common to the intervals that meet $[t, H]$. Here $\Lambda(T-u)$ is a constant plus an element of the translates of λ , because A^{-1} is a polynomial in A by the Cayley–Hamilton theorem, and $\lambda(T-u)T = \lambda(T-u)(T-t) + t\lambda(T-u)$. Both lie in $S^+ + \mathcal{E}_1(\cdot - t)$, because a block containing λ contains $c e^{Ax}y$ for every y , by differentiation and controllability, and $\mathcal{E}_1$ contains $x\lambda(x)$. With the version from (a), $f(t, \cdot)$ has a version in $\mathcal{M}_t(\mathcal{E}_1)$. Under (ii), on each I_m the cross term is a constant plus $c e^{AT}y_m(t) + Tc e^{AT}z_m(t)$, which lies in the added span.

(d) is (b) for a genuine step factor with $\rho \neq 0$. □

The proposition is conditional on the cited calculus, with the scope of Proposition 34 otherwise. Unlike Proposition 34(d), it does not locate an open set of times at which no version exists. Not covered: block factors with an exponent zero, whose polynomial part lets the front end absorb affine parts of the jump; several block or step factors; stochastic volatilities, treated for predictable bounded scales in Proposition 41; curve jumps; and consistency from every initial time.

3.32 The block's restrictions survive the uncorrelated splice

Fix $\beta > 0$, integers $n_1, n_2 \geq 0$, and the biexponential-polynomial family of [53] §2.2,

$$F(x, z) = \left(\sum_{\mu=0}^{n_1} z_{1,\mu} x^\mu \right) e^{-\beta x} + \left(\sum_{\mu=0}^{n_2} z_{2,\mu} x^\mu \right) e^{-2\beta x}, \quad (3.159)$$

with parameters $z \in \mathbb{R}^{n_1+n_2+2}$ indexed by (i, μ) . For a real vector b and a symmetric nonnegative definite matrix a , both indexed like z , the consistency equation of [53] §2.1, quoting [24] Proposition 3.2, is

$$\partial_x F(x, z) = \sum_i b^i \partial_{z_i} F(x, z) + \sum_{i,j} a^{ij} \left[\frac{1}{2} \partial_{z_i z_j}^2 F(x, z) - \partial_{z_i} F(x, z) \int_0^x \partial_{z_j} F(\eta, z) d\eta \right]. \quad (3.160)$$

Lemma 36. If (z, a, b) satisfies (3.160) for every $x \geq 0$, then $a_{(i,\mu),(j,v)} = 0$ unless $i = j = 1$ and $\mu, v \leq \min(n_1, \lfloor n_2/2 \rfloor)$. So a process consistent with the family has at most $\min(n_1, \lfloor n_2/2 \rfloor) + 1$ components with nonzero diffusion, all among the $Z^{1,\mu}$.

This is the first proposition of [53] §4.2; the proof below is ours.

Proof. F is linear in z , so the second derivatives vanish, and the two sides of (3.160) are exponential polynomials with exponents $-\beta, -2\beta, -3\beta, -4\beta$ and no exponent zero apart from constants multiplying exponentials. Since $\int_0^x \eta^v e^{-c\eta} d\eta = v!/c^{v+1} - e^{-cx}(x^v/c + \text{lower powers})$, the term $-a_{(i,\mu),(j,v)} \partial_{z_{i,\mu}} F \int_0^x \partial_{z_{j,v}} F$ contributes $a_{(i,\mu),(j,v)} x^{\mu+v} e^{-(i+j)\beta x} / (j\beta)$ plus lower powers. Only the pairs $i = j = 2$ produce $e^{-4\beta x}$. If some $a_{(2,k),(2,k)} \neq 0$, take k largest; nonnegative definiteness gives $a_{(2,\mu),(2,v)} = 0$ when μ or v exceeds k , so the coefficient of $x^{2k} e^{-4\beta x}$ is $a_{(2,k),(2,k)} / (2\beta) \neq 0$, which contradicts (3.160). Hence every diagonal entry with $i = 2$ vanishes, and with it every entry in its row and column. The coefficient of $e^{-2\beta x}$ then receives, from a , the terms $a_{(1,\mu),(1,v)} x^{\mu+v} / \beta$ plus lower powers, while $\partial_x F$ and the b -terms contribute degree at most

n_2 . If some $a_{(1,k),(1,k)} \neq 0$ with $2k > n_2$, take k largest; the coefficient of $x^{2k}e^{-2\beta x}$ is $a_{(1,k),(1,k)}/\beta \neq 0$, a contradiction. So $a_{(1,\mu),(1,\mu)} = 0$ for $\mu > \lfloor n_2/2 \rfloor$, and nonnegative definiteness removes the rest. $\square$

Now splice the family onto a step-plus-ramp front end. Keep the dates and horizon of Section 3.25, let $W = (W^S, W^B)$ be a standard Brownian motion in $\mathbb{R}^{d_S+d_B}$ with the cited calculus of Section 3.25 and Itô's formula ([4] §3.5), and let

$$f(t, T) = \sum_m \mathbf{1}_{I_m}(T) [L_m(t) + C_m(t)(T - T_m)] + F(T - t, Z_t), \quad 0 \leq t \leq T \leq H, \quad (3.161)$$

where each L_m and C_m is a continuous Itô process driven by W^S alone, and $dZ = b dt + \sigma_Z dW^B$ with σ_Z predictable with locally square-integrable paths and b progressively measurable with locally integrable paths; put $a = \sigma_Z \sigma_Z^\top$.

Proposition 37 (conditional). Suppose the drift and volatility that Itô's formula gives for (3.161) satisfy Assumption 2.1.

(a) Then for almost every t , almost surely, (Z_t, a_t, b_t) satisfies (3.160) for every $x \geq 0$, not only for $x \leq H - t$, and the front end satisfies Assumption 2.1 for its own drivers.

(b) Hence Lemma 36 applies to Z : at most $\min(n_1, \lfloor n_2/2 \rfloor) + 1$ components of Z have nonzero diffusion. Conversely, if (Z, a, b) solves (3.160) for every $x \geq 0$ and the front end satisfies Assumption 2.1 for its own drivers, as a front end of Theorem 11 driven by W^S does, then the curve (3.161) satisfies Assumption 2.1. So the uncorrelated splice imposes on the block exactly the restrictions of the block alone.

(c) The front end absorbs no part of (3.160), because the functions $x^k e^{-i\beta x}$, $i = 1, \dots, 4$, are linearly independent of the polynomials on every interval, and the splice has no cross terms, because the two groups of drivers are independent.

[34] §3 splices by independence without discussing the block's restrictions. So, in the uncorrelated splice, the exponent restrictions of [53] survive unchanged inside the block.

Proof. For $T \in I_m$ the front-end part is an Itô process with volatility σ^S in the W^S coordinates and drift $D^S(t, \cdot)$ affine on I_m . Itô's formula for $(t, z) \mapsto F(T - t, z)$, which is smooth, gives the drift $D^B = -\partial_x F + \sum_i b^i \partial_{z_i} F + \frac{1}{2} \sum_{i,j} a^{i,j} \partial_{z_i z_j}^2 F$ and the volatility $\sigma^B = \sum_i \partial_{z_i} F \sigma_Z^i$ in the W^B coordinates, at $(T - t, Z_t)$. Because the driver groups are disjoint, $|\int \sigma|^2 = |\int \sigma^S|^2 + |\int \sigma^B|^2$, and differentiating (2.3) in T gives $D^S + D^B = \sigma^S \cdot \int_t^T \sigma^S + \sigma^B \cdot \int_t^T \sigma^B$ for almost every T . With $x = T - t$, $\sigma^B \cdot \int \sigma^B = \sum_{i,j} a^{i,j} \partial_{z_i} F(x, Z) \int_0^x \partial_{z_j} F(\eta, Z) d\eta$, so the residual $R(t, x) = D^B - \sigma^B \cdot \int \sigma^B$ of (3.160) equals $-(D^S - \sigma^S \cdot \int \sigma^S)(t, t + x)$. The right side is a polynomial in x on each $I_m \cap [t, H]$, while $R(t, \cdot)$ is an exponential polynomial with exponents $-\beta, \dots, -4\beta$ whose coefficients are polynomials; by linear independence both vanish on each such interval, hence on $[0, H - t]$, and by analyticity $R(t, x) = 0$ for all $x \geq 0$. This is (a) and (c); the vanishing of the right side is the front end's own Assumption 2.1. Lemma 36 gives the bound in (b). For the converse, (3.160) and the front end's condition give the differentiated identity, and integrating in T gives (2.3). $\square$

The proposition is conditional on the cited calculus. It assumes Assumption 2.1 for the drift of the representation (3.161); identifying it with the drift of another Itô decomposition of $f(\cdot, T)$ uses uniqueness of Itô decompositions, which is standard but not among the cited inputs. Not covered: correlated splices, where cross terms appear (Propositions 34 and 35 treat deterministic volatilities); blocks with terms of exponent zero, such as the constant of the Svensson and Nelson–Siegel families, where the front end absorbs the part of (3.160) that is affine in T (Proposition 38); the extended Svensson families of [53] §5 as such; the front end's own consistency when its volatility is stochastic; and existence of a parameter process

with nontrivial factors. Sufficient and exact conditions for $\min(n_1, \lfloor n_2/2 \rfloor) + 1$ nondegenerate factors of the block alone are Proposition 39.

3.33 Blocks with constant or linear terms

Keep the uncorrelated splice (3.161) of Section 3.32 with two changes. The block is any family linear in its parameters, $F(x, z) = \sum_{i=1}^N z_i \varphi_i(x)$, with fixed real-analytic φ_i linearly independent on every interval. The front end is of the type of Theorem 11: each C_m has finite variation, so the front-end volatility is constant on each I_m , while the drifts of L_m and C_m are free. The block residual is

$$R(t, x) = \sum_i b^i \varphi_i(x) - \sum_{i,j} a^{ij} \varphi_i(x) \int_0^x \varphi_j(\eta) d\eta - \partial_x F(x, Z_t), \quad (3.162)$$

the difference of the two sides of (3.160), since $\partial_z^2 F = 0$.

Proposition 38 (conditional). Suppose the drift and volatility that Itô's formula gives for (3.161) satisfy Assumption 2.1.

(a) Then for almost every t , almost surely, $R(t, \cdot)$ is affine: $R(t, x) = c_0(t) + c_1(t)x$ for all $x \geq 0$. Conversely, if $R(t, \cdot)$ is affine and the front-end drifts are chosen so that $D^S(t, T) - \sigma^S(t, T) \cdot \int_t^T \sigma^S(t, u) du = -R(t, T - t)$ on each I_m , the splice satisfies Assumption 2.1. If the block contains no nonzero affine function this gives $R = 0$, as in Proposition 37.

(b) For the Nelson–Siegel family with fixed exponent $\beta > 0$,

$$F(x, z) = z_1 + (z_2 + z_3 x)e^{-\beta x}, \quad (3.163)$$

$R(t, \cdot)$ is affine if and only if a is supported on the single entry a^{11} , $b^2 = z_3 - \beta z_2$ and $b^3 = -\beta z_3$, with $a^{11} \geq 0$ and b^1 arbitrary. Then $R = b^1 - a^{11}x$.

(c) Consequently, in the uncorrelated splice only the level z_1 of a Nelson–Siegel block may carry a factor. A move of z_1 is the same curve move as adding one common constant to every L_m , a parallel shift that the front-end family already contains, driven by W^B ; the front-end drifts absorb its convexity, C_m acquiring the ramp a^{11} , so that the front end is then not consistent on its own. The block's own parameters z_2 and z_3 carry no factor, as when the block stands alone.

Alone, a consistent Itô process for the Nelson–Siegel family is deterministic, with a constant level and z_2, z_3 following the drifts in (b) ([24] Proposition 3.2 and Theorem 4.1). Proposition 38 shows that the scheduled-step front end changes this only through the overlap of the two families, the constants.

Proof. (a) As in the proof of Proposition 37, Assumption 2.1 gives $R(t, T - t) = -(D^S - \sigma^S \cdot \int \sigma^S)(t, T)$ for almost every $T \in [t, H]$. The front-end volatility is constant on each I_m and D^S is affine there, so the right side is affine in T , hence in $x = T - t$, on each $I_m \cap [t, H]$, almost everywhere and then everywhere by continuity. $R(t, \cdot)$ is real-analytic, because the primitive of a real-analytic function is real-analytic, so it is affine on $\mathbb{R}$. Conversely, if $R = c_0 + c_1 x$, take the drift of L_m to be $-(c_0 + c_1(T_m - t))$ plus the value of $\sigma^S \cdot \int \sigma^S$ at T_m , and the drift of C_m to be $-c_1$ plus the slope of that term on I_m ; then the differentiated drift condition holds and integrating in T gives (2.3).

(b) Here $\varphi_1 = 1$, $\varphi_2 = e^{-\beta x}$, $\varphi_3 = xe^{-\beta x}$, and $\int_0^x \eta^k e^{-\beta \eta} d\eta = k!/\beta^{k+1} - P_k(x)e^{-\beta x}$ for polynomials P_k with $x^k = \beta P_k - P'_k$. Only the pairs $i, j \in \{2, 3\}$ produce $e^{-2\beta x}$, with coefficient $\sum a^{ij} x^{i-2} P_{j-2} = \beta S - S'/2$, where $S = \sum a^{ij} P_{i-2} P_{j-2}$ and a is symmetric. If R is affine this coefficient vanishes; the polynomial solutions of $\beta S - S'/2 = 0$ are zero, so $S = 0$. The $\{2, 3\}$ block of a is nonnegative definite, so S is a sum of squares

of elements of the span of P_0 and P_1 , which are independent; hence that block vanishes, and nonnegative definiteness gives $a^{12} = a^{13} = 0$. With only a^{11} left, the $e^{-\beta x}$ part of R is $(b^2 - z_3 + \beta z_2) + (b^3 + \beta z_3)x$, which must vanish, and the exponent-zero part is $b^1 - a^{11}x$. The converse is the same computation.

(c) a supported on a^{11} means σ_Z has no loading on z_2 or z_3 , and moving z_1 by c moves $F(\cdot, z)$ by the constant c , as does adding c to every L_m ; the ramp is the $c_1 = -a^{11}$ of (a)'s converse. $\square$

The proposition is conditional on the cited calculus, with Assumption 2.1 for the drift of the representation (3.161) as in Proposition 37. Not covered: a varying Nelson–Siegel exponent; the Svensson family, whose second exponential needs its own computation (Proposition 40 for fixed exponents); blocks whose polynomial part has degree two or more, where only part of the exponent-zero equation is absorbed (Proposition 45); front ends with stochastic ramp volatility, which absorb polynomials of degree up to three; correlated splines; and existence of a process with the stated coefficients.

3.34 Correlated factors for the biexponential-polynomial family

Keep the family (3.159) and the consistency equation (3.160) of Section 3.32, and put $n = \lfloor n_2/2 \rfloor$. Assume the conditions and conventions of [53] §4.2: (C1) $n_1 \geq n$; (C2) $z_{1,i} \neq 0$ for $0 \leq i \leq n_1$; (C3) for $0 \leq i \leq 2n$, $z_{2,i} \neq 0$ or $z_{2,i+1} \neq 0$; and $z_{2,2n} \neq 0$. Let $K_0 = \{k \leq 2n : z_{2,k} = 0\}$; as in the source's reduced system, a coefficient $z_{2,k}$ with $k \in K_0$ is structurally absent, not a parameter, and its drift $b_{2,k}$ is zero. By (C3), K_0 contains no two consecutive integers. Put $r_k = (k+1)z_{2,k+1}$ for $k \in K_0$. By Lemma 36, a is supported on the block $A = (a_{(1,\mu),(1,\nu)})_{0 \leq \mu, \nu \leq n}$, and $n+1$ nondegenerate factors means A positive definite. Let P_ν be the polynomials with $P_\nu(x)e^{-\beta x} = \int_x^\infty \eta^\nu e^{-\beta \eta} d\eta$, and for a polynomial $S = \sum_{j \leq 2n} s_j x^j$ put

$$t_k(S) = \beta s_k - \frac{1}{2}(k+1)s_{k+1}, \quad S_A = \sum_{\mu, \nu \leq n} a_{\mu\nu} P_\mu P_\nu. \quad (3.164)$$

Proposition 39. Under these conditions:

(a) (z, a, b) satisfies (3.160) for every $x \geq 0$ if and only if the drifts b are given by explicit formulas in (z, A) and

$$t_k(S_A) = r_k \quad \text{for every } k \in K_0. \quad (3.165)$$

(b) A positive definite A satisfying (a) exists if and only if there is a polynomial S of degree at most $2n$ with $S > 0$ on $\mathbb{R}$, $s_{2n} > 0$ and $t_k(S) = r_k$ for every $k \in K_0$. Equivalently, there is no $\lambda \neq 0$ in $\mathbb{R}^{K_0}$ such that the coefficient sequence ℓ of $\sum_k \lambda_k t_k$ has a positive semidefinite Hankel matrix $(\ell_{i+j})_{0 \leq i, j \leq n}$ and $\sum_k \lambda_k r_k \leq 0$.

(c) If $n_2 = 2n$ or $z_{2,2n} \neq 0$, a positive definite A exists for every z satisfying the conditions: (3.160) has a solution with $n+1$ nondegenerate, in general correlated, factors at every such parameter point.

(d) If $n_2 = 2n+1$ and $z_{2,2n} = 0$, then $z_{2,2n+1} > 0$ is necessary, even for a nonnegative definite A with $a_{nn} > 0$. For $n = 0$ it is sufficient. For $n = 1$ the exact condition is $z_{2,3} > 0$, together with $3z_{2,3} + 4\beta^2 z_{2,1} > 0$ when also $z_{2,0} = 0$.

(e) At $\beta = 1$, $n_1 = 0$, $n_2 = 1$, $z_1 = (1)$, $z_2 = (0, -1)$, which satisfies the conditions, no nonnegative definite a and no b satisfy (3.160); and no path whose coordinate $Z^{2,1}$ is right-continuous at a time where it is negative satisfies (3.160) at almost every later time near it.

Part (e) shows that the central proposition of [53] §4.2, which asserts $\min(n_1, \lfloor n_2/2 \rfloor) + 1$ nontrivial factors under these conditions, is false as the source's reduced system states it, with $z_{2,k}$ structurally absent and $b_{2,k} = 0$ for $k \in K_0$; if every $b_{2,k}$ were free, the proposition would hold trivially. Part (c) shows that its

conclusion does hold, with correlated rather than diagonal factors, whenever $2n \notin K_0$. The source's proof treats only diagonal A , and its reduced system carries the wrong sign on the $e^{-2\beta x}$ terms in a . The identity behind (3.165) and the reduction to positive polynomials are ours; that nonnegative polynomials in one variable are sums of squares, and the Hankel description of the dual cone, are classical.

Proof. (a) Since $\int_0^x \eta^\nu e^{-\beta\eta} d\eta = \nu!/\beta^{\nu+1} - P_\nu(x)e^{-\beta x}$, the a -term of (3.160) is $-\sum a_{\mu\nu}(\nu!/\beta^{\nu+1})x^\mu e^{-\beta x} + [\sum a_{\mu\nu}x^\mu P_\nu]e^{-2\beta x}$. The coefficient of $x^k e^{-\beta x}$ fixes $b_{1,k}$, since every $z_{1,k}$ is a parameter by (C2); that of $x^k e^{-2\beta x}$ fixes $b_{2,k}$ for $k \notin K_0$, and for $k \in K_0$ it reads $r_k = [x^k] \sum a_{\mu\nu}x^\mu P_\nu$; nothing arises for $k > 2n$. Writing $A = \sum_i \varepsilon_i v_i v_i^\top$ with $\varepsilon_i = \pm 1$, $q_i = \sum_\mu v_{i\mu}x^\mu$ and $Q_i = \sum_\nu v_{i\nu}P_\nu$, differentiation gives $q_i = \beta Q_i - Q'_i$, so $\sum a_{\mu\nu}x^\mu P_\nu = \sum_i \varepsilon_i (\beta Q_i^2 - Q_i Q'_i) = \beta S_A - S'_A/2$, which is (3.165).

(b) Since $P_\nu = x^\nu/\beta + \text{lower terms}$, $A \mapsto S_A$ is the Gram map $M \mapsto m^\top M m$, $m = (1, x, \dots, x^n)$, composed with an invertible congruence. The Gram map sends the positive semidefinite cone onto the nonnegative polynomials of degree at most $2n$, because these are sums of squares, and, being linear and surjective, sends the positive definite cone onto the interior $U = \{S > 0 \text{ on } \mathbb{R}, s_{2n} > 0\}$. This is the first form. For the second, separate the open convex cone U from the affine space $\{t_k(S) = r_k, k \in K_0\}$: a separating functional is $\sum_k \lambda_k t_k$, nonnegative on the nonnegative polynomials, which means that its coefficient sequence has a positive semidefinite Hankel matrix, and $\lambda \cdot r \leq 0$; the t_k are independent because no two indices in K_0 are consecutive.

(c) It suffices to find $S_h \in U$ with $t_k(S_h) = 0$ on K_0 : then $S_p + N S_h \in U$ for a particular solution S_p and large N . By (b) with $r = 0$, this fails only if some $\lambda \neq 0$ gives a positive semidefinite Hankel matrix. Here $\ell_j = \beta \lambda_j [j \in K_0] - \frac{j}{2} \lambda_{j-1} [j-1 \in K_0]$, and a zero diagonal entry kills its row, so either $\ell = \ell_0 \delta_0$ or $\ell_{2n} > 0$. The first forces $0 \in K_0$, $1 \notin K_0$, $\ell_1 = -\lambda_0/2 = 0$, hence $\ell_0 = 0$. In the second, $2n \notin K_0$ forces $2n-1 \in K_0$, and descending induction with the 2×2 principal minors gives $\ell_{2j} > 0$ and $2j-1 \in K_0$ for $j = n, \dots, 1$; at $j = 1$, $\ell_0 > 0$ needs $0 \in K_0$ next to $1 \in K_0$, a contradiction. For $n = 0$, $K_0 = \emptyset$ and $S_h = 1$.

(d) $\lambda = \delta_{2n}$ gives $\ell = \beta \delta_{2n}$, whose Hankel matrix is positive semidefinite, so (b) requires $r_{2n} = (2n+1)z_{2,2n+1} > 0$; for nonnegative definite A with $a_{nn} > 0$ the leading coefficient a_{nn}/β^2 of S_A gives the same. For $n = 0$, $U = \{s_0 > 0\}$ and the condition is $\beta s_0 = r_0$. For $n = 1$, $U = \{s_2 > 0, 4s_0 s_2 > s_1^2\}$; with $K_0 = \{2\}$ the condition is $r_2 > 0$, and with $K_0 = \{0, 2\}$ the quadratic condition on s_1 has a solution if and only if $s_2/\beta + 4r_0 > 0$, which is the stated inequality.

(e) Here $n = 0$, $K_0 = \{0\}$, and (3.165) reads $a_{00}/\beta = z_{2,1} = -1$, impossible for $a_{00} \geq 0$. A right-continuous $Z^{2,1}$ that is negative at a time stays negative on an interval of positive length after it, where (3.160) would have to hold almost everywhere. $\square$

The proposition is pointwise algebra at one parameter point, as is the source's; existence of an Itô process with these coefficients along its path is not shown. Not covered: the explicit form of the condition in (b) for $n \geq 2$ when $2n \in K_0$, which is a semidefinite feasibility condition; degenerate nonnegative definite A with positive diagonal but rank below $n+1$; structurally absent $z_{1,k}$, and coefficients that vanish at isolated times rather than structurally. With Proposition 37 the conclusions transfer to the uncorrelated splice, but they are stated here for the block alone.

3.35 The Svensson block with fixed exponents

Keep the setting of Section 3.33, with the Svensson family with fixed exponents $0 < \beta_1 \neq \beta_2$,

$$F(x, z) = z_1 + (z_2 + z_3 x) e^{-\beta_1 x} + z_4 x e^{-\beta_2 x}, \quad (3.166)$$

and basis $\varphi_1 = 1$, $\varphi_2 = e^{-\beta_1 x}$, $\varphi_3 = xe^{-\beta_1 x}$, $\varphi_4 = xe^{-\beta_2 x}$. Alone means $R(t, \cdot) = 0$ in (3.162); in the splice means $R(t, \cdot)$ affine, which by Proposition 38(a) is consistency of the splice. The statements hold at one parameter point (z, a, b) with a nonnegative definite.

Proposition 40. (a) If $\beta_2 \neq 2\beta_1$ and $z_4 \neq 0$, no a and no b make R affine. No path whose coordinate Z^4 is right-continuous at a time where it is nonzero makes R affine at almost every later time near it, alone or in the splice.

(b) Let $\beta_2 = 2\beta_1 =: 2\beta$. Then R is affine if and only if every entry of a involving z_3 or z_4 vanishes, $a_{22} = \beta z_4$ (so $z_4 \geq 0$), $b_2 = z_3 + z_4 - \beta z_2 - a_{12}/\beta$, $b_3 = a_{12} - \beta z_3$, $b_4 = -2\beta z_4$, and $a_{11}a_{22} \geq a_{12}^2$, with b_1 arbitrary; then $R = b_1 - a_{12}/\beta - a_{11}x$. Alone, in addition $a_{11} = a_{12} = b_1 = 0$.

(c) Consequently the splice adds only the level z_1 as a factor, a parallel shift already in the front-end family, correlated with z_2 up to the Cauchy–Schwarz bound $a_{12}^2 \leq a_{11}\beta z_4$; for $z_4 > 0$ a correlated splice point with $a_{12} \neq 0$ exists. The block’s own nontrivial factor is z_2 , with variance βz_4 , as alone.

The Svensson family alone is treated in [25] §9.2: by its Theorems 8.1 and 8.2 only $\beta_1 = 2\beta_2$ or $\beta_2 = 2\beta_1$ can carry a nontrivial process, and in the second case its equations (30) and (31) are the case alone of (b). Part (a) makes explicit that the first case also needs $z_4 = 0$. Parts (b) and (c) in the splice follow the affine reduction of Proposition 38.

Proof. Write R as a sum of polynomials times $e^{-\kappa x}$, $\kappa \in \{0, \beta_1, \beta_2, 2\beta_1, \beta_1 + \beta_2, 2\beta_2\}$, using $\int_0^x \eta^k e^{-\gamma \eta} d\eta = k!/ \gamma^{k+1} - P_k^{(\gamma)}(x)e^{-\gamma x}$ with $P_0^{(\gamma)} = 1/\gamma$ and $P_1^{(\gamma)} = x/\gamma + 1/\gamma^2$. Polynomials times exponentials with distinct real exponents are linearly independent on every interval, so every coefficient with $\kappa \neq 0$ vanishes, in the splice as alone.

The only pair with exponent $2\beta_2$ is φ_4 with $\int \varphi_4$, contributing $a_{44}(x^2/\beta_2 + x/\beta_2^2)e^{-2\beta_2 x}$, unless $2\beta_2 = \beta_1$. In the first case $a_{44} = 0$. If $2\beta_2 = \beta_1$, the group $e^{-\beta_1 x}$ also receives $-a_{13}x^2e^{-\beta_1 x}$, and the group $e^{-2\beta_1 x}$ has x^2 -coefficient a_{33}/β_1 and no other source; so $a_{33} = 0$, hence $a_{13} = 0$ and $a_{44} = 0$. Either way row 4 of a vanishes by nonnegative definiteness.

(a) With row 4 zero, the terms with exponent β_2 are $-z_4(1 - \beta_2 x)e^{-\beta_2 x}$ and $b_4 xe^{-\beta_2 x}$, together with terms from φ_2, φ_3 only if $2\beta_1 = \beta_2$. Otherwise the constant coefficient $-z_4$ must vanish. A right-continuous Z^4 nonzero at a time stays nonzero on an interval of positive length after it, where the condition must hold almost everywhere.

(b) Row 4 vanishes since $4\beta \neq \beta$. The $x^2e^{-2\beta x}$ coefficient is $a_{33}/\beta - a_{14}$, so $a_{33} = 0$ and row 3 vanishes. The remaining $e^{-2\beta x}$ terms are $-z_4(1 - 2\beta x)$, $b_4 x$ and a_{22}/β , giving $a_{22} = \beta z_4$ and $b_4 = -2\beta z_4$. The $e^{-\beta x}$ terms are $-(z_3 - \beta z_2 - \beta z_3 x)$, $b_2 + b_3 x$ and $a_{12}/\beta - a_{12}x - a_{22}/\beta$, giving b_2 and b_3 . The exponents 3β and 4β involve rows 3 and 4 only, and the exponent-zero part is $b_1 - a_{11}x - a_{12}/\beta$. Conversely these values make every part with $\kappa \neq 0$ vanish. Alone, the exponent-zero part vanishes too, so $a_{11} = 0$, hence $a_{12} = 0$ and $b_1 = 0$.

(c) The bound is nonnegative definiteness of the $\{1, 2\}$ block, and $a = vv^\top$ with $v = (1, \sqrt{\beta z_4}, 0, 0)$ gives $a_{12} \neq 0$; the level is a parallel shift as in Proposition 38(c). $\square$

The proposition is pointwise algebra, with the pathwise part (a); existence of a process with these coefficients is not shown. Not covered: varying exponents; correlated splices; a front end with stochastic ramp volatility, which absorbs polynomials of degree up to three; the case $z_4 = 0$ is Proposition 38(b).

3.36 The correlated splice with stochastic scales

Keep the setting of Section 3.31, including the quasi-exponential $\lambda = c e^{Ax}b$ with A invertible, (A, b) controllable and $c \neq 0$, the blocks and consistency, and replace the deterministic volatilities by predictable bounded processes:

$$\sigma^S(u, T) = s_m(u) \ (T \in I_m), \quad \sigma^B(u, T) = \psi_u \lambda(T - u), \quad dW^S = dB^1, \quad dW^B = \rho_u dB^1 + \sqrt{1 - \rho_u^2} dB^2, \quad (3.167)$$

with $s_0, \dots, s_N$, ψ and ρ predictable and bounded, $|\rho| \leq 1$, and $\Delta_m = s_m - s_{m-1}$. The drift is (3.153) with ρ replaced by ρ_u , path by path; it satisfies Assumption 2.1 with respect to the independent drivers. The initial curve lies in $\mathcal{M}(\mathcal{E})$.

Proposition 41 (conditional). Under the cited calculus:

(a) For $t < T_m$, on every path, the jump across T_m of the analytic extensions of the cross term is

$$c e^{AT} (A^{-1} + (T - T_m)I) v_m(t) - c A^{-1} b \int_0^t \rho_u \psi_u \Delta_m(u) du, \quad v_m(t) = \int_0^t \rho_u \psi_u \Delta_m(u) e^{-Au} b du. \quad (3.168)$$

For every t and T , almost surely, the random part of $f(t, T)$ equals $\sum_m \mathbf{1}_{I_m}(T) \int_0^t s_m dW^S + c e^{AT} \int_0^t \psi_u e^{-Au} b dW_u^B$, one version for all T .

(b) If the curve is consistent with $\mathcal{M}(\mathcal{E})$ for some block $\mathcal{E}$, then almost surely $\rho_u \psi_u \Delta_m(u) = 0$ for almost every $u < T_m$, for every m .

(c) (i) If $f(0, \cdot) \in \mathcal{M}(\mathcal{E}_1)$ and almost surely $\rho_u \psi_u \Delta_m(u) = 0$ for almost every $u < T_m$, for every m , the curve is consistent with $\mathcal{M}(\mathcal{E}_1)$. (ii) For all predictable bounded scales, the curve is consistent with the indicator-enlarged family of Proposition 35(c)(ii).

(d) If, with positive probability, $\rho_u \psi_u \Delta_m(u) \neq 0$ on a set of u of positive measure for some m , the curve is consistent with $\mathcal{M}(\mathcal{E})$ for no block $\mathcal{E}$.

So with predictable bounded scales on the step factor, the block and the correlation, the independence splice and the indicator-times-quasi-exponential terms remain the only two ways; a correlation that is switched off wherever the step factor has maturity steps is the independence splice locally in time. [12] eqs. 35 to 41 construct the indicator-times-exponential model with Heston scales and do not claim that it is forced; [11] has neither breakpoints nor a splice.

Proof. The integrands are predictable and bounded, hence admissible, and the drift is bounded, so its time integral is a pathwise Lebesgue integral. For a fixed path and u , (3.167) is the volatility of Section 3.31 with the constants replaced by the numbers $s_m(u)$, $\psi_u b$ and ρ_u , so Assumption 2.1 holds as there, and the cross part of the drift is $\rho_u \psi_u$ times the integrand of Proposition 35(a); integrating in u gives (3.168) on every path. The random part follows by linearity, with finitely many random variables. For (b), fix $t < T_m$; on one event of probability one the version in $\mathcal{M}_t(\mathcal{E})$ agrees with the version from (a) plus the drift parts, the parts other than the cross term lie in $\mathcal{M}_t(\mathcal{E})$, and the argument of Proposition 35(b), which is pathwise, gives $v_m(t, \omega) \in \mathcal{K}$. Intersecting over rational $t < T_m$ and using continuity of $v_m(\cdot, \omega)$ and closedness of $\mathcal{K}$, $v_m(t, \omega) \in \mathcal{K}$ for every $t < T_m$; then the derivative $\rho_u \psi_u \Delta_m(u) e^{-Au} b$ lies in $\mathcal{K}$ for almost every u , and controllability, as in Proposition 35(b), forces $\rho_u \psi_u \Delta_m(u) = 0$ almost everywhere. Part (c) is Proposition 35(c) path by path; the version is measurable because predictable scales are jointly measurable in (u, ω) , and in (i) it is set to zero on a measurable null set. For (d), on the event named there some rational $t < T_m$ has $v_m(t) \notin \mathcal{K}$ with positive probability, and (b)'s argument excludes a version at that t . $\square$

The proposition is conditional on the cited calculus, with controllability, $c \neq 0$ and A invertible as standing hypotheses as in Proposition 35. Not covered: a block volatility that depends on the block's own state, where the jump carries a random vector in place of $\psi_u b$ (Proposition 43); several step or block factors; unbounded or square-root scales, such as the Heston scale of [12]; a diffusive front-end ramp, which the front end's own Assumption 2.1 excludes (Theorem 11, Proposition 28(c)); and existence of processes beyond the integrals themselves.

3.37 The correlated splice with a state-dependent block volatility

Keep the splice setting of Section 3.33, with independent drivers $B = (B^1, \dots, B^k)$ and the cited calculus with Itô's formula. The front end is of the type of Theorem 11, $L_m(t) + C_m(t)(T - T_m)$ on I_m with C_m of finite variation and free drifts, and L_m has the noise $s_m(t)H^S(t) \cdot dB$ with s_m predictable and bounded, $\Delta_m = s_m - s_{m-1}$, and a predictable bounded row vector H^S . The block is a quasi-exponential family in minimal form,

$$F(x, z) = c e^{Ax} z, \quad z \in \mathbb{R}^r, \quad (3.169)$$

with A invertible and (A, c) observable, that is, $c e^{Ax} y \equiv 0$ only for $y = 0$; every finite-dimensional shift-invariant family of exponential polynomials without an exponent-zero part has this form. The curve is

$$f(t, T) = \sum_m \mathbf{1}_{I_m}(T) [L_m(t) + C_m(t)(T - T_m)] + c e^{A(T-t)} Z_t, \quad (3.170)$$

with $dZ = b dt + H^Z dB$, H^Z a predictable bounded $r \times k$ matrix and b locally integrable, so the block volatility depends on the block's own state through H^Z . Put

$$w(u) = H^Z(u) H^S(u)^\top \in \mathbb{R}^r, \quad (3.171)$$

the covariation density of the block's noise with the step noise. All correlation enters through the weights: the independence splice has H^S and the rows of H^Z on disjoint drivers, and a correlation ρ_u with a single block direction $\sigma_Z(u)$ gives $w = \rho_u \sigma_Z(u)$. Assumption 2.1 is assumed for the drift and volatility that Itô's formula gives for (3.170).

Lemma 42. For $v \in \mathbb{R}^r$ and real T_m , $c e^{AT} (TI + A^{-1} - T_m I) v \equiv 0$ only if $v = 0$.

Proof. With $k(T) = c e^{AT} A^{-1} v$, the function is $(T - T_m)k'(T) + k(T) = \frac{d}{dT} [(T - T_m)k(T)]$. If it vanishes, $(T - T_m)k(T)$ is constant and zero at T_m , so $k \equiv 0$, and observability gives $A^{-1} v = 0$. $\square$

Proposition 43 (conditional). (a) Almost surely $\Delta_m(u)w(u) = 0$ for almost every $u < T_m$, for every m : the block's noise is orthogonal to the step noise wherever the step volatility has a maturity step.

(b) If $\Delta_m w = 0$ almost everywhere, almost surely, the integrated cross term has no jump across the meetings, and the curve satisfies Assumption 2.1 if and only if the cross term together with the block residual is affine in the time to maturity and the front-end drifts absorb that affine part, as in Proposition 38(a).

(c) If $\Delta_m w \neq 0$ on a set of positive $dt \otimes dQ$ measure, no choice of the drifts makes (3.170) satisfy Assumption 2.1; on each maturity interval the jump to be absorbed has the form $c e^{AT} y + T c e^{AT} z$, which a family can only absorb by indicator-times-exponential terms as in [12] eq. 36.

Proposition 35 is the case $w(u) = \rho\psi/b$ with (A, b) controllable, and Proposition 41 its extension to random scales; here observability of (A, c) replaces controllability and makes the subspace of Proposition 35(b) trivial. [12] eqs. 35 to 41 make each factor's volatility state-dependent through its Heston variance but keep the factors independent; [11] Propositions 4.11, 4.12 and 5.1 concern scale states that move the shapes, which are fixed here.

Proof. The block volatility is $c e^{A(T-u)} H^Z(u)$ and the front-end volatility $s_m(u) H^S(u)$, both in driver form, so the differentiated drift condition is $D^S + D^B = \sigma \cdot \int \sigma$ with $\sigma = \sigma^S + \sigma^B$. With $S^S(u, T) = s_m(u)T + \kappa_m(u)$ on I_m , the cross term is $s_m(u) c A^{-1} (e^{A(T-u)} - I) w(u) + c e^{A(T-u)} w(u) (s_m(u)T + \kappa_m(u))$, the integrand of Proposition 35 with ρb replaced by $w(u)$. Integrating over $u \in [0, t]$ path by path, the front-end parts are affine in T on each $I_m \cap [t, H]$, the block parts are real-analytic and the same on every interval, and the cross term $X_{t,m}$ is entire and depends on m . So $X_{t,m} - X_{t,m-1} = c e^{AT} (A^{-1} + (T - T_m)I) v_m(t) - c A^{-1} \int_0^t \Delta_m w du$, with $v_m(t) = \int_0^t \Delta_m(u) e^{-Au} w(u) du$, must be affine; its exponential part has only nonzero exponents and vanishes, and Lemma 42 gives $v_m(t) = 0$. This holds for rational $t < T_m$ on one event, hence for all $t < T_m$ by continuity, and differentiation gives (a). Under (a)'s condition the jump vanishes, and (b) is the affine reduction of Proposition 38(a) with the cross term added to the block residual. For (c), on an event of positive probability some rational t gives a jump that is not affine. $\square$

The proposition is conditional on the cited calculus. Not covered: blocks with an exponent-zero part, such as the Nelson–Siegel or Svensson level, whose covariation with the step noise enters the affine part; several step factors (Proposition 44); unbounded volatilities and the Heston scale; and existence of the processes Z and L_m with given coefficients.

3.38 The correlated splice with several step and block factors

Keep the setting of Section 3.37 in driver form, now with n_S step factors: on I_m the front-end noise is $\sum_{\ell=1}^{n_S} s_{\ell,m}(t) H_\ell^S(t) \cdot dB$, with $s_{\ell,m}$ predictable and bounded and predictable bounded row vectors H_ℓ^S . Put $\Delta_{\ell,m} = s_{\ell,m} - s_{\ell,m-1}$ and define the aggregate maturity-step integrand at T_m ,

$$G_m(u) = \sum_{\ell=1}^{n_S} \Delta_{\ell,m}(u) H_\ell^S(u) \in \mathbb{R}^{1 \times k}. \quad (3.172)$$

The block is a sum of quasi-exponential factors,

$$F(x, z) = \sum_{j=1}^J c_j e^{A_j x} z_j = \mathcal{C} e^{\mathcal{A} x} z, \quad \mathcal{A} = \text{diag}(A_1, \dots, A_J) \text{ invertible, } \mathcal{C} = (c_1, \dots, c_J), \quad (3.173)$$

with $dZ = b dt + H^Z dB$ as in Section 3.37 and H_j^Z the rows of H^Z belonging to z_j . Assumption 2.1 is assumed as in Proposition 43.

Proposition 44 (conditional). (a) If $(\mathcal{A}, \mathcal{C})$ is observable, then almost surely

$$H^Z(u) G_m(u)^\top = 0 \quad \text{for almost every } u < T_m, \text{ every } m : \quad (3.174)$$

the block's noise is orthogonal to the front end's aggregate maturity-step noise. Conversely, under (3.174) the integrated cross term has no jump across the meetings, so (3.174) is exactly the condition for a jump-free cross term; the curve then satisfies Assumption 2.1 if and only if that cross term together with the block residual is affine in the time to maturity and the front-end drifts absorb that affine part, as in

Proposition 38(a) applied to the residual plus the cross term. The cross term need not be affine, so (3.174) is necessary for consistency but does not characterize it; Proposition 52(b)(iv) and (e) make the converse explicit and show that, for diagonalizable $\mathcal{A}$, consistency also requires $H^\zeta V_{j(u)}^\top = 0$.

(b) If each (A_j, c_j) is observable and the spectra of the A_j are pairwise disjoint, then $(\mathcal{A}, \mathcal{C})$ is observable, and (3.174) holds block by block: $H_j^Z(u)G_m(u)^\top = 0$ for each j .

(c) In general, a minimal realization $F(x, z) = \mathcal{C}' e^{\mathcal{A}'x} Pz$ with $(\mathcal{A}', \mathcal{C}')$ observable, $\mathcal{A}'$ invertible and $P\mathcal{A} = \mathcal{A}'P$ exists, and (3.174) holds for the reduced noise PH^Z : blocks with shared exponents act as one effective factor.

(d) If (3.174), or its reduced form in (c), fails on a set of positive $dt \otimes dQ$ measure, no choice of the drifts makes the curve satisfy Assumption 2.1, and indicator-times-exponential terms are forced as in Proposition 43(c).

We count (3.174) as the independence splice in vector form; this is a classification convention, not a result. Condition (3.174) can hold even when single step factors are correlated with the block, as long as the block is uncorrelated with the aggregate step, and the conclusion of Sections 3.36 to 3.38 that there are only two ways depends on this convention. A reader who counts such cancellation separately obtains a third construction, in which correlations of single step factors with the block cancel in the aggregate, and Proposition 44 characterizes by (3.174) exactly when the cross term has no jumps; consistency then needs, in addition, the affine condition on the block residual plus the cross term. With one step factor it is Proposition 43(a). [34] §3.1.3 decomposes correlated deterministic factors into independent ones; (3.174) is the condition that survives that decomposition at the scheduled maturities. Observability of block-diagonal pairs with disjoint spectra and minimal realizations are standard linear systems theory; the arguments are written out below.

Proof. (a) The front-end volatility on I_m is $\sum_\ell s_{\ell,m} H_\ell^S$ and the block volatility $\mathcal{C} e^{\mathcal{A}(T-u)} H^Z$, so the cross term of $\sigma \cdot \int \sigma$ is the sum over ℓ of the cross terms of Proposition 43 with $w_\ell = H^Z(H_\ell^S)^\top$. The jump across T_m of the integrated cross term is therefore $\mathcal{C} e^{\mathcal{A}T}(\mathcal{A}^{-1} + (T - T_m)I)v_m(t) - \mathcal{C}\mathcal{A}^{-1} \int_0^t W_m du$ with $W_m = \sum_\ell \Delta_{\ell,m} w_\ell = H^Z G_m^\top$ and $v_m(t) = \int_0^t e^{-\mathcal{A}u} W_m(u) du$, and the argument of Proposition 43(a), with Lemma 42 for $(\mathcal{A}, \mathcal{C})$, gives $W_m = 0$ almost everywhere, almost surely. The converse is Proposition 43(b).

(b) The set $\{y : \mathcal{C} e^{\mathcal{A}x} y \equiv 0\}$ is invariant under every polynomial in $\mathcal{A}$. Let p_j be the characteristic polynomial of A_j and $q_j = \prod_{i \neq j} p_i$; disjoint spectra make p_j and q_j coprime. By the Cayley–Hamilton theorem $q_j(\mathcal{A})$ annihilates every block but the j -th, and a Bézout identity $\alpha p_j + \gamma q_j = 1$ shows that $q_j(\mathcal{A})$ is invertible on the j -th block. So for y in the set, $(\gamma q_j)(\mathcal{A})y$ lies in it and has y_j as its only nonzero block, whence $c_j e^{A_j x} y_j \equiv 0$ and $y_j = 0$ by observability of (A_j, c_j) . The components of $H^Z G_m^\top$ along z_j are $H_j^Z G_m^\top$.

(c) The unobservable subspace N is $\mathcal{A}$ -invariant, and $\mathcal{A}N = N$ since $\mathcal{A}$ is invertible, so $\mathcal{A}$ induces an invertible $\mathcal{A}'$ on $\mathbb{R}^r/N$; with P the quotient map, $P\mathcal{A} = \mathcal{A}'P$ passes through the exponential series and gives $\mathcal{C} e^{\mathcal{A}x} z = \mathcal{C}' e^{\mathcal{A}'x} Pz$. The curve is unchanged with $(\mathcal{A}', \mathcal{C}', PZ)$, and (a) applies.

(d) As in Proposition 43(c), with W_m in place of $\Delta_m w$. □

The proposition is conditional on the cited calculus and is proved on one path. Not covered: blocks with an exponent-zero part; unbounded scales and the Heston scale of [12]; front ends with a diffusive ramp; and existence of the processes.

3.39 Polynomial parts in the uncorrelated splice

Keep the uncorrelated splice of Section 3.33 with a front end of the type of Theorem 11, which absorbs exactly the part of the block residual that is affine in x (Proposition 38(a)). The block is an exponential-polynomial family in the sense of [25] eq. 5 with a full polynomial part,

$$F(x, z) = p_0(x) + \sum_{i=1}^K p_i(x) e^{-\kappa_i x}, \quad p_0(x) = \sum_{\mu=0}^d z_{0,\mu} x^\mu, \quad (3.175)$$

where every coefficient $z_{0,0}, \dots, z_{0,d}$ is a parameter, the p_i are polynomials with parameter coefficients, and the exponents κ_i are fixed or are themselves parameters, positive at the point considered. Z is driven by W^B with $a = \sigma_Z \sigma_Z^\top$, and R is the residual of the consistency equation; by [25] Remark 8.4 it extends from $x \leq H - t$ to all $x \geq 0$ by analyticity. Alone means $R = 0$, in the splice means R affine. The statements hold at one parameter point.

Proposition 45. Let $m = \lfloor (d - 1)/2 \rfloor$.

(a) Let $d \geq 1$. Then (a, b) makes R affine if and only if (a, b') makes $R = 0$, where b' agrees with b except in $b_{0,0}$ and $b_{0,1}$. So the admissible covariances, and all drifts other than those two, are the same alone and in the splice: the front end changes none of the block's restrictions.

(b) For $d \geq 1$, alone and in the splice, $a_{(0,\mu),(0,\nu)} = 0$ unless $\mu, \nu \leq m$: at most $\lfloor (d + 1)/2 \rfloor$ polynomial coefficients have nonzero diffusion.

(c) If the block is polynomial only ($K = 0$), every nonnegative definite a supported on $\{(0, \mu), (0, \nu) : \mu, \nu \leq m\}$ is admissible with suitable b , alone and in the splice, so exactly $\lfloor (d + 1)/2 \rfloor$ possibly correlated polynomial factors can occur: $z_{0,0}$ for $d = 1, 2$, $z_{0,0}$ and $z_{0,1}$ for $d = 3, 4$, and so on.

(d) For $d = 0$, a pure level, $R = 0$ forces $a_{(0,0),(0,0)} = 0$, while in the splice the level is free, as in Proposition 38. This is the only case in which the front end changes the restrictions of a full polynomial part.

The case $d = 1$ of (c) is the nontrivial diffusion for $z_0 + z_1 x$ noted in [25] Remark 8.3. [9] §7 shows that the full exponential-polynomial family with varying exponents is inconsistent under deterministic volatility and treats fixed exponents, but gives no count of polynomial factors; we found the count $\lfloor (d + 1)/2 \rfloor$ stated in neither source nor in [26].

Proof. Sort R by exponential factor. For an exponential parameter with exponent $\kappa > 0$, its derivatives carry $e^{-\kappa x}$, and $\int_0^x$ of such a derivative is a constant c_j minus $e^{-\kappa x}$ times a polynomial; so terms involving only exponential parameters have positive exponents. The exponent-zero part of R comes from the polynomial part:

$$\sum_{k \leq d} \left[b_{0,k} - (k + 1) z_{0,k+1} - \sum_{j \text{ exp}} a_{(0,k),j} c_j \right] x^k - \sum_{\mu, \nu \leq d} a_{(0,\mu),(0,\nu)} \frac{x^{\mu+\nu+1}}{\nu + 1}, \quad (3.176)$$

with $z_{0,d+1} = 0$, since second derivatives in z_0 vanish and $x^\mu \int_0^x \eta^\nu d\eta = x^{\mu+\nu+1}/(\nu + 1)$.

(a) The parts with positive exponents are the same alone and in the splice, because the front end absorbs only polynomials and $x^k e^{-\kappa x}$, $\kappa > 0$, are independent of the polynomials on every interval. In (3.176) the coefficient of x^k , $k \leq d$, contains $b_{0,k}$ with coefficient one, while for $k \geq d + 1 \geq 2$ it involves no drift. The front end absorbs exactly the coefficients of x^0 and x^1 , which alone are set to zero by $b_{0,0}$ and $b_{0,1}$.

(b) Let μ^* be the largest μ with $a_{(0,\mu),(0,\mu)} > 0$; nonnegative definiteness kills every row beyond it, so the coefficient of $x^{2\mu^*+1}$ in (3.176) is $-a_{(0,\mu^*),(0,\mu^*)}/(\mu^* + 1) \neq 0$. If $2\mu^* + 1 > d$ this coefficient has no drift and degree above one, a contradiction; so $\mu^* \leq m$.

(c) With $K = 0$ and a supported on $\mu, \nu \leq m$, every term of the double sum has degree at most $2m + 1 \leq d$, so each coefficient of (3.176) contains a drift that can be chosen to make it zero, or is left free for $k \leq 1$ in the splice.

(d) For $d = 0$, (3.176) is $b_{0,0} - \sum_j a_{(0,0),j} c_j - a_{(0,0),(0,0)} x$, whose coefficient of x has no drift; alone it vanishes, and in the splice it is absorbed. $\square$

The proposition is conditional on the cited calculus through the reduction of Proposition 38(a); the statements themselves are algebra at a parameter point. Not covered: polynomial parts that are not full, where a structurally absent coefficient of degree at most one removes a drift and the front end lifts a constraint; front ends with a diffusive ramp, which absorb polynomials of degree up to three; correlated splices with a polynomial part; and existence of Z .

3.40 Varying exponents in the uncorrelated splice

Keep the uncorrelated splice (3.161) of Section 3.32 with a general front end, L_m and C_m Itô processes driven by W^S , which absorbs on each I_m exactly a polynomial in T of degree at most three. The block is the bounded exponential-polynomial family of [25] Definition 3.1,

$$F(x, z) = \sum_{i=1}^K p_i(x, z) e^{-z_{i,n_i+1} x}, \quad p_i(x, z) = \sum_{\mu \leq n_i} z_{i,\mu} x^\mu, \quad (3.177)$$

in which every exponent z_{i,n_i+1} is a parameter that may vary, and Z , driven by W^B , stays in the set of parameters with $\sup_x |F(x, z)| < \infty$. Assumption 2.1 is assumed for the drift of the representation, as in Proposition 37. Hypothesis (P) says that $dt \otimes dQ$ -almost every (t, ω) has every exponent Z_t^{i,n_i+1} positive.

Proposition 46 (conditional). (a) At every point with all exponents positive at which the splice satisfies Assumption 2.1, Z satisfies the consistency equation of [25] Proposition 2.2 for the block alone: the front end absorbs no part of it.

(b) Under (P), and conditional on the cited conclusions of [25] Theorem 3.2 and Corollaries 3.4 and 3.5 for a process satisfying that equation almost everywhere, those conclusions hold for Z in the splice. In particular the diffusion and drift of the exponent Z^{i,n_i+1} vanish where $p_i(Z) \neq 0$, respectively where $p_i(Z)$ and the polynomial of the paired exponent do not vanish, so the exponents are constant there, and under the conditions of Corollary 3.5 the block is quasi-deterministic.

So the exponent restrictions of [25] for the block alone hold unchanged in the uncorrelated splice. Proposition 37 is the case of fixed exponents. Hypothesis (P) also requires positivity where $p_i(Z) = 0$, where the source lets the exponent be an arbitrary Itô process (its Lemma 5.1), so (b) covers a narrower class of processes than Theorem 3.2 does.

Proof. (a) As in the proof of Proposition 37, Assumption 2.1 gives $R(t, x) = -(D^S - \sigma^S \cdot \int \sigma^S)(t, t + x)$, a polynomial of degree at most three in x on each $I_m \cap [t, H]$, where R is the residual of the consistency equation, with the parameter derivatives of F including those in the exponents. At a point with exponents $\beta_i > 0$, every term of R carries a factor $e^{-\beta_i x}$ or $e^{-(\beta_i + \beta_j)x}$, since $\int_0^x \eta^\mu e^{-\beta \eta} d\eta = \mu! / \beta^{\mu+1} - e^{-\beta x}(\text{polynomial})$. So $R(t, \cdot)$ is a finite sum of $x^k e^{-\kappa x}$ with $\kappa > 0$, independent of the polynomials on every interval; it vanishes on $[0, H - t]$, hence for all $x \geq 0$ by analyticity. (b) By (a) and (P) the consistency equation holds $dt \otimes dQ$ -almost everywhere, which is the hypothesis under which the cited results are stated; their proofs use the equation at fixed points, the support condition and lemmas about the Itô process Z alone, none of which involves the front end. $\square$

The proposition is conditional on the cited calculus and, for (b), on the cited results of [25]. Not covered: points where an exponent vanishes, such as the Nelson–Siegel and Svensson level, where the front end can absorb part of an exponent-zero group (Propositions 38, 40 and 45 treat fixed exponents); correlated splices with varying exponents; and existence of Z .

3.41 Exponent-zero block parts in the correlated splice

Keep the driver-form setting of Section 3.37, with one step factor, and let the block have a full polynomial part and an observable quasi-exponential part,

$$F(x, z) = \sum_{\mu=0}^d z_{0,\mu} x^\mu + \mathcal{C} e^{\mathcal{A}x} \zeta, \quad (3.178)$$

with $\mathcal{A}$ invertible and $(\mathcal{A}, \mathcal{C})$ observable. $Z = (z_0, \zeta)$ is an Itô process in driver form with predictable bounded integrand rows $H^{0,\mu}$ for $z_{0,\mu}$ and H^ζ for ζ , and the covariation densities with the step integrand are

$$w_\mu(u) = H^{0,\mu}(u)H^S(u)^\top \in \mathbb{R}, \quad w_\zeta(u) = H^\zeta(u)H^S(u)^\top \in \mathbb{R}^r. \quad (3.179)$$

Assumption 2.1 is assumed for the drift of the representation, as in Proposition 43.

Proposition 47 (conditional). (a) Almost surely, for every m and almost every $u < T_m$, $\Delta_m(u)w_\mu(u) = 0$ for $\mu = 1, \dots, d$ and $\Delta_m(u)w_\zeta(u) = 0$. No condition is imposed on w_0 , the covariation of the level.

(b) Under these conditions the jumps of the cross term across the meetings are affine in T and come only from the level, and the front end absorbs them. The curve satisfies Assumption 2.1 if and only if the block residual plus the jump-free cross terms is affine in the time to maturity, as in Propositions 38(a) and 45; the bound and the admissible covariances of Proposition 45 carry over, with only the drifts changed.

(c) Every block direction outside the constants, that is every polynomial coefficient of degree at least one and every exponential part, must be uncorrelated with the step noise wherever the step volatility jumps, or no choice of drifts satisfies Assumption 2.1.

The level is a constant in calendar maturity, a direction the front-end family already contains (Proposition 38(c)), so its correlation with the step noise is internal to the front end. We count a correlated level as part of the front end rather than as a third construction; this is a classification convention, not a result. A reader who counts it separately obtains a third construction, which Proposition 47 characterizes exactly: only the level may correlate with the step noise where the step volatility jumps. With this convention the answer is that of Sections 3.30 to 3.38, the overlap of the two families being exempt. [12] has no level component correlated with meeting steps, and [34] §3 splices by independence.

Proof. The block volatility is $\sum_\mu (T-u)^\mu H^{0,\mu}(u) + \mathcal{C} e^{\mathcal{A}(T-u)} H^\zeta(u)$, so the cross part of $\sigma \cdot \int \sigma$ splits by component: for $z_{0,\mu}$ it is $w_\mu(u)[s_m(u)(T-u)^{\mu+1}/(\mu+1) + (T-u)^\mu(s_m(u)T + \kappa_m(u))]$, and the exponential part gives the cross term of Proposition 43 with w_ζ . As in Proposition 43(a), the jump across T_m of the integrated cross term must be affine in T . With $\kappa_m - \kappa_{m-1} = -\Delta_m T_m$ it is $J = \sum_\mu J_\mu + J_\zeta$, where $J_\mu(T) = \int_0^t w_\mu \Delta_m [(T-u)^{\mu+1}/(\mu+1) + (T-u)^\mu(T - T_m)] du$ is a polynomial of degree at most $\mu+1$ with leading coefficient $\frac{\mu+2}{\mu+1} \int_0^t w_\mu \Delta_m du$, and J_ζ is the jump of Proposition 43. A quasi-exponential with nonzero exponents that equals a polynomial on an interval vanishes, so the exponential part of J_ζ vanishes and Lemma 42 gives $\Delta_m w_\zeta = 0$. For $d \geq 1$ the coefficient of T^{d+1} comes from J_d alone and must vanish, so $\int_0^t w_d \Delta_m = 0$ for rational and then all $t < T_m$, and $w_d \Delta_m = 0$ almost everywhere; descending through $\mu = d-1, \dots, 1$, whose

leading degree is at least two, gives (a). $J_0(T) = \int_0^T w_0 \Delta_m(2T - u - T_m) du$ is affine for every w_0 . For (b), $J = J_0$ is absorbed by the front end, and the remaining cross terms form one analytic function joining the block residual. For the bound, $w_\mu \neq 0$ needs $a_{(0,\mu),(0,\mu)} \neq 0$ by Cauchy–Schwarz, so the extra cross terms have degree at most $\mu^* + 1$, below the top degree $2\mu^* + 1$ of Proposition 45(b) when $\mu^* \geq 1$. Part (c) is (a) with (b). $\square$

The proposition is conditional on the cited calculus and is proved on one path. Not covered: polynomial parts that are not full; several step factors, treated in Proposition 52; front ends with a diffusive ramp; varying exponents in the correlated splice; and existence of the processes.

3.42 The compensating shape of a scheduled jump

Proposition 9 treats shapes that are affine on each interval. This section removes that restriction on the interval after one scheduled date. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{G} \subseteq \mathcal{F}$ a sub-sigma-algebra and $L \in (0, \infty]$. Let X be a real random variable, the level jump, and $h : [0, L) \times \Omega \rightarrow \mathbb{R}$ a $\mathcal{B}([0, L)) \otimes \mathcal{G}$ -measurable function, the compensating shape, with $\int_0^\tau |h(u, \omega)| du < \infty$ for every $\tau < L$ and every ω . The curve jump at maturity $T_n + u$ and its primitive are

$$\xi(u) = X + h(u), \quad H(\tau) = \int_0^\tau h(u) du. \quad (3.180)$$

Let κ be a regular conditional distribution of X given $\mathcal{G}$, and put

$$M(\omega, \tau) = \int e^{-\tau x} \kappa(\omega, dx) \in (0, \infty], \quad K(\omega, \tau) = \log M(\omega, \tau); \quad (3.181)$$

where $M(\omega, \tau) < \infty$, let $\tilde{E}_\tau[X]$ and $\tilde{V}_\tau[X]$ be the mean and variance of the tilted law $e^{-\tau x} \kappa(\omega, dx) / M(\omega, \tau)$. Hypothesis (H) is Assumption 2.2 for this jump, with $\mathcal{G} = \mathcal{F}_{T_n-}$ and L the distance to the next scheduled date:

$$E\left[\exp\left(-\int_0^\tau \xi(u) du\right) \mid \mathcal{G}\right] = 1 \quad \text{a.s., for every } \tau \in [0, L). \quad (3.182)$$

Proposition 48. (a) (H) holds if and only if there is one $\mathcal{G}$ -measurable null set off which $M(\omega, \tau) < \infty$ and $H(\omega, \tau) = K(\omega, \tau)$ for every $\tau \in [0, L)$.

(b) Under (H), off that null set, $K(\omega, \cdot)$ is infinitely differentiable on $(0, L)$ with $K'(\tau) = -\tilde{E}_\tau[X]$ and $K''(\tau) = \tilde{V}_\tau[X] \geq 0$, and K'' vanishes somewhere if and only if $\kappa(\omega, \cdot)$ is a Dirac mass. The shape is fixed by the law of the level jump:

$$h(\omega, \tau) = -\tilde{E}_\tau[X](\omega) \quad \text{for almost every } \tau \in (0, L) \text{ and every right-continuity point } \tau, \quad (3.183)$$

so $h(\omega, \cdot)$ agrees almost everywhere with a nondecreasing function, strictly increasing unless the law is a Dirac mass; if $h(\omega, \cdot)$ is right-continuous at 0, then X is conditionally integrable and $h(\omega, 0) = -\int x \kappa(\omega, dx)$. Conversely, (H) holds if $M(\omega, \cdot) < \infty$ on $[0, L)$ and (3.183) holds almost everywhere, for almost every ω . Two level jumps satisfying (H) with the same $\mathcal{G}$ and h have the same conditional law, and every conditional law with finite Laplace transform on $[0, L)$ has a measurable locally integrable shape satisfying (H).

(c) (i) An affine shape $h(u) = c + yu$ with $\mathcal{G}$ -measurable c, y satisfies (H) if and only if, almost surely, $y \geq 0$ and $\kappa(\omega, \cdot) = N(-c, y)$; with $c = 0$ this is the first interval of Proposition 9, and with $y = 0$ the

one-date form of Lemma 5. (ii) For trivial $\mathcal{G}$, $L = \infty$ and $X = \Delta \neq 0$ with probability $p \in (0, 1)$ and 0 otherwise, the continuous shape is

$$h(\tau) = -\Delta q(\tau), \quad q(\tau) = \frac{pe^{-\tau\Delta}}{1 - p + pe^{-\tau\Delta}}, \quad (3.184)$$

with $h' = \Delta^2 q(1-q) > 0$; it is strictly concave on $[0, \infty)$ if and only if $\Delta(1-2p) \geq 0$, and otherwise convex before and concave after $\tau^* = \log(p/(1-p))/\Delta$, and never affine. The Gaussian level jump with the same mean and variance has the ramp $h(0) + h'(0)\tau$, the tangent of h at 0; for $p = \frac{1}{2}$ the gap $g = h(0) + h'(0)\tau - h(\tau)$ satisfies

$$0 \leq g(\tau) \leq \frac{|\Delta|^4 \tau^3}{48}, \quad 0 \leq \frac{1}{\tau} \int_0^\tau g \leq \frac{|\Delta|^4 \tau^3}{192}. \quad (3.185)$$

(d) Let $t < T_n$, $\mathcal{G} = \mathcal{F}_{T_n-} \supseteq \mathcal{F}_t$, r_t $\mathcal{F}_t$ -measurable, J $\mathcal{F}_{T_n-}$ -measurable, and $f(t, \cdot)$ a $\mathcal{B} \otimes \mathcal{F}_t$ -measurable locally integrable curve with $f(t, T) = r_t$ for $T \in [t, T_n)$. In the model in which the curve is frozen off T_n and jumps there to $r_t + J$ on $[T_n, T_n + L)$, (H) holds if and only if, almost surely, $E[e^{-\tau J} \mid \mathcal{F}_{T_n-}] < \infty$ for $\tau \in [0, L)$ and

$$f(t, T) = r_t + \frac{E_{T_n-}[Je^{-\tau J}]}{E_{T_n-}[e^{-\tau J}]}, \quad \tau = T - T_n, \text{ for almost every } \tau \in (0, L), \quad (3.186)$$

with the expectations computed from the regular conditional distribution of J given $\mathcal{F}_{T_n-}$. Under (H), the conditional laws of J given $\mathcal{F}_{T_n-}$ and given $\mathcal{F}_t$ agree, so (3.186) is (5.1) as printed; if $f(t, \cdot)$ is right-continuous it holds for every $\tau \in [0, L)$, at $\tau = 0$ as $f(t, T_n) = r_t + E_t[J]$ with $E_t|J| < \infty$; and $P(t, T) = e^{-r_t(T-t)} E_t[e^{-(T-T_n)J}]$ for $T \in [T_n, T_n + L)$. For a general initial curve, $f(t, T)$ is the tilted conditional expectation of the post-meeting short rate r_T , and (5.1) holds for every τ if and only if $r_{T_n-} = r_t$ and the short rate is constant after T_n .

(e) On a later interval, if $E[e^{-S} \mid \mathcal{G}] = 1$ for the exponent S accumulated before it, the shape there is the derivative of the conditional cumulant generating function of its level jump under the measure $e^{-S} \cdot P$, which is the mechanism of Proposition 9 on later intervals.

The Gaussian case of (c)(i) is the compensator of [30] Example 3.10, and the Gaussian case of the bond formula in (d) is [41] Proposition 1; the characterization with one null set for general shapes, the uniqueness of the law and the exact form and range of (5.1) are ours. The identity of a forward rate with a forward-measure expectation of the short rate is textbook, and (5.1) is its specialization. As a numerical check, not part of the statement, the gap in (3.185) at $\Delta = 25$ basis points is of order 10^{-11} basis points at six weeks and 10^{-8} at one year, against about 12.5 basis points for h itself, so a two-point meeting law and its Gaussian ramp are indistinguishable at pricing precision, as Section 3.9 found for the ramps.

Proof. For fixed τ , $\exp(-\int_0^\tau \xi) = e^{-\tau X} e^{-H(\tau)}$ with $e^{-H(\tau)}$ positive and $\mathcal{G}$ -measurable, so (3.182) at τ is $M(\cdot, \tau) = e^{H(\cdot, \tau)}$ almost surely. (a) Take the union of the exceptional sets over rational τ . Off it $M(\omega, \cdot)$ is finite at the rationals, hence by convexity finite and continuous on $(0, L)$, and $H(\omega, \cdot)$ is continuous, so the identity extends to every τ . (b) Domination of $|x|^j e^{-sx}$ by exponentials at nearby arguments gives $M \in C^\infty((0, L))$ with $M^{(j)}(s) = \int (-x)^j e^{-sx} \kappa(dx)$, hence $K' = -\tilde{E}_\tau[X]$ and $K'' = \tilde{V}_\tau[X]$; the tilted law and κ are equivalent, so K'' vanishes only for a Dirac mass. Since $H = K$, the Lebesgue differentiation theorem gives (3.183) almost everywhere, and right-continuity gives it at the point; at 0, Fatou and dominated convergence give integrability and $h(0) = -\int x \kappa$. The converse integrates K' . For uniqueness, tilt both

laws at some $\tau_0 \in (0, L)$: the tilted Laplace transforms are finite and equal on an open interval containing 0, so the laws agree ([19]). For existence, K' is jointly measurable and locally integrable. (c)(i) $K(\tau) = c\tau + \frac{1}{2}y\tau^2$ forces $y = K'' \geq 0$, and the tilted law has the Laplace transform of a Gaussian, so by uniqueness $\kappa = N(-c, y)$; conversely $N(-c, y)$ has $K = H$. (ii) Differentiate $K = \log(1 - p + pe^{-\tau\Delta})$; the sign of $h'' = -\Delta^3 q(1 - q)(1 - 2q)$ follows the monotone q , and for $p = \frac{1}{2}$, $g = (|\Delta|/2)(z - \tanh z)$ with $z = |\Delta|\tau/2$ and $0 \leq z - \tanh z \leq z^3/3$. (d) The jump is $\xi(T) = r_t + J - f(t, T)$, which is (3.180) with $X = J$ and $h(u) = r_t - f(t, T_n + u)$, so (a) and (b) give (3.186). Under (H), $E[e^{-\tau J} | \mathcal{F}_t] = e^{H(\tau)}$ by the tower property, and uniqueness identifies the two conditional laws; right-continuity and the statement at 0 are (b); the bond price is $\exp(-r_t\tau + H(\tau))$ times the pre-meeting discount. For a general initial curve the tilt by $\exp(-\int_{T_n}^T r_s ds)$ is $e^{-\tau X}$ times a positive $\mathcal{G}$ -measurable factor, which gives the tilted expectation, and comparing it with (5.1) at every τ gives the stated equivalence. (e) The measure $e^{-S} \cdot P$ agrees with P on $\mathcal{G}$, and the abstract Bayes formula reduces the condition to (a) under it. $\square$

The proposition treats one scheduled date. Its model in (d) has zero drift and volatility off T_n ; the assembly into a model satisfying Assumptions 2.1 and 2.2 on all of $[0, \infty)$, the induction over several meetings for general shapes, and a characterization of which functions are cumulant generating functions are not covered. The implication in (d) fails if the laws are taken given $\mathcal{F}_t$ while information about J arrives on (t, T_n) .

3.43 Meeting variances with a diffusion between meetings

The identification results of Sections 3.13 to 3.21 use a model in which the curve moves only at meetings. This section adds a parallel Gaussian diffusion between meetings and an initial curve. Keep the jumps $Z_1, \dots, Z_N \sim N(0, v_n)$ of Section 3.10, $N \geq 3$, and let W be an independent Brownian motion, with the usual augmentation of the filtration they generate at their revelation times. Fix a partition $0 = c_0 < c_1 < \dots < c_p$, variances $u_p \geq 0$ with $\sigma(s) = \sqrt{u_p}$ on $[c_{p-1}, c_p)$ and $\sigma = 0$ after c_p , and a bounded piecewise continuous initial curve. With $Y_t = \int_0^t \sigma dW$, the curve is

$$f(t, T) = f(0, T) + \sum_{T_n \leq t} [Z_n + v_n(T - T_n)] + Y_t + \int_0^t \sigma(s)^2(T - s) ds, \quad (3.187)$$

and the unknown parameter is $\theta = (v, u) \in [0, \infty)^{N+P}$. The contracts are the ideal futures and European cash calls of Section 3.16. With $w(s) = (b-s)^+ - (a-s)^+$, $d(s) = ((b-s)^{+2} - (a-s)^{+2})/2$, $h(s) = d(s) + w(s)^2/2$, $A = \int_a^b f(0, u) du$ and the coefficients of (3.71), put

$$p = \sum_i h_i v_i + \int_0^b \sigma^2 h, \quad z = \sum_i j_i v_i + \int_0^S \sigma(s)^2 w(s)(S - s) ds, \quad q = \sum_i k_i v_i + \int_0^S \sigma^2 w^2, \quad m = e^{A+p-z}. \quad (3.188)$$

For a panel of complete surfaces at expiries $S_1 < \dots < S_L$ before their accrual windows, let $G_\ell = (S_{\ell-1}, S_\ell]$, $S_0 = 0$, be the gaps, E the set of meeting-free gaps, $\lambda_p(J)$ the length of $J \cap [c_{p-1}, c_p)$, and $Q(S) = \sum_{T_i \leq S} v_i + \int_0^S \sigma^2$.

Proposition 49 (conditional). (a) The curve (3.187) satisfies the regularity conditions and Assumptions 2.1 and 2.2, and every discounted bond is a true martingale, the product of $P(0, T)$, a Gaussian stochastic exponential and the discounted bond of the model of Section 3.10.

(b) $G(a, b)$ is an integrable martingale with $G_0 = e^{A+p}$, the call price is $P(0, S)$ times the Black formula (3.72) with forward m and variance q , and the complete surface at one expiry determines exactly $P(0, S)$, m and q . For $S < a$,

$$q(S) = \delta^2 Q(S), \quad z(S) = \delta \int_0^S Q(x) dx, \quad (3.189)$$

so the initial curve does not enter q or z ; the futures quote adds $z = \log G_0 - \log m$, and p only when the bond curve is known.

(c) The panel $(Q(S_\ell))_\ell$ equals $\mathcal{A}\theta$ with $\mathcal{A}_{\ell,i} = \mathbf{1}\{T_i \leq S_\ell\}$ and $\mathcal{A}_{\ell,N+p} = \lambda_p((0, S_\ell])$. It identifies θ on $[0, \infty)^{N+P}$ if and only if $\text{rank } \mathcal{A} = N + P$, and otherwise every strictly positive θ has a distinct strictly positive θ' with the same panel. Moreover

$$\text{rank } \mathcal{A} = N + P \iff T_N \leq S_L, \text{ no gap contains two meetings, and } \text{rank}(\lambda_p(G_\ell))_{\ell \in E, p \leq P} = P, \quad (3.190)$$

so identification needs at least P meeting-free gaps; the diffusion is read off the meeting-free gaps and each meeting variance off its own gap.

(d) On the listed calendar of Proposition 24, under its assumptions (A1) to (A4), exactly eight of the 24 gaps are meeting-free. With a constant diffusion variance ($P = 1$) the sixteen meeting variances and u_1 are identified. No identifying partition has more than eight cells, and eight are attained, for example with cell boundaries at the meetings of 28 January, 18 March, 17 June and 16 September 2026 and 27 January, 17 March and 28 July 2027. A partition by calendar quarter (rank 23 of 24) and a partition at every meeting (rank 24 of 33) do not identify, and explicit pairs of parameter vectors with equal panels exhibit this; with the futures quotes the stacked matrix of (Q, z) has full column rank for both, and even with one cell per gap.

(e) With $P = 1$ on that calendar, whose first gap is meeting-free, an error ε in the at-the-money premium C/δ perturbs $Q(S)$ to first order by

$$\eta(S) = \frac{2\varepsilon \sigma_\Sigma(S)}{P(0, S) m \varphi(\delta \sigma_\Sigma(S)/2)}, \quad \sigma_\Sigma(S)^2 = Q(S), \quad (3.191)$$

and the estimator $\hat{u} = Q(S_1)/\lambda_1$, $\hat{v}_i = Q(S_\ell) - Q(S_{\ell-1}) - \lambda_\ell \hat{u}$ for $T_i \in G_\ell$, with $\lambda_\ell = |G_\ell|$, has the sharp first-order error bound

$$|\Delta \hat{v}_i| \leq \eta_\ell + \eta_{\ell-1} + \frac{\lambda_\ell}{\lambda_1} \eta_1, \quad \eta_\ell = \eta(S_\ell), \quad \eta_0 = 0. \quad (3.192)$$

The model is that of Remark 12 with a diffusion, the contracts those of Proposition 19 and the calendar that of Proposition 24; the form $q = \delta^2(\sum v + \int \sigma^2)$ is the standard Gaussian computation ([41] Proposition 1 has its bond-price analogue). [58] separates event from background variance in one-day options by learning the background where there is no event and subtracting it where there is one, the statistical counterpart of (3.190), without an identification statement for a fixed calendar of expiries. The failing partitions in (d) do not fail because each meeting trades against the diffusion over its own gap: a cell whose diffusion enters no meeting-free gap trades against the meetings of every gap it overlaps. The cell [18 Mar, 29 Apr 2026) overlaps the gap of the 18 March meeting by 23 days and that of 29 April by 19, so the meeting-date pair moves both meetings and u on that cell, with every other variance fixed; the quarterly

pair moves the last three meetings of 2027 and u on the last quarter. The futures-assisted identification of (d) is formal only: the meeting-date pair changes every $z(S_\ell)$ by at most about 1.9×10^{-9} , while an error of half a basis point in a futures rate moves $\log G_0$ by about 1.3×10^{-5} . As a numerical check, not part of the statement, the number of meetings practically identified in the sense of Proposition 24, (3.192) below s_0^2 , always the first ones in date order, is, with zero initial curve:

σ (bp per year)	$s_0 = 10, \varepsilon = 0.25$	10, 0.5	15, 0.25	15, 0.5	25, 0.25	25, 0.5
none	16	4	16	9	16	16
50	2	0	12	2	16	13
100	0	0	3	0	16	5

Two effects combine: the accumulated standard deviation includes the diffusion, which dominates the meeting variances at two years, and the diffusion variance is learned only from meeting-free gaps, best from the first two-week gap, with its error entering each meeting weighted by that meeting's gap length. In every case of the table, (3.192) coincides with the smallest worst-case bound over all linear estimators exact on the model, computed by linear programming, which is also a check. With a nonzero initial curve, η is multiplied by $1/(P(0, S)m)$, a few percent away from one at current rate levels.

Proof. (a) The drift $\sigma(s)^2(T - s)$ is the differentiated HJM drift, so (2.3) holds at every (ω, t) , and the jump condition is the computation of Remark 12, since Z_n is independent of $\mathcal{F}_{T_n-}$. Itô's product rule for the deterministic integrand $T - s$ and Fubini split $\int_0^t r + \int_t^T f(t, \cdot)$ into the initial-curve part, the jump part of Remark 12 and $\int_0^t (T - s)\sigma dW + \frac{1}{2} \int_0^t \sigma^2(T - s)^2$; the Gaussian factor is a true martingale, and the two factors have independent increments. (b) Stochastic Fubini gives $\int_a^b Y = \int_0^b \sigma w dW$, and conditioning on $\mathcal{F}_t$ leaves independent Gaussians, so $\log G_t$ is explicit and $G_0 = e^{A+P}$. Under the discount measure $\log G_S$ is Gaussian with variance q and mean shifted by the covariance $-z$ with $-\int_0^S r$, which gives the Black formula; the surface returns $P(0, S)$ and $P(0, S)m$ as limits at zero strike and q at the strike m , as in Proposition 19. For $S < a$, $w = \delta$ on $[0, S]$, which is (3.189). (c) Differencing the rows gives, on a gap with no meeting, $\sum_p u_p \lambda_p(G_\ell)$, and on a gap with one meeting v_i plus that sum; a meeting after S_L has a zero column, two meetings in one gap give a kernel vector, and a kernel vector of the meeting-free block extends to one of $\mathcal{A}$, while conversely the rank condition on the meeting-free block and the meeting rows force a kernel vector to vanish. (d) is exact computation on the dates of Proposition 24. (e) The at-the-money price is $P(0, S)m(2\Phi(\sqrt{q}/2) - 1)$, and differentiating in $q = \delta^2 Q$ gives (3.191); the error of $\hat{v}_i$ is a combination of the errors in $Q(S_\ell)$, $Q(S_{\ell-1})$ and $Q(S_1)$ with coefficients 1, -1 and $-\lambda_\ell/\lambda_1$, the last two merging for the first meeting, which lies in G_2 , and its worst case, attained with aligned signs, is (3.192). $\square$

Parts (a) and (b) are conditional on the Wiener-integral facts the proof uses: with deterministic integrands, Wiener integrals are linear, jointly Gaussian and independent of the jumps, Y is continuous and satisfies the stochastic Fubini identity, and the filtration has the stated independence properties; no Brownian motion is constructed. Parts (c) to (e) are unconditional; the table, the linear-programming comparison and the size of the z -difference are numerical checks. The proposition does not treat an exponentially decaying diffusion factor, under which the diffusion's contribution to q is reweighted and (3.189) changes. Part (e) is first order in ε , and practical identification is the convention of Proposition 24, not a statistical statement; nonlinear estimators are not considered. The listed contracts' settlement, exercise and daily compounding are outside the model, as in Propositions 19 and 24.

3.44 A maturity-dependent diffusion between meetings

The diffusion of Section 3.43 moves every forward rate by the same amount. Keep that section's setting, notation and contracts, with one change: the diffusion volatility is $\sigma(s)\phi(s, T)$ for $T \geq s$, where σ is as before and the shape ϕ is a known bounded Borel function on $\{0 \leq s \leq T\}$. With $\Phi(s, T) = \int_s^T \phi(s, x) dx$ and the differentiated drift, the curve is

$$f(t, T) = f(0, T) + \sum_{T_n \leq t} [Z_n + v_n(T - T_n)] + \int_0^t \sigma(s)\phi(s, T) dW_s + \int_0^t \sigma(s)^2 \phi(s, T) \Phi(s, T) ds, \quad (3.193)$$

and $\phi \equiv 1$ is (3.187). The meeting shocks stay parallel. Two shapes are treated in detail: (S1) exponential decay, $\phi(s, T) = e^{-\kappa(T-s)}$ with $\kappa > 0$; and (S2) no loading before the next meeting, $\phi(s, T) = \mathbf{1}\{T \geq T_{j(s)+1}\}$ with $T_{N+1} = \infty$, the loading of the first meeting-indexed component of [12] §3.1, eqs. (4) to (7). For a window $[a, b]$ and $s < b$ put $\tilde{w}(s) = \int_{a \vee s}^b \phi(s, x) dx$, $\tilde{d}(s) = (\Phi(s, b)^2 - \Phi(s, a \vee s)^2)/2$, $\tilde{h} = \tilde{d} + \tilde{w}^2/2$, the window weight $\beta(s) = \tilde{w}(s)/\delta$ for $s < a$, and, with the coefficients of (3.71),

$$\tilde{p} = \sum_i h_i v_i + \int_0^b \sigma^2 \tilde{h}, \quad \tilde{z} = \sum_i j_i v_i + \int_0^S \sigma(s)^2 \Phi(s, S) \tilde{w}(s) ds, \quad q = \sum_i k_i v_i + \int_0^S \sigma^2 \tilde{w}^2, \quad m = e^{A + \tilde{p} - \tilde{z}}. \quad (3.194)$$

For a panel of expiries S_ℓ with their own windows $[a_\ell, b_\ell]$, $S_\ell < a_\ell$, and weights β_ℓ , put $D_{\ell, p} = \int_{(0, S_\ell] \cap [c_{p-1}, c_p)} \beta_\ell^2$, $D_{0, p} = 0$ and $\tilde{\Lambda}_{\ell, p} = D_{\ell, p} - D_{\ell-1, p}$.

Proposition 50 (conditional). (a) For every bounded Borel shape, the curve (3.193) satisfies the regularity conditions and Assumptions 2.1 and 2.2, the process $s \mapsto \int_0^s \sigma(v)\phi(v, s) dW_v$ has a jointly measurable version, and with it every discounted bond is the true martingale $P(0, T) \exp(-\int_0^t \sigma \Phi(\cdot, T) dW - \frac{1}{2} \int_0^t \sigma^2 \Phi(\cdot, T)^2)$ times the discounted bond of Section 3.10.

(b) Proposition 49(b) holds with $\tilde{p}$, $\tilde{z}$, q and m of (3.194), and for $S < a$

$$Q(S) := \frac{q(S)}{\delta^2} = \sum_{T_i \leq S} v_i + \int_0^S \sigma^2 \beta^2, \quad \tilde{z}(S) = \delta \left[\sum_{T_i \leq S} (S - T_i) v_i + \int_0^S \sigma(s)^2 \Phi(s, S) \beta(s) ds \right]. \quad (3.195)$$

(c) The panel $(Q(S_\ell))_\ell$ is $\mathcal{A}\theta$, with the meeting columns of Proposition 49(c) and the diffusion columns $D_{\cdot, p}$. For any real diffusion columns,

$$\text{rank } \mathcal{A} = N + P \iff T_N \leq S_L, \text{ no gap contains two meetings, and } \text{rank}(\tilde{\Lambda}_{\ell, p})_{\ell \in E, p \leq P} = P, \quad (3.196)$$

and every other conclusion of Proposition 49(c) holds with $\tilde{\Lambda}$ in place of $\lambda_p(G_\ell)$; in particular $P \leq |E|$ for every shape.

(d) On the listed calendar of Proposition 24, of the eight meeting-free gaps exactly two, (11 Dec 2026, 15 Jan 2027] and (11 Jun, 16 Jul 2027], lie between expiries on different futures.

(d1) With $P = 1$, every shape with $D_{1,1} > 0$ identifies the sixteen meeting variances and u_1 ; this holds for (S1) at every $\kappa > 0$ and for (S2), where $D_{1,1} = 14/360$.

(d2) For every shape, the eight-cell partition $\mathcal{P}_8$ of Proposition 49(d) gives a lower-triangular $\tilde{\Lambda}_E$, whose six rows between expiries on the same future have only the diagonal entry $\int_{G_t} \beta_t^2$; $\mathcal{P}_8$ identifies if and only if the eight diagonal entries are nonzero.

(d3) Under (S2), $\beta_t = 1$ except on (9 Dec, 11 Dec 2026] and (9 Jun, 11 Jun 2027] for the expiries ending them, where $\beta = 49/91$, and on (8 Dec, 10 Dec 2027] for the last expiry, where $\beta = 0$. The matrix $\tilde{\Lambda}_E(\mathcal{P}_8)$ is diagonal with entries 14, 28, 28, 28, $35 + w$, $28, 35 + w, 28$, divided by 360, $w = 2(1 - (49/91)^2)$, so $\mathcal{P}_8$ identifies and no identifying partition has more than eight cells. The partition by calendar quarter does not identify, as the pair $v_i = 100$, $u_p = 2500$ against the same except $v = 86, 72, 74$ for the meetings of 15 September, 27 October and 8 December 2027 and $u = 2860$ on the last cell shows; merging the last two quarters identifies; and the partition at every meeting does not identify, with the pair of Proposition 49(d).

(d4) Under (S1), with $\mathcal{P}_8$, the six same-window diagonal entries are positive and the two others both equal $(g(\kappa)/2\kappa)F(\kappa/360)$, where $g(\kappa) = ((1 - e^{-\kappa\delta})/\kappa\delta)^2$ and, with exponents in days,

$$F(x) = e^{-122x} - e^{-364x} - e^{-10x} + e^{-182x}. \quad (3.197)$$

F has exactly one positive zero x^* , and $\mathcal{P}_8$ identifies if and only if $\kappa \neq \kappa^* := 360x^* \in (0.828, 0.864)$.

(d5) With $P = 1$ and any shape of (d1), take instead the twelve expiries listed on 2 January 2026: the four nearest serials, 16 January, 13 February, 10 April and 15 May 2026, and the quarterlies of Proposition 24 through 10 December 2027. Their gaps contain 0, 1, 0, 1, 1, 0, 2, 3, 1, 3, 1, 3 meetings in date order. For strictly positive θ , a linear functional of θ is determined by the panel if and only if it lies in the span of u_1 and the gap sums $\sum_{T_i \in G_t} v_i$, which has dimension 10, so $\text{rank } \mathcal{A} = 10$ of 17. The panel identifies u_1 and, individually, the meetings of 28 January, 18 March and 29 April 2026 and of 27 January and 28 July 2027; it identifies the other eleven only through the four sums over 17 June and 29 July 2026, over 16 September, 28 October and 9 December 2026, over 17 March, 28 April and 9 June 2027, and over 15 September, 27 October and 8 December 2027, and no other combination of them.

The diffusion now loads on each window with the weight β , and a meeting-free gap measures

$$\tilde{\Lambda}_{\ell,p} = \int_{[c_{p-1}, c_p) \cap (0, S_{t-1}]} (\beta_t^2 - \beta_{t-1}^2) + \int_{[c_{p-1}, c_p) \cap G_t} \beta_t^2, \quad (3.198)$$

a combination of the diffusion rates that includes cells wholly before the gap when the two expiries have different windows. Under (S1) a later window has the smaller weight, $\beta_t(s)^2 = g(\kappa)e^{-2\kappa(a_t-s)}$, and that term is negative; under (S2) the cell (9 Dec, 11 Dec 2026) has the positive entry $(2/360)(1 - (49/91)^2)$ in the row of the gap ending 15 January 2027. This is why (d4) turns on the two window changes, whose day counts (61, 182, 5, 91) coincide and give the same F . Part (d5) depends only on which gap holds each meeting. The 2027 meeting dates are tentative, and (d5) is unchanged if any of them moves within its gap; every decision day is a Wednesday, the next business day a Thursday and every expiry a Friday, so using effective dates in place of decision days changes no gap assignment, here or for the 24 expiries of (d1) to (d4). Proposition 24(e) is the special case of the two gaps from 12 June to 11 December 2026. Parts (d1) and (d4) are exact statements about rank, not about practical identification: at $\kappa = 50$, $D_{1,1} \approx 10^{-9}$ days, and above $\kappa^* \approx 0.8527$ the two window-change entries are negative. The identity $\tilde{z}/\delta = \int_0^S Q$ of (3.189) survives exactly when $\int_0^S \sigma(s)^2 [\Phi(s, S)\beta(s) - (S - s)\beta(s)^2] ds = 0$, which $\phi \equiv 1$ satisfies. Proposition 49 is the case $\phi \equiv 1$, and the proof of (3.190) never uses the form of the diffusion columns; (S2) is the loading of [12] §3.1, where no identification from option panels is made; [58] is the statistical counterpart, as in Section 3.43.

As a numerical check, not part of the statement, redo Proposition 49(e) with $P = 1$, $\hat{u} = Q(S_1)/D_{1,1}$ and $\hat{v}_i = Q(S_\ell) - Q(S_{\ell-1}) - \tilde{\Lambda}_{\ell,1}\hat{u}$. Under (S2) the counts of practically identified meetings are those of the table in Section 3.43, since $\beta = 1$ almost everywhere on the calendar; under (S1) they are never smaller and often larger, because the diffusion loads on each window with $\beta < 1$, and less as the window moves out:

(S1), σ (bp per year)	$s_0 = 10, \varepsilon = 0.25$	10, 0.5	15, 0.25	15, 0.5	25, 0.25	25, 0.5
$\kappa = 0.1, 50$	2	0	14	2	16	15
$\kappa = 0.1, 100$	0	0	4	0	16	6
$\kappa = 0.5, 50$	4	0	16	4	16	16
$\kappa = 0.5, 100$	0	0	6	1	16	9
$\kappa = 1, 50$	4	1	16	4	16	16
$\kappa = 1, 100$	2	0	10	2	16	12

Each estimator's first-order bound agrees with the linear-programming optimum over linear estimators exact on the model. In basis points of standard deviation, an error Δv in a meeting variance of reference size s_0^2 is, to first order, $\Delta v/(2s_0)$, and a meeting is practically identified exactly when this is below $s_0/2$. For the first and the sixteenth meetings it is 11.9 and 40.7 basis points under (S2) with $\sigma = 100$, $s_0 = 10$ and $\varepsilon = 0.5$, and 1.3 and 5.6 under (S1) with $\kappa = 1$, $\sigma = 50$, $s_0 = 25$ and $\varepsilon = 0.25$. The condition number of the 24×17 matrix $\mathcal{A}$ with $P = 1$, time in years, is 94 for the parallel diffusion, 109, 103 and 102 under (S1) at $\kappa = 0.1, 0.5$ and 1, and 93 under (S2).

Proof. (a) For fixed t , $x \mapsto \Phi(t, x)$ is absolutely continuous with derivative $\phi(t, x)$ almost everywhere, so $\int_t^T \sigma^2 \phi^2 = \sigma^2 \Phi(t, T)^2/2$, which is (2.3); the jump condition is that of Proposition 49(a). Stochastic Fubini for the bounded measurable integrand $\sigma(v)\phi(v, s)\mathbf{1}\{v \leq s\}$ gives the jointly measurable version and turns the diffusion part of $\int_0^t r + \int_t^T f(t, \cdot)$ into $\int_0^t \sigma \Phi(\cdot, T) dW$, and Fubini with $\partial_x \Phi^2/2 = \phi \Phi$ turns the drift part into $\frac{1}{2} \int_0^t \sigma^2 \Phi(\cdot, T)^2$. (b) Stochastic Fubini gives $\int_a^b \int_0^x \sigma(s)\phi(s, x) dW_s dx = \int_0^b \sigma \tilde{w} dW$, and the rest is the argument of Proposition 49(b), with variance $\int_0^S \sigma^2 \tilde{w}^2$ and, by (a) at $t = T = S$, covariance $-\tilde{z}$; for $S < a$, $\tilde{w} = \delta\beta$ on $[0, S]$. (c) The proof of (3.190) uses only that the meeting columns are prefix indicators and that differencing maps row ℓ to $(\sum_{T_i \in G_\ell} e_i^\top, \tilde{\Lambda}_{\ell, \cdot})$; (3.198) is the definition of D split at $S_{\ell-1}$. (d1) The first gap is meeting-free and $D_{0,1} = 0$. (d2) Row ℓ involves only $(0, S_\ell]$, the k -th meeting-free gap lies in cell k of $\mathcal{R}_8$, and cell $k+1$ starts after it; in a same-window row the first term of (3.198) vanishes. (d3) $\beta < 1$ only where the first meeting after S_ℓ falls inside the window or does not exist, which on the calendar happens at the three stretches listed, each inside the cell of its row's gap; the pairs are kernel vectors. (d4) The window-change entry is $g[\int_c^{S_\ell} e^{-2\kappa(a_\ell-s)} ds - \int_c^{S_{\ell-1}} e^{-2\kappa(a_{\ell-1}-s)} ds]$ with c the start of the cell. With $G(x) = e^{10x}F(x)$ and $H(x) = e^{112x}G'(x) = -112 - 172e^{-60x} + 354e^{-242x}$, $H(0) = 70$, $H \rightarrow -112$ and H' changes sign once, so H has one zero, G rises and then falls to -1 , and F has one positive zero; $F(0.0023) > 0 > F(0.0024)$ gives the bracket. (d5) After differencing, the first row is $(0, D_{1,1})$ with $D_{1,1} > 0$, so u_1 lies in the row space, and subtracting multiples of that row from the others leaves the sums over the nine gaps with meetings; these ten vectors have disjoint supports, so the rank is 10 and v_i lies in the span exactly when T_i is alone in its gap. A functional outside the row space changes along a kernel vector, and a small step along it keeps a strictly positive θ positive. $\square$

Parts (a) and (b) are conditional on the Wiener-integral facts of Proposition 49 and on stochastic Fubini for bounded jointly measurable deterministic integrands with a jointly measurable version; no Brownian motion is constructed. Parts (c) and (d) are unconditional; the value of κ^* , the table, the standard-deviation errors, the linear-programming comparison and the condition numbers are numerical checks. Part (d4) is proved for $\mathcal{R}_8$ only. In (S2) the model has no meeting after 8 December 2027; adding the January 2028

meeting would change only the last coefficient of the quarterly pair and no rank. If the jump at T_i loaded on window ℓ with a known factor $\omega_{\ell,i}$, the meeting columns would no longer be prefix indicators, (3.196) would not apply, and a surface would identify $\omega_{\ell,i}^2 v_i$; that case is not treated.

3.45 Approximation by recurrent loadings on a finite horizon

Propositions 30 and 33 show that loadings satisfying no linear recurrence admit no realization serving every schedule. On a finite horizon a model with recurrent loadings can stand in for such a model, each keeping its own drift. Keep the notation $i(s, T)$ of Section 3.27, one Brownian factor W with the cited calculus of Section 3.25, a horizon $H > 0$, a finite schedule $0 < T_1 < \dots < T_N < H$, and a Borel shape λ on $[0, H]$, not identically zero, with $\Lambda = \sup_{[0, H]} |\lambda| < \infty$; a constant scale is absorbed into λ . Take original loadings $a_0, \dots, a_N$ and approximating loadings $\tilde{a}_0, \dots, \tilde{a}_N$, the only ones that occur on $[0, H]$, and put

$$\varepsilon = \max_{i \leq N} |a_i - \tilde{a}_i|, \quad \bar{a} = \max_{i \leq N} \max(|a_i|, |\tilde{a}_i|). \quad (3.199)$$

Each model has the volatility (3.138) and the differentiated drift generated from its own volatility, $\sigma(s, T) = a_{i(s, T)} \lambda(T - s)$ and $\alpha(s, T) = \sigma(s, T) \int_s^T \sigma(s, u) du$, with $\tilde{\sigma}, \tilde{\alpha}$ likewise from $\tilde{a}$. With a common bounded Borel initial curve f_0 , $F_0 = \sup |f_0|$, a common driver and no curve jumps, the curves are $f(t, T) = f_0(T) + \int_0^t \alpha(s, T) ds + \int_0^t \sigma(s, T) dW_s$ and $\tilde{f}$ likewise, for $0 \leq t \leq T \leq H$.

Proposition 51 (conditional). (a) For $0 \leq s \leq T \leq H$, $|\sigma - \tilde{\sigma}| \leq \varepsilon \Lambda$ and $|\alpha - \tilde{\alpha}| \leq 2\varepsilon \bar{a} \Lambda^2 (T - s)$. For $0 \leq t \leq T \leq H$, with the deterministic $m(t, T) = \int_0^t (\alpha - \tilde{\alpha})(s, T) ds$,

$$E|f(t, T) - \tilde{f}(t, T)|^2 = m(t, T)^2 + \int_0^t (\sigma - \tilde{\sigma})(s, T)^2 ds \leq \varepsilon^2 [\Lambda^2 t + 4\bar{a}^2 \Lambda^4 (tT - t^2/2)^2], \quad (3.200)$$

$$\sup_{0 \leq t \leq T \leq H} E|f(t, T) - \tilde{f}(t, T)|^2 \leq \varepsilon^2 (\Lambda^2 H + \bar{a}^2 \Lambda^4 H^4). \quad (3.201)$$

For $a \equiv \bar{a}$, $\tilde{a} \equiv \bar{a} - \varepsilon$ and $\lambda \equiv \Lambda$, $E|f - \tilde{f}|^2 = \varepsilon^2 [\Lambda^2 t + (2\bar{a} - \varepsilon)^2 \Lambda^4 (tT - t^2/2)^2]$, so (3.200) is attained as $\varepsilon/\bar{a} \rightarrow 0$.

(b) Assume (H1). Each curve $f(t, \cdot)$ has a version measurable in (ω, u) and bounded on $[t, H]$, which defines $P(t, T) = \exp(-\int_t^T f(t, u) du)$ and $\tilde{P}$, and for $0 \leq t \leq T \leq H$

$$E|\log P - \log \tilde{P}|^2 \leq \varepsilon^2 (\Lambda^2 H^3 + \frac{1}{4} \bar{a}^2 \Lambda^4 H^6), \quad E|P - \tilde{P}|^2 \leq 2\sqrt{3} e^{2HF_0 + 5\bar{a}^2 \Lambda^2 H^3} \varepsilon^2 (\Lambda^2 H^3 + \frac{1}{4} \bar{a}^2 \Lambda^4 H^6). \quad (3.202)$$

(c) Assume (H1), with A of size r , and (H2) for the approximating loadings, $\tilde{a}_i = uM^i v$ with M of size p , and take $f_0 = 0$. For every schedule in $(0, H)$, the approximating model with its own drift $\tilde{\alpha}$ is realized by Proposition 31 with $\psi \equiv 1$, with the state (Ξ, Π, Θ) of dimension $m + m(m+1)/2 + m$, $m = pr$, the same for every schedule and every H , of which only Ξ is random; and its realized curve and bond price satisfy (3.201) and (3.202) with $F_0 = 0$ against the original model.

The constants depend only on $\Lambda, \bar{a}, H$ and, for prices, F_0 : not on the number or spacing of the meetings, on the initial curve, which cancels, or on λ beyond Λ . The schedule enters only through the indices compared, since ε and $\bar{a}$ are maxima over $0, \dots, N$. So once bounds on $\sup_i |a_i|$ and $\sup_i |\tilde{a}_i|$ are fixed, the bounds are free of the number and spacing of meetings. For $\tilde{a}_i = uM^i v$ with (u, M, v) minimal, the sequence is bounded if and only if the spectral radius of M is below one, or equal to one with every eigenvalue of

modulus one semisimple (standard linear systems theory); otherwise $\bar{a}$, and with it the factor $e^{5\bar{a}^2\Lambda^2H^3}$, grows with N , and the comparison holds on each schedule with constants that are not schedule-free. The price bound needs $Ee^{-4X} < \infty$ for $X = -\log P$ in the zero-curve model; X is Gaussian with $|EX| \leq \frac{1}{2}\bar{a}^2\Lambda^2H^3$ and $\text{Var } X \leq \bar{a}^2\Lambda^2H^3$, so $Ee^{-4X} \leq e^{10\bar{a}^2\Lambda^2H^3}$, automatically because the volatility is deterministic and bounded; at $\bar{a}\Lambda \approx 0.01$ per square-root year and $H = 3$ years, $5\bar{a}^2\Lambda^2H^3 \approx 0.014$. Part (c) is a comparison conditional on a supplied recurrent approximation. Every $a_0, \dots, a_N$ is matched exactly by a recurrence of order $N+1$, whose dimension grows with N ; that is interpolation, not compression, and nothing here says which loading sequences have accurate recurrent approximations of small order. Only the supremum of pointwise mean-square errors is bounded, not an expected supremum over the curve.

Parts (a) and (b) are the standard stability argument for Gaussian HJM models with deterministic volatility, through the isometry and the locally Lipschitz drift map $\sigma \mapsto \sigma \int \sigma$, and the standard passage from Gaussian log-prices to prices through exponential moments. What is specific is the schedule-free constant in the ordinal-loading parametrization and the pairing with Proposition 31, which realizes the approximating model with its own drift in a dimension independent of the schedule. The nearest precedent, [7] Theorem 4.1, approximates forward-curve dynamics in commodity markets by arbitrage-free models with finite-dimensional state through a Riesz basis, with rates of convergence; its forward prices are martingales, so there is no drift map to control, and it has no schedule. As a numerical check, not part of the statement, on six schedules with 2 to 200 meetings, evenly spaced and random, the largest ratio of the mean-square error to (3.201) is 0.115.

Proof. (a) $\sigma - \tilde{\sigma} = (a_i - \tilde{a}_i)\lambda$, and $\alpha - \tilde{\alpha} = (\sigma - \tilde{\sigma})S + \tilde{\sigma}(S - \tilde{S})$ with $S = \int_s^T \sigma(s, u) du$, $|S| \leq \bar{a}\Lambda(T-s)$ and $|S - \tilde{S}| \leq \varepsilon\Lambda(T-s)$. Since f_0 cancels, $f - \tilde{f} = m + \int_0^t (\sigma - \tilde{\sigma}) dW$, and the isometry and the mean-zero property give the equality in (3.200); the bounds follow, with $tT - t^2/2 \leq H^2/2$. In the example, $\sigma - \tilde{\sigma} \equiv \varepsilon\Lambda$ and $\alpha - \tilde{\alpha} = (2\bar{a} - \varepsilon)\varepsilon\Lambda^2(T-s)$. (b) Fix t and split $[0, t]$ at the meetings into pieces with right ends τ_l ; on each the number n_l of meetings in $(s, t]$ is constant and $i(s, u) = n_l + i(t, u)$ for $u \geq t$, so by (H1) $\int_0^t \sigma(s, u) dW_s = \sum_l a_{n_l+i(t, u)} c e^{A(u-\tau_l)} \zeta_l$ with $\zeta_l = \int_{\text{piece } l} e^{A(\tau_l-s)} b dW_s$. This defines the version, the same ζ_l serve $\tilde{f}$, and by Lemma 32 they are jointly Gaussian. So $X = \int_t^T (f(t, u) - f_0(u)) du$ and $\tilde{X}$ are jointly Gaussian, with $EX = \frac{1}{2} \int_0^t (S(s, T)^2 - S(s, t)^2) ds$ and $\text{Var } X = \int_0^t U^2, U(s) = \int_t^T \sigma(s, u) du$, and $D = X - \tilde{X}$ has $|ED| \leq \frac{1}{2}\varepsilon\bar{a}\Lambda^2H^3$ and $\text{Var } D \leq \varepsilon^2\Lambda^2H^3$, which is the first bound. For the second, $|e^{-X} - e^{-\tilde{X}}| \leq |D| \max(e^{-X}, e^{-\tilde{X}})$, and Cauchy-Schwarz with $ED^4 \leq 3(ED^2)^2$ and $Ee^{-4X} = e^{-4EX+8\text{Var } X}$ gives the constant. (c) Proposition 31 with $\psi \equiv 1$ and loadings $\tilde{a}$ has the volatility $\tilde{\sigma}$ and the drift $\tilde{\alpha}$. Its realized curve is jointly measurable and agrees with $\tilde{f}(t, t+x)$ almost surely for each x , hence, by Fubini, for almost every x almost surely, so the realized bond price is $\tilde{P}$ almost surely; (ii) is (a) and (b). $\square$

Part (a) is conditional on the cited calculus of Section 3.25, parts (b) and (c) in addition on W being a Brownian motion, which gives the Gaussian law of Lemma 32, and part (c) on the inputs of Proposition 31. For a bounded Borel λ that is not quasi-exponential, (3.202) would follow from a jointly measurable version with stochastic Fubini, which is not shown. A realization from a nonzero initial curve is not given, since Proposition 31 starts from the zero curve; (a) and (b) hold for any common bounded initial curve. Stochastic scales, several factors, fixed-maturity curve jumps, the choice of recurrence, approximation rates, minimal state dimension and option-payoff errors are not treated.

3.46 The unified correlated splice

Proposition 44 treats several step factors and blocks without an exponent zero, and Proposition 47 a full polynomial part with one step factor. This section combines them and states the result without a counting

convention. Keep the driver-form setting of Section 3.38: independent drivers B with the cited calculus, a front end of the type of Theorem 11 whose level on I_m has the noise $V_m \cdot dB$, $V_m = \sum_{\ell} s_{\ell,m} H_{\ell}^S$, and the aggregate maturity-step integrand G_m of (3.172). The block is (3.178), with $\mathcal{A}$ invertible and $(\mathcal{A}, \mathcal{C})$ observable, and $Z = (z_0, \zeta)$ in driver form with predictable bounded rows $H^{0,\mu}$ and H^{ζ} ; Assumption 2.1 is assumed for the drift and volatility of the representation (3.161), as in Propositions 43, 44 and 47. Write φ_k for the block's coordinate functions (x^{μ} for $z_{0,\mu}$, the components of $\mathcal{C} e^{\mathcal{A}x}$ for ζ), $\Phi_k = \int_0^x \varphi_k$, H^k for the corresponding rows, $a = (H^k \cdot H^l)$ and R for the residual (3.162). Let

$$\bar{\omega}_k(u) = H^k(u) V_{j(u)}(u)^{\top}, \quad H_{\#}^{0,0} = H^{0,0} + V_{j(u)}, \quad H_{\#}^k = H^k \text{ otherwise}, \quad (3.203)$$

the covariation of each block coordinate with the current front-end noise and the effective level, and let $a_{\#} = (H_{\#}^k \cdot H_{\#}^l)$ and $R_{\#}$ be R with $a_{\#}$ in place of a .

Proposition 52 (conditional). (a) If the curve satisfies Assumption 2.1, then almost surely, for every m and almost every $u < T_m$,

$$H^{0,\mu}(u) G_m(u)^{\top} = 0 \quad (\mu = 1, \dots, d), \quad H^{\zeta}(u) G_m(u)^{\top} = 0; \quad (3.204)$$

no condition is imposed on the level's covariation $H^{0,0} G_m^{\top}$.

(b) Assume (3.204). (i) The jump across T_m of the cross term between front end and block comes from the level only and is the affine function $T \mapsto (H^{0,0} G_m^{\top})(u) (2T - u - T_m)$. (ii) For every $m \geq j(u)$ the non-level part of the cross term on I_m is one function of $x = T - u$,

$$K(u, x) = \sum_{\mu=1}^d \bar{\omega}_{\mu}(u) \frac{\mu+2}{\mu+1} x^{\mu+1} + \mathcal{C} [x e^{\mathcal{A}x} + \mathcal{A}^{-1}(e^{\mathcal{A}x} - I)] \bar{\omega}_{\zeta}(u), \quad (3.205)$$

and the level part is affine in T on each I_m . (iii) For every u and x ,

$$R(u, x) - K(u, x) = R_{\#}(u, x) + (\text{an affine function of } x) : \quad (3.206)$$

on the current interval the front end's noise moves the curve by a constant, exactly as the level does. (iv) The curve satisfies Assumption 2.1 if and only if (3.204) holds and, for $dt \otimes dQ$ -almost every (u, ω) , $x \mapsto R_{\#}(u, x)$ is affine on $[0, \infty)$; the front-end drifts then absorb that affine part, the level's cross terms including the jumps of (i), and $\sigma^S \cdot \int \sigma^S$, as in Proposition 38(a).

(c) If $\mathcal{A}$ is diagonalizable over $\mathbb{C}$, then $R_{\#}$ affine forces, for $dt \otimes dQ$ -almost every (u, ω) ,

$$H^{\zeta} H^{0,\mu \top} = 0 \quad (\mu = 1, \dots, d), \quad H^{\zeta} (H^{0,0} + V_{j(u)})^{\top} = 0, \quad (3.207)$$

and then $R_{\#}$ affine is the exponent-zero condition (3.176) with $a_{\#}$ in place of a and without polynomial-exponential constants, together with the nonzero-exponent condition of the block alone, which involves only $a_{\zeta\zeta}$, b_{ζ} and ζ ; pairs of eigenvalues with $\lambda_i + \lambda_j = 0$ contribute constants to the exponent-zero part, which are affine.

(d) Condition (3.204) alone is not sufficient: for $d = 1$, no exponential part, a block uncorrelated with the front end and $a_{(0,1),(0,1)} > 0$, $R_{\#}$ contains $-\frac{1}{2} a_{(0,1),(0,1)} x^3$, which no drift reaches.

(e) With $d = 0$ and $H^{0,0} = 0$, which changes no curve, (3.204) is (3.174), (b)(iv) is the converse of Proposition 44(a) with $K = \mathcal{C} [x e^{\mathcal{A}x} + \mathcal{A}^{-1}(e^{\mathcal{A}x} - I)] H^{\zeta} V_{j(u)}^{\top}$, and for diagonalizable $\mathcal{A}$ consistency also

requires $H^\zeta V_{j(u)}^\top = 0$: the block noise must be orthogonal to the current front-end noise, not only to the aggregate step. With one step factor, (a) and (b) are Proposition 47(a) and (b) with K and the effective level explicit. In the uncorrelated splice $K = 0$ and $R_\# = R$ up to an affine function. Proposition 45 holds with $a_\#$: the bound (b) is unchanged, since the entries with $\mu, \nu \geq 1$ are the same in $a_\#$; for a polynomial-only block every bounded predictable choice of $H^{0,\mu}$, zero for $\mu > [(d-1)/2]$ and satisfying (3.204), is admissible with suitable drifts, whatever its covariation with the current front-end noise; and for $d \geq 1$ the splice is consistent if and only if (3.204) holds and $(a_\#, b')$ makes the block-alone residual vanish, b' differing from b only in $b_{0,0}$ and $b_{0,1}$.

The step factors enter (3.204) only through G_m , so single step factors may covary with any block direction as long as their maturity steps cancel in the aggregate; two step factors on the same noise with $\Delta_{1,m} = -\Delta_{2,m}$ give $G_m = 0$ and no restriction. The current front-end noise is constrained only through the effective level. If $(\mathcal{A}, \mathcal{C})$ is not observable, the minimal realization of Proposition 44(c) leaves the curve unchanged, and (a) to (c) hold with PH^ζ in place of H^ζ . Relative to the uncorrelated splice, the drift $b_{0,\mu+1}$ shifts by $\frac{\mu+2}{\mu+1}\bar{\omega}_\mu$ for $1 \leq \mu \leq d-1$, and $b_{0,0}$ and b_ζ absorb the constant and the $\mathcal{C}\mathcal{A}^{-1}e^{\mathcal{A}x}\bar{\omega}_\zeta$ part of K ; the term $\mathcal{C}xe^{\mathcal{A}x}\bar{\omega}_\zeta$ cancels, for diagonalizable $\mathcal{A}$, only through (3.207). For a non-diagonalizable $\mathcal{A}$, such as the Nelson–Siegel term $xe^{-\beta x}$, a drift reaches $x^ke^{\lambda x}$, (3.207) is not forced by this argument, and the condition stays “ $R_\#$ affine”. The aggregate integrand and observability are those of Proposition 44, the descending-degree argument and the level exemption those of Proposition 47, the affine absorption that of Proposition 38 and the polynomial bound that of Proposition 45. New are the effective-level identity (3.206), which turns the correlated splice’s residual condition into the block-alone condition for $a_\#$ and makes the carry-over of Proposition 45 exact, and (c), which sharpens Proposition 44’s converse. [34] §3.1.3 decomposes correlated deterministic factors into independent ones, [12] has no level correlated with meeting steps, and [25] and [9] §7 treat exponential-polynomial blocks alone; the drift of a two-factor Gaussian model with a correlated level and an exponential factor, which produces $xe^{-\kappa x}$, is standard in form, and its reading as an effective level is what (3.206) adds.

Proof. Itô’s formula for the representation and Assumption 2.1 at rational maturities, extended by continuity, give for $dt \otimes dQ$ -almost every (u, ω) that $R(u, T-u) - c_m(u, T)$ is affine in T on each $I_m \cap [u, H]$, where c_m is the cross part of $\sigma \cdot \int \sigma$ on I_m ; everything below is at such a point. With $\Sigma^S(u, T) = \sum_\ell (s_{\ell,m}T + \kappa_{\ell,m})H_\ell^S$, $\kappa_{\ell,j(u)} = -s_{\ell,j(u)}u$ and $\kappa_{\ell,m} - \kappa_{\ell,m-1} = -\Delta_{\ell,m}T_m$, the jump is $c_m - c_{m-1} = \sum_k (H^k G_m^\top)[\varphi_k(x)(T - T_m) + \Phi_k(x)]$. (a) Its part with nonzero exponents, $\mathcal{C}e^{\mathcal{A}x}[(x+u-T_m)I + \mathcal{A}^{-1}]H^\zeta G_m^\top$, must vanish, and Lemma 42 gives $H^\zeta G_m^\top = 0$; in the polynomial part the μ -th term has degree $\mu+1$ and leading coefficient $\frac{\mu+2}{\mu+1}H^{0,\mu}G_m^\top$, and descending from $\mu = d$ to $\mu = 1$ gives (3.204), while the level term $(H^{0,0}G_m^\top)(2x+u-T_m)$ is affine. (b) (i) is this computation under (3.204). For (ii), under (3.204) the sums $\sum_\ell s_{\ell,m'}H^k H_\ell^{S^\top}$ equal $\bar{\omega}_k$ and $\sum_\ell \kappa_{\ell,m'}H^k H_\ell^{S^\top} = -u\bar{\omega}_k$ for every later interval and non-level k , so the non-level cross part is $\sum_k \bar{\omega}_k[x\varphi_k + \Phi_k]$, which is (3.205). For (iii), $a_\# - a$ vanishes outside the level row and column, where it is $\bar{\omega}_k$ and $2\bar{\omega}_0 + |V_{j(u)}|^2$, so $R - R_\# = \sum_{k \neq 0} \bar{\omega}_k(\Phi_k + x\varphi_k) + (2\bar{\omega}_0 + |V_{j(u)}|^2)x$. For (iv), $R - K$ is affine on the current interval, which has positive length, and is real-analytic in x , hence affine on $[0, \infty)$; conversely the drifts are chosen interval by interval as in Proposition 38(a). (c) In a complex eigenbasis with eigenvalues $\lambda_i \neq 0$ and $c_i \neq 0$, a term $x^k e^{\lambda_i x}$ with $k \geq 1$ comes only from the pair (μ, ζ_i) , with coefficient $-c_i[a_{\#,(k,i)}/\lambda_i + a_{\#,(k-1,i)}/k]$; descending from $k = d+1$ gives (3.207). (d) The coefficient of x^3 in (3.176) is $-a_{(0,1),(0,1)}/2$, and no drift has degree 3. (e) is substitution, with the top-degree argument of Proposition 45(b), which uses only entries unchanged in $a_\#$. $\square$

The proposition is conditional on the cited calculus through Itô’s formula for the representation, and on Assumption 2.1 for that representation; the statements are algebra at a point, and no process Z , L_m or

C_m is constructed. Not covered: varying exponents, polynomial parts that are not full, front ends with a diffusive ramp, a non-diagonalizable exponential part beyond " $R_\#$ affine", and unbounded coefficients, which Proposition 53 treats.

3.47 The unified splice beyond bounded coefficients

Proposition 52 assumes bounded predictable integrands. Keep its setting on the horizon H and replace boundedness by the class (L):

- (L1) each front-end noise V_m and each block row H^k is predictable with $\int_0^H |\cdot|^2 du < \infty$ almost surely;
- (L2) the block drifts b_k and the drifts of L_m and C_m are progressively measurable with $\int_0^H |\cdot| du < \infty$ almost surely, C_m having no noise.

Condition (L1) is placed on the sums $V_m = \sum_\ell s_{\ell,m} H_\ell^S$, not on their factors: local square integrability is not closed under products, since $s(u) = h(u) = u^{-1/3}$ are square integrable on $(0, 1]$ and sh is not. One locally bounded factor times a square-integrable one suffices; for example, a Heston-type scale $\sqrt{v_u}$ with v continuous, adapted and nonnegative, times a bounded row. The aggregate G_m enters only through the pointwise products $H^k G_m^\top$ and needs no integrability of its own.

Proposition 53 (conditional). (a) Under (L), if the curve satisfies Assumption 2.1, then (3.204) holds, Q -almost surely, for every m and almost every $u < T_m$; nothing is imposed on the level.

(b) Parts (b) to (e) of Proposition 52 hold verbatim under (L), with the quantifier "for $dt \otimes dQ$ -almost every (u, ω) ".

(c) Let Z be of class (L), satisfy (3.204), and have $R_\#(u, \cdot)$ affine for $dt \otimes dQ$ -almost every (u, ω) . Then the front-end drifts of Proposition 52(b)(iv) are locally integrable, and the processes L_m and C_m defined from arbitrary initial values by these drifts and the noises V_m give a curve that satisfies the regularity conditions of Section 2 and Assumption 2.1.

No stopping argument is used, and the quantifiers are those of Proposition 52. After Itô's formula, the proof of Proposition 52 is algebra at a fixed (u, ω) and never uses a bound on the coefficients, and Itô's formula and the stochastic integral are already cited for locally square-integrable integrands and locally integrable drifts ([4] §3.3 and §3.5). Localizing times $\tau_n = \inf\{t : \int_0^t (\sum_m |V_m|^2 + \sum_k |H^k|^2 + \sum_k |b_k| + \text{front-end drifts}) du \geq n\} \wedge H$ would reduce (L) to square integrability in expectation, but they give no pointwise bound, which needs locally bounded coefficients; local boundedness implies local square integrability, not conversely ([4] p. 45). Stopping the parameters is also not stopping the curve: after τ the block $F(T - t, Z_\tau)$ still moves through the drift $-\partial_x F$, and a stopped consistent model must continue with $z(t) = S_{t-\tau} Z_\tau$, where $F(x + h, z) = F(x, S_h z)$ with S_h the binomial shift on z_0 and $e^{\mathcal{A}h}$ on ζ ; under that continuation the curve is frozen. Localization of stochastic integrals is textbook; what the proposition adds is that no bound is needed after Itô's formula, that the drifts of the converse stay integrable, and where the integrability hypothesis belongs. It extends Proposition 52 to Heston-type scales, conditional on their existence.

Proof. For $T \in I_m$, $\sigma(u, T) = V_m(u) + \sum_k \varphi_k(T - u) H^k(u)$ is square integrable in u , a finite sum with bounded continuous multipliers, and $\alpha(\cdot, T)$ is locally integrable: it collects the drifts of L_m and C_m , the terms $b_k \varphi_k(T - u)$ and $-\partial_x F(T - u, Z_u)$, which is continuous in u because Z has continuous paths. So Itô's formula applies to the representation as in Proposition 52. By (L1) the coefficients are finite for almost every u , almost surely, and the rest of the proof of Proposition 52 uses only their values at a fixed (u, ω) ; Fubini restores the stated quantifiers. (c) On each I_m the chosen drifts are the value at T_m and the slope of

$-(R - c_m) + \sigma^S \cdot \int \sigma^S$, finite combinations, with coefficients continuous in (u, T_m) , of the b_k , the products $H^k \cdot H^l$, $\tilde{\omega}_k$, $H^{0,0} \cdot V_{m'}$ and $V_m \cdot V_{m'}$, and the components of Z_u ; $|x \cdot y| \leq (|x|^2 + |y|^2)/2$ makes them locally integrable. The regularity of Section 2 follows from $\sup_{T \leq H} |\sigma(u, T)|^2 \leq C(\sum_m |V_m(u)|^2 + \sum_k |H^k(u)|^2)$ and the same bookkeeping for $\sup_{T \leq H} |\alpha(u, T)|$. $\square$

The proposition is conditional on the cited calculus and on the existence of Z and of the front-end scales with the stated integrability; in particular neither existence nor nonnegativity of the square-root variance processes of [12] eqs. 29 and 30 is shown, and the remark on $\sqrt{v}$ assumes a continuous v . The true-martingale property of the discounted bonds is not addressed, since local integrability gives no exponential moments. Coefficients outside (L) have no stochastic integral in the cited calculus, and no finite-dimensional realization for general stochastic scales is given.

4. Related work

Drift conditions with stochastic discontinuities. [29] Theorem 3.7 gives necessary and sufficient conditions for a risk-neutral measure in a multi-curve HJM model whose forward rates carry a pure-jump term with jumps only at fixed dates and whose bond prices integrate against a measure that may have atoms at those dates; its conditions (iii) and (iv) are the ones specific to that setting. [30] Theorem 3.5 is the single-curve version with an overnight numéraire and separate sets of roll-over dates and expected jump dates; condition (ii) is the drift condition, condition (iv) the jump condition, and Remark 3.9 with Example 3.10 is the no-atoms case that Section 2 adopts. [40] Proposition 6.11 eq. 74 and Corollary 6.12 give the drift condition in affine-semimartingale form with a deterministic càdlàg clock. [33] introduced the atoms-at-predictable-times device, in credit. [30] §3 notes that a risk-neutral measure ensures no asymptotic free lunch with vanishing risk for the large market of bonds of all maturities, citing [18], whose Theorem 3.3 makes that condition equivalent to the existence of an equivalent separating measure and whose Section 6 shows that in a large market it need not give a σ -martingale measure. The conditions of Section 2 are therefore sufficient for absence of arbitrage in that sense; nothing here shows them necessary.

Jumps at fixed dates in short-rate and affine models. [40] §6 (Examples 6.15 and 6.16) has Gaussian jumps at fixed dates in Vasiček-type models; [30] §4.3 and §4.4 treat Hull–White with jumps at scheduled dates (Lemma 4.3, Propositions 4.5 and 4.6), and its Lemma 4.11 and Proposition 4.12 derive the Gaussian forward-measure tilt and a forward-looking caplet price without studying which meeting variances a chosen set of strikes and expiries identifies or what meeting-variance bounds follow when those prices are known only within intervals; [32] a CIR process with jumps at deterministic dates; [13] short-rate models with fixed-time random-size jumps by a PDE approach; [20] affine jump-diffusions with Skellam-distributed scheduled jumps and closed-form characteristic functions; [41] a Gaussian affine model with jumps of the whole state at deterministic times, its Proposition 1 a backward recursion across the jump dates; [51] a target rate that is a pure jump process with intensity elevated on meeting days, so jumps at random rather than deterministic times; [21] §4.2 likewise lets the jump probability of a Poisson–Gaussian model of the Fed funds rate depend on FOMC meeting days (Table 7); and [2] models spikes and hikes in the overnight rate for SOFR caplets. In the scalar Markov construction of [30] Example 3.10, eq. 3.14, the Gaussian variance at each scheduled date is a free deterministic model parameter; after those parameters are fixed, the vector of meeting variances is independent of the current scalar state. None poses the consistency question for a curve family.

Consistency and finite-dimensional realizations. [9] Theorem 4.1 characterizes invariance of a curve manifold under Wiener-driven HJM, Propositions 5.1 to 5.3 show Nelson–Siegel inconsistent with Ho–Lee and Hull–White, and Theorem 6.1 extends the criterion to marked-point-process jumps with an

intensity-form compensator. [24] Proposition 3.2 and Theorem 4.1 show that no nontrivial Itô process is consistent with Nelson–Siegel; spliced onto a scheduled-step front end with independent drivers, a Nelson–Siegel block with fixed exponent can carry only a level factor (Proposition 38). [25] Theorem 3.2 constrains the exponents of a consistent bounded exponential-polynomial family to be constant, and Corollary 3.5 and Theorem 8.2 require an exponent-doubling pair for a nontrivial process; these restrict exponents only and permit polynomial factors, such as $\lambda(x) = xe^{-\kappa x}$, and exponent sets such as $\lambda, 2\lambda, 4\lambda$. [10] gives Lie-algebra conditions for finite-dimensional realizations with a fixed state-to-curve map and autonomous state equations (its Definition 3.1). For a deterministic volatility $\lambda(x)$ in time to maturity its Proposition 5.1 and Corollary 5.1 give a realization if and only if λ is quasi-exponential, and Proposition 6.1 the same for $\gamma(r)\lambda(x)$; Proposition 6.2, with running time as a state, realizes every calendar-fixed shape that is C^1 and nowhere zero, with no scheduled dates. [55] Corollary 8.1 characterizes affine realizations by the quasi-exponential condition, and Corollaries 7.1 and 7.3 treat volatility of constant direction; in these results the scale is a constant or a functional of the curve. [11] lets the volatility depend on a separate Markov diffusion y : for $\sigma = \varphi(r, y)\lambda(x)$ with scalar y , its Proposition 5.3 gives a realization if and only if λ is quasi-exponential when the curve and scale noises are not perfectly correlated, and Propositions 5.1 and 5.2 give the necessary form under orthogonal noise and sufficiency of quasi-exponential shapes. None of these covers a scheduled maturity breakpoint. [53] bounds the number of nontrivial factors for a biexponential-polynomial family by $\min(n_1, \lfloor n_2/2 \rfloor) + 1$ (its first proposition of §4.2, proved here as Lemma 36), states a sufficient condition for that many factors, and proposes extensions of the Svensson family. Its central proposition of §4.2, that sufficient condition, is not established by its proof, whose reduced system carries the wrong sign on the $e^{-2\beta x}$ terms in the diffusion matrix and which treats only diagonal diffusion matrices; as the reduced system states it the proposition is false (Proposition 39(e)), and with correlated factors its conclusion holds whenever $z_{2,2n} \neq 0$ or $n_2 = 2n$, Proposition 39(b) giving the exact condition. [39] Theorems 2.5 and 2.7 and Corollary 2.6 classify the curve shapes that the Svensson family can keep under consistent dynamics with fixed exponents, complex shapes disappearing after a deterministic horizon; it counts no factors, does not address the sufficient condition of [53] §4.2, and has no scheduled breakpoints. [56] gives affine realizations for Lévy-driven HJM, [23] gives necessary and sufficient conditions for invariance of finite-dimensional submanifolds with boundary under HJM-type equations driven by a Wiener process and a compensated Poisson random measure, whose jumps are not at scheduled dates, [31] treats multi-curve HJM, [28] and [27] give existence and positivity for HJM-type SPDEs with Wiener and Poisson noise, and [26] is the monograph of the framework. [7] Theorem 4.1 approximates forward-curve dynamics in commodity markets by arbitrage-free models with finite-dimensional state, with rates of convergence; its forward prices are martingales, so no drift map enters, and there are no scheduled dates (compare Proposition 51). Every jump in this literature is stochastically continuous.

Separable and stochastic volatility. [12] §3.5 takes a one-factor volatility $\sigma(t, T) = \chi(t)\varphi(T)$ with χ stochastic (eq. 20), exponential φ and the accumulated-variance state (eq. 28), obtaining with a Heston variance (eq. 29) the three-state system of eq. 30 and the curve (Appendix A.1); §3.6 repeats this per meeting factor with the indicators of eq. 36, and eqs. 35 to 41 make the drift and the diffusion part of a short-rate meeting jump depend on the Heston variance, with within-factor leverage. Its drift is taken from the HJM result and its bond prices are obtained by integrating the curve, so, given solutions of its variance equations, Assumption 2.1 holds by construction and Assumption 2.2 trivially for every parameter value, and the only cross-meeting consequences are the fixed-parameter ones of Propositions 14, 16 and 26. The paper does not prove that its square-root variance equations have solutions on a given basis, nor that its discounted bonds are true rather than local martingales. [43] gives Laplace-transform uniqueness and transition laws for continuous-state branching processes, and [32] Lemma A.3 the support of a continuous image; they supply tools, not SOFR variance restrictions. [12] footnote 28 attributes the Markov property under separable

volatility to [14], whose Section 1 and Proposition 2.1(iii) assume deterministic bond-price volatilities and stationarity in time to maturity. [30] Example 3.10 shows that the deterministic-volatility Cheyette class, with separable calendar volatility, stays finite-dimensional and time-inhomogeneous Markov under Gaussian scheduled jumps when the jump shape matches the diffusion shape; it has no maturity breakpoints. Proposition 28 extends these calendar-shape results to meeting restarts, stochastic scales and continuous shapes and is not new in substance; it is a coefficient representation, not a realization.

Stochastic-calculus inputs. The cited calculus of Section 5.1 comes from [4]: predictability of adapted left-continuous processes (p. 14), the Itô integral with its isometry and local-martingale property (§3.1 to §3.3), the multivariate Itô formula (§3.5), strong existence and pathwise uniqueness under Lipschitz coefficients (§5.2), convergence of left-endpoint sums (p. 48) and an exponential-martingale criterion under a deterministic quadratic-variation bound (pp. 60–61); the Markov property of solutions with coefficients Borel in time is [57] Theorem 5.2.2. A textbook account of this calculus is [52]. None of these treats the square-root equation (3.62): [1] Corollary 2.19 proves its pathwise uniqueness but not existence, [47] Proposition 1 quotes the transition law with positive mean reversion from [17], and [22] Lemmas 8 and 9 solve its forward equation, zero mean reversion included, without constructing a solution on a given filtered space, so the solutions of (3.62) and (3.54) remain hypotheses. The Laplace-transform uniqueness of Section 3.14 is [43] Theorem 1.1, and [46] gives closed-form solutions of the square-root equation with time-dependent coefficients, on which nothing here depends.

SOFR and meeting-date models. [34] builds a short-rate model with step volatilities at meeting dates (eq. 26), derives the risk-neutral solution with its ramp drift (eq. 28), states informally in §3.1.5 and §5 that piecewise-constant short-rate paths constrain the initial curve, and splices the step block onto a Vasiček block by independence (§3). The remark after its eq. 35 distinguishes diffusion-factor count from full curve-state dimension: factor values frozen at successive meetings remain in the state. [12] continues it with the ramp drift eqs. 18 and 19, the stochastic-volatility finite-dimensional model of §3.5 and §3.6, principal-component dimension reduction (§4.1.1), joint calibration to futures and options (§4.2), and the drift term structure of Appendix C. Its eqs. 35 to 41 permit time-dependent calibration functions, but it does not characterize the meeting-variance vectors attainable as those scales vary or the restrictions that survive every such recalibration. For the deterministic-volatility family of Proposition 13, Proposition 25 characterizes the attainable cone at fixed decay and loadings and its surviving inequalities; no such result is established here for stochastic volatility. Neither [34] nor [12] asks whether its way of attaching the long-end block is forced. With deterministic or predictable bounded volatilities, one or several step and block factors, and fixed exponential-polynomial blocks, Propositions 34, 35, 41, 43 and 44 show that the independence splice of the first and the indicator-times-exponential terms of the second (eq. 36) are the only two constructions, provided a block uncorrelated with the aggregate maturity-step noise is counted as the independence splice; in the uncorrelated splice the front end frees only the level direction that the two families share (Propositions 38, 40 and 45). [37] fits a static piecewise-flat curve between meeting dates to futures, and [35] §5, eqs. (5) to (8), unwinds monthly-average federal funds futures meeting by meeting into the expected rate after each of the next two meetings; (3.105) and Proposition 49(c) unwind option variances across expiry gaps in the same triangular way, for second moments and within a stated model. [45] extends the LIBOR market model to backward-looking in-arrears rates, with the compounded rate of (3.70) (its eq. 5), a volatility that decays to zero over the accrual period (eqs. 13 and 14) and the futures convexity as an integrated drift (§6.1, eq. 23); its rates are diffusions, without scheduled jumps. [3] is a short-rate model with expected jumps at scheduled dates whose sizes have a discrete law controlled by a state (§2.3) and unexpected jumps, its curve computed by pricing rather than assumed as a family. [54] fits

arbitrage-free Nelson–Siegel models to SOFR futures and finds the convexity correction significant only beyond two years. [50] is a statistical model under the physical measure.

Bounds from finitely many prices. [8] bounds option prices and moments of one underlying from finitely many call prices, and [6] gives model-independent bounds from complete call surfaces of one asset at several dates; the panels of Sections 3.15 to 3.19 have several accrual underlyings, discounting and expiry-dependent forward measures, and a variance functional as target, so neither supplies their bounds.

Empirical side of the restrictions. [58] and [59] estimate event variance by specified empirical procedures, [41], [54] and [16] price jump risk, futures convexity and unspanned volatility, [38], [44], [15], [48] and [42] price options on futures and American options around scheduled events, and [12] Table 2 does not publish the inputs needed to reproduce a calibrated meeting-variance matrix; none of them proves or refutes that equal specified SOFR option prices give equal event variances.

5. Outlook

This section states what is not done. Section 5.1 lists what the results rest on; Sections 5.2 to 5.5 take each group of results in turn, with what is known from the literature, what remains open, and the conjectures; Section 5.6 is the empirical programme, which is not started. Assumptions 2.1 and 2.2 are in force throughout.

5.1 What the results rest on

Standing assumptions. The drift condition and the jump martingale condition of Section 2 are taken from [30] Theorem 3.5 as hypotheses. No result proves that they characterize absence of arbitrage in the present setting beyond what that theorem states.

Cited stochastic calculus. Propositions 14, 16, 26 to 31, 33 to 35, 37, 38, 41, 43, 44, 46, 47 and 51 to 53, and Lemma 32, use stochastic integration, Itô’s formula, strong solutions of Lipschitz equations, predictability of continuous adapted processes, the Markov property of solutions, left-endpoint approximation of the Itô integral or an exponential-martingale criterion. These are cited from [4] and [57], and Proposition 46(b) also cites [25] Theorem 3.2 and Corollaries 3.4 and 3.5. They are not proved here. In the formalization each enters as a field of a hypothesis structure, checked against its source. Every such structure has only a degenerate instance, a model of every field on a trivial base, or, where the fields assume Brownian drivers, a vacuous one, accompanied by separate checks that the premises and conclusions are inhabited. Neither kind of instance constructs a nontrivial model of the fields, so these results are conditional on the cited inputs, in the text and in Lean alike. Only the standing-hypothesis structure of Section 2 has a nonzero instance.

Solutions taken as given. Propositions 14, 16 and 26 assume nonnegative solutions of their variance equations, and Proposition 26 a moment bound; the square-root equations are not solved here. Where a parameter process Z of a block appears, as in Propositions 37 to 40 and 45 to 47, its existence with the stated coefficients is not shown; those propositions are statements about any process that has them, or algebra at one parameter point.

Unconditional results. Theorem 1, Lemmas 3 to 7, Propositions 8 to 10, Theorem 11 and Remark 12 use only the standing assumptions. Propositions 15, 17, 19, 20 and 23 to 25, Corollaries 21 and 22 and Lemma 18 are about explicit Gaussian models and ideal payoffs, and Proposition 48 uses only conditional expectations and the uniqueness of laws from Laplace transforms, and Lemmas 36 and 42 and Propositions 39, 40 and 45 are algebra at a parameter point with, in two cases, a statement along right-continuous paths.

Proposition 13 is proved in the text from Wiener integrals of deterministic integrands; its formalization assumes the resulting variance-integral representation (Appendix A). Propositions 49(a), 49(b), 50(a) and 50(b) rest on the same Wiener-integral facts together with stochastic Fubini for deterministic integrands; their formalizations take them as one explicit hypothesis each, whose only instance has no diffusion, and the remaining parts are unconditional.

5.2 The step-plus-ramp family

The step-plus-ramp family (2.5) is a consistent front-end curve for carry and roll-down.

Theorem 11 answers the deterministic-volatility classification: consistency is equivalent to the differentiated HJM drift and the tilted-Gaussian jump conditions, and Proposition 10 gives a nonzero multi-date instance. Condition (ii) of the theorem reduces to "Gaussian jump laws with matching covariances", eq. (3.43), when the level jumps are jointly Gaussian; that form is a natural guess, and Proposition 9(d) shows that its necessity direction fails, since the jump martingale condition does not force joint Gaussianity. Explicit instances of the drift condition are [34] eq. 28, a member of S^+ with deterministic pairwise coefficients $C_{qi}(t) = -\xi_q \xi_i \sum_j \lambda_{qj} \lambda_{ij} (t \wedge x_q \wedge x_i)$ for constant step volatilities and without curve jumps, and [12] eq. 18 with time-dependent parameters; the jump instances in [30] Example 3.10 eq. 3.14, [40] Examples 6.15 and 6.16 and [41] Proposition 1 are all jointly Gaussian. The representation of the forward rate before a scheduled date as the exponentially tilted mean of the jump,

$$f(t, T) = r_t + \frac{E_t[J_n e^{-\tau J_n}]}{E_t[e^{-\tau J_n}]}, \quad \tau = T - T_n, \quad (5.1)$$

is Proposition 48(d): it holds exactly in the model in which the jump at T_n is the only randomness up to the next date, with the conditional law of the jump given $\mathcal{F}_{T_n-}$, which (H) makes equal to the law given $\mathcal{F}_t$. It is implicit in [30] Propositions 4.2 and 4.5 and in [41] Proposition 1 for Gaussian jumps. By Proposition 48(b) the compensating shape is the derivative of the conditional cumulant generating function of the level jump, so ramps linear in T force a Gaussian level jump (Proposition 48(c)), with the tilt on later intervals as the correction (Proposition 9). Jumps in multiples of a tick, as in [20], are excluded by (ii); the difference between a two-point jump law and the Gaussian ramp with the same mean and variance is at most $|\Delta|^4 \tau^3 / 48$ at time to maturity τ past the meeting, for a symmetric jump of size Δ (Proposition 48(c), (3.185)), of order 10^{-8} basis points at one year for Δ of 25 basis points, and convexity effects of this kind are below quote precision ([12] Appendix C; [54]), so the distinction is a matter of classification, not of pricing. With multiplicative stochastic step volatility $\sigma(t, T) = \sum_n s_n \sqrt{v(t)} \mathbf{1}\{T \geq T_n\}$, s_n deterministic, the drift is again a ramp with the realized coefficient $C_{nm}(t) = \int_0^t \langle s_n, s_m \rangle v(s) ds$, so S^+ stays closed with C adapted and of finite variation in place of deterministic. For constant s_n and a predictable v with $\int_0^H v(s) ds < \infty$ almost surely, the local form of this is Proposition 28(c) with $\varphi_j = 1$, $\chi_j = \sqrt{v}$ and $a_{j,k,m} = \sum_{n \leq m} s_n s_j$, conditional on the cited Itô calculus: the slope on I_m is the initial slope plus $|\sum_{n \leq m} s_n|^2 \int_0^t v(s) ds$. Time-dependent step volatilities and the pairwise coefficients C_{nm} themselves, of which only the sums attached to a date are identifiable, are not covered there.

5.3 Cross-meeting restrictions on event variances

The programme behind this paper sought, besides the consistent front-end curve, cross-meeting restrictions on option-implied event variances usable as a tradeable signal: deviations from a restriction forced by consistency would be event-volatility calendar spreads and flies. The results of Sections 3.10 to 3.23 say

that consistency theory does not supply such a signal. Remark 12 rules out a restriction inferred from state dimension alone: scalar-state members of the jointly Gaussian class fill the nonnegative orthant of meeting variances as their parameters vary, so any restriction has to come from a fixed model's parameters and from the payoffs observed. Proposition 13 gives the fixed-parameter Gaussian cone when loadings, decay and constant scales are fixed, and shows why meeting-dependent scale freedom removes every common linear equality; Proposition 14, conditional on its stochastic-solution hypotheses, gives the affine variance map of the stochastic-volatility specification, and Proposition 16 makes its support and affine hull exact at zero variance mean reversion, with affine-hull dimension at most two for the printed constant parameters. Proposition 26 extends the exact support to nonnegative mean reversion and piecewise-constant volatility of volatility: all three factors in the printed time-dependent column are active, while the affine-hull dimension remains the rank of its incompletely specified loading matrix. These are restrictions of a fixed model, not of consistency, and they change under recalibration. Proposition 25 makes exact what survives recalibration of the scales: the attainable set is a polyhedral cone whose facets are the surviving inequalities; with constant ordinal loadings they are the consecutive-meeting ratio bounds set by the largest decay, they survive joint recalibration of scales and bounded decays, and free loadings or unbounded decays leave only nonnegativity.

For deterministic volatility, ramp coefficients equal the risk-neutral conditional covariances of the meeting jumps plus inter-meeting diffusion, so curve convexity and risk-neutral event variance are the same quantity; this is implicit in [34] eqs. 28 and 32, [30] Proposition 4.6 and [41] Proposition 1, and is not stated as an identity in any source. Under multiplicative stochastic volatility it is not an identity: the local slope is the initial convexity plus realized accumulated variance, a finite-variation process (Proposition 28(c), for loadings constant between meetings and conditional on the cited Itô calculus), while expected remaining variance has a martingale part, so the difference is a quantity of its own (the martingale part of expected remaining variance is not proven).

At a fixed observation time, Lemma 7 and $-\log P(t, T) = \int_t^T f(t, u) du$ show that three exact positive bond prices at distinct maturities inside one scheduled interval recover its current affine slope. This does not identify a future short-rate event variance from a listed panel.

Propositions 15 and 17 show in the scalar Gaussian model that a full ideal bond-call surface at one expiry identifies only cumulative variance, that expiries separating successive meetings identify every event variance, and that deterministic price intervals give an exact feasibility test and sharp bounds. Proposition 19 adds ideal compounded futures and European cash calls: equality of a finite panel is a matrix equation in the event variances, full column rank is necessary and sufficient for identification, and two fixed strikes per contract suffice (Lemma 18); its positive pair shows that every later-window quote and surface can leave individual variances unidentified. Corollaries 21 and 22 show that exact discrete compounding and American cash exercise, after the last meeting or across exactly one unrevealed meeting, do not remove such a pair, and Proposition 20 shows why: every contract settled after a date, with American exercise across any number of unrevealed meetings, sees the revealed variances only through their sum, initial futures quotes add the date-weighted sum, and two quotes leave exactly a polytope of earlier variances with sharp coordinate bounds. Proposition 23 shows that futures-style margining removes the early-exercise value, as is known for futures-style options in general, and lets the surfaces identify each unrevealed variance inside the window, but leaves the same polytope of revealed variances. Proposition 24 closes the listed-calendar question for the ideal payoffs: on the calendar of every serial and quarterly expiry the surfaces identify every meeting variance of 2026 and 2027, the limit being precision rather than rank, since a tick-size premium error propagates in proportion to the accumulated standard deviation; with only the nearest serials listed, identification collapses to sums from mid-2026 on. Proposition 49 adds a piecewise-constant

diffusion between meetings and an arbitrary initial curve: the surfaces do not depend on the initial curve, the diffusion is identified from the expiry gaps without a meeting, and on the listed calendar a constant diffusion is identified with every meeting variance, but at 50 to 100 basis points a year its error leaves only the first meetings practically identified at small reference sizes. Proposition 50 keeps these answers for a known maturity shape: exactly for a loading that starts at the next meeting, and for exponential decay at every rate but one, with somewhat better precision. Whether a crossed variance is itself unidentified, what happens with two unrevealed meetings or accrual across a meeting, and whether identification survives listed settlement and exercise rules, an unknown diffusion shape, meeting shocks that are not parallel and noisy prices, are open; a transfer to listed instruments needs the contract family with its accrual state, exercise payoff, discounting and allowed curve and parameter variation specified before the question is posed.

Conjecture 1. For a fixed consistent model and a specified collection of listed SOFR futures and options with their settlement, exercise and discounting rules, equal prices among admissible states and allowed parameter choices imply equal short-rate event-variance vectors.

[34] (remark after eq. 35) notes that full curve-state dimension need not equal diffusion-factor count. [12] reduces the diffusion factors by principal components and calibrates jointly to futures and options without reporting a cross-meeting restriction. [58] and [59] estimate event variance under specified empirical procedures but do not prove the listed-contract implication. An infeasible restriction would reject the model; it would not by itself be an arbitrage certificate.

The listed-market gap. Every identification result of Sections 3.13 to 3.21, 3.24, 3.43 and 3.44 concerns ideal payoffs priced exactly in a stated model. Propositions 24, 49 and 50 take the expiry calendar and tick size of the listed contracts as inputs, but not their settlement, margining, exercise into futures or accrual conventions, nor noisy quotes; only Propositions 49 and 50 allow an unknown initial curve. None of these results is a statement about listed SOFR instruments; the transfer is Conjecture 1, and it is open.

5.4 Stochastic volatility and finite-dimensional closure

With separable factor volatilities $\sigma_j(t, T) = \chi_j(t)\varphi_j(T)$, arbitrary continuous deterministic maturity shapes and meeting-dependent loadings, Proposition 28 shows, conditional on the cited Itô calculus, that the drift condition holds and the current curve is the initial curve plus at most $2d(N + 1)$ coefficients on the shape pairs $(\varphi_j, \varphi_j\Phi_{j,m})$ restarted at each meeting, the drift-generated coefficient being realized accumulated variance of finite variation. In [12] eq. 30 the state Φ has no Brownian term, and its Appendix A.1 shows that its variance state enters neither the curve nor the bond price. For exponential φ , the drift-generated shape gives the exponent pair $(\lambda, 2\lambda)$ that [25] Theorem 8.2 requires for a nontrivial bounded family. Proposition 28 is a representation of the current curve, not a Markov realization. Proposition 29 adds a Lipschitz scale state and obtains a finite-dimensional, time-inhomogeneous Markov realization with running time as a state, an extension of known calendar-maturity realizations; it is not the realization in time to maturity of Conjecture 2. Time homogeneity is untouched, and the separable form $\chi_j(t)\varphi_j(T)$ cannot carry a classification of shapes. With running time as a state every calendar-fixed shape that is C^1 and nowhere zero has a realization in the time-varying sense of [10] Proposition 6.2 (Section 4). A single nowhere-zero shape whose translates agree up to scalar normalization, $\varphi(a + x)/\varphi(a)$ independent of a , is exponential by an elementary calculation, so imposing translation invariance on a shape anchored at the current time decides the answer by the setting, not by a theorem.

The question that remains uses the realizations in time to maturity defined at the start of Section 3.27, which serve every schedule. Without meetings and with a constant scale the answer is known: a realization

exists exactly for quasi-exponential λ_j ([10] Proposition 5.1 and Corollary 5.1, [55] Corollary 8.1). So exponential shapes are not the only admissible ones, already without meetings: $\lambda(x) = xe^{-kx}$ is quasi-exponential and not exponential ([10] Corollary 5.1).

The loadings matter as well. Proposition 30 shows that with the constant shape $\lambda \equiv 1$, which is quasi-exponential, a realization serving every schedule forces the ordinal loadings to satisfy a linear recurrence, and that harmonic cumulative loadings admit none. So quasi-exponential shapes alone do not suffice under this definition, already for an exponential shape; the obstruction comes from the loadings.

Conjecture 2 (quasi-exponential shapes and recurrent loadings). Under this definition, a realization exists for every schedule and every scale exactly when each λ_j is quasi-exponential and each sequence of ordinal loadings satisfies a linear recurrence (3.141).

For one factor, Proposition 31 gives sufficiency for every bounded Lipschitz function of an autonomous Lipschitz scale state, with any correlation, and Propositions 30 and 33 give necessity of the recurrence for every nonzero quasi-exponential shape with a constant scale; so the one-factor constant-scale case of the conjecture's recurrence condition is settled (Proposition 33(c)). Necessity of the recurrence under a stochastic scale, necessity of quasi-exponential shapes when meetings are present, several factors, and whether the requirement that one realization serve schedules with arbitrarily many meetings is the right scope for the SOFR use, are open. On a finite horizon, Proposition 51 gives a quantitative substitute: a supplied recurrent approximation with loading error ε gives a realized model whose forward-curve and bond-price mean-square errors are of order ε^2 , with schedule-free constants when the loadings are bounded; which loading sequences have accurate recurrent approximations of small order is open. Without meetings, a scale driven by a separate scalar Markov diffusion gives the same quasi-exponential answer: [11] Proposition 5.3 proves it under its smoothness and non-degeneracy assumptions when the curve and scale noises are not perfectly correlated, and gives sufficiency when they are.

Proposition 27 disproves the proposed exclusion of every bounded piecewise exponential-polynomial family when the variance state enters the curve. Its three-exponent family closes because the new 4λ coefficient has finite variation, even though the drift-generated 2λ coefficient has a martingale part. It does not classify the exponent sets that can close or settle Conjecture 2. [12] §3.5 and §3.6 is the exponential instance relevant to that conjecture. The Wiener-driven criteria of [10] and [55], the Lévy-driven ones of [56] and the multi-curve ones of [31] have no predictable jump times.

5.5 The splice

Two constructions attach a smooth long-end block to a scheduled-step front end. [34] §3 (eqs. 4 to 13 and 48 to 53) drives the step block and a Vasiček block by independent Brownian motions, so forward rates and bond prices factorize; independence is assumed there. [12] eq. 36 puts the meeting indicator and the exponential decay in the same factor, so bond prices (their eq. 43) carry exponential integrals restarted at each meeting, and their §5.2 (eqs. 84 to 87) shows the piecewise meeting weights can mimic an exponential mean-reversion profile. [30] Example 3.10 gives the matching condition between continuous and jump shapes for a Cheyette block to remain one-dimensional, and [9] Theorem 6.1 condition 78 is the general closure-under-jumps requirement for intensity-form jumps.

Conjecture 3. Beyond the scope of Propositions 34 to 47, 52 and 53, in particular for varying exponents in the correlated splice and for unbounded scales outside the class of Proposition 53, a fixed-size exponential-polynomial block attached to a scheduled-step front end is consistent if and only if either the block's noise, apart from its level, is uncorrelated with the front end's aggregate maturity-step noise wherever the step volatility jumps, or the family includes the indicator-times-exponential terms.

What is proved. In the following scope the statement of Conjecture 3 is proved, the “if” direction by the two constructions and the “only if” direction by the propositions cited: deterministic volatilities (Propositions 34 and 35), predictable bounded scales on the step factor, the block and the correlation (Proposition 41), a block volatility driven by the block’s own state for an observable block (Proposition 43), and several step and block factors (Proposition 44), for quasi-exponential blocks without an exponent zero and, in the correlated splice, for blocks with a full polynomial part, with one step factor (Proposition 47) and with several (Proposition 52), the last also for locally square-integrable noises, which include Heston-type scales, conditional on their existence (Proposition 53). Whether the block’s own restrictions survive the splice is answered for the uncorrelated splice: the front end absorbs nothing of a biexponential-polynomial block’s equation (Proposition 37), or of an exponential-polynomial block’s with positive varying exponents (Proposition 46), and only the affine part of a block with constant or linear terms (Propositions 38, 40 and 45), so that only a level is freed.

Conventions. Three parts of this classification are counting conventions, not results. A block whose noise is orthogonal to the aggregate maturity-step noise of several step factors is counted as the independence splice, although single step factors may be correlated with it (Proposition 44); a block level correlated with the step noise is counted as part of the front end, since constants lie in both families (Proposition 47); and a correlation switched off wherever the step factor has maturity steps is counted as the independence splice locally in time (Proposition 41). A reader who counts any of these separately obtains a further construction, which the cited proposition characterizes, for Proposition 44 only up to the affine condition on the block residual plus the cross term in its part (a); with the conventions there are only the two. Proposition 52 states the correlated splice with several step factors and a full polynomial part without these conventions, naming the covariance and drift restrictions directly.

What is not done. The indicator-times-exponential branch is shown consistent here for deterministic and bounded predictable scales (Propositions 34, 35 and 41, part (c)(ii)). For the Heston-scale model of [12] eqs. 35 to 41 its realizability is taken from that paper and not verified here: that paper proves neither that its square-root variance equations have solutions nor that its discounted bonds are true martingales (Section 4), and making the model rigorous would need sourced existence and martingale inputs, which are not pursued. Also open are correlated splices with varying exponents, polynomial parts with structurally absent low-degree coefficients, front ends with a diffusive ramp compensated by the block, unbounded scales outside the class of Proposition 53, and consistency from every initial time rather than from time zero. The use is splice-zone relative value.

5.6 Empirical programme

Not started. Planned: fit the family of Theorem 11 to SOFR futures and options, report residuals against the step-plus-spline construction, and test Conjecture 1 for a specified contract collection. Confounds to handle: quarter-end and month-end SOFR spikes, which are scheduled-date effects and fit the framework; the FF–SOFR basis; futures margin funding; compounded-rate settlement; transaction costs of a quarter to half a basis point.

Appendix A. Machine verification

The results through Remark 12 have formal statements and proofs in Lean 4 against Mathlib, except Corollary 2. Propositions 13 and 14 have the conditional formalizations described below; their stochastic derivations are not formalized. Proposition 15 has a full formalization for its stated scalar Gaussian model and ideal European bond calls. Proposition 16 is fully formalized, with the standard stochastic calculus results it

uses taken as fields of audited hypothesis structures, and the solutions of its variance equations as hypotheses. Proposition 17 is fully formalized for Proposition 15's ideal prices, deterministic interval feasibility, sharp individual bounds and local inverse stability. Proposition 19 is fully formalized for its continuously compounded futures and ideal European cash calls. Corollary 21 is fully formalized for exact discrete bond-implied compounding and its late American cash exercise problem. Corollary 22 is fully formalized for the exercise window crossing exactly one unrevealed meeting. Proposition 20 is fully formalized for its stated payoffs of the post-window state. Proposition 23 is fully formalized for the futures-style contract on the compounded rate. Proposition 24 is fully formalized on its calendar, with the assumptions (A1) to (A4) entering as the calendar, tick sizes and reference sizes. Proposition 25 is fully formalized on the deterministic cone of Proposition 13. Proposition 26 is fully formalized in the same way, with the predictability result also taken as a field, the solutions of its variance equations as hypotheses, and the given noise and forward-loading integrands assumed in the admissible Itô domain; it covers the conditional source process, its regular conditional laws and the three-factor printed specialization, and existence of the variance-state solutions and calibrated rank three are not proved. Proposition 27 is formalized under the stated two-filtration calculus inputs, from which it derives the stopped strong solution, actual discounted-bond martingales, variance identities and nonzero short-rate jump; existence of the calculus structures on the Brownian basis is not proved. Proposition 28 is formalized under the supplied calculus, except that "semimartingale" for $M_{j,k}$ is paper-only. Proposition 29 is formalized in parts (a) to (e) under the supplied calculus and the audited strong-existence, predictability and time-dependent Markov inputs; the paragraph after its statement is descriptive. Proposition 30 is formalized under the supplied calculus, with only the parts of the realization notion its proof uses. Proposition 31 is formalized under the supplied calculus and predictability input, for any adapted scale state with continuous paths. Lemma 32 and Proposition 33 are formalized under the supplied calculus; the sufficiency half of Proposition 33(c) is formalized in the one-time sense of the rank bounds, the full realization being Proposition 31. Proposition 34 is formalized under the supplied calculus, Proposition 35 under the supplied calculus with controllability and $c \neq 0$ as standing hypotheses, Lemma 36 as a deterministic statement, Propositions 37 and 38 under the supplied calculus, Propositions 39 and 40 as deterministic statements with pathwise parts, Proposition 41 under the supplied calculus with the standing hypotheses of Proposition 35, Lemma 42 and Propositions 43 and 44 under the supplied calculus, Proposition 45 at a parameter point, Proposition 46 under the supplied calculus and, for (b), the cited exponent restrictions, Proposition 47 under the supplied calculus, Proposition 48 without conditions, Proposition 49 with (a) and (b) under an explicit Gaussian-law hypothesis and (c) to (e) without conditions, Proposition 50 with (a) and (b) under the corresponding hypothesis with a shape and (c) and (d) without conditions, Proposition 51 with its deterministic estimates without conditions, (a) under the supplied calculus and (b) and (c) also assuming a Brownian driver, Proposition 52 with Assumption 2.1 as an explicit hypothesis and no hypothesis structure, and Proposition 53 in the same way. The hypothesis structures behind Propositions 27, 29 and 46(b) have vacuous instances, because their fields assume Brownian drivers that the trivial base lacks, with separate checks that their premises and conclusions are inhabited; the other calculus structures have degenerate instances, and HJMScheduled a nonzero one. Neither kind gives a nontrivial model of its fields, so every result resting on them is conditional on the cited inputs. Each formal statement is a proposition in a module under `lean/StandAlone/`, which imports `Mathlib` only and can be read without the proofs, and is proved in a module under `lean/Novel/`. The axiom audit of every Lean proof reports only `propext`, `Classical.choice` and `Quot.sound`; no proof uses `sorry`. Corollary 2 uses Fubini's theorem and the vanishing of the Itô integral of a null integrand and is not formalized.

Statements live in the namespace `StandAlone` and proofs in the namespace `Novel`. For each module `X` in the table, the statement is `X.statement` in `lean/StandAlone/X.lean`, which imports `Mathlib` only

and can be read without the proofs, and its proof is `XProof.x` in `lean/Novel/XProof.lean`, where `x` is `X` with a lower-case first letter. The first module of each row is the main declaration and the others are the targets it builds on. The proof modules `MGF` (Lemma 4) and `MGFUniqueness`, `CondCalculus`, `JumpLawCore` and `JumpLawExample` (Proposition 9) have no separate statement.

Result	Lean modules
Lemma 3	LemmaA
Lemma 4	LemmaB
Lemma 5	StepJumpsVanish
Lemma 6	StepDriftVanish
Theorem 1	Theorem1
Lemma 7	PiecewiseAffineIntegral
Proposition 8	RampDriftIdentified
Proposition 9	JumpLawTiltedGaussian
Proposition 10	D2Instance
Theorem 11	Theorem2a
Remark 12	D3EventVariances
Proposition 13	GaussianMeetingVarianceCone
Proposition 14	StochasticMeetingVariance
Proposition 15	BondOptionMeetingVariances
Proposition 16	ZeroMeanReversionUpstreamBridge, ZeroMeanReversionVarianceSupport
Proposition 17	BondOptionPriceIntervals
Lemma 18	CompoundedFuturesIdentification (within Proposition 19)
Proposition 19	CompoundedFuturesIdentification
Proposition 20	PreWindowVarianceAggregates
Corollary 21	LateAmericanExercise
Corollary 22	CrossMeetingAmerican
Proposition 23	FuturesStyleExercise
Proposition 24	ListedSr3Identification
Proposition 25	SurvivingScaleRestrictions
Proposition 26	PositiveMeanReversionSupport
Proposition 27	BoundedVarianceCompletion, BoundedVarianceBond, BoundedVarianceExistence, BoundedVarianceIndependence, BoundedVarianceIntegralComparison, BoundedVarianceIto, BoundedVarianceMartingale, BoundedVarianceState, BoundedVarianceUniqueness
Proposition 28	SeparableMeetingAssembly, SeparableMeetingShapes, SeparableMeetingCoefficients, SeparableMeetingIntegrals, SeparableMeetingRepresentation
Proposition 29	SeparableMeetingMarkovSeveral, SeparableMeetingScaleVersion, SeparableMeetingMarkovCoefficients, SeparableMeetingMarkovSolution, SeparableMeetingMarkovRealization, SeparableMeetingMarkovProperty
Proposition 30	MeetingLoadingObstruction, MeetingLoadingHankel, MeetingLoadingDimension, MeetingLoadingLaw, MeetingLoadingRank, MeetingLoadingCurve
Proposition 31	ExternalScaleAssembly, ExternalScaleDrift, ExternalScaleStateP, ExternalScaleStateQ, ExternalScaleConditions, RecurrentLoadingAssembly, RecurrentLoadingAlgebra, RecurrentLoadingDiffusion, RecurrentLoadingDrift, RecurrentLoadingRestartP, RecurrentLoadingRestartQ, RecurrentLoadingStateP, RecurrentLoadingStateQ, RecurrentLoadingXi, RecurrentLoadingXiEquation
Proposition 33	RecurrentLoadingCharacterization, RecurrenceNecessityAssembly, RecurrenceNecessityAlgebra, RecurrenceNecessityReduction, RecurrenceNecessityCurve, RecurrenceNecessityRank
Lemma 32	RecurrenceNecessityGaussian

Result	Lean modules
Proposition 34	SpliceCrossTermAssembly, SpliceCrossTermDrift, SpliceCrossTermCurve, SpliceCrossTermAlpha, SpliceCrossTermAnalytic, SpliceCrossTermNecessity, SpliceCrossTermOpen, SpliceCrossTermUncorrelated, SpliceCrossTermConsistency, SpliceCrossTermSufficiency
Proposition 35	SpliceQuasiExponentialAssembly, SpliceQuasiExponentialCross, SpliceQuasiExponentialCurve, SpliceQuasiExponentialBlock, SpliceQuasiExponentialAlgebra, SpliceQuasiExponentialKey, SpliceQuasiExponentialConsistency
Lemma 36	SharefFilipovicMaxFactors
Proposition 37	SharefFilipovicAssembly, SharefFilipovicIndependence, SharefFilipovicResidual, SharefFilipovicSplit, SharefFilipovicIto, SharefFilipovicPartA, SharefFilipovicPartB
Proposition 38	SpliceAffineOverlapAssembly, SpliceAffineOverlapAbsorb, SpliceAffineOverlapNS, AnalyticPrimitive
Proposition 39	CorrelatedFactorsAssembly, CorrelatedFactorsReduction, NonnegPolySOS, CorrelatedFactorsGram, CorrelatedFactorsExists, CorrelatedFactorsHankel, CorrelatedFactorsAlways, CorrelatedFactorsSign, CorrelatedFactorsSmall, CorrelatedFactorsProcess
Proposition 40	SvenssonSplice
Proposition 41	SpliceRandomScalesAssembly, SpliceRandomScalesPath, SpliceRandomScalesCurve, SpliceRandomScalesNecessity, SpliceRandomScalesSufficiency
Lemma 42	SpliceStateBlockObservability
Proposition 43	SpliceStateBlockAssembly, SpliceStateBlockCross, SpliceStateBlockAX01
Proposition 44	SpliceSeveralFactorsAssembly, SpliceSeveralFactorsCore, SpliceSeveralFactorsAX01, SpliceSeveralFactorsSufficiency, SpliceSeveralFactorsBlocks, SpliceSeveralFactorsMinimal
Proposition 45	SplicePolynomialParts
Proposition 46	SpliceVaryingExponentsAssembly, SpliceVaryingExponents, SpliceVaryingExponentsRestrictions
Proposition 47	SpliceExponentZeroAssembly, SpliceExponentZeroQuasiPoly, SpliceExponentZeroCross, SpliceExponentZeroAX01, SpliceExponentZeroResidual
Proposition 48	JumpShapeAssembly, JumpShapeCond, JumpShapeCondB, JumpShapeKernel, JumpShapeProfile, JumpShapeAffine, JumpShapeOrigin, JumpShapeExist, JumpShapeTwoPoint, JumpShapeForward, JumpShapeGeneral, JumpShapeLater
Proposition 49	DiffusionMeetingAssembly, DiffusionMeetingGauss, DiffusionMeetingConsistency, DiffusionMeetingPricing, DiffusionMeetingIdentities, DiffusionMeetingFutures, DiffusionMeetingRank, DiffusionMeetingRankValue, DiffusionMeetingInversion, DiffusionMeetingCalendar, DiffusionMeetingPrecision, DiffusionMeetingInstance
Proposition 50	MaturityShapeAssembly, MaturityShapeGauss, MaturityShapeConsistency, MaturityShapeIdentities, MaturityShapePricing, MaturityShapeRank, MaturityShapeCalendar, MaturityShapeS1, MaturityShapeS2, MaturityShapeListing, MaturityShapeInstance
Proposition 51	RecurrentApproxAssembly, RecurrentApproxEstimates, RecurrentApproxMeanSquare, RecurrentApproxPriceBounds, RecurrentApproxBond, RecurrentApproxRealization
Proposition 52	UnifiedSpliceAssembly, UnifiedSpliceAlgebra, UnifiedSpliceNecessity, UnifiedSpliceConverse, UnifiedSpliceStep0, UnifiedSpliceQuantifiers, UnifiedSpliceConverseAX01, UnifiedSpliceExpPoly, UnifiedSpliceDiag, UnifiedSpliceSpecial
Proposition 53	SpliceLocalizationAssembly, SpliceLocalization, SpliceLocalizationConverse, SpliceLocalizationRegularity

The declarations for Propositions 13 and 14 are conditional formalizations, described below.

The hypothesis structures imported by proof modules, with the results that use them:

- `lean/Upstream/HJMScheduled.lean`: Theorem 1, Proposition 10, Theorem 11 and Remark 12
- `lean/Upstream/ItoCalculus.lean`: Propositions 16 and 26 to 29
- `lean/Upstream/Predictability.lean`: Propositions 26, 27 and 29
- `lean/Upstream/LipschitzSDE.lean`: Propositions 27 and 29, and Proposition 46 (Brownian premise only)
- `lean/Upstream/LipschitzMarkov.lean`: Proposition 29, for its test-function predicate only
- `lean/Upstream/LipschitzMarkovTime.lean`: Proposition 29
- `lean/Upstream/ExpPolyConsistency.lean`: Proposition 46
- `lean/Upstream/ExponentialMartingale.lean`: Proposition 27
- `lean/Upstream/IntegralApproximation.lean`: Proposition 27

Conventions of the formal statements. The dates are a function $\tau : \mathbb{N} \rightarrow \mathbb{R}$ with $\tau(0) = 0$, strictly increasing on $\{0, \dots, N\}$, with $I_k = [\tau(k), \tau(k+1))$ and $I_N = [\tau(N), \infty)$, so no extended real enters; a step or piecewise-affine function is any function with the prescribed values on each I_k ; $\mathbb{R}^d$ is `EuclideanSpace  $\mathbb{R}$  (Fin d)`; integrals are Mathlib’s interval integrals against Lebesgue measure; conditional expectations are Mathlib’s, which vanish on non-integrable functions, so a conditional-expectation identity forces integrability. From Section 3.25 on, the time interval I_k is often represented by $(T_k, T_{k+1}]$, which changes no integral, and the last date by the horizon.

Theorem 1 to Remark 12. Lemmas 3 to 7 are stated as in the text, with their parts bundled; the proofs of Lemmas 5 and 6 avoid the constants c_k, A_k, C_k . Theorem 1 restates the regularity conditions and Assumptions 2.1 and 2.2 as the fields of `Upstream.HJMScheduled`, with an abstract sub-sigma-algebra in place of $\mathcal{F}_{T_n-}$; a second declaration, `theorem1_of_HJMScheduled`, applies it to an instance of the structure. That structure has a nonzero deterministic instance, `HJMScheduled.nonvacuityInstance`, outside the step family. Corollary 2 uses Fubini’s theorem and the vanishing of the Itô integral of a null integrand and is not formalized. Proposition 8 asserts the cycle $(A) \Rightarrow (A') \Rightarrow (B) \Rightarrow (A)$ with $(B) \Leftrightarrow (B')$; the computation on the parametrization (2.5) after it is not formalized. Proposition 9 encodes the $[0, \infty]$ -valued conditional identities by integrals over the sets of $\mathcal{G}$ and uses no regular conditional distributions; (3.28) is proved through a one-sided uniqueness theorem for moment generating functions, `MGFUniqueness.map_eq_of_mgf_eqOn_Ico`. Proposition 10 verifies every regularity field and both assumptions for the product of Gaussian coordinate laws; the reconstruction of the curve on a Brownian extension at the end of Section 3.7 is not formalized. Theorem 11 formalizes the three equivalences, and a second declaration applies necessity to the standing-hypothesis structure; the coefficient calculation after it is not formalized. Remark 12 constructs the standing-hypothesis instance on the product Borel measure with the uncompleted coordinate filtration; completion, the path-space Markov assertion, the bank-account state assertion, the option factorization and the multivariate Gaussian packaging are proved in the text only.

Propositions 13 to 15. Proposition 13 is formalized after assuming its variance-integral representation; the Brownian construction, the maturity limits, the conditional-variance derivation and the Gaussian law are not formalized. Proposition 14 takes its stochastic ingredients (the centered representation, conditional isometry, cross-factor orthogonality and conditional first-moment identity) as explicit hypotheses and derives Mathlib’s conditional variance formula and the deterministic parts; it constructs no Brownian motion

or solution. Proposition 15 is fully formalized on the finite product Gaussian space of Remark 12 with the actual bank account and bonds, with no pricing premise.

Proposition 16. The first layer proves the distributional consequences from explicit premises: the transition law, product kernel, Laplace uniqueness, image support, affine hull, covariance, rank and atom. The second layer discharges those premises from Upstream.ItoCalculus, for an adapted nonnegative solution of (3.62) with almost surely continuous paths, taken as a hypothesis; it derives the Itô representation, localization, conditional transforms, isometry and orthogonality, the centered representation and the regular conditional law, and identifies the meeting jumps with the actual short-rate jumps. Not covered: existence of the solutions, the second-moment bound of Section 3.12, and anything about calibration or listed options.

Propositions 17, 19, 20 and 23 to 25 and Corollaries 21 and 22. Each is formalized in its stated ideal model on the finite Gaussian product space with its completed reveal filtration, from the actual accrual, discounting and exercise definitions, with no pricing, measurability, stopping-time, optimizer or Gaussian-law premise; Lemma 18 is part of the declaration of Proposition 19, and Proposition 23 reuses those of Propositions 19 and 20 and Corollary 21. American values are suprema over all stopping times of the completed filtration valued in the exercise window. Corollary 22 is stated with the number of earlier meetings as its index, and on the single adapted version (3.92). In Proposition 20 the inclusion of the cash calls of Corollaries 21 and 22 in (c) and (d) is argued in the text only. Proposition 24 checks the calendar by decision, as day numbers from 1 January 2026 with actual-over-360 accrual, and takes (A1) to (A4) as the calendar, tick sizes and reference sizes; the forward is taken as one in the exact evaluation. Proposition 25 reuses Proposition 13's matrix and Proposition 16's closedness lemma; the sentence of (d) on stochastic volatility claims nothing and is not formalized. Listed settlement, margin, exercise into futures, unknown initial curves, noisy prices, calibration and arbitrage conclusions are outside every one of these statements.

Proposition 26. Its inputs are Upstream.ItoCalculus, Upstream.Predictability, the driver covariations, nonnegative adapted solutions of (3.54) with almost surely continuous paths, initial state one, the supremum-square moment bound and admissibility of the original integrands. It derives the transforms, laws, conditional independence and the exact support, affine hull, mean, covariance, rank and vertex mass. It does not construct the solutions, prove rank three for the incomplete calibration, or cover time-dependent mean reversion.

Proposition 27. Its inputs are the two calculus structures on one Brownian basis with their filtrations and covariations and the ItoCalculus, LipschitzSDE, Predictability, ExponentialMartingale and IntegralApproximation interfaces. It derives the stopped strong solution and its uniqueness, the identification of the integrals between filtrations, the variance identities, the true martingale property of the discounted bonds and the nonzero short-rate jump. The LipschitzSDE instance is vacuous, and no instance constructs the two structures on a Brownian basis.

Proposition 28. The drivers are all coordinates of an ItoCalculus instance, and the shapes are continuous on $\mathbb{R}$. The statement proves the coefficients (3.125) on each interval, (2.3) on every path, (3.128) for each fixed maturity on one event for all times without a stochastic Fubini theorem, the start and freeze properties, (3.129) and the regularity of $A_{j,k}$ and $V_{j,m}$. The word "semimartingale" for $M_{j,k}$ is paper-only: the local-martingale property of the integral of an integrand with finite path energy is not a field of the calculus.

Proposition 29. Its inputs are ItoCalculus with the Brownian premise and solution class of LipschitzSDE, Predictability, and LipschitzMarkovTime, whose field is the Markov property at deterministic times for one solution; the instances of LipschitzSDE and LipschitzMarkovTime are vacuous, with separate checks. The several-time property is proved for any adapted process with the

one-time property, by products and a π - λ argument, since Mathlib's conditional-independence lemmas need a standard Borel space. The paragraph after the statement has no formal counterpart.

Propositions 30 to 33. Only what the proofs use of a realization is assumed: the state at one time in a set $Z \subseteq \mathbb{R}^q$, not necessarily measurable, and the curve map at finitely many maturities, locally Lipschitz on Z . The driver is a Brownian motion of Mathlib with its Gaussian law and independent increments, and the calculus enters through the standalone restatement of its fields. The dimension step uses the Hausdorff dimension of a Lipschitz image. Proposition 31 is stated for any adapted scale state with continuous paths, with continuous-adapted predictability as a hypothesis, and (3.146) in the version form of (3.139); its constant-scale case is also formalized separately (`RecurrentLoadingAssembly`), with the characterization for $\lambda \equiv 1$ stated in Proposition 33(c) (`RecurrentLoadingCharacterization`). Lemma 32 is proved for bounded Borel integrands continuous almost everywhere. In Proposition 33 the reduction uses the Krylov matrix and the companion matrix of the characteristic polynomial, and the sufficiency half of (c) is stated in the one-time sense of the rank bounds.

Propositions 34, 35, 41, 43 and 44. Consistency is in the version form: a random element with measurable evaluations agreeing with the curve almost surely at every maturity. The proofs are pathwise in the drift, through the linear independence of polynomials times exponentials with distinct exponents and analytic continuation across the meetings. Propositions 35 and 41 take controllability, $c \neq 0$ and invertibility of A as standing hypotheses; the reduction to that case is not formalized. Propositions 43 and 44 are stated on one path, with Assumption 2.1 for almost every time and every maturity; Proposition 44(b) is proved with the Cayley–Hamilton theorem and a Bézout identity. Not formalized, as remarks: that the enlarged family of Proposition 34(c)(ii) does not depend on t , the sentences on what an enlargement absorbing the jump must contain, and, for Proposition 35, the open set of times.

Lemma 36 and Propositions 37 to 40 and 45 to 47. Lemma 36 is the source's proposition proved as a lemma of this paper, with no hypothesis structure. Propositions 37, 38 and 46 take Assumption 2.1 for the drift that Itô's formula gives for the representation (3.161). Proposition 38 uses the lemma that the primitive of a real-analytic function is real-analytic, and Proposition 39 the lemma that nonnegative polynomials in one variable are sums of squares, both proved here since Mathlib lacks them. Propositions 39, 40 and 45 are statements at a parameter point, with the pathwise parts of 39(e) and 40(a) for paths right-continuous at the point. Proposition 46(b) restates `Upstream.ExpPolyConsistency`, whose fields are the cited conclusions of [25] Theorem 3.2 and Corollaries 3.4 and 3.5, under the Brownian premise, an adapted driver-form Itô process with $\mathcal{F}_0$ -measurable initial value, boundedness of the family and the consistency equation almost everywhere; its instance is vacuous, with separate checks. Proposition 47 is stated on one path.

Proposition 48. The formal statement takes a regular conditional distribution of X given $\mathcal{G}$ as a hypothesis with its defining property, states (H) through set integrals over $\mathcal{G}$, so that no integrability is assumed, and allows $L = \infty$. It proves (a) with one $\mathcal{G}$ -measurable null set, (b) with the derivatives of Mathlib's cumulant generating function at $-\tau$, the uniqueness of the law through the moment-generating-function uniqueness lemma used for Proposition 9, and existence with joint measurability through parametric kernel integrals; (c)(i) and (c)(ii) with (3.184) and (3.185); (d) with its equivalence, the agreement of the conditional laws, the case of a right-continuous curve and the bond formula, and the general-curve form with its equivalence; and (e). The numerical values of the gap in the discussion after the proposition are not formalized; the bound (3.185) is.

Proposition 49. The probabilistic parts of (a) and (b) are proved under an explicit hypothesis, not a field of the calculus structures: the Wiener integrals of deterministic integrands are linear, jointly Gaussian with the stated covariances and independent of the jumps, Y is continuous and satisfies the stochastic Fubini identity, and the filtration has the stated measurability and independence properties. The formal

model allows any bounded measurable σ^2 and initial curve. The hypothesis is shown satisfiable, with nondegenerate Gaussian jumps, only by the model of Section 3.10, which has no diffusion; no Brownian motion is constructed, so with $\sigma \neq 0$ the parts (a) and (b) are conditional. The deterministic identities of (b), including (3.189), the rank criterion (3.190) with the inversion formulas and the strictly positive pairs, the calendar facts of (d) with both non-identifying pairs and the rank of the stacked matrices, and the error formula (3.191) with the sharp bound (3.192) are proved without conditions. The table of practically identified meetings, the linear-programming comparison and the size 1.9×10^{-9} of the z -difference are not formalized.

Proposition 50. As for Proposition 49, the probabilistic parts of (a) and (b) are proved under an explicit hypothesis, the Gaussian law of Proposition 49 with the process $\int_0^t \sigma(s) \phi(s, x) dW_s$ in place of Y : it is the Wiener integral of $\mathbf{1}_{[0,t]} \phi(\cdot, x)$, integrable along the diagonal and in x , and satisfies stochastic Fubini in the two forms the proof uses. It is shown satisfiable, for every bounded Borel shape, only by the model of Section 3.10 without diffusion; no Brownian motion is constructed. The formal model allows any bounded measurable σ^2 and initial curve and any bounded Borel shape. The deterministic identities of (a) and (b), including (3.195), the criterion (3.196) with its consequences, and (d1) to (d5) are proved without conditions: (d2) for every bounded Borel shape, (d3) in exact arithmetic with both pairs, which prove non-identification without stating the ranks, (d4) with the uniqueness of the zero of (3.197) and its bracket through exact Taylor bounds on the exponential, and (d5) with its gap counts, the span and rank for every diffusion column with $D_{1,1} \neq 0$, the identified meetings, and, in day numbers, the weekdays and the unchanged gap assignments under effective dates. That the business day after each decision is not a holiday is a check, since the holiday calendar is not modelled. The condition for $\tilde{z}/\delta = \int_0^S Q$, the decimal value of κ^* , the table, the standard-deviation errors, the linear-programming comparison and the condition numbers are not formalized.

Proposition 51. The deterministic estimates are kept apart from the probability and proved without conditions: part (a)'s bounds on $\sigma - \tilde{\sigma}$ and $\alpha - \tilde{\alpha}$, the bounds on m and $\int (\sigma - \tilde{\sigma})^2$, the bounds on the mean and variance of the log-prices, the Gaussian fourth moment, which is derived from the moment-generating function, and the price inequality for jointly Gaussian log-prices. Part (a)'s mean-square identity and (3.200) and (3.201) assume a standard driver of the supplied calculus, for any bounded Borel initial curve. Parts (b) and (c) assume in addition that the driver is a Brownian motion, the hypothesis of Lemma 32 that gives the Gaussian laws; no model with such a driver is constructed. The version of (b) is constructed, with $[0, t]$ split by the number of meetings in $(s, t]$, and stochastic Fubini for that split kernel is proved from finite linearity, so (3.202) holds with the stated constants. Part (c) identifies the approximating model with the realization of Proposition 31 at $\psi \equiv 1$, whose volatility and drift are $\tilde{\sigma}$ and $\tilde{\alpha}$ by definition, and proves by Tonelli's theorem that the realized bond price equals $\tilde{P}$ almost surely; the state dimension is that of Proposition 31. The exact mean-square error in the example of (a) is formalized. Not formalized: the characterization of bounded recurrences, the growth of the constants for growing recurrences, the remark on shapes that are not quasi-exponential, and the numerical check.

Proposition 52. The result is pathwise algebra together with the passage from Assumption 2.1, which enters as an explicit hypothesis in its integrated form; no hypothesis structure is used. Because the block is linear in its state, the drift and volatility of the representation are written out, as the product rule gives them, rather than derived from the calculus structure, and the existence of Z , L_m and C_m is not constructed. Formalized: the identity (3.206), the general jump and (3.205); (a) with the level jump, through Lemma 42 and the descending-degree argument; (b)(iv) at a point and, through the rational maturities, continuity and Fubini, with the proposition's quantifiers; the converse back to Assumption 2.1 with explicit front-end drifts, whose measurability as processes is not addressed and is not needed by the formal assumption;

(3.207) for $\mathcal{A}$ diagonalizable over $\mathbb{C}$, through a proved linear independence of exponential polynomials, under every resonance; the counterexample (d); the aggregation example and the reduction given a minimal realization; and in (e), the sharpening of Proposition 44, $K = 0$ in the uncorrelated splice and the entries of $a_\#$. Not formalized: the split in (c) of " $R_\#$ affine" into the two conditions; the existence of the minimal realization and the block-by-block form of Proposition 44(b) and (c); the carry-over of Proposition 45 in (e) beyond the entries of $a_\#$, including the drift shifts; the remark on non-diagonalizable $\mathcal{A}$; and the symbolic check.

Proposition 53. As for Proposition 52, Assumption 2.1 is an explicit hypothesis and no hypothesis structure is used. The formal statements of Proposition 52, including the passage from Assumption 2.1 and the converse back to it, put no bound, measurability or integrability hypothesis on the coefficients at any (u, ω) , so (a) and (b) hold verbatim for the class (L), with the same quantifiers; the class appears in Lean only in (c). There the drifts of the converse are written out, with the front end's primitive proved by induction over the meetings, Assumption 2.1 holds with them, and they are integrable on $[0, H]$ on every path whose noises V_m and block rows are square integrable and whose block drifts and state are integrable. Maturity-uniform envelopes for σ and α , square integrable and integrable on $[0, H]$, give the integrability part of the regularity of Section 2. The product counterexample, the stability under a bounded factor, the integrability of products of square-integrable rows and the shift invariance with the frozen curve are formalized. Not formalized: the construction of L_m and C_m as stochastic integrals in (c), whose converse is proved as Assumption 2.1 at each (u, ω) with the explicit drifts; progressive measurability in (t, T) ; the localizing times; the Heston-type remark; and the scripted checks.

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